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Road Design Manual

Volume 6: Traffic Control Facilities & Communication Systems Design

Part 2: Traffic Signs

Chief Engineer (Roads)

State Department for Roads
Ministry of Roads and Transport
Transcom House, Ngong Road
P.O. Box 30260 – 00100
NAIROBI

Principal Secretary

State Department for Roads
Ministry of Roads and Transport
Works Building
Ngong Road
P.O. Box 30260 – 00100
NAIROBI

Foreword

This manual was developed by the Ministry pursuant to The Fourth Schedule of the Constitution which assigns to the National Government the functions and powers of setting standards for the construction and maintenance of all public roads including those under the County Governments.

It is part of a series of manuals that replace the first generation of road manuals developed in the first and second decades after independence. This second generation of the road manuals were developed to cover the entire road project cycle covering planning, appraisal, design, contracts, construction, maintenance, operations and monitoring. The series incorporates best practices, climate change considerations, and recent technologies to enable the provision of road infrastructure that is safe, secure, and efficient.

Under the Kenya Vision 2030 long term plan, infrastructure expansion and modernisation are some of the foundations for the realisation of economic, social and political transformation of Kenya into a rapidly industrialising middle-income country. The plan envisages an integrated, safe and efficient transport and communication infrastructure network consisting of roads, railways, ports, airports, waterways, and telecommunications infrastructure.

The strategies to be pursued under the Vision 2030 plan to improve infrastructure services and to maximise the economic and social impacts of infrastructure development and management include: Strengthening of the institutional framework for infrastructure development and maintenance; Raising efficiency and quality of infrastructure projects; Enhancing local content of identified infrastructure projects to minimise import content; Benchmarking infrastructure facilities and services provision with globally acceptable performance standards; and, Implementing infrastructure projects that will stimulate demand in hitherto marginalised areas.

The first three 5-year Medium Term Plans (MTP) under the Vision 2030 from 2008 to 2022 targeted construction of 1,950 km, 5,500 km and 10,000 km of new paved roads under MTP I, II and III, respectively, totalling 17,450 km. This was a massive infrastructure development program intended to double the paved road network in 10 years compared to 8,600 km developed from independence in 1963 to 2008.

Implementation of MTP I to III resulted in the construction of 14,000 km of paved roads, which extended the paved road coverage to Arid and Semi-Arid regions, that had been previously neglected. However, some key milestones of the Vision 2030 goals have not been realised. This has been due to internal and external challenges. External challenges included: climate change – prolonged droughts; the emergence of COVID-19 pandemic; global supply chain disruptions; exchange rate volatility; and rising interest rates in the leading economies.

The internal challenges included: inadequate road maintenance equipment; pavement overloading by heavy goods vehicles; huge maintenance backlog of the road network; low contracting and supervision capacity particularly in the Counties; poor quality control and assurance of works; congestion in urban areas; encroachment on road reserves; high costs and delays in payments of land acquisition; lack of harmonisation of cross-border transport regulation and operational procedures; rapid urbanisation; increased traffic volume with exponential growth of motorcycle traffic; high cost/delays in relocation of utilities and services along and across road reserves; inadequate funding of projects and programs; and, delay or default in payments for goods, services and works.

The implementation of MTP III came to an end on 30th June 2023, ushering in the implementation of the Fourth Medium Term Plan (MTP IV), which has been aligned to the aspirations of the Kenya Vision 2030 and the Kenya Kwanza Government's Bottom-Up Economic Transformation Agenda (BETA) planning approach and its key priorities.

BETA is the Government's transformation agenda geared towards economic turnaround through a value chain approach. BETA has targeted sectors with the highest impact to drive economic recovery and growth. This will be achieved through bringing down the cost of living; eradicating hunger; creating jobs; expanding the tax base; improving foreign exchange balances; and inclusive growth. BETA ensures rational resource allocation by eliminating wastage of resources occasioned by duplication, overlaps, fragmentation and ineffective coordination in the implementation of programmes and projects.

The Fourth Medium Term Plan key priorities are clustered under five key sectors, namely: Finance and Production; Infrastructure; Social; Environment and Natural Resources; and Governance and Public Administration. The infrastructure sector seeks to: enhance transport connectivity by constructing 6,000 km of new roads, maintaining rural and urban roads, rail, air and seaport facilities and services; expand communication and broadcasting systems; and promote the development of energy generation and distribution by increasing investments in green energy (geothermal, wind, solar and hydro). The infrastructure gap is expected to be bridged by promoting economic participation of the private sector through public private partnerships in the financing, construction, development, operation, and maintenance of infrastructure.

The plan entails a shift of focus to fundamentals in project planning and implementation which include: respect for technical input, regulations and standard practices; adherence to project life cycle i.e., planning, feasibility studies and design before procurement of works; public and stakeholder consultation; procurement within budgetary ceilings; shifting focus during project implementation from the finished product 'black top' to the construction of the foundation; building local capacity particularly MSMEs by ensuring prompt payments; and capacity building at all levels to enable internalisation of policies and processes.

The first generation of the road manuals were used for 35 to 45 years. It is my sincere hope that the second generation of the road standards which have been developed in alignment with the Government's strategy will provide guidance in solving most of the above challenges and those expected to emerge in the next 50 years. Implementation of the manuals will enable achievement of the Government aspirations which include inclusive growth; creation of sustainable employment; building of MSMEs; climate change adaptation and realisation of the UN SDGs; enhanced efficiency in management of infrastructure and transport system; and, laying the foundation for the next national long-term plan at the end of the Vision 2030.

On behalf of the Government of Kenya, I would wish to thank the European Union for financing the development of the first drafts of the manuals in 2009 and the African Development Bank for the financial support in the review and updating of the manuals. I would also like to thank the members of the National Steering Committee and the Technical Task Force for their input. The Technical Administrators, and the Kenya National Highways Authority (KeNHA) for the procurement and able administration of the consultancy Contract. I also thank the Consultant, TRL Limited for their role in providing technical expertise that was essential for the success of the manuals updating exercise. I also wish to express my deepest appreciation to our stakeholders and all those who have contributed to this process and the staff of the Ministry for their continued input.

Hon. Davis K. Chirchir, E.G.H

Cabinet Secretary, Ministry of Roads and Transport

Preface

The Manual is intended to assist users in the proper installation of road signs. The primary four types of road signs: regulatory, warning, informatory and road works signs are described. The siting, orientation, material, retro-reflectivity of signs are explained. Colour, size, letters, visibility are elaborated. The size of regulatory signs to be used are presented in terms of class of road and approach speed of vehicles. Warrants for installing two of the most important signs – Stop and Give Way are presented along with siting details. Mandatory and prohibitive regulatory signs are presented.

The size of warning signs to be used are also given in terms of class of road and approach speed. Colour scheme of Informatory signs for directional purpose are given. For what type of road junction directional signs are to be used is explained. Height of letters, dimensions of stroke for sign face design are explained in detail. Other Information signs included are signs for geographic location, facilities, diversions, tourist facilities and truck escape ramps. A separate chapter deals with signs for persons with disabilities. Signs to be used on Expressways are presented in a separate chapter. Sign face design for Expressway signs is explained along with placing of signs. A chapter on signs for urban roads discusses the signs to be used in urban areas. Road work signs are presented in detail. Traffic control for various scenarios of road work is given.

Eng. Joseph M. Mbugua, CBS

Principal Secretary, State Department for Roads

Document Management

Document Status

This document has the status of a Manual. Users shall apply the contents there-in to fully satisfy the requirements set out. The content of the manual is based on current practice in Kenya and latest practices in the road sector, both regionally and internationally.

Sources of the Document

Copies of the document can be obtained from:

The Principal Secretary, State Department for Roads, Ministry of Roads and Transport, Works Building, Ngong Road, P.O. Box 30260 - 00100, NAIROBI Email: ps@road.go.ke

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While all care and consideration has been applied in the compilation of this document, the Ministry accepts no responsibility for failure in any way related to the application of this manual or any reference documents cited in it.

Requests for edits and corrections can be freely sent to the following address:

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Amendments Request Form

Request No.	Name	Organisation	Chapter	Page	Section/ Clause	Ref. to: Figure/ Table/	Type of Request	Request

Type of request: General – G; Editorial – E; Technical - T

Amendments to Date

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Abbreviations

ACM	Aluminium Composite Material
ASTM	American Society of Testing and Materials
BS	British Standards
CIE	International Commission on Illumination
EAS	East Africa Standard
EN	European Norm (standard)
FRP	Fibre Reinforced Plastic
GI	Galvanised Iron
ISA	International Symbol of Accessibility
ISO	International Organisation for Standardisation
KeNHA	Kenya National Highways Authority
KeRRA	Kenya Rural Roads Authority
Kmph, km/h	Kilometre Per Hour
KS	Kenya Standard
KURA	Kenya Urban Roads Authority
M25	Grade of concrete
mm	Millimetre
MoR	Ministry of Roads and Transport
PwD	Persons With Disabilities
s/w	Stroke Width
TER	Truck Escape Ramp
vpd	Vehicles Per Day

Glossary of Terms

Acceleration lane	An auxiliary lane to enable a vehicle to increase its speed so that it can merge safely with through traffic.
Arterial	A highway designed to move relatively large volumes of traffic at high speeds over long distances. Arterials offer little or no direct access to abutting properties.
Auxiliary Lane	Part of the roadway adjoining the carriageway for parking, speed change, turning, storage for turning, weaving, truck climbing, and for other purposes supplementary to through traffic movement.
Capacity	The maximum sustainable flow rate at which vehicles or persons can reasonably be expected to traverse a point or uniform segment of a lane or roadway during a specified time period under given roadway, geometric, traffic, environmental, and control conditions; usually expressed as vehicles per hour, passenger cars per hour, or persons per hour.
Carriageway	Portion of the roadway including the various physically contiguous traffic lanes and auxiliary lanes, serving one or both directions of traffic, and not including shoulders.
Collector	A standard of road that is characterised by an approximately even distribution of access and mobility functions.
Connector	A collective term for interchange links, link roads, ramps and loops.
Design Speed	An index which links road function, traffic flow and terrain to the design parameters of sight distance and curvature to ensure that a driver is presented with a reasonably consistent speed environment. It is now defined as the 85th percentile speed of passenger cars travelling in free flow conditions. In practice, most roads will only be constrained to minimum parameter values over short sections or on specific geometric elements.
Diverging	The opposite of merging. When a traffic stream splits into two or more streams.
Expressway	A multi-lane, divided highway with a minimum of two lanes for the exclusive use of traffic in each direction and full control of access without traffic interruption.
Footway / footpath	A path adjacent to the carriageway for use by pedestrians.
Grade Separation	A crossing of two highways or roads, or a road and a railway, at different levels.
Gradient	Rate of rise or fall on any length of road, with respect to the horizontal. It is typically expressed as a percentage or as the vertical rise or fall in m/100 m. In the direction of increasing stake value, upgrades are taken as positive and downgrades as negative.
High Speed	Speeds above 80km/h.
Horizontal Clearance	Lateral clearance between the edge of shoulder and obstructions.
Interchange	A system of interconnecting roads (referred to as ramps) in conjunction with one or more grade separations providing for the movement of traffic between two or more roadways which are at different levels at their crossing point.
Kerb	Concrete, often precast, element adjacent to the travelled way and used for drainage control, delineation of the pavement edge or protection of the edge of surfacing. Usually applied only in urban areas.

Glossary of Terms *(continued)*

Mainline	The primary through roadway as distinct from ramps, additional lanes, and collector roads.
Median	The area in the middle of a roadway separating the opposing traffic flows. The median thus includes the inner shoulders.
Merging	Movement of a vehicle or vehicles from one or more lanes into a traffic stream.
Nose	A paved area, approximately triangular in shape, between a connector road (ramp) and the mainline at a merge or diverge, suitably marked to discourage drivers from crossing it.
Ramp	A one-way, often single-lane, road providing a link between two roads that cross each other at different levels.
Right Turn Lane	Auxiliary lane to accommodate deceleration and storage of right- turning vehicles at junctions.
Roadside	A general term denoting the area beyond the shoulder breakpoints.
Road Reserve	Strip of land legally awarded to the Roads Authority, specifically for the provision of public right of way, in which the road is, or will be, situated and where no other work or construction may take place without permission from the Roads Authority.
Roadway	The area normally travelled by vehicles and consisting of traffic lanes, including auxiliary lanes and shoulders.
Shoulder	Part of the road outside the carriageway, but at substantially the same level, for accommodation of stopped vehicles for emergency use, for lateral support of the carriageway and for use by pedestrians and cyclists when no other facility has been provided.
Sight Distance	Distance visible to the driver of a passenger car measured along the normal travel path of a roadway to the roadway surface or to a specified height above the roadway surface, when the view is unobstructed by traffic.
Speed Hump (Bump)	Device for controlling the speed of vehicles, consisting of a raised area or recess on the roadway.
Taper	Transition length between a passing place, additional lane or passing lane and the standard roadway.
Traffic Flow	Number of vehicles or persons that pass a specific point in a stated time, in both directions unless otherwise stated.
Traffic Lane	Part of a carriageway way intended for a single stream of traffic in one direction, which has normally been demarcated as such by road markings.
Traffic Island	Central or subsidiary area raised or marked on the roadway, generally at a road junction, shaped and placed so as to direct traffic movement.
Trunk Road	International Trunk Road linking centres of international importance and crossing international boundaries or terminating at international ports.

Contents

	Foreword	iii
	Preface	v
	Document Management	vi
	Acknowledgements	vii
	Abbreviations	viii
	Glossary of Terms	ix
	Contents	xi
	List of Figures	xvi
	List of Tables	xxii
1	Introduction	1
	1.1 General	1
	1.2 Objective	3
	1.3 Scope	3
	1.4 Organisation of the Manual	3
	1.5 Design Process	3
2	General	5
	2.1 Background	5
	2.2 The Road Hierarchy	5
	2.3 Classification of Road Signs	7
	2.3.1 <i>Regulatory Signs</i>	7
	2.3.2 <i>Warning Signs</i>	7
	2.3.3 <i>Informatory Signs</i>	7
	2.3.4 <i>Road Works Signs</i>	7
	2.4 Siting of Signs	8
	2.5 Orientation of Signs	9
	2.6 Material for Signs	11
	2.6.1 <i>Concrete</i>	11
	2.6.2 <i>Reinforcing Steel</i>	11
	2.6.3 <i>Bolts, Nuts and Washers</i>	11
	2.6.4 <i>Plates and Supports</i>	11
	2.6.5 <i>Substrate</i>	11
	2.6.6 <i>Fibre Reinforced Plastic</i>	11
	2.6.7 <i>Plate Thickness</i>	12
	2.6.8 <i>Retro-Reflective Sheeting</i>	12
	2.6.9 <i>Selection of Retro-reflective Sheeting</i>	13
	2.6.10 <i>Messages / Borders</i>	15
	2.6.11 <i>Warranty and Durability</i>	15
	2.7 Posts and Mountings for Signs	15
	2.8 Shape and Colour of Signs	16
	2.9 Size of Signs	16
	2.10 Visibility of Signs	16
	2.11 Size of Letters	16
	2.12 Maintenance of Signs	18
	2.13 Testing for Retro-reflectivity and Quality	18
	2.14 Bollards	18

3	Regulatory Signs	19
	3.1 Introduction	19
	3.2 Priority Signs	19
	3.2.1 STOP Sign	19
	3.2.2 GIVE WAY sign	21
	3.3 Mandatory Signs	22
	3.4 Prohibitory Signs	23
	3.5 Speed Limit Signs	24
	3.5.1 Maximum Speed Limit	24
	3.6 Parking Restriction Signs	25
	3.7 Signs for Cycle Facilities	26
	3.7.1 Cycle Tracks	26
	3.7.2 Cycleway	26
	3.7.3 Shared Pedestrian/Cycle Facilities	26
	3.8 School Warden's Stop Sign	27
	3.9 Pedestrianised Streets	27
	3.9.1 Prohibition on Entry to or Parking in Certain Streets	27
	3.10 Weight and Height Restrictions	27
	3.11 No Overtaking Sign	29
	3.12 Bus Lane Sign	29
	3.12.1 With-Flow Bus Lanes	29
	3.12.2 Contra-Flow Bus Lanes	30
	3.12.3 Other Bus Signs	30
	3.13 Other Prohibitive Signs	32
4	Warning Signs	33
	4.1 Introduction	33
	4.2 Junction Ahead	33
	4.2.1 Junctions with Roads of Lesser Importance	34
	4.2.2 Junctions with Roads of Equal Importance	35
	4.2.3 Junctions with Roads of Greater Importance	35
	4.3 Merging Traffic	35
	4.4 Roundabouts	35
	4.5 Sharp Turns and Bends	36
	4.6 Other Warning Signs	37
	4.6.1 Sharp Change of Direction	37
	4.6.2 Road Narrowing	38
	4.6.3 Divided Carriageway	38
	4.6.4 Steep Gradient	39
	4.6.5 Restricted Headroom	39
	4.6.6 Overhead Electric Cables	40
	4.6.7 Level Crossings	40
	4.6.8 Rise and Dips	41
	4.6.9 Slippery Road	42
	4.6.10 River, Canal or Quayside	42
	4.6.11 Traffic Signals	42
	4.6.12 Schools and Children	42
	4.6.13 Animal Crossing	43
	4.6.14 Cross Winds	44
	4.6.15 Pedestrians	44
	4.6.16 Tunnel	44
	4.6.17 Falling Rocks	44
	4.6.18 Low Flying Aircraft	45
	4.6.19 Drive on the Left	45
	4.7 Roadside Delineators	45

5	4.8	Mounting of Warning Signs	46
	4.8.1	<i>Mounting More Than One Sign on a Post</i>	46
	4.9	Additional Warning Signs	47
		Informatory Signs	49
	5.1	Introduction	49
	5.2	General Principles of Design	50
	5.2.1	<i>The Road Hierarchy and Colour Coding</i>	50
	5.2.2	<i>Provision of Directional Signs</i>	50
	5.2.3	<i>Terminal Destinations</i>	51
	5.2.4	<i>Other Destinations</i>	51
6	5.2.5	<i>Alphabets</i>	51
	5.2.6	<i>Common Dimensions</i>	51
	5.2.7	<i>Letter and Block Spacing</i>	52
	5.2.8	<i>Patches and Legend Panels</i>	52
	5.3	Sign Face Design	53
	5.3.1	<i>Advance Sign Design</i>	53
	5.3.2	<i>Stack Type Design</i>	53
	5.3.3	<i>Route Direction Signs</i>	55
	5.3.4	<i>Map Type Signs for Roundabouts</i>	56
	5.3.5	<i>Lane Destination Signs</i>	60
	5.3.6	<i>Direction Signs</i>	61
	5.3.7	<i>Direction Signs for Arterial or Trunk (Class A, B & UA) Roads</i>	62
	5.3.8	<i>Direction Signs for Collector and Local roads</i>	63
	5.3.9	<i>Route Confirmatory Signs</i>	65
		Other Informatory Signs	67
	6.1	Introduction	67
	6.2	Geographical Information Sign	68
	6.2.1	<i>Town or Village Signs</i>	68
	6.2.2	<i>County Boundary Signs</i>	68
	6.3	Advisory Informatory Signs	69
	6.3.1	<i>Advance Sign for Low Clearances at Bridges</i>	69
	6.3.2	<i>Slow Lanes</i>	69
	6.3.3	<i>No Through Way</i>	70
	6.4	Signs Indicating Facilities Ahead	70
	6.4.1	<i>Parking Signs</i>	70
	6.4.2	<i>Hospital Signs</i>	71
	6.4.3	<i>Airport Signs</i>	71
	6.4.4	<i>Ferry Terminal Signs</i>	72
	6.4.5	<i>Industrial Estate Signs</i>	72
	6.5	Signs to Tourist Attractions	73
	6.6	Additional Facilities Signs	73
	6.7	Signs for Escape Ramp	74
7		Signs for Persons with Disabilities	75
	7.1	The International Symbol of Accessibility	75
	7.2	Size, Shape and Colour	75
	7.3	Parking Information	75
	7.4	Ramped Entrance to Subway/Over Bridge	75
	7.5	Telephone and Toilet Facility	75
	7.6	Way Finding Sign for Disabled	75
	7.7	Old / Disabled People Crossing	76
	7.8	Signs for Visually Challenged and Persons with Hearing Impairment	76

8	Expressway Signs	79
	8.1 Introduction	79
	8.2 General Principles	80
	8.2.1 Colour Coding	80
	8.2.2 The Selection of Destinations	81
	8.2.3 Alphabets	81
	8.3 Sign Face Design	81
	8.3.1 Signs on Trunk or Collector Roads Indicating Expressways	81
	8.3.2 Patches	82
	8.3.3 Legend Panels	83
	8.3.4 Direction Signs	85
	8.4 Signs on Expressway	85
	8.4.1 Gantry Signs on Expressways	89
	8.4.2 Non-Lane-Drop Signs	90
	8.4.3 Lane-Drop Signs	90
9	Guidelines for Signs on Urban Roads	91
	9.1 General	91
	9.2 Colour of Signs	91
	9.3 Messages on Overhead and Shoulder Mounted Signs	91
	9.4 Warrants for Sign Installation	91
	9.5 Size of Signs	91
	9.6 Siting of the Signs with Respect to Carriageway	91
	9.7 Size of Letters	91
	9.8 Guidelines for Informatory Signs Installation	92
	9.8.1 Gantry Mounted Signs	92
10	Signs for Roadworks	93
	10.1 Introduction	93
	10.2 Recommended Signs for Road works	93
	10.2.1 Roadworks Ahead	94
	10.2.2 Uneven Surface	94
	10.2.3 Slippery Road Ahead	94
	10.2.4 Road Narrows	94
	10.2.5 Manual Traffic Control Ahead	95
	10.2.6 Traffic Signals Ahead	95
	10.2.7 Two-way Traffic Ahead	95
	10.2.8 Signs for Lane Closures Ahead	95
	10.2.9 Major Road Works Ahead	96
	10.2.10 No Through Road	96
	10.2.11 Road Closed	96
	10.3 Delineation of Roadworks	97
	10.3.1 Drums	97
	10.3.2 Cones	98
	10.3.3 Barricades	99
	10.3.4 Lamps	99
	10.3.5 Keep Left / Keep Right / Pass Either Side	100
	10.3.6 Chevron	100
	10.4 Protection of Pedestrians at Roadworks	100
	10.4.1 Pedestrians Signs	101
	10.5 Traffic Control at Road Works	101
	10.5.1 Temporary Traffic Signals	102
	10.6 Diversions	102
	10.7 Deployment of Roadwork Signs	104
	10.7.1 Procedure for Setting Out Signs	104

10	10.8	Urban Roadworks	104
	10.9	Rural Roadworks	105
	10.10	Fixed Works	105
	10.10.1	<i>Choosing a Layout</i>	105
	10.11	Margin for Safety	105
	10.12	Lighting	105
	10.13	Reflectorising	106
	10.14	Colour	106
	10.15	One Way Operations	106
	10.16	Implementing Layout Guidelines	106
	10.17	Mobile Works	107
	10.18	Four Levels of Warning	108
	10.19	Detours	108
11		Appendices	109
	Appendix 1:	Letter Sizes and Siting	109
	Appendix 2:	Tourists Signs	114
	Appendix 3:	Expressway Signs	117
	Appendix 4:	Traffic Control for Roadworks	119
	Appendix 5:	Catalogue of Signs	143
	Appendix 6:	Specifications & Standards	159

List of Figures

Figure 2.1	Siting of Signs with respect to the Carriageway (<i>Height and Clearance</i>)	9
Figure 2.2	Siting of Signs with respect to the Carriageway (<i>Height and Clearance</i>)	10
Figure 2.3	Orientation of Sign	10
Figure 2.4	Height and Gap Between Bollards	18
Figure 3.1	Stop, Sign R1	19
Figure 3.2	Visibility Funnel	20
Figure 3.3	Location of STOP Sign	20
Figure 3.4	GIVE WAY, Sign R2	21
Figure 3.5	Location of GIVE WAY sign	22
Figure 3.6	Left Turn, Sign M1	22
Figure 3.7	Straight Ahead, Sign M2	22
Figure 3.8	Right Turn, Sign M3	22
Figure 3.9	Straight Ahead or Turn Left, Sign M4	23
Figure 3.10	Keep Left, Sign M5	23
Figure 3.11	Keep Right, Sign M6	23
Figure 3.12	Turn Left Ahead, Sign M7	23
Figure 3.13	Turn Right Ahead, Sign M8	23
Figure 3.14	Pass Either Side, Sign M9	23
Figure 3.15	No Entry, Sign P1	23
Figure 3.16	No Right Turn, Sign P2	23
Figure 3.17	No Left Turn, Sign P3	23
Figure 3.18	No U Turn, Sign P4	23
Figure 3.19	Built Up Area, Speed Limit 50 km/h, Sign P5	24
Figure 3.20	Special Area, Speed Limit 70 km/h, Sign P6	24
Figure 3.21	Special Area, Speed Limit 80 km/h, Sign P7	24
Figure 3.22	Expressway, Speed Limit 120 km/h, Sign P8	24
Figure 3.23	End of Maximum Speed Limit (<i>other than General Speed Limit</i>), Sign P9	24
Figure 3.24	End of Minimum Speed Limit, Sign P10	24
Figure 3.25	Parking Permitted, Sign P11	25
Figure 3.26	Restriction Plate, Sign P12	25
Figure 3.27	Parking Prohibited, Sign P13	25
Figure 3.28	No Stopping Sign, Sign P14	25
Figure 3.29	Disc Parking Plate, Sign P15	25
Figure 3.30	Taxi Stand, Sign P16	25
Figure 3.31	Commencement of Cycle Track, Sign P17	26
Figure 3.32	End of Cycle Track, Sign P18	26
Figure 3.33	Commencement of Cycleway, Sign P19	26
Figure 3.34	Shared Pedestrian / Cycle Track, Sign P20	26
Figure 3.35	School Warden's Stop Sign, Sign P21	27
Figure 3.36	Pedestrianised Street, P22	27
Figure 3.37	Weight Restriction, Sign P23	28
Figure 3.38	Vehicle Weight Restriction, Sign P24	28
Figure 3.39	Axle Weight Restriction, Sign P25	28
Figure 3.40	Advance Sign Indicating Weight Restriction Ahead, Sign P26	28
Figure 3.41	Indicating Route Subject to Restriction, Sign P27	28

List of Figures

Figure 3.42	Sign Indicating Route Subject to Restriction, Sign P28	28
Figure 3.43	Height Restriction, Sign P29	28
Figure 3.44	Parking Restriction for Heavy Vehicles, Sign P30	28
Figure 3.45	End of Restriction, Sign P31	28
Figure 3.46	No Overtaking, Sign P32	29
Figure 3.47	Advance Sign for With-Flow Bus Lane, Sign P33	29
Figure 3.48	With-Flow Bus Lane, Sign P34	29
Figure 3.49	Information Plate, Sign P35	29
Figure 3.50	Advance Sign for Offside Bus Lane, Sign P36	30
Figure 3.51	Offside Bus Lane, Sign P37	30
Figure 3.52	Information Plate, Sign P38	30
Figure 3.53	Contra Flow Bus Lane, Sign P39	30
Figure 3.54	Signing for 'With-Flow' Bus Lane	31
Figure 3.55	Signing for 'Contra-Flow' Bus Lanes	31
Figure 3.56	Bus Lane, Sign P40	31
Figure 3.57	Bus Lane, Sign P41	31
Figure 3.58	Bus Lane, Look Left, Sign P42	31
Figure 3.59	Bus Lane, Look Left, Sign P43	31
Figure 3.60	Buses Prohibited, Sign P44	32
Figure 3.61	Cars Prohibited, Sign P45	32
Figure 3.62	Trucks Prohibited, Sign P46	32
Figure 3.63	Tractors Prohibited, Sign P47	32
Figure 3.64	Motorcycles Prohibited, Sign P48	32
Figure 3.65	Cycles Prohibited, Sign P49	32
Figure 3.66	Horns Prohibited, Sign P50	32
Figure 3.67	Do Not Block Intersection, Sign P51	32
Figure 4.1	Standard Shape of Warning Signs	33
Figure 4.2	Crossroads, Sign W1	34
Figure 4.3	Side Road, Sign W2	34
Figure 4.4	T Junction, Sign W3	34
Figure 4.5	Y Junction, Sign W4	34
Figure 4.6	Staggered Crossroads, Sign W5	34
Figure 4.7	Crossroads, Sign W6	34
Figure 4.8	Side Road, Sign W7	34
Figure 4.9	T Junction, Sign W8	34
Figure 4.10	Staggered Crossroads, Sign W9	34
Figure 4.11	Y Junction, Sign W10	34
Figure 4.12	Crossroads, Sign W11	34
Figure 4.13	T Junction, Sign W12	34
Figure 4.14	Crossroads with Dual Carriageway, Sign W13	34
Figure 4.15	T Junction with Dual Carriageway, Sign W14	34
Figure 4.16	Merging Traffic, Sign W15	34
Figure 4.17	Merging / Diverging Traffic, Sign W16	34
Figure 4.18	Roundabout Ahead, Sign W17	35
Figure 4.19	Mini Roundabout Ahead, Sign W18	35

List of Figures

Figure 4.20	Sharp Turn Ahead (Left / Right), Sign W19	36
Figure 4.21	Sharp Bend Ahead (Left / Right), Sign W20	36
Figure 4.22	Series of Sharp Curves Ahead, Sign W21	36
Figure 4.23	Series of Bends Ahead, Sign W22	36
Figure 4.24	Distance Plates, Sign W23	36
Figure 4.25	Chevrons, Sign W24	37
Figure 4.26	Road Narrows on One Side, Sign W25	38
Figure 4.27	Road Narrows on Both Sides, Sign W26	38
Figure 4.28	Road Divides, Sign W27	38
Figure 4.29	End of Divided Carriageway, Sign W28	38
Figure 4.30	Two Way Traffic, Sign W29	38
Figure 4.31	Signing at Transition To and From Divided Carriageway	38
Figure 4.32	Steep Descent Ahead, Sign W30	39
Figure 4.33	Steep Ascent Ahead, Sign W31	39
Figure 4.34	Restricted Headroom, Sign W32	39
Figure 4.35	'Goal Post' Marking at Hazard, Sign W33	39
Figure 4.36	Overhead Electric Cables, Sign W34	40
Figure 4.37	Safe Height Plate, Sign W35	40
Figure 4.38	Level Crossing Ahead, Unguarded by Gates or Lifting Barriers, Sign W36	40
Figure 4.39	Level Crossing Ahead, Guarded by Gates or Lifting Barriers, Sign W37	40
Figure 4.40	Level Crossing Ahead with Lights and Barriers, Sign W38	40
Figure 4.41	Stop When Red Lights Show, Sign W39	41
Figure 4.42	Slow Automatic Level Crossing, Sign W40	41
Figure 4.43	Spacing and Distance of Signs from Level Crossing	41
Figure 4.44	Sharp Rise Ahead e.g., Hump-Back Bridge, Sign W41	41
Figure 4.45	Sharp Dip Ahead, Sign W42	41
Figure 4.46	Series of Bumps and Dips Ahead, Sign W43	41
Figure 4.47	Slippery Road Ahead, Sign W44	42
Figure 4.48	Unprotected Canal, River or Quay Ahead, Sign W45	42
Figure 4.49	Traffic Signals Ahead, Sign W46	42
Figure 4.50	School Ahead, Sign W47	42
Figure 4.51	School Children Crossing Ahead, Sign W48	42
Figure 4.52	Children Crossing (in residential areas), Sign W49	42
Figure 4.53	Accompanied Horses or Ponies, Sign W50	43
Figure 4.54	Cattle or Farm Animals, Sign W51	43
Figure 4.55	Sheep, Sign W52	43
Figure 4.56	Wild Animals, Sign W53	43
Figure 4.57	Crosswinds, Sign W54	44
Figure 4.58	Pedestrian Crossing Ahead, Sign W55	44
Figure 4.59	Tunnel Ahead, Sign W56	44
Figure 4.60	Falling Rocks, Sign W57	44
Figure 4.61	Low Flying Aircraft, Sign W58	45
Figure 4.62	Keep Left, Sign W59	45
Figure 4.63	Delineators on Road with/without Shoulder	45
Figure 4.64	Assembly Combinations of Warning Signs and Supplementary Plates	46

List of Figures

Figure 4.65	Road Widens, Sign W60	47
Figure 4.66	Narrow Bridge Ahead, Sign W61	47
Figure 4.67	Gap in the Median, Sign W62	47
Figure 4.68	Built Up Area, Sign W63	47
Figure 4.69	Speed Breaker, Sign W64	47
Figure 4.70	Rumble Strip, Sign W65	47
Figure 4.71	Ferry, Sign W66	47
Figure 4.72	Queue Ahead, Sign W67	47
Figure 4.73	Object Hazard (Left), Sign W68	47
Figure 4.74	Object Hazard (Right), Sign W69	47
Figure 4.75	Two Way Hazard Marker, Sign W70	47
Figure 4.76	Old/ Disabled People Crossing, Sign W71	47
Figure 5.1	Advance Directional Sign	49
Figure 5.2	Direction Sign	49
Figure 5.3	Route Confirmatory Sign	49
Figure 5.4	Sample Text	52
Figure 5.5	Legend Panel Design	53
Figure 5.6	Patch Design	53
Figure 5.7	Design and Orientation of Arrows	53
Figure 5.8	Design of Standard Stack Type Sign (<i>Sign I1-A</i>)	54
Figure 5.9	Alternative Design of Stack Type Sign (<i>Sign I1- B</i>)	55
Figure 5.10	Alternative Design of Stack Type Sign (<i>Sign I1- C</i>)	55
Figure 5.11	Design of Route Direction Sign (<i>Sign I2-a</i>)	56
Figure 5.12	Design of Route Direction Sign (<i>Sign I2-b</i>)	56
Figure 5.13	Forming Break in Roundabout Symbol	57
Figure 5.14	Design of Roundabout Map Type Sign (<i>Sign I3-a</i>)	58
Figure 5.15	Design of Roundabout Map Type Sign (<i>Sign I3-b</i>)	59
Figure 5.16	Design of Roundabout Map Type Sign (<i>Sign I3-c</i>)	59
Figure 5.17	Design of Lane Destination Sign (<i>Sign I4-a</i>)	60
Figure 5.18	Design of Lane Destination Sign (<i>Sign I4-b</i>)	61
Figure 5.19	Design of Direction Sign for Class A Road	62
Figure 5.20	Design of Directional Sign for Class C Sign I5-b	63
Figure 5.21	Design of Directional Sign for Class C Showing Distance Sign I5-c	63
Figure 5.22	Design of Direction Sign for Class C Road (<i>Two Destination</i>) Sign I5-d	64
Figure 5.23	Design of Direction Sign for Class C Road Sign I5-e	64
Figure 5.24	Design of Route Confirmatory Sign with One Destination (<i>Sign I6-a</i>)	66
Figure 5.25	Design of Route Confirmatory Sign with Three Destinations (<i>Sign I6-b</i>)	66
Figure 6.1	Town or Village, Sign I7	67
Figure 6.2	Alternative Route for High Vehicle, Sign I8	67
Figure 6.3	Car Park with Facilities for Disabled, Sign I9	67
Figure 6.4	Tourist Facility, Sign I10	67
Figure 6.5	Escape Ramp, Sign E1	67
Figure 6.6	Town or Village, Sign I7	68
Figure 6.7	County Boundary, Sign I10	68
Figure 6.8	Advance Sign for Low Clearance Sign I11	69

List of Figures

Figure 6.9	Slow Lane Sign I12	69
Figure 6.10	No Through Way Sign I13	69
Figure 6.11	Lay-by Ahead Sign I14	70
Figure 6.12	Lay-by Sign I15	70
Figure 6.13	Car Park Sign I16	70
Figure 6.14	Disabled Car Park Sign I17	70
Figure 6.15	Hospital Ahead Sign I18	71
Figure 6.16	Hospital Sign I19	71
Figure 6.17	Airport Symbol Sign I20	71
Figure 6.18	Directional Sign Incorporating Airport Symbol I20	71
Figure 6.19	Ferry Symbol Sign I21	72
Figure 6.20	Directional Sign Incorporating Ferry Symbol I21	72
Figure 6.21	Industrial Estate Symbol Sign I22	72
Figure 6.22	Directional Sign Incorporating Industrial Estate Symbol, Sign I22	72
Figure 6.23	Eating Place, Sign I23	73
Figure 6.24	Light Refreshments, Sign I24	73
Figure 6.25	Resting Place, Sign I25	73
Figure 6.26	Toilet, Sign I26	73
Figure 6.27	Fuel Pump, Sign I27	73
Figure 6.28	Pedestrian Over Bridge, Sign I28	73
Figure 6.29	Pedestrian Subway, Sign I29	73
Figure 6.30	Picnic Site, Sign I30	73
Figure 6.31	Camp Site, Sign I31	73
Figure 6.32	Escape Ramp Sign at Ramp, Sign I32	74
Figure 6.33	Direction of Ramp, Sign I33	74
Figure 7.1	International Symbol of Accessibility, Sign P1	75
Figure 7.2	Parking Information, Sign PD2	75
Figure 7.3	Parking Areas, Sign PD3	76
Figure 7.4	Ramped Entrance, Sign PD4	76
Figure 7.5	Telephone Facilities, Sign PD5	76
Figure 7.6	Toilet Facilities, Sign PD6	76
Figure 7.7	Wayfinding, Sign PD7	76
Figure 7.8	Old People Crossing, Sign W71	76
Figure 7.9	Sign for Visually Challenged	77
Figure 8.1	Advance Direction, Sign IM1	79
Figure 8.2	Direction Sign, Sign IM2	79
Figure 8.3	Route Confirmation, Sign IM3	80
Figure 8.4	End of Expressway, Sign IM4	80
Figure 8.5	Expressway Symbol, Sign IM4	81
Figure 8.6	Sign Incorporating Expressway Patch, Sign IM5	82
Figure 8.7	Sign Incorporating Expressway Patch, Sign IM6	82
Figure 8.8	Intermediate Junction Signing from Connecting National Route, Sign IM7	83
Figure 8.9	Signing from a Terminal Roundabout, Sign IM8	83
Figure 8.10	Expressway Status Sign on National Route, Sign IM10	84
Figure 8.11	Direction Sign from National or Regional Road, Sign IM11	85

List of Figures

Figure 8.12	Expressway Ahead, Sign IM12	85
Figure 8.13	Entry to Expressway, Sign IM13	85
Figure 8.14	Advance Direction Sign for Intermediate Expressway Junction Sign IM 14	86
Figure 8.15	Final Advance Direction Sign for Intermediate Expressway Junction Sign IM 15	86
Figure 8.16	300 m Countdown Marker	86
Figure 8.17	200 m Countdown Marker	86
Figure 8.18	100 m Countdown Marker	86
Figure 8.19	Direction Sign for Expressway Exit Slip Road Sign IM19	87
Figure 8.20	Position of Expressway Signs at an Intermediate Junction	87
Figure 8.21	Map Type Advance Direction Sign on Expressway Exit Slip Road, Sign IM 20	88
Figure 8.22	Route Confirmation Sign for Placement Beyond Entry Slip Road Sign IM 21	88
Figure 8.23	Countdown Marker Arrangement for Both Sides Sign IM 22	88
Figure 8.24	Sign Arrangement for Entry to Roads of Different Classification at Grade Sign IM 23	89
Figure 8.25	End of Expressway 1km, Sign IM24	89
Figure 8.26	End of Expressway 500m, Sign IM25	89
Figure 8.27	End of Expressway, Sign IM26	89
Figure 8.28	Non Lane-Drop Signing – sign on the bottom Sign IM 27	90
Figure 8.29	Lane-Drop Signing Sign IM 28	90
Figure 10.1	Roadworks Ahead, Sign RW 1	94
Figure 10.2	Uneven Surface, Sign RW 2	94
Figure 10.3	Slippery Road Ahead, Sign RW 3	94
Figure 10.4	If narrowing is from left side, the left line should be shown as narrowing, Sign RW4	94
Figure 10.5	Road Narrows, Sign RW5	94
Figure 10.6	Manual Traffic Control Ahead, Sign RW 6	95
Figure 10.7	Traffic Signals Ahead, Sign RW 7	95
Figure 10.8	Two-Way Traffic Ahead, Sign RW8	95
Figure 10.9	Left Lane Closed for Two Lanes in One Direction, Sign RW 9	95
Figure 10.10	Left Lane Closed for Three lanes in One Direction, Sign RW10	95
Figure 10.11	Centre Lane Closed for Three-lanes in One Direction, Sign RW11	96
Figure 10.12	Right Lane Closed for Three-lanes in One Direction, Sign RW12	96
Figure 10.13	Left Lane Closed Two Right Lanes Diverted, Sign RW13	96
Figure 10.14	Left Through Centre Lane, Closed Right Lane Diverted, Sign RW14	96
Figure 10.15	Major Road Works Ahead, Sign RW15	96
Figure 10.16	No Through Road, Sign RW16	96
Figure 10.17	Road Closed, Sign RW17	96
Figure 10.18	Drum	97
Figure 10.19	Do Not Weigh Down a Drum Using a Block	97
Figure 10.20	Do Not Place a Plank Across Drums	98
Figure 10.21	Traffic Cones	98
Figure 10.22	Standard Barricades	99
Figure 10.23	Keep Left, Sign RW21	100
Figure 10.24	Keep Right, Sign RW22	100
Figure 10.25	Pass On Either Side, Sign RW23	100
Figure 10.26	Chevron, Sign RW24	100

Figure 10.27	Protective Pedestrian Barrier, Sign RW25	101
Figure 10.28	Crossing Not in Use, Sign RW26	101
Figure 10.29	Pedestrians Keep Right, Sign RW27	101
Figure 10.30	Pedestrians Look Left, Sign RW28	101
Figure 10.31	Stop/Go Sign RW29	102
Figure 10.32	Diversion ('m') Ahead, Sign RW30	102
Figure 10.33	Diversion Right, Sign RW31	102
Figure 10.34	Diversion Left, Sign RW32	102
Figure 10.35	Diverted Traffic Straight Ahead, Sign RW33	103
Figure 10.36	Diverted Traffic Right, Sign RW34	103
Figure 10.37	End of Diversion, Sign RW35	103
Figure 10.38	Crossover to Left, Sign RW36	103
Figure 10.39	Crossover to Right, Sign RW37	103
Figure 10.40	Divided Road Ahead, Sign RW38	103
Figure 10.41	End of Divided Road, Sign RW39	103
Figure 10.42	No Overtaking, Sign RW40	103
Figure 10.43	End of Construction, Sign RW41	103
Figure A1-1	Terminal Destinations for Class A and B Roads	110
Figure A1-2	Upper Case Alphabet (<i>Tile width expressed as a percentage of X-Height</i>)	111
Figure A1-3	Lower Case Alphabet (<i>Tile width expressed as a percentage of X-Height</i>)	112
Figure A1-4	Numerical (<i>Tile width expressed as a percentage of X-Height</i>)	112
Figure A1-5	Numerical for Expressway (<i>Tile width expressed as a percentage of X-Height</i>)	113
Figure A1-6	Examples of Coloured Sign Faces	113
Figure A2-1	Tourist Advance Direction Sign	114
Figure A2-2	Tourist Attraction Direction Sign	114
Figure A2-3	Sign Showing Distance to Tourist Attraction	114
Figure A2-4	Signs to Approved Tourist Information Points	114
Figure A2-5	Directional Tourist Signs	115
Figure A2-6	Youth Hostel	115
Figure A2-7	Pedestrian Sign to Car Park	115
Figure A3-1	Examples of Signs with Patches	117
Figure A3-2	Examples of Signs for Expressways	118
Figure A4-1	Roadworks Partially Obstructing Footpath	121
Figure A4-2	'Give and Take' Operation on Lightly Trafficked Minor Roads (<i>Unobstructed Carriageway Width <5.5 m</i>)	122
Figure A4-3	One-Way shuttle Working Using Flagmen/ Batten-men (<i>Suitable for Daylight only</i>)	123
Figure A4-4	One-Way shuttle Working Using Temporary Traffic Signals (<i>Suitable for Daylight and Darkness</i>)	124
Figure A4-5	Two-Way works Forcing Traffic Over Centre Line	125
Figure A4-6	Three Lane One-way Street, Works Inside Lane	126
Figure A4-7	Three Lane One-Way Street, Works in Centre Lane	127
Figure A4-8	Four Lane Street, Works in Two Lanes at Junction	128
Figure A4-9	Four Lane Street, Works In Two Lanes Near Junction	129
Figure A4-10	Four Lane Street, Works in Inner City Bound Lane, Morning Peak	130
Figure A4-11	Four Lane Street, Works in Inner City Bound Lane, Off Peak and Evening Peak	131
Figure A4-12	Four Lane Street, Works in Outer City Bound Lane, Morning Peak	132

Figure A4-13	Four Lane Street, Works in Outer City Bound Lane, Off Peak and Evening Peak	133
Figure A4-14	Two-Way Traffic Works in Margin/Shoulder/Footpath Only	134
Figure A4-15	Two-Way Traffic Carriageway Encroachments	135
Figure A4-16	Use of Road Shoulder by Traffic	136
Figure A4-17	One-Way Shuttle Working Using Flagmen/ Batten-men (<i>Suitable for Daylight Only</i>)	137
Figure A4-18	One-Way Shuttle Working Using Temporary Traffic Signals (<i>Suitable for Daylight and Darkness</i>)	138
Figure A4-19	One-Way Shuttle Working at a 'T' Junction	139
Figure A4-20	Two-Way Traffic Carriageway Encroachments	140
Figure A4-21	Dual Carriageway, Works in Slow Lane	141
Figure A4-22	Dual Carriageway, One Carriageway Closed	142

List of Tables

Table 1.1	Road Design Manual (RDM) Coding Structure	2
Table 2.1	Road Classification in Kenya	6
Table 2.2	Specifications for Aluminium Composite Material (ACM)	12
Table 2.3	Suggested Guideline for Usage of Retro-Reflective Sheeting	13
Table 2.4	Chromaticity Coordinates (Daytime)	17
Table 2.5	Daytime Luminance Factor (Y %)	17
Table 2.6	Colour Scheme for Directional Signs	17
Table 3.1	Sizes of Stop signs	19
Table 3.2	Visibility Funnel Dimensions with Speed	20
Table 3.3	Sizes of GIVE WAY sign	21
Table 3.4	Sizes of Bus Lane Signs	31
Table 4.1	Sizes of Warning Signs and their Siting Details	33
Table 4.2	Spacing of Single Chevron Signs	37
Table 5.1	Colour Schemes for Directional Signs	50
Table 5.2	When to Provide Directional Signs	50
Table 5.3	Destinations to be shown	51
Table 5.4	Spacing Between Words, Route Numbers and Distances	52
Table 5.5	Dimensions for Stack Type Signs	54
Table 5.6	Dimensions for Route Direction Signs	56
Table 5.7	Dimensions for Roundabout Route Symbols	57
Table 5.8	Dimensions for Map Type Signs of Roundabouts	58
Table 5.9	Dimensions for Lane Destination Signs	60
Table 5.10	Dimensions for Designing Direction Signs	62
Table 5.11	Dimensions for Route Confirmatory Signs	65
Table 10.1	Colour Shades for Roadwork Signs	93
Table 10.2	Description of Roadwork Figures	105
Table 10.3	Level of Warning	108
Table A1-1	Letter Sizes and Siting Details of Directional Information Sign	109
Table A1-2	Terminal Destinations for International Trunk Roads (Class A)	111
Table A1-3	Terminal Destinations for National Trunk Roads (Class B)	111
Table A4-1	Lead-in Taper for Urban Situations	119
Table A4-2a	Siting of Roadworks Signs in Urban Situations	119
Table A4-2b	Siting of 'End of Works' Sign for Urban Situations	119
Table A4-3	Siting of Roadworks Signs in Rural Situations	119
Table A4-4	Lead-in Taper for Rural Situations	119
Table A4-5	Siting 'End of Works' Sign for Rural Situations	120
Table A6-1	Materials and Specifications	159
Table A6-2	List of Standards / Specifications	160

1 Introduction

1.1 General

This Manual was prepared by the Ministry as part of the Roads Manual that attends to the project cycle for the provision, maintenance, and operation of roads in Kenya.

The Manual is applicable to roads at national, municipal and county levels and incorporates best practices, climate change considerations, and modern technologies thereby enabling the provision of road infrastructure that is safe, secure and efficient.

The Roads Manual is composed of the following documents:

Project Cycle Stage	Manual: Volume or Part/Chapter	Code
A. General	Procedures and Standards Manual	PSM
	1. General	
	2. Policies	
	3. Procedures Guidance	
	4. Codes of Practice	
	5. Guidelines	
B. Planning	Network and Project Planning Manual	NPM
	1. Road Classification	
	2. Route/Corridor Planning	
	3. Route/Corridor Planning	
	4. Highway Capacity	
	5. Project Planning	
C. Appraisal	Project Appraisal Manual	PAM
	1. Environmental Impact Assessment and Audit	
	2. Social Impact Assessment	
	3. Traffic Impact Assessment	
	4. Road Safety Audits	
	5. Project Appraisal	
D. Design	Road Design Manual	RDM
	1. Geometric Design	
	2. Hydrology and Drainage Design	
	3. Materials and Pavement Design for New Roads	
	4. Bridges and Retaining Structures Design	
	5. Pavement Maintenance, Rehabilitation and Overlay Design	
	6. Traffic Control Facilities and Communication Systems Design	
E. Contracts	Works and Services Contracts Manual	WSCM
	1. Forms of contracts	
	2. Standard Specification for Road and Bridge Construction	
	3. Bills of Quantities	
F. Construction	Road Construction Manual	RCM
	1. Construction Management	
	2. Project Management	
	3. Site Supervision	
	4. Quality Assurance	
	5. Quality Control	

This table continues onto the next page...

Project Cycle Stage	Manual: Volume or Part/Chapter	Code
G. Maintenance	Road Asset Management Manual	RAAM
	1. Maintenance Management	
	2. General Maintenance	
	3. Pavement Maintenance	
	4. Bridges and Structures Maintenance	
H. Operations	Road Operation Manual	ROM
	1. Traffic Management	
	2. Vehicle Load Control	
	3. Emergency Services	
	4. Tolling	
I. Monitoring & Evaluation	Road Design Manual	MEM
	1. Performance Monitoring Manual	
	2. Technical Audits	
	3. Poverty, Gender Equality and Social Inclusion Monitoring	

This Road Design Manual, Volume 6, Part 2 – Traffic Signs is part of the Roads Design Manual made up of a series of volumes and shown below:

Table 1.1 Road Design Manual (RDM) Coding Structure

Vol.	Manual Title	Part Name	Code
1	Road Design Manual: Vol. 1 Geometric Design	Part 1 - Topographic Survey	RDM 1.1
		Part 2 – Traffic Surveys	RDM 1.2
		Part 3 – Geometric Design of Highways, Rural and Urban Roads	RDM 1.3
2	Road Design Manual: Vol. 2 Hydrology & Drainage Design	Part 1 – Hydrological Surveys	RDM 2.1
		Part 2 – Drainage Design	RDM 2.2
3	Road Design Manual: Vol. 3 Materials & Pavement Design for New Roads	Part 1 – Ground Investigations and Material Prospecting	RDM 3.1
		Part 2 – Materials Field and Laboratory Testing	RDM 3.2
		Part 3 – Pavement Foundation and Materials Design	RDM 3.3
		Part 4 – Flexible Pavement Design	RDM 3.4
		Part 5 – Rigid Pavement Design	RDM 3.5
4	Road Design Manual: Vol. 4 Bridges & Retaining Structures Design	Part 1 – Geotechnical Investigation and Design	RDM 4.1
		Part 2 – Bridge and Culvert Design	RDM 4.2
		Part 3 – Retaining Structures Design	RDM 4.3
		Part 4 – Bridges and Structures Condition Survey	RDM 4.4
		Part 5 – Bridge Maintenance Design	RDM 4.5
5	Road Design Manual: Vol. 5 Pavement Maintenance, Rehabilitation & Overlay Design	Part 1 – Pavement Condition Survey	RDM 5.1
		Part 2 – Pavement Maintenance, Rehabilitation and Overlay Design	RDM 5.2
6	Road Design Manual: Vol. 6 Traffic Control Facilities & Communication Systems Design	Part 1 – Road Marking	RDM 6.1
		Part 2 – Traffic Signs	RDM 6.2
		Part 3 – Traffic Signals and Communication System	RDM 6.3
		Part 4 – Other Traffic Control Devices	RDM 6.4
7	Road Design Manual: Vol. 7 Road Lighting Design	Part 1 – Grid-connected Road Lighting	RDM 7.1
		Part 2 – Solar Road Lighting	RDM 7.2

This Manual must be applied sensibly and flexibly in conjunction with the skill and judgement of the designer. Compliance with the guidance given in the Manual does not relieve designers of the responsibility for establishing that their design is suitable, appropriate, safe, and adequate for the purpose stated in the project requirements.

1.2 Objective

The objective of the Manual is to guide the users to design and install proper types of traffic signs so that safe, efficient traffic management is possible.

1.3 Scope

The Manual includes four major types of road signs: regulatory, warning, informatory and road works signs. The siting, orientation, material, retro-reflectivity of signs is explained. Colour, size, letters, visibility are elaborated. The sizes of regulatory signs to be used are presented in terms of class of road and approach speed of vehicles. Warrants for installing two of the most important signs – Stop and Give Way are presented along with siting details. Mandatory and prohibitive regulatory signs are presented.

The sizes of warning signs to be used are also given in terms of class of road and approach speed. Colour schemes of Informatory signs for directional purpose are given. For what type of road junction directional signs are to be used is explained. Height of letters, dimensions of stroke for sign face design is explained in detail. Other Informatory signs included are signs for geographic location, facilities, diversions, tourist facilities and truck escape ramps. A separate chapter deals with signs for persons with disabilities. Signs to be used on Expressways are presented in a separate chapter. Sign face design for Expressway signs are explained along with placing of signs. A chapter on signs for urban roads discusses the signs to be used in urban areas. Road work signs are presented in detail. Traffic control for various scenario of road work is given.

1.4 Organisation of the Manual

The Manual has 10 chapters. Chapter 1 gives scope of the manual, organisation of the manual and design process. Chapter 2 gives a background of signs, hierarchy of roads, classification of signs, siting, orientation, material, post, mounting, maintenance and also on testing of signs. Chapter 3 is about regulatory signs- mandatory, prohibitive and priority. Chapter 4 describes the warning signs. Informatory signs are given in chapter 5 and other Informatory signs like facilities signs are given in chapter 6. Chapter 5 also details design of letters, block, stroke width, border to design sign faces. Signs for persons with disabilities are given in chapter 7. Chapter 8 is on Expressway signs and guidelines for urban roads are detailed in chapter 9. Chapter 10 elaborates signs for roadworks.

This manual is to be read in conjunction with RDM Vol 1, Part 3: Geometric Design for Highways and Rural and Urban Roads, RDM Vol 6, Part 1: Road Marking and RDM Vol 6, Part 3: Traffic Signals and Communication Systems.

1.5 Design Process

The process for design and installation of road sign is:

1. Study the need for sign and the type of sign to be used based on purpose.
2. Choose the size of sign based on the class of road and the approach speed of vehicles.
3. Determine the siting location based on the approach speed of vehicles, space available and road geometry.
4. Prepare plans, specification, Bill of Quantities for installation.

2 General

2.1. Background

The Vienna Convention on Road Signs and Signals, 1968 and subsequent amendments of 1993 and 2006 helped to achieve a degree of uniformity in traffic signing in various countries. Most countries all over the world follow the signing convention, to a great extent, to avoid confusion of drivers.

The legal framework for traffic signage (including signs, devices, notices, and markings) is contained in the Kenya Traffic Act, Chapter 403, revised 2015.

The Ministry of Roads and Transport (MoR) has overall responsibility for the planning and supervision of roadworks and signage. Under the MoR, various authorities such as KeNHA, KeRRA or KURA are responsible for different classes of roads.

Clear and effective signing is essential for the efficient operation of the road network, for the enforcement of traffic regulations, and for enhancing road safety.

Signing includes signs on posts, carriageway markings, and traffic signals. All of these must give road users their message clearly and at the correct time and must be uniform throughout the country.

This manual provides a comprehensive guide to the types of signs to be used in the country.

This manual also sets out the technical standards to be followed in the provision of the various types of signs, including temporary signs for use in connection with road works, and those used in emergencies by the Police.

To be effective, a road sign should meet five basic requirements:

1. Fulfil a need.
2. Command attention.
3. Convey a clear and simple meaning.
4. Command respect from road users.
5. Give adequate time for response.

Design, placement, operation, maintenance, and uniformity are aspects that should be carefully considered in order to maximise the ability of a road sign to meet these five basic requirements.

2.2 The Road Hierarchy

The foreground and background colours of directional information signs depend on the classification of the road and the destination.

The present road classifications according to the Kenya Roads Acts 2007 (2012 edition) are given [Table 2.1](#).

Table 2.1 Road Classification in Kenya

Class	Description
Part A National Roads	
Class A	International trunk roads linking centres of international importance and crossing international boundaries or terminating at international ports.
Class B	National trunk roads linking nationally important centres.
Class C	Primary roads linking provincially important centres to each other or to two higher class roads.
Part B Rural Roads	
Class D	Secondary roads linking locally important centres to each other, to more important centres or to higher class roads.
Class E	Any link to minor centres.
Class F	Forest roads.
Class G	Roads serving government institutions
Class K	Roads accessing coffee (kahawa) growing areas.
Class L	Roads accessing settlement scheme areas.
Class P	National Park roads.
Class R	Roads accessing rural areas.
Class S	Roads accessing sugar growing areas.
Class T	Roads accessing tea growing areas.
Class U	Unclassified rural roads, including mining roads, etc.
Class W	Roads accessing wheat growing areas. Shopping streets.
Part C Urban Roads	
Class UA	Urban Arterials.
Class UC	Urban collectors including primary distributors, district distributors.
Class UL	Urban local roads including minor distributors, local streets, residential stand accesses, commercial and industrial stand accesses.

As per functional classification, for rural roads, class A and B roads can be considered as major and minor arterials. Class C and D roads can be considered as major and minor collectors. Class E, F, G and the rest can be considered as local roads.

For urban roads, class UA roads are arterial roads. UC roads are collectors and distributors, and UL roads are local roads.

2.3 Classification of Road Signs

The four types of signs are:

1. Regulatory.
2. Warning.
3. Informatory.
4. Road works.

2.3.1 Regulatory Signs

Regulatory traffic signs indicate the existence of road regulations or implement such regulations, or both. Regulatory signs may be either mandatory, prohibitory or priority.

The mandatory signs give instructions to drivers about what they must do. Examples are: KEEP LEFT, STOP, GIVE WAY. Most mandatory signs such as the KEEP LEFT sign are circular with white symbols and border on a blue background. Others, such as the GIVE WAY sign have black lettering or symbols on a white background, with a red border.

The prohibitory signs give instructions to drivers about what they must not do. Examples are: NO ENTRY. Speed restriction signs are further examples. Priority signs tell the driver who has priority or right of way. Examples are Give Way and Stop signs.

2.3.2 Warning Signs

Warning signs are triangular in shape with a red border and black symbols on a white background. They are used to caution the road users to potential hazards on or adjacent to the roadway. These signs indicate a need for caution and may require a reduction in speed or some other manoeuvre. Some examples of these signs are Hairpin Bend, Narrow Bridge, Gap in Median, School Ahead etc.

2.3.3 Informatory Signs

Informatory signs give road users information about routes, places, and facilities. Most informatory signs are rectangular, but direction signs have one end pointed.

The colours used on informatory signs depend on the class of road:

1. Signs for Expressway have white lettering, symbols, and borders on a blue background.
2. Signs indicating International and National Trunk Roads have white lettering, symbols, and borders on a green background.
3. Signs indicating other routes have black lettering, symbols, and borders on a white background.

In addition, facilities of interest to tourists are shown with white lettering, symbols, and borders on a brown background.

2.3.4 Road Works Signs

These signs indicate that road works are in progress. These signs are either rectangular or triangular in shape and are coloured yellow with a black border and symbols. These signs indicate that extra care is necessary.

2.4 Siting of Signs

Road signs are the means of communication to the road users. Signs are to be placed such that the drivers can recognise them easily and in time. Normally the signs shall be placed on the left-hand side of the road. For two lane roads, the signs are placed on the left side of the carriageway, repeated on the other side of the carriageway. For multi-lane divided roads, the signs can be placed on the left side of each carriageway. In the case of hill roads, the signs are generally installed on the valley side of the road, unless traffic and road conditions warrant these to be placed on the hill side.

On all roads with or without kerb and with or without shoulder, signs are placed on the extreme edge. Signposts are at a distance of 600 mm to 3 m from the carriageway or paved shoulder edge, depending upon the local conditions. For roads with kerbs, it shall not be less than 300 mm away from the kerb line. In no case shall any part of the sign be in the way of vehicular traffic.

Gantry mounted signs should be mounted on columns, preferably 7 m or more from the nearest traffic lane. The minimum lateral offset is intended to keep it away from vehicles that may use the shoulder. If there is a situation where this lateral clearance cannot be maintained, the column of the gantry sign shall be shielded with a W-Beam crash barrier. Lesser clearances may be used on connecting roadways or ramps at interchanges. However, the clearance should not be less than 1.8 m.

On kerbed roads, the bottom edge of the lowest sign shall not be less than 2.1 m and not more than 2.5 m above the kerb. On roads without a kerb, the bottom edge of the lowest sign shall not be less than 2 m and not more than 2.5 m above the crown of the pavement. Where signs are erected above footpaths or in areas likely to be used by pedestrians, a minimum headroom of 2.1 m is to be provided.

Where a sign would be ineffective if placed on the left-hand side shoulder of a road with dual carriageway, it may be placed on the median instead. To improve the visibility of the signs on multi-lane roads, the minimum height of the lower edge of the sign should be kept at 2.5 m above the highest point of the carriageway.

On multi-lane roads, the signs may be mounted overhead, as this would ensure better visibility. Overhead signs may be used in addition to the ground signs where warranted. The following conditions determine the need of overhead signs:

1. Traffic volume at or near capacity.
2. Complex interchange design.
3. Three or more lanes in each direction.
4. Restricted visibility.
5. High speed traffic.
6. Insufficient space for ground mounted signs.
7. Large percentage of commercial vehicles.
8. Closely spaced interchanges.

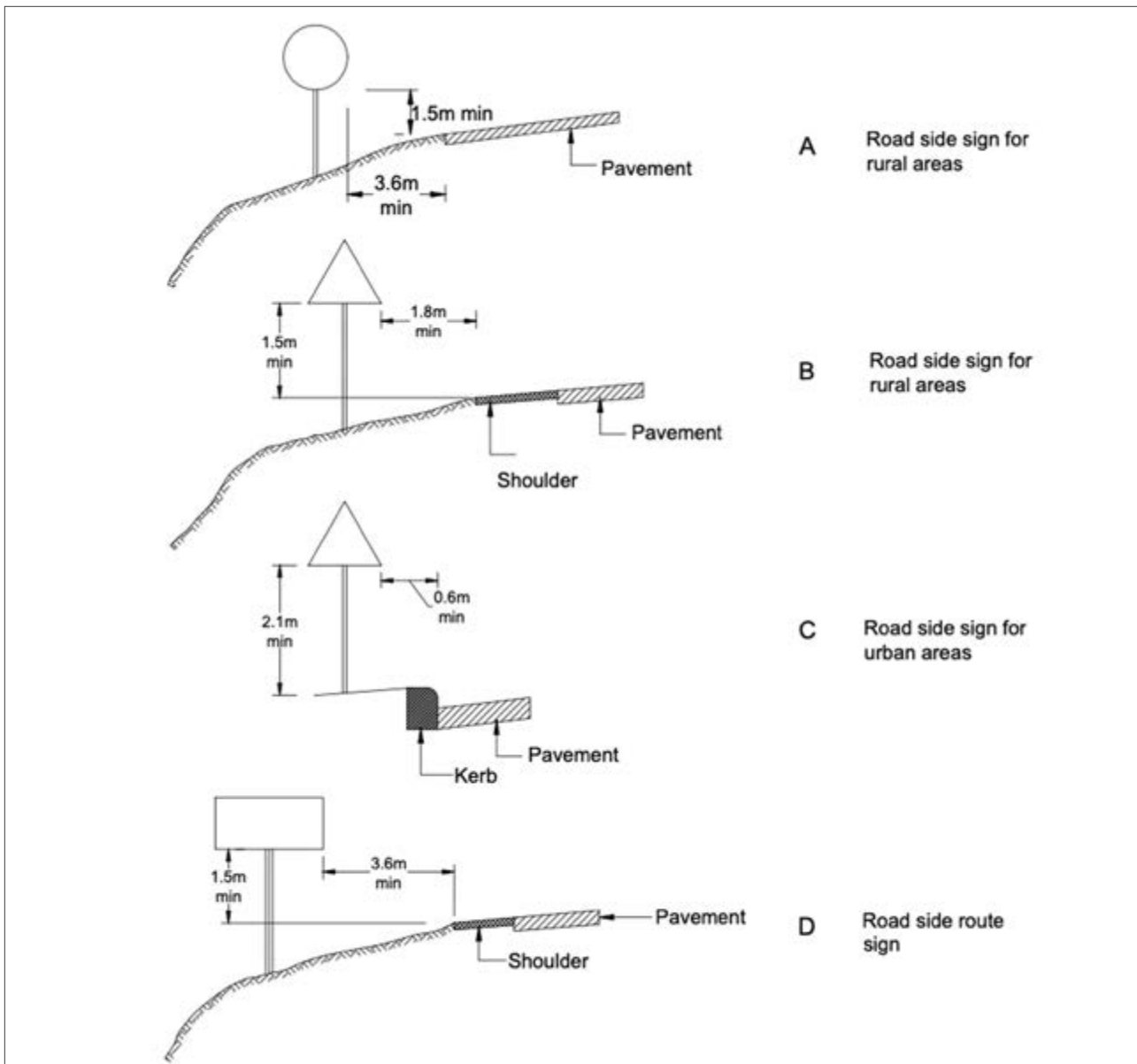
Overhead signs shall be mounted on bridge structures wherever possible. Overhead signs shall provide a vertical clearance of not less than 5.5 m. Where overhead sign supports cannot be placed at a safe lateral distance, it should be protected by a guard rail. The placement of signs, and the required heights and clearances are given in and [Figure 2.1](#) and [Figure 2.2](#).

Regulatory signs on traffic islands are to be mounted so that the bottom edge is about 1 m above the paved surface. When several signs have to be placed along the same section of road, they should not obscure each other. The signs should be located at a minimum of $0.6 \times V$ m apart (where V is the 85th percentile speed in km/h).

Speed limit signs should be placed at locations where they are clearly visible. They shall be placed at the start, middle and the end of the speed limit section. Speed limit signs should be placed after major junctions within 500m. Repeater signs shall be determined by the design engineer depending on the prevailing conditions and future projections.

Signs should be placed such that those can be read and understood in 6 seconds. The distance of the placement of the sign ahead of the hazard should be decided by the Engineer based on the approach speed. Guidance on minimum visibility distance is provided in [Appendix 1](#).

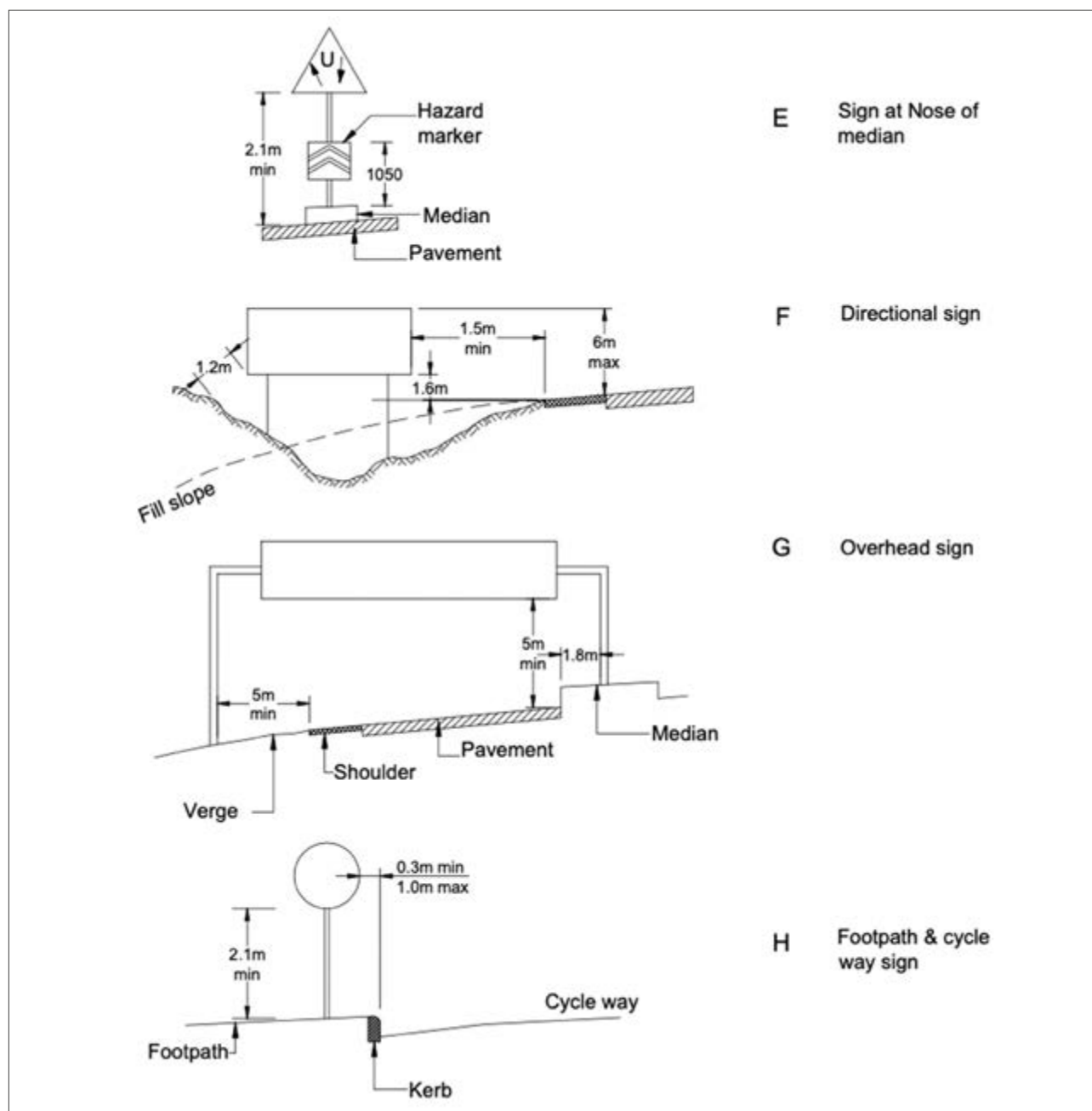
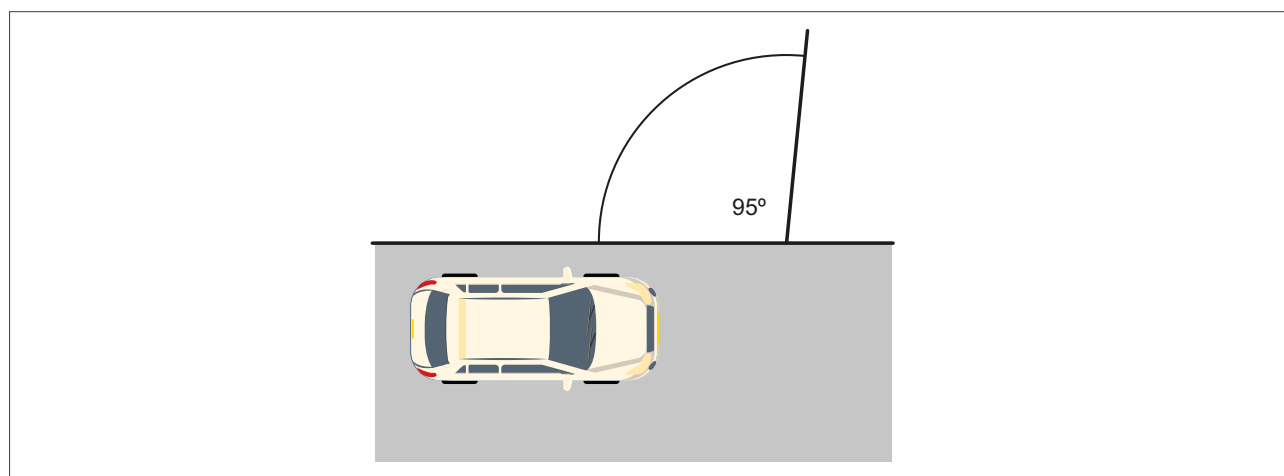
Figure 2.1 Siting of Signs with respect to the Carriageway (Height and Clearance)



2.5 Orientation of Signs

Signs shall be placed at 95° angle to the line of the approaching traffic to eliminate specular reflection. Signs relating to parking should also be placed at 95° to the carriageway so as to give better visibility.

Where light reflection from a sign face reduces legibility, it should be turned slightly away from the road as shown in [Figure 2.32](#). On horizontal curves, the sign should not be fixed normal to the carriageway. The angle of placement should be determined with regard to the approaching traffic. Sign faces are normally vertical. On gradients it may be desirable to tilt a sign forward or backward from the vertical to make it normal to the line of sight.

Figure 2.2 Siting of Signs with respect to the Carriageway (Height and Clearance)**Figure 2.3** Orientation of Sign

2.6 Material for Signs

The various materials and fabrication of road signs are given in the following section. The relevant standards/ specifications are given in [Appendix 6](#).

2.6.1 Concrete

Concrete shall be used for footing and support of signs. It shall be of M25 grade as per KS 95.

2.6.2 Reinforcing Steel

Reinforcing steel shall be used for support structures of signs. It shall conform to the requirements of KS EAS 412-2 unless otherwise specified.

2.6.3 Bolts, Nuts and Washers

Bolts, nuts and washers shall be used to fix signs to the substrate and support. It shall conform to KS ISO 6362, ASTM A193, ASTM A194 and ASTM F436.

2.6.4 Plates and Supports

Plates and support sections for the signposts shall conform to KS 572 or any other stated specification.

2.6.5 Substrate

Stainless steel substrate shall be as per ASTM A 240. Carbon steel, Mild steel plates shall be as per ASTM A 516, grade 60 & 70.

Aluminium sheeting or Aluminium Composite Material (ACM) conforming to following sub-sections:

Aluminium Sheeting

Aluminium sheets used for sign boards shall be of smooth, hard and corrosion resistant aluminium alloy conforming to KS ISO 6362.

Aluminium Composite Materials (ACM)

ACM sheets used for sign boards are a sandwiched construction with a thermoplastic core of 'Low Density Polyethylene' (LDPE) between two thick skins/sheets of aluminium. As shown in [Table 2.2](#), ACM is specified in sheets of 3 mm or 4 mm thickness, with aluminium skin thickness of 0.25 - 0.3 mm on both sides (for the 3 mm sheet) and 0.4 - 0.5 mm on both sides for the 4 mm sheet.

The retro-reflective sheeting must be applied on the top surface (i.e., the aluminium surface) with recommended surface preparation from the sheeting manufacturer. A fluorocarbon coating may be applied over the exposed surface of aluminium to ensure corrosion resistance and weather proofing and this shall conform to relevant ASTM. The mechanical properties of ACM and that of its aluminium skin shall conform to the requirements given in [Table 2.3](#) when tested in accordance with the test methods given in the table.

2.6.6 Fibre Reinforced Plastic

Fibre reinforced plastic (FRP) is a lightweight material that can be used for signs and is less prone to vandalism compared to metallic signs. The material should conform to ISO 527: 2019 standard. The thickness of FRP sheets shall be 3 – 5 mm.

Table 2.2 Specifications for Aluminium Composite Material (ACM)

S. No.	Description	Specification for 4 mm		Specification for 3 mm
		Standard Test	Acceptable Value	Acceptable Value
A. Mechanical Properties of ACM				
1	Peel off strength with retro reflective sheeting. (Drum Peel Test)	ASTM D903	Min. 4 N/mm	Min. 4 N/mm
2	Tensile strength	ASTM E638	Min. 40 N/mm ²	Min. 30 N/mm ²
3	0.2 % Proof Stress	ASTM E638	Min. 34 N/mm ²	Min. 34 N/mm ²
4	Elongation	ASTM E638	Min. 6 %	Min. 5 %
5	Flexural strength	ASTM C393	Min. 130 N/mm ²	Min. 120 N/mm ²
6	Shear strength will punch shear test	ASTM D732	Min. 18 N/mm ²	Min. 18 N/mm ²
B. Properties of Aluminium Skin				
1	Tensile strength (Rm)	ASTM E8	Min. 150 N/mm ²	Min. 130 N/mm ²
2	Modulus of elasticity	ASTM E8	Min. 70,000 N/mm ²	Min. 70,000 N/mm ²
3	Elongation	ASTM E8	A50 Min. 2 %	A50 Min. 2 %
4	0.2% Proof Stress	ASTM E8	Min. 110 N/mm ²	Min. 110 N/mm ²

2.6.7 Plate Thickness

Shoulder mounted ground signs with a maximum side dimension of 600 mm shall not be less than 1.5 mm thick when made with aluminium sheeting. It shall not be less than 3 mm thick with aluminium composite material. All other aluminium signs shall be at least 2 mm thick and aluminium composite material signs 4 mm thick. Mild steel sheets should be 5 – 10 mm thick. The thickness of the sheet shall be related to the size of the sign and its support and shall be such that the sign does not bend or deform under prevailing wind and other loads. All overhead signs made with aluminium composite material shall be a minimum of 4 mm thick to withstand wind and other loads without deformation.

2.6.8 Retro-Reflective Sheeting

The retro-reflective sheeting used on signs shall consist of white or coloured sheeting having a smooth outer surface which has the property of retro-reflection over its entire surface. It shall be weather resistant and exhibit colour fastness. It shall be new and unused and show no evidence of cracking, scaling, pitting, blistering, edge lifting or curling and shall have negligible shrinkage or expansion. The manufacturer must provide certificates that the sheeting has been tested for: coefficient of retro-reflection, daytime colour and luminance, shrinkage, flexibility, liner removal, adhesion, impact resistance, specular gloss and fungus resistance, and 3 years of outdoor weathering. Certification shall be obtained from relevant International/Government Laboratory/ Institute by the manufacturer of the sheeting. Where the certificate is obtained from an international agency, it should also be obtained from the relevant Kenyan agency within 3 years of launching the product in Kenya. Alternatively, a certificate conforming to ASTM Specification (D 4956-09) on artificial accelerated weathering requirements from a reputed laboratory in Kenya can be accepted provisionally. In such a situation, the Employer/Client, if they so desire, could seek for a performance guarantee which would be released after receipt of a certificate meeting the requirement of 3 years of outdoor weathering.

Retro-reflective sheeting is divided into three classes:

- **Class A Sheeting:** Engineering and Super Engineering Grade Sheeting as per KS 814 and ASTM D 4956-09 Type I and II.
- **Class B Sheeting:** High Intensity and High Intensity Prismatic grade sheeting as per KS 814 and ASTM D 4956-09 Type III and IV.
- **Class C Sheeting:** All Micro Prismatic grade sheets as per KS 814 and ASTM D 4956-09 Type VIII, IX and XI.

ASTM D4956 Type XI is to be adopted for permanent works.

2.6.9 Selection of Retro-Reflective Sheeting

'Retro-reflectivity' is the extent to which a surface, material, or device can reflect a large proportion of light back to the vicinity of the original light source. In effect, a retro-reflective traffic sign receives light from a car's headlamps and reflects a 'cone of light' back to the car, allowing the driver to see the sign.

The 'coefficient of retro-reflection' is a measure of the amount of light returned and is determined from laboratory tests that use certain specified entrance and observation angles for the light source. By assessing a material's retro-reflectivity using the same angles and light source, the performance of different materials can be compared.

Headlights emit light, measured in candelas. When the beam from a headlight strikes the sign surface, the amount of light meeting the surface is measured in lux. When this light is reflected towards an observer, the brightness is measured in candelas per square metre (cd/m^2).

The coefficient of retro-reflection (RA) is the ratio of this luminance to the illuminance (i.e., $\text{RA} = \text{cd}/\text{m}^2/\text{lux}$).

The coefficient of retro-reflection is a measure of the retro-reflective efficiency of a surface at a specific set of observation and entrance angles and is expressed in candelas per square metre per lux.

Micro prismatic sheeting is preferred for gantry mounted overhead signs. Type IV micro prismatic sheeting may be used for delineator posts.

Table 2.3 gives a general guideline for selection of sheeting based on the performance characteristics, class of roads, economic considerations and visibility requirements in the Kenyan context.

Table 2.3 Suggested Guideline for Usage of Retro-Reflective Sheeting

Class of Sheeting	Type of Sheeting (ASTM)	Expressway	Arterial or Trunk Road (Class A, B & UA roads)	Collector Road (Class C, D & UC roads)	Local Road (Class E to W & UL roads)
Class A	Type I	No	No	Yes	Yes
	Type II	No	No	Yes	Yes
Class B	Type III *	No	Yes	Yes	Yes
	Type IV	No	Yes	Yes	Yes
Class C	Type VIII	Yes	Yes	No	No
	Type IX	Yes	Yes	No	No
	Type XI	Yes	Yes	No	No

*Type III sheeting is available both as glass beaded and micro prismatic technology as per ASTM D4956-09. The light reflecting efficiency of glass beaded sheeting is lower than micro prismatic sheeting.

The various types of sheeting are:

Class A (Engineering grade sheeting)

- Type I engineering grade sheeting: This sheeting shall be of enclosed lens glass bead type consisting of microscopic lens elements embedded beneath the surface of a smooth, flexible, transparent, water-proof plastic, resulting in a non-exposed lens optical reflecting system.
- Type II super engineering grade sheeting: This is enclosed lens glass-bead type consisting of microscopic lens elements embedded beneath the surface of a smooth, flexible, transparent, water-proof plastic, resulting in a non-exposed lens optical reflecting system.

Class B (High intensity grade sheeting)

- Type III high intensity grade: This high intensity retro-reflective sheeting shall be of encapsulated lens type consisting of spherical glass lens, elements adhered to a synthetic resin and encapsulated by a flexible, transparent waterproof plastic having a smooth surface or as a non-metallic micro prismatic reflective material element.
- Type IV High intensity micro-prismatic grade sheeting (HIP): This sheeting shall be of high intensity retro-reflective sheeting made of micro-prismatic retro-reflective element material coated with pressure sensitive adhesive.

Class C (Micro prismatic grade sheeting)

- Type VIII Micro prismatic grade sheeting: Retro-reflective sheeting is typically manufactured as a cube corner. The reflective sheeting shall be retro-reflective sheeting made of micro prismatic retro-reflective material.
- Type IX Micro prismatic grade sheeting: Retro-reflective sheeting is typically manufactured as a cube corner. The reflective sheeting shall be retro-reflective sheeting made of micro prismatic retro-reflective material.
- Type XI Micro Prismatic Grade Sheetting: Retro-reflective sheeting is typically manufactured as a cube corner. The reflective sheeting shall be retro-reflective sheeting made of micro prismatic retro-reflective material.

Adhesives

The sheeting shall have a pressure-sensitive adhesive of the aggressive-tack type that does not require heat, solvent or other preparation for adhesion. It should be applied on a smooth clean surface, in a manner recommended by the sheeting manufacturer. The adhesive shall be protected by an easily removable liner (removable by peeling without soaking in water or other solvent) and shall be suitable for the type of material of the base plate used for the sign. The adhesive shall form a durable bond with the smooth, corrosion and weather resistant surface of the base plate such that it shall not be possible to remove the sheeting from the sign base in one piece by use of a sharp instrument.

Fabrication

The surface to be reflectorised shall be effectively prepared to receive the retro-reflective sheeting. The aluminium sheeting shall be de-greased either by acid or hot alkaline etching and all scale/dust removed to obtain a smooth plain surface before the application of retro-reflective sheeting. If the surface is rough, approved surface primer may be used. After cleaning, metal shall not be handled, except by suitable device or clean canvas gloves, between all cleaning and preparation operation and application of reflective sheeting/primer. There shall be no opportunity for metal to come in contact with grease, oil or other contaminants prior to the application of retro-reflective sheeting. Complete sheets of the material shall be used on the signs except where it is unavoidable. At splices, sheeting with pressure-sensitive adhesives shall be overlapped not less than 5 mm. Where screen printing with transparent colours is proposed, only butt joint shall be used. The material shall cover the sign surface evenly and shall be free from twists, cracks and folds. Cut-outs to produce legends and borders shall be bonded with the sheeting in the manner specified by the manufacturer.

2.6.10 Messages / Borders

The messages (legends, letters, numerals, etc.) and borders shall either be printed or cut-out from durable transparent overlay or cut-out from the same type of reflective sheeting for the warning and regulatory sign boards. Printing shall be processed and finished with materials and in a manner specified by the sheeting manufacturer. For the informatory and other sign boards, the messages (legends, letters, numerals etc.) and borders shall be cut-out from durable transparent overlay film or cut-out from the same reflective sheeting only. Cut-outs shall be from durable transparent overlay materials as specified by the sheeting manufacturer and shall be bonded with the sheeting in the manner specified by the manufacturer. Cut-out messages and borders, wherever used, shall be either made out of retro-reflective sheeting or made out of durable transparent overlay except those in black which shall be of non-reflective sheeting or opaque in the case of durable transparent overlay. Creating coloured areas by means of screen-printing with ink shall not be permitted. Printing shall follow the procedures set out in KS 814.

2.6.11 Warranty and Durability

The retro-reflective sheeting types 'A', 'B' and 'C' shall be covered respectively under 5, 7 and 10-year warranty issued for field performance, including the printed areas and cut-out sheeting and cut-out durable transparent overlay film. The contract shall indicate the minimum retro-reflectivity of the signs at the end of the warranty period.

The parts of the sign, material to be used and the standard / specification are given in [Appendix 6](#).

2.7 Posts and Mountings for Signs

The traffic signs shall be mounted on support posts, which may be of galvanised iron (GI) pipes conforming to KS 572, for either circular hollow section or square hollow section. In the case that signs are supported on two or more posts, bracing may also be provided. Signposts, their foundations and sign mountings shall be so constructed as to hold these in a proper and permanent position against the normal storm wind loads or displacement by vandalism. Normally, signs with an area up to 0.9 square metres shall be mounted on a single post, and for greater area, two or more supports shall be provided. Sign supports may be of mild steel, reinforced concrete (M25) or galvanised iron (GI Post) sections. Ends shall be firmly fixed to the ground by means of a properly designed foundation. The work of foundation conforms to relevant specifications.

All components of signs and supports, other than the reflective portion of GI posts, shall be thoroughly descaled, cleaned, primed and painted with two coats of epoxy paint. Any part of Mild Steel (MS) post below ground shall be painted with three coats of red paint for corrosion resistance.

The signs shall be fixed to the posts by welding in the case of steel posts and by bolts and washers of suitable size in the case of reinforced concrete or GI posts. After the nuts have been tightened, the tails of the bolts shall be furred over with a hammer to prevent removal.

For overhead signs, the support system should be properly designed, to safely sustain the dead load, live load and wind load. The overhead signs shall be designed to withstand a wind loading of 150 kg/m² normal to the face of the sign and 30 kg/ m² transverse to the face of the sign. In addition to the dead load of the structure, walkway loading of 250kg concentrated live load shall also be considered for the design of the overhead sign structure.

2.8 Shape and Colour of Signs

The general convention for shapes of signs is:

1. Triangular sign with red border on white background is used as a warning sign. It denotes a hazard ahead and warns the user.
2. Round sign with red border on white background is used for regulatory and prohibitive signs. Examples are signs for no entry or no right turn. These types of signs are also used for showing the posted speed limit. Octagonal sign with red border on white background is a unique type of sign, used for stop control.
3. Square shaped signs are used for giving information. Rectangular shaped signs are used for giving direction and for general information about an area.

Signs shall be provided with retro-reflective sheeting and/or overlay film. The reverse side of all signs shall be painted grey. The signposts shall be painted in 250 mm wide bands, alternately black and white. The lowest band next to the ground shall be in black.

The colour of the material shall be located within the area defined by the chromaticity coordinates in Table 2.4 and comply with the luminance factor given in Table 2.5 when measured as per ASTM D: 4956-09. Chromaticity is the objective specification of the quality of a colour regardless of its luminance, that is, as determined by its hue and colourfulness (or saturation/chroma, or intensity).

The four pairs of chromaticity coordinates determine the acceptable colour in terms of the CIE 1931 Standard Colourimetric System measured with CIE Standard Illuminant D65. The saturation limit of green and blue may extend to the border of the CIE chromaticity locus for spectral colours.

The colours shall be durable and uniform in acceptable hue when viewed in daylight or under normal headlights at night.

Colour pattern for direction informatory signs is given in Table 2.6.

2.9 Size of Signs

As a general rule, there shall be three sizes of signs: small (600 mm), normal (750 mm) and large (900 mm), for regulatory and warning signs. For Expressways, as a general rule, signs shall be of 1200 mm.

General dimensions of different categories of sign are given in the respective sections.

2.10 Visibility of Signs

In order to make signs more visible and legible at night, they shall be lighted or provided with reflective material. Care should, however, be taken that this does not result in road users becoming dazzled.

2.11 Size of Letters

The legibility of traffic signs is of prime importance. Target value is an assessment of how well a road user can identify that there is a road sign ahead. It depends on the colour and size of the sign. The use of symbols only is an effective sign layout. Where lettering has to be used, the message should be very short.

The factors that determine the distance over which a sign message is legible are:

1. The size of lettering or logo.
2. The number of messages to be scanned.
3. The lateral distance of the sign from the edge of the road.
4. The speed of the approaching vehicle.

Table 2.4 Chromaticity Coordinates (Daytime)

Colour	1		2		3		4	
	x	y	x	y	x	y	x	Y
White	0.303	0.300	0.368	0.366	0.340	0.393	0.274	0.329
Yellow	0.498	0.412	0.557	0.442	0.479	0.520	0.438	0.472
Green	0.026	0.399	0.166	0.364	0.286	0.446	0.207	0.771
Red	0.648	0.351	0.735	0.265	0.629	0.281	0.565	0.346
Blue	0.140	0.035	0.244	0.210	0.190	0.255	0.065	0.216
Orange	0.558	0.352	0.636	0.364	0.570	0.429	0.506	0.404
Brown	0.430	0.340	0.610	0.390	0.550	0.450	0.430	0.390
Fluorescent Yellow-Green	0.387	0.610	0.369	0.546	0.428	0.496	0.460	0.540
Fluorescent Yellow	0.479	0.520	0.446	0.483	0.512	0.421	0.557	0.442
Fluorescent Orange	0.583	0.416	0.535	0.400	0.595	0.351	0.645	0.355

Table 2.5 Daytime Luminance Factor (Y %)

Colour	Non-Metallic Portion		Metallic Portion	
	Min	Max	Min	Max
White	27	—	15	—
Yellow	15	45	12	30
Green	3.0	12	2.5	11
Red	2.5	15	2.5	11
Blue	1.0	10	1.0	10
Orange	10	30	7.0	25
Brown	1.0	9.0	1.0	9.0
Fluorescent Yellow-Green	60	—	—	—
Fluorescent Yellow	40	—	—	—
Fluorescent Orange	20	—	—	—

Table 2.6 Colour Scheme for Directional Signs

Class of Road	Background Colour	Colour of Text, Borders, Arrows and Chevrons	Colour of Route Number
Expressways	Blue	White	White
Class A, B & UA	Green	White	Yellow
Class C, D & UC	White	Black	Black
Class E to W & UL	White	Black	Black

2.12 Maintenance of Signs

Signs must be maintained to preserve their effectiveness. It is essential that periodic inspections of signs be carried out in order to ensure their prompt repair or replacement. Illuminated or reflectorised signs should be regularly inspected after dark.

All signs require regular cleaning. Materials used on the sign-face should be cleaned in accordance with guidelines issued by the manufacturer.

2.13 Testing for Retro-reflectivity and Quality

The retro-reflectivity of signs deteriorates over time. Measurements need to be taken annually or every other year to check that retro-reflectivity is within acceptable limits. Hand-held devices for testing are available in the market.

Measurements are defined in ASTM D 4956-09 (0.2-degree observation angle and 4.0° entrance angle). The measuring devices can store data, are easy to use and have rechargeable batteries.

Refer to Part 1 of EN 12899 for testing of complete sign assemblies, signs, plates and other major components (retroreflective sheeting, supports and luminaires).

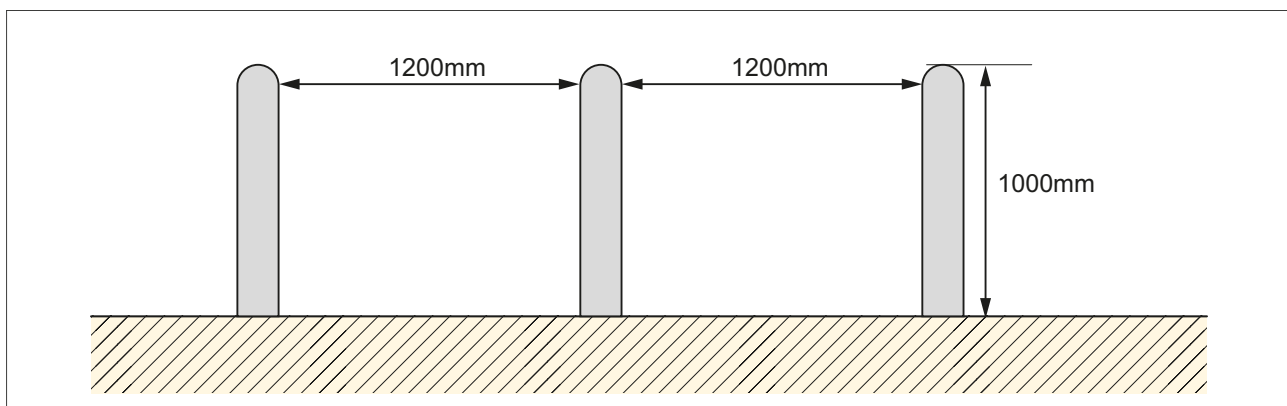
Tests for FRP can be referred to in ASTM D3039 for tensile test, ASTM D 256 for Impact test and ASTM D 790 for flexure test.

2.14 Bollards

Bollards are solid vertical posts that provide protection from unwanted impacts, such as vehicle collisions. These can be installed at the side of busy roads to protect pedestrians against kerb-mounting vehicles, as well as in front of commercial and residential properties to restrict vehicle access. These can also be used to direct traffic. Common use of bollards is to keep pedestrians away from vehicles on busy roads. Bollards should be placed such that they do not hinder wheelchair users and not obstruct movement of persons with visual limitations. Bollards of minimum 1000 mm height should be identified by using contrasting colours with provision of reflective marking / tapes. Bollards are also used to prevent cycle and motorcycle movement on pedestrian ways, by providing a restrictive gap of 1200 mm. [Figure 2.4](#) shows placement of bollards.

For details of placement, refer to RDM Volume 1, Part 3: Geometric Design of Highways, Rural and Urban Roads.

Figure 2.4 Height and Gap Between Bollards



3 Regulatory Signs

3.1 Introduction

Regulatory traffic signs indicate road regulation or implement a regulation, or both. They also indicate provision of an enactment relating to road traffic.

Regulatory signs are either priority, mandatory or prohibitory signs.

Priority signs are used to indicate which vehicular movement has priority. Examples are STOP and GIVE WAY signs. STOP signs are octagonal in shape with a red background, with a white border and white lettering. GIVE WAY signs have black lettering on a white background, with a red border.

Mandatory signs must be obeyed by the drivers. Examples are signs for KEEP LEFT, KEEP RIGHT. These signs are circular with a white symbol and border, on a blue background.

Prohibitory signs give instructions to drivers on prohibited moves. Examples are signs banning turns or entry. Speed restriction signs, clearway signs and signs for weight restrictions are further examples. Most are circular, have a red border, and a cancellation bar.

All recommended regulatory signs given in this manual start with R and are followed by a number. For example, STOP sign is R1. Mandatory signs start with M and Prohibitive signs start with P.

3.2 Priority Signs

Priority signs indicate the order in which vehicles should pass. The general principle that establishes who has the right to go first is called 'right of way' or 'priority'. It establishes who has the right to use the conflicting part of the road and who has to wait until the other does so. The vehicle that does not need to wait is said to 'have the right of way' or to 'have priority.' Examples of Priority Signs are STOP sign and GIVE WAY sign.

3.2.1 STOP Sign

The STOP sign (R1) shown in Figure 3.1, one imposes a requirement on all approaching traffic to stop. It is generally provided in association with a STOP line (refer to Road Markings Manual). Vehicles must stop at or ahead of a STOP line. In the absence of a STOP line, it should stop at or ahead of a STOP sign.

At priority intersections, traffic on the minor road is expected to give way to traffic on the major road. Priority intersections are usually controlled by GIVE WAY signs. Where visibility is less than that recommended for GIVE WAY control in Table 3.3, an intersection should be controlled by a STOP sign.

Three sizes of the STOP sign are recommended: small (600 mm), normal (750 mm) and large (900 mm).

Figure 3.1 Stop, Sign R1



Table 3.1 Sizes of Stop signs

85 percentile Approach Speed (km/h)	Class of Road	Size of Sign (mm)
More than 80	A	900
65 - 80	B, C, UA	750
50 - 65	D, UC	750
30 - 50	E, UL	600
Less than 30	F - W	600

Note: Higher size can be used where greater emphasis is required by site conditions or by the crash history.

Warrants for installation

The STOP sign should be used on a minor road at its intersection with a major road where conditions are considered to be unduly hazardous due to restricted visibility, bad alignment or high crash rate. Generally, if the visibility funnel shown in [Figure 3.2](#) is not obstruction free, the STOP sign shall be installed. The visibility funnel for different situations is given in [Table 3.2](#) Visibility funnel dimensions with speed.

Figure 3.2 Visibility Funnel

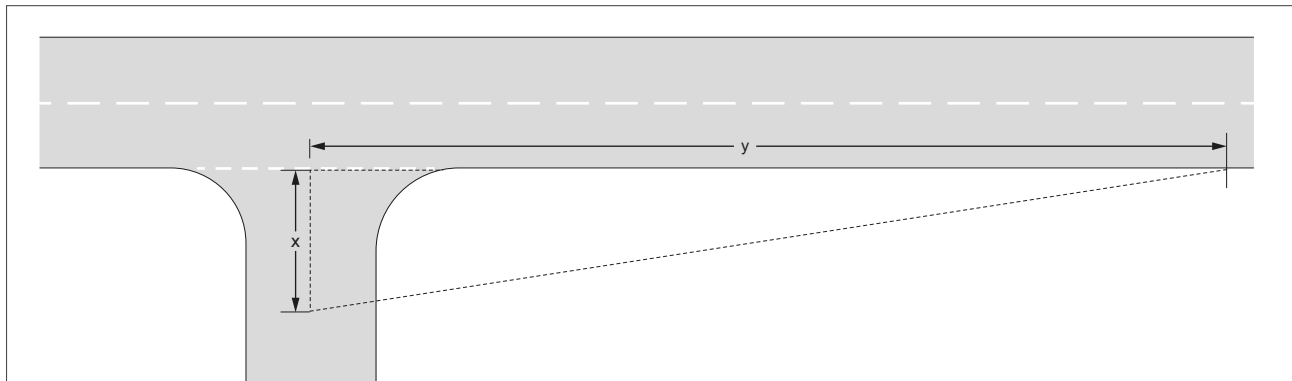


Table 3.2 Visibility Funnel Dimensions with Speed

Speed on Major Road (km/h)	Visibility Distance on Major Road (y)	Distance Measured along the Centre Line of Minor Road from Edge of Major Road (x)
Up to 30	15 m	X = 4.5 m for medium traffic and 3 m for light traffic on the minor road
31 - 50	30 m	
51 - 65	45 m	
66 - 80	70 m	
81 - 100	90 m	
More than 100	120 m	

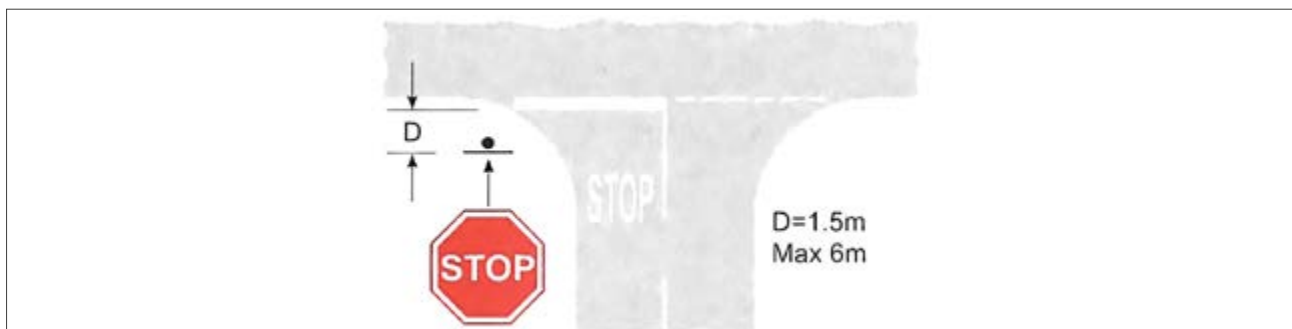
Note:

1. Visibility distance (y) is measured along the nearer edge of the major road from a point 1.05 m above the central line of the minor road (representing the driver's eye position.)
2. Distance x is measured from the edge of the major road carriageway along the centre-line of the minor road.
3. Light traffic is less than 1500 vehicles per day (vpd).

Siting the STOP Sign

The STOP sign should be sited as close as possible to the STOP line without impairing visibility along the major road. It will be placed between 1.5 and 6 m in advance of the STOP line with the former measurement being the ideal. The sign should always be sited on the left-hand side of the road but can be duplicated on the right-hand side for greater emphasis. Duplication should be normal on a wide one-way road and where a central refuge exists in the mouth of the minor road. Location of the STOP sign is shown in [Figure 3.3](#).

Figure 3.3 Location of STOP Sign



Associated Road Markings

A STOP sign should be accompanied by associated road markings. See Road Design Manual: Volume 6, Part 1 – Road Markings Manual. The absence of road markings does not, however, invalidate the requirement to stop.

3.2.2 GIVE WAY Sign

The GIVE WAY sign R2, is shown in Figure 3.4. It imposes an obligation on approaching traffic to give right of way to traffic on the major road.

At GIVE WAY sign-controlled priority intersections a driver approaching on a minor road should have sufficient visibility along the major road to make a manoeuvre without stopping.

Figure 3.4 GIVE WAY Sign R2



Table 3.3 Sizes of GIVE WAY Sign

85 percentile Approach Speed (km/h)	Class of Road	Size of Sign (mm)
More than 80	A	900
65 - 80	B, C, UA	750
50 - 65	D, UC	750
30 - 50	E, UL	600
Less than 30	F - W	600

Note: Higher size can be used where greater emphasis is required by site conditions or by the crash history.

Warrants for installation

The GIVE WAY sign shall be used on a minor road at the entrance to an intersection. It shall be used where it is necessary to assign right-of-way to the major road but where a stop is not necessary at all times. The sign shall also be used on hill roads to assign right-of-way to vehicles climbing uphill. The GIVE WAY sign, together and its associated road markings, shall be used as follows:

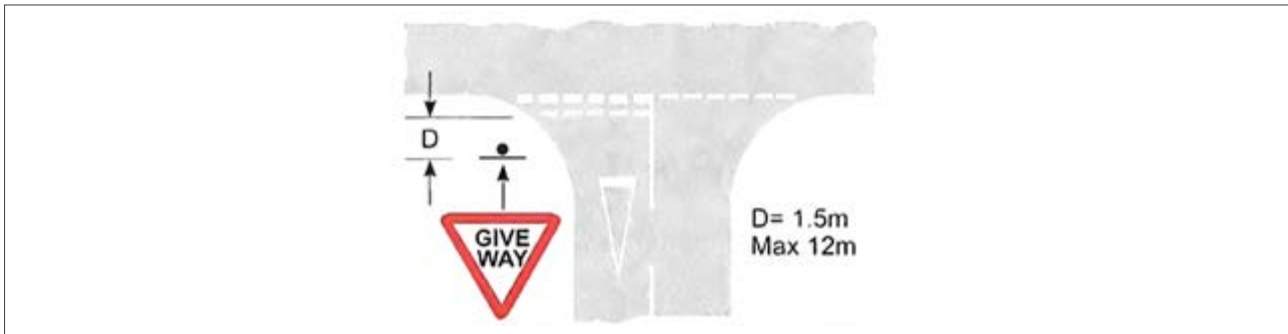
1. In non-urban areas at all junctions of minor roads with a main highway, it should be placed on the minor road.
2. In urban areas generally at the junction of a minor road with a trunk or principal road. If the minor road is a residential or local street, it is advisable to provide a STOP sign.
3. At other junctions and roundabouts where the traffic regulatory authority considers it desirable on account of traffic speeds or volumes.

The visibility funnel already depicted in Figure 3.2 shall be obstruction free while installing a GIVE WAY sign.

Siting the GIVE WAY Sign

The GIVE WAY sign should be sited as near as possible to its associated transverse road markings as long as it does not impair visibility.

The sign should be placed on the left-hand side but it can be duplicated on the right-hand side if greater emphasis is required. Duplication should be considered where there is a refuge in the mouth of the minor road and should be the norm for wide one-way streets. Location of GIVE WAY sign is shown in Figure 3.5.

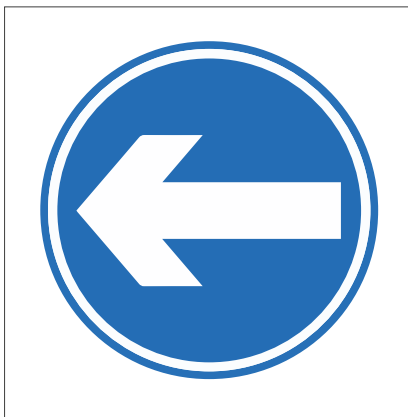
Figure 3.5 Location of GIVE WAY Sign

3.3 Mandatory Signs

The 'Turn Left' sign, **Figure 3.6**, indicates that vehicular traffic must turn left at that location. It is used for urban traffic management. It can be placed on the central island of roundabouts and on the median at junctions on dual carriageways where there is no median break.

The 'Straight Ahead' sign, **Figure 3.7**, indicates that vehicular traffic must proceed straight ahead only.

The 'Turn Right' sign, **Figure 3.8**, indicates that vehicular traffic must turn right at that location.

Figure 3.6 Left Turn, Sign M1**Figure 3.7** Straight Ahead, Sign M2**Figure 3.8** Right Turn, Sign M3

The 'Turn Left Ahead' sign, **Figure 3.9**, indicates that vehicular traffic must turn left at the junction ahead. If the arrow is reversed, the sign represents turn right ahead.

The 'Keep Left' sign, **Figure 3.10** indicates that vehicular traffic must keep to the left of the sign. The 'Keep Right' sign **Figure 3.11** indicates that vehicular traffic must keep to the right of the sign. This sign is most commonly used at road works sites. **Figure 3.12** Turn Left Ahead and **Figure 3.13** Turn Right Ahead indicates that the traffic must turn left or right at the junction ahead. **Figure 3.14** indicates that vehicular traffic may reach the same destination by passing either side of the sign. The sign should not be used at junctions or on traffic islands where traffic streams diverge to reach different destinations.

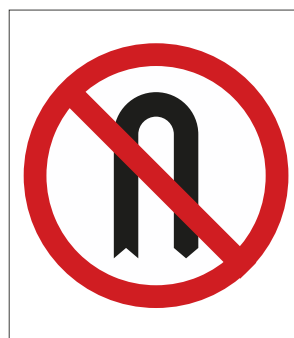
Three sizes are prescribed: small, normal and large. Small is of 600 mm for single carriageway with approach speed of 50 km/h. Normal size is of 750 mm for single carriageway with approach speed of more than 50 km/h. Large size is of 900 mm for dual carriageway.

Figure 3.9 Straight Ahead or Left Turn, (turn right when arrow is reversed) Sign M4**Figure 3.10** Keep Left, Sign M5**Figure 3.11** Keep Right, Sign M6**Figure 3.12** Turn Left Ahead, Sign M7**Figure 3.13** Turn Right Ahead, Sign M8**Figure 3.14** Pass Either Side, Sign M9

3.4 Prohibitory Signs

The signs shown in **Figure 3.15** to **Figure 3.18** indicate that traffic is prohibited from proceeding in the direction indicated by the arrows. The NO ENTRY sign is usually associated with one-way streets and should be accompanied by appropriate road markings (Refer to Road Marking Manual, RDM Volume 6, Part 1).

The 'No U-Turn' sign is usually used in connection with dual carriageways or other roads having a central reservation. It should be mounted on the central reservation as close as practicable to the junction. It should face traffic approaching from the direction or directions to which the prohibition applies. Where there is no central island, a sign should be mounted on the left-hand side of the road and duplicated on the right-hand side.

Figure 3.15 No Entry, Sign P1**Figure 3.16** No Right Turn, Sign P2**Figure 3.17** No Left Turn, Sign P3**Figure 3.18** No U Turn, Sign P4

Three sizes are prescribed: small, normal and large. Small is of 600 mm for single carriageway with approach speed of 50 km/h. Normal size is of 750 mm for single carriageway with approach speed of more than 50 km/h. Large size is of 900 mm for dual carriageway.

3.5 Speed Limit Signs

3.5.1 Maximum Speed Limit

Maximum Speed limit signs indicate the maximum allowable speed on the road. Four types of maximum speed limit are to be used:

1. Built-up area speed limit (50 km/h).
2. Special speed limit (70 km/h or 80 km/h).
3. General speed limit (100 km/h).
4. Expressway speed limit (120 km/h).

Speed limit signs are shown in Figure 3.19 to Figure 3.22.

Figure 3.19 Built Up Area, Speed Limit 50 km/h, Sign P5



Figure 3.20 Special Area, Speed Limit 70 km/h, Sign P6



Figure 3.21 Special Area, Speed Limit 80 km/h, Sign P7



Figure 3.22 Expressway, Speed Limit 120km/h, Sign P8



The sign for the end of built-up area or end of special speed limit and commencement of general speed limit is shown in Figure 3.23. End of minimum speed limit is shown in Figure 3.24.

Figure 3.23 End of Maximum Speed Limit (other than General Speed Limit), Sign P9

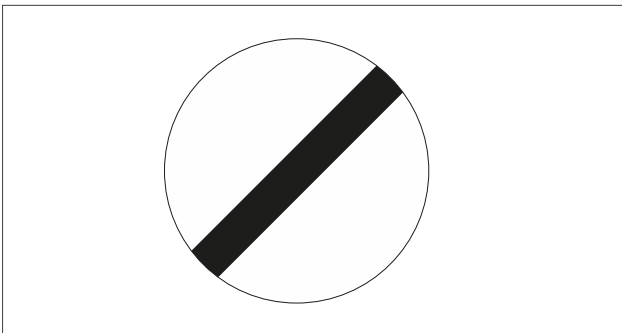


Figure 3.24 End of Minimum Speed Limit, Sign P10



Speed limit signs should be placed at locations where they are clearly visible. They shall be placed at the start, middle and the end of the speed limit section. Speed limit signs should be placed after major junctions within 500m. Repeater signs shall be determined by the design engineer depending on the prevailing conditions and future projections.

Three sizes are prescribed: small, normal and large. Small is of 600 mm for single carriageway with approach speed of 50 km/h. Normal size is of 750 mm for single carriageway with approach speed of more than 50 km/h. Large size is of 900 mm for dual carriageway.

3.6 Parking Restriction Signs

Parking restrictions are applied by a range of methods. This section mentions controls applied by the provision of roadside signs.

The sign shown in Figure 3.26 indicates that parking is allowed, subject to restrictions. Such restrictions should be shown on information plates (see Figure 3.27), which accompany the sign. The information plate and the sign should be mounted on the same post or on the same plate. Other parking related signs are shown and explained in Figure 3.28 to Figure 3.29.

Figure 3.25 Parking Permitted, Sign P11



Figure 3.26 Restriction Plate, Sign P12



Figure 3.27 Parking Prohibited, Sign P13



Figure 3.28 No Stopping Sign, Sign P14

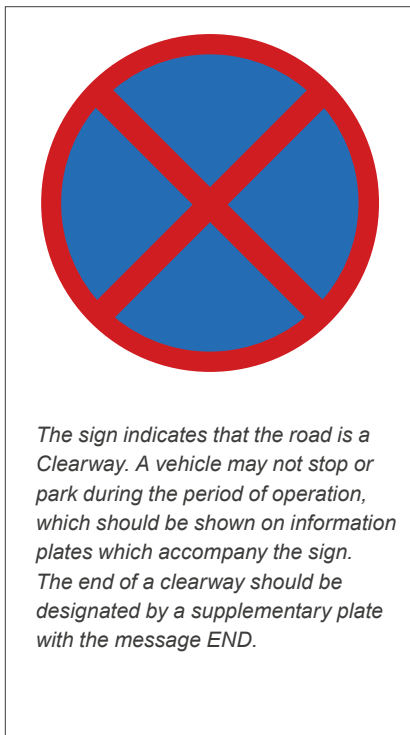


Figure 3.29 Disc Parking Plate, Sign P15



Figure 3.30 Taxi Stand, Sign P16



All signs in this section should be erected parallel to the carriageway, facing the road.

3.7 Signs for Cycle Facilities

3.7.1 Cycle Tracks

A cycle track is part of a road reserved for the use by cyclists. All other vehicles, except wheelchairs, are prohibited from entering a cycle track except for the purposes of access or emergency.

Figure 3.31 Commencement of Cycle Track, Sign P17

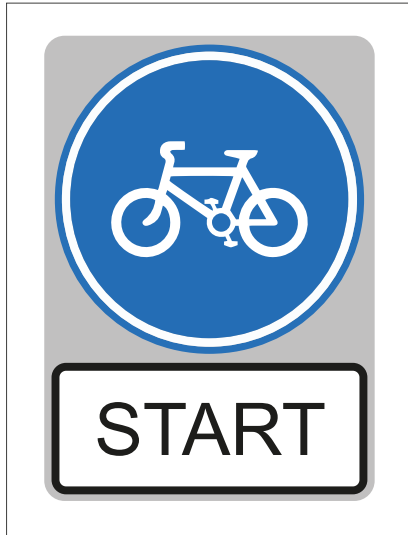


Figure 3.32 End of Cycle Track, Sign P18

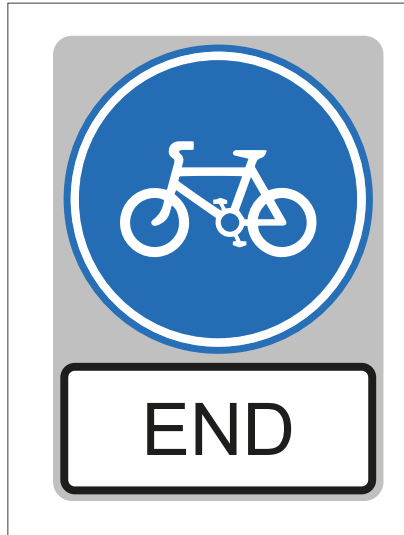


Figure 3.33 Commencement of Cycleway, Sign P19



The sign shown in **Figure 3.32** indicates the commencement point of a cycle track. **Figure 3.33** shows the sign to indicate the termination point of a cycle track. These signs are used in association with road markings for cycle tracks. Refer to Road Markings Manual (RDM Volume 6, Part 1) for appropriate markings.

3.7.2 Cycleway

A cycleway is part of a public road reserved for the exclusive use of cyclists. **Figure 3.34** indicates a cycleway.

3.7.3 Shared Pedestrian/Cycle Facilities

Where there is a shared facility for cycles and pedestrians on a cycle track or cycleway the sign shown in **Figure 3.35** should be used.

Figure 3.34 Shared Pedestrian / Cycle Track, Sign P20



3.8 School Warden's Stop Sign

The school warden's STOP sign is illustrated in Figure 3.36. The sign is double-sided. Only the 450 mm diameter size is prescribed for the sign.

Figure 3.35 School Warden's Stop, Sign P21



Figure 3.36 Pedestrianised Street, Sign P22



3.9 Pedestrianised Streets

3.9.1 Prohibition on Entry to or Parking in Certain Streets

Figure 3.37 shows the sign to be used on pedestrian only roads. It should be accompanied by information plates showing the hours/days of operation.

The 'Pedestrianised Street' sign should usually be provided on both sides of the street and orientated to face oncoming traffic. On very narrow pedestrianised streets the entrances may be marked by only one sign as long as it can be seen clearly by all approaching drivers.

3.10 Weight and Height Restrictions

The sign shown in Figure 3.37 is to be used to indicate that any vehicle with the weight specified on the sign is prohibited from entering. The sign should be located on both sides of the road and should face approaching traffic.

For bridges where restrictions apply, the signs shown in Figure 3.39 and/or Figure 3.40 should be placed at each end of the bridge or on the approach roads adjacent to the bridge.

The advance sign indicating the restriction and showing the distance should be provided on all approaches to the bridge as shown in Figure 3.41.

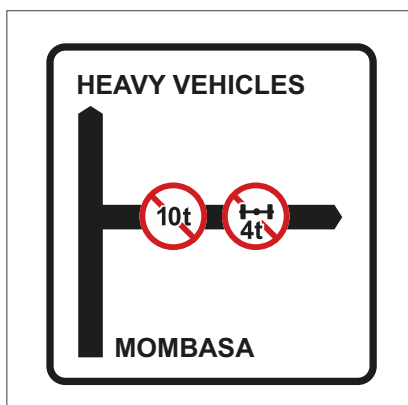
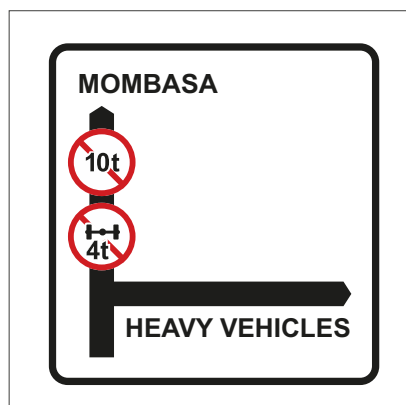
The signs shown on Figure 3.42 and Figure 3.43 should be provided at appropriate locations to indicate alternative routes for heavy vehicles.

1

2

3

Regulatory Signs

Figure 3.37 Weight Restriction, Sign P23**Figure 3.38** Vehicle Weight Restriction, Sign P24**Figure 3.39** Axle Weight Restriction, Sign P25**Figure 3.40** Advance Sign Indicating Weight Restriction Ahead, Sign P26**Figure 3.41** Indicating Route Subject to Restriction, Sign P27**Figure 3.42** Sign Indicating Route Subject to Restriction, Sign P28

The sign illustrated in **Figure 3.43** is used to restrict the height of vehicles on a road. The sign should be placed at the entrance to the road with information regarding suitable detours for vehicles with heights greater than specified on the sign.

Figure 3.43 Height Restriction, Sign P29**Figure 3.44** Parking Restriction for Heavy Vehicles, Sign P30**Figure 3.45** End of Restriction, Sign P31

The sign shown in **Figure 3.44** indicates beginning of a restriction zone on parking of vehicles exceeding a specific weight e.g., 3 tonne as shown in the figure. The sign shown in **Figure 3.45** indicates the end of the restriction zone. The sign should be placed at all exit points.

Figure 3.46 No Overtaking, Sign P32

3.11 No Overtaking Sign

The 'No Overtaking' sign shown in Figure 3.46 is used to prohibit overtaking where it is hazardous to do so. It can be used along with Road Works signs.

A 'No Overtaking' sign should be erected on both sides of the road at both ends of the affected length of road to face traffic approaching from each direction. The signs should be supplemented by the distance plate. Distances less than 1 kilometre should be shown in metres. Greater distance should be shown rounded to the nearest whole kilometre.

The siting and size of the 'No Overtaking' sign is governed by the approach. The end of the restriction should be indicated by the sign as in Figure 3.46 with a supplementary plate with the message 'END'.

3.12 Bus Lane Sign

Bus lanes reserve sections of road for buses at designated times. They are designed to minimise the delay to buses. Cycles may be allowed in the bus lane. All other road users are prohibited from using bus lanes at the specified times.

Several signs are prescribed for use with bus lanes. Most have white symbols and borders on a blue background. They should be used in conjunction with the appropriate road markings which are defined in the Road Marking Manual (RDM Volume 6, Part 1). Bus lanes may be either:

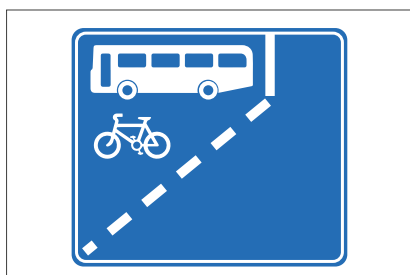
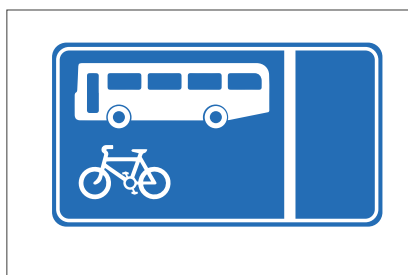
1. With-flow, which run in the same direction as the traffic using the adjacent lane
2. Contra-flow, which run in the opposite direction to traffic using the adjacent lane. No other traffic may use contra-flow lanes.

3.12.1 With-Flow Bus Lanes

A with-flow bus lane is usually provided on the nearside of the carriageway although it can be positioned on the off side if necessary.

Advance signing of a with-flow bus lane should be provided by a sign as shown in Figure 3.47. Note that the inclusion of the cycle symbol indicates that cyclists are permitted to use the bus lane. This sign should be positioned in advance of the dotted road marking (taper) indicating the start of the lane.

The regulatory sign, shown in Figure 3.48 should be sited facing oncoming traffic at the point where the prohibition begins after the taper. It should be repeated along the bus lane and should be sited after each side road. The distance between the signs should not normally exceed 300 m. When junctions are further apart, additional signs should be erected. Information plate as shown in Figure 3.49 may be placed to show the time and days of operation.

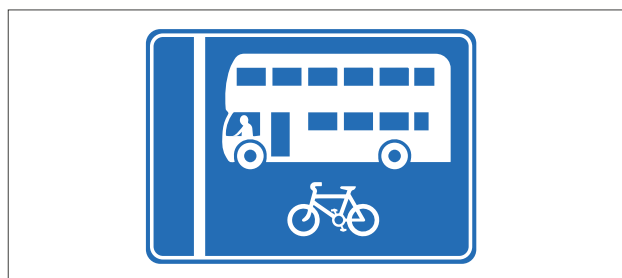
Figure 3.47 Advance Sign for With Flow Bus Lane, Sign P33**Figure 3.48** With-Flow Bus Lane, Sign P34**Figure 3.49** Information Plate, Sign P35

For offside bus lanes, the signs shown in [Figure 3.50](#) and [Figure 3.51](#) should be used instead of those shown in [Figure 3.48](#) and [Figure 3.49](#). The signing for a with-flow bus lane is shown in [Figure 3.54](#) and signing for contra-flow bus lane is shown in [Figure 3.55](#).

Figure 3.50 Advance Sign for Offside Bus Lane, Sign P36



Figure 3.51 Offside Bus Lane, Sign P37



3.12.2 Contra-Flow Bus Lanes

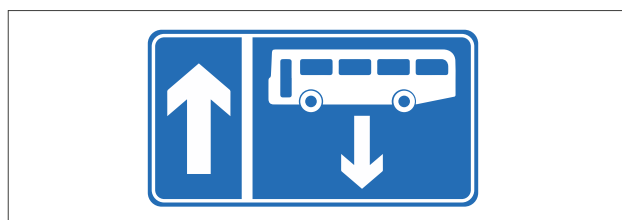
Cyclists are not permitted contra-flow bus lanes. At the start of a contra-flow lane 'No Entry' signs as defined in [section 3.4](#) should be positioned on either side of the bus lane carriageway. Black on white plates as shown in [Figure 3.52](#) should be located immediately below the 'No entry' sign to indicate that the prohibition does not apply to buses.

The regulatory sign shown in [Figure 3.53](#) should be erected facing traffic entering the street. These signs should also be repeated facing the direction of the main traffic flow after every side road junction, on whichever side the side road is located.

Figure 3.52 Information Plate, Sign P38



Figure 3.53 Contra Flow Bus Lane, Sign P39



3.12.3 Other Bus Signs

Other signs that are used in association with Bus Lanes and are shown in [Figure 3.56](#) to [Figure 3.59](#). These signs are not regulatory but inform other road users and pedestrians of the presence of the bus lane.

Every side road carrying traffic proceeding towards the bus lane should be provided with the appropriate sign as illustrated in [Figure 3.56](#) and [Figure 3.57](#).

The signs shown in [Figure 3.58](#) and [Figure 3.59](#) should be used along the bus lane to warn pedestrians to look in the direction of oncoming buses.

For each bus lane sign there are two sizes prescribed as shown in [Table 3.4](#). The smaller signs should be used on roads with 85th percentile approach speeds of 50 km/h or less. The larger signs should be used on all other roads.

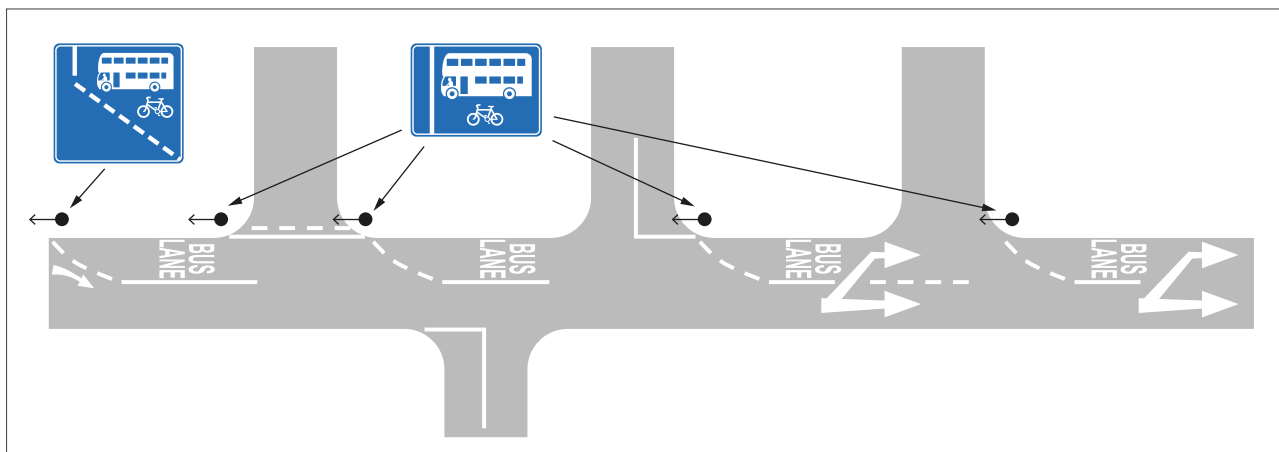
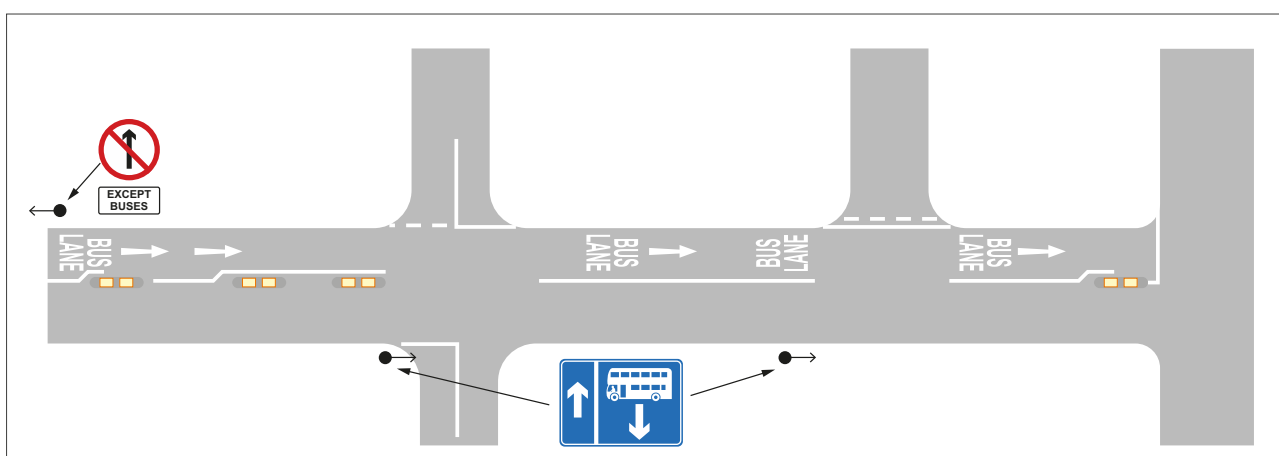
Figure 3.54 Signing for 'With-Flow' Bus Lane**Figure 3.55** Signing for 'Contra-Flow' Bus Lanes**Figure 3.56** Bus Lane, Sign P40**Figure 3.57** Bus Lane, Sign P41**Figure 3.58** Bus Lane, Look Right, Sign P42**Figure 3.59** Bus Lane, Look Left, Sign P43**Table 3.4** Sizes of Bus Lane Signs

Figure No	Sign No	Width (mm)	Height (mm)
3-47	P33	750 (1000)	750 (1000)
3-48	P34	675(800)	550 (650)
3-49	P35	750 (1000)	750 (1000)
3-50	P36	750 (1000)	750 (1000)
3-51	P37	800 (1000)	600 (700)
3-52	P38	500 (600)	600 (700)
3-53	P39	450 (600)	375 (500)
3-55	P40	450 (600)	375 (500)
3-56	P41	450 (600)	375 (500)
3-57	P42	450 (600)	375 (500)
3-58	P43	450 (600)	375 (500)

3.13 Other Prohibitive Signs

The other prohibitive signs that shall be used are shown in [Figure 3.60](#) to [Figure 3.66](#). These signs are:

1. Buses prohibited, [Figure 3.60](#), Sign P44.
2. Cars prohibited, [Figure 3.61](#), Sign P45.
3. Trucks prohibited, [Figure 3.62](#), Sign P46.
4. Tractors prohibited, [Figure 3.63](#), Sign P47.
5. Motorcycles prohibited, [Figure 3.64](#), Sign P48.
6. Cycles prohibited, [Figure 3.65](#), Sign P49.
7. Horns prohibited, [Figure 3.66](#), Sign P50.
8. Do Not Block Intersection, [Figure 3.67](#), Sign P51. This sign is to be placed along with yellow box marking, ahead of the road intersection.

These signs are to be used in appropriate places.

Figure 3.60 Buses Prohibited, Sign P44



Figure 3.61 Cars Prohibited, Sign P45



Figure 3.62 Trucks Prohibited, Sign P46



Figure 3.63 Tractors Prohibited, Sign P47



Figure 3.64 Motorcycles Prohibited, Sign P48



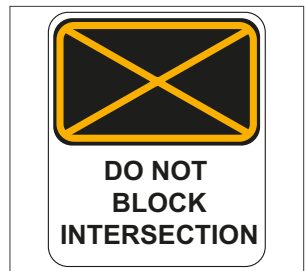
Figure 3.65 Cycles Prohibited, Sign P49



Figure 3.66 Horns Prohibited, Sign P50



Figure 3.67 Do Not Block Intersection, Sign P51



4 Warning Signs

4.1 Introduction

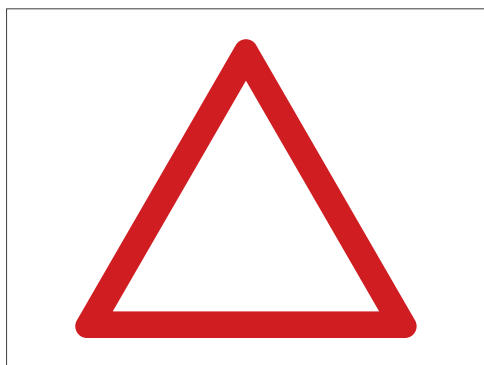
Warning signs are used to alert drivers of danger ahead. They indicate a need for extra caution and may require a consequent manoeuvre or a reduction in speed.

Most warning signs are triangular in shape with a red border encompassing a black symbol on a white background. The black symbol is usually a pictorial representation of the hazard. Supplementary plates showing a word or phrase can be mounted either separately or combined with the sign on a common backing plate.

Warning signs are prescribed in three sizes: small (600 mm), normal (750 mm) and large (1000 mm). The large size is for use on the higher speed roads and should be sited at greater distances in advance of hazards. This should allow sufficient time for the warning message to be absorbed and necessary manoeuvres to be completed before the hazard is met. The standard shape of warning signs is illustrated in Figure 4.1.

There must always be a distance clear of obstructions ahead of a sign. This is known as the clear visibility distance and it varies according to the speed of the traffic. Table 4.1 summarises all the available sign sizes and gives the appropriate siting distance and clear visibility distance for each category of road. The sizes of supplementary plates should be determined by choosing the appropriate 'x' - height size of lettering which is also indicated in Table 4.1. The text should be arranged using the same tiles that are used for directional informatory signs.

Figure 4.1 Standard Shape of Warning Signs



All warning signs proposed in this manual start with W. *Example: Sign W1 for Crossroads.*

Use of signs for road works is described in a separate chapter.

Table 4.1 Sizes of Warning Signs and their Siting Details

85 percentile Approach Speed (km/h)	Class of Road	Size of Sign (mm)	Siting Distance of Sign from Hazard (m)	Advisable Clear Visibility Distance (m)	Advisable 'x' – height of Supplementary Plates
More than 65	A, B, UA	900	1000	200 - 300	100
50 - 65	C, D, UC	750	900	100 - 200	75
Less than 50	E, UL, F - W	600	600	50 - 100	60

4.2 Junction Ahead

These signs provide advance warning of a junction and illustrate the priority route at the junction by use of varying widths of the road symbol arms.

'Junction Ahead' warning signs can be categorised into three groups showing:

1. A junction ahead with a road of lesser importance.
2. A junction ahead with a road of equal importance.
3. A junction ahead with a road of greater importance.

4.2.1 Junctions with Roads of Lesser Importance

The roads of less importance are indicated by arms of lesser widths. The five signs to be used are shown in Figure 4.2 to Figure 4.6.

Sign for Y junction is shown in Figure 4.5. The sign shown in Figure 4.6 should only be used where the distance between two consecutive junctions does not exceed 60 metres. Otherwise, two versions of the sign shown in Figure 4.3 should be used (one of which should be reversed).

Figure 4.2 Crossroads, Sign W1

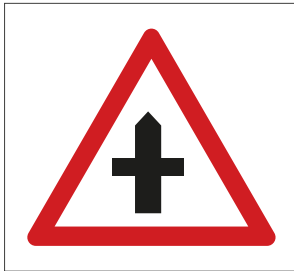


Figure 4.3 Side Road, Sign W2

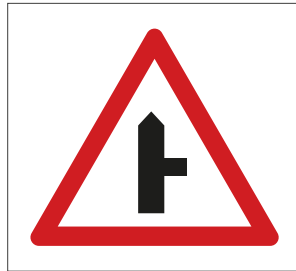


Figure 4.4 T Junction, Sign W3



Figure 4.5 Y Junction, Sign W4



Figure 4.6 Staggered Crossroads, Sign W5

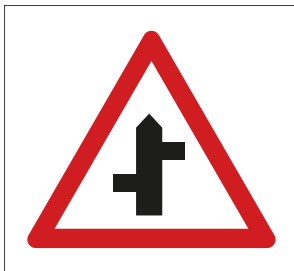


Figure 4.7 Crossroads, Sign W6

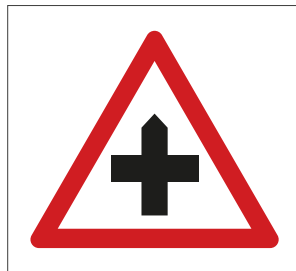


Figure 4.8 Side Road, Sign W7

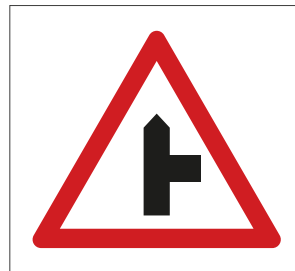


Figure 4.9 T Junction, Sign W8

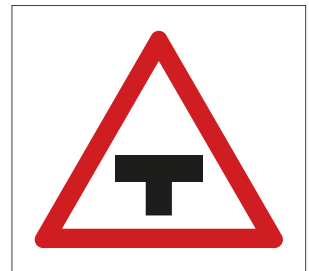


Figure 4.10 Staggered Crossroads, Sign W9

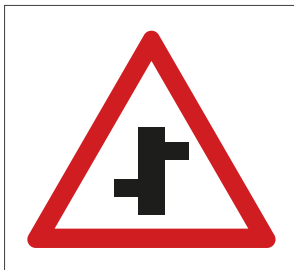


Figure 4.11 Y Junction, Sign W10

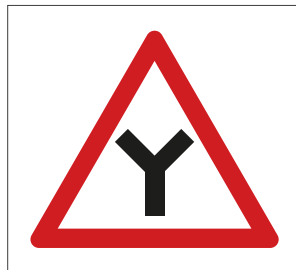


Figure 4.12 Crossroads, Sign W11



Figure 4.13 T Junction, Sign W12

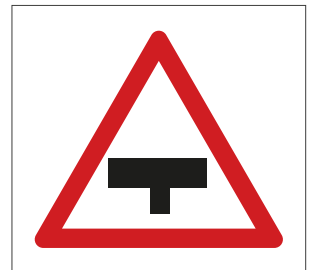


Figure 4.14 Crossroads with Dual Carriageway, Sign W13

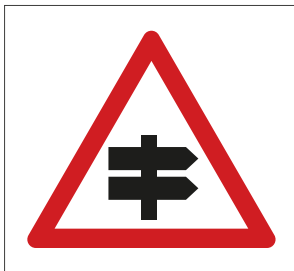


Figure 4.15 T – Junction with Dual Carriageway, Sign W14

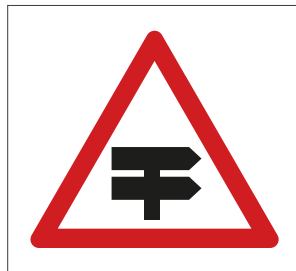


Figure 4.16 Merging Traffic, Sign W15

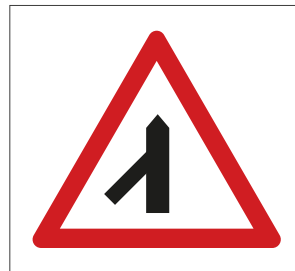


Figure 4.17 Merging / Diverging Traffic, Sign W16



4.2.2 Junctions with Roads of Equal Importance

Junctions with roads of equal importance can be indicated by the signs shown in Figure 4 7 to Figure 4 9. These show the same junction layouts as the signs in Figure 4 2 to Figure 4 6 except that all route symbol components have equal widths.

The "Staggered Crossroads Ahead" sign shown in Figure 4 10 should be used where consecutive junctions are within 60 m of each other.

4.2.3 Junctions with Roads of Greater Importance

Figure 4.12 to Figure 4.15 show the signs that are permitted for use on roads approaching a junction with a road of greater importance. Figure 4.12 and Figure 4.13 show junction layouts where the road of greater importance is a single carriageway.

When a road of lesser importance has a junction with a road of greater importance with dual carriageway, the signs to be used are shown in Figure 4.14 and Figure 4.15.

The 'Junction Ahead' signs give advance warning of a major road ahead. It is marked at the junction either by a STOP sign or GIVE WAY sign. Placement of STOP and GIVE WAY signs has been explained in Chapter 3.

4.3 Merging Traffic

Figure 4.16 illustrates the warning sign used to indicate merging traffic.

This sign should not be used where there is no need for either traffic stream to give priority. It should not be used when an additional lane is present for merging. When used, the sign should not obstruct the driver's view of vehicles entering the roadway.

When traffic merges to join the same section of carriageway and then diverges shortly afterwards the sign to be used is shown in Figure 4.17.

Figure 4.18 Roundabout Ahead, Sign W17



Figure 4.19 Mini Roundabout Ahead, Sign W18



4.4 Roundabouts

The sign shown in Figure 4.18 should be used to indicate that there is a roundabout ahead. These signs are internationally recognised for roundabouts.

On dual carriageways, two roundabout warning signs should be used in addition to the map-type sign. One should be sited on the central reserve and the other on the verge.

The size and siting of this sign are given in Table 4.1.

The roundabout warning sign should be used for proper roundabouts. This excludes advance warning of one way working round a triangular or 'Y' junction or at the entry to a large one-way system.

The sign shown in Figure 4.19 should be used to indicate a mini-roundabout.

4.5 Sharp Turns and Bends

There are four signs prescribed to indicate sharp turns or bends. They are shown in [Figure 4.20](#) to [Figure 4.23](#). Sharp turns require a reduction in speed. All the signs should be used sparingly.

The symbols should indicate whether the change of direction is to the left or right. Each sign has a variant in the reverse direction. The distinction between a sharp turn and a bend is at the discretion of the Engineer. Usually, bends are flatter than sharp curves.

The degree of danger at a sharp turn depends on five factors:

1. The speed of approach.
2. The radius of curvature.
3. The super elevation.
4. The skid resistance of the road surface.
5. Forward visibility.

Sharp turns in left and right direction are shown in [Figure 4.20](#) and [Figure 4.21](#).

The signs showing series of curves / bends in [Figure 4.22](#) and [Figure 4.23](#) should only be used where turns follow in close proximity. The black on white distance plate shown in [Figure 4.24](#) is prescribed for use with both signs where roads are hazardous for longer distances. If a supplementary plate is provided it should not be necessary to sign individual corners or bends occurring within the distance shown.

Distances below 1km should be indicated in metres. Greater distances should be indicated in kilometres.

Figure 4.20 Sharp Turn Ahead (Left / Right), Sign W19



Figure 4.21 Sharp Bend Ahead (Left / Right), Sign W20



Figure 4.22 Series of Sharp Curves Ahead, Sign W21



Figure 4.23 Series of Bends Ahead, Sign W22



Figure 4.24 Distance Plates, Sign W23



4.6 Other Warning Signs

4.6.1 Sharp Change of Direction

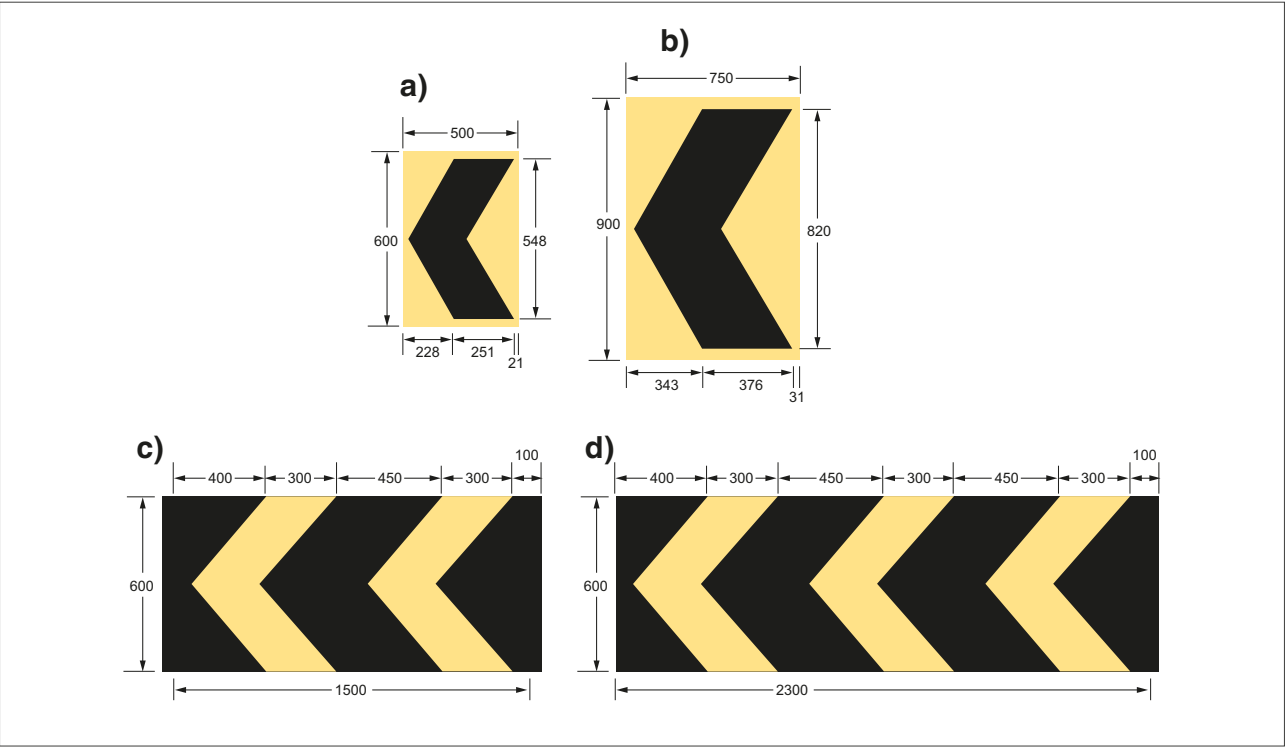
At the curved alignment of a roadway, chevron signs shall be used to inform drivers about a sharp curve. The chevron sign shall be a vertical rectangle board and shall always be installed on the outside of a turn or curve. It should be approximately at right angles to approaching traffic. Spacing of chevron signs should be such that the road user always has at least two signs in view. The spacing is given in Table 4.2. Chevron signs should be visible for a sufficient distance to provide the road user with adequate time to react to the change in alignment.

Table 4.2 Spacing of Single Chevron Signs

Curve Radius (m)	Distance Between Single Chevron (m)	
	On Curve	Before Curve
50	15	30
100	20	40
200	30	60
300	45	90
400	60	120
500	70	140
>500	80	150

Depending on the sharpness of the curve, Single Chevron (Figure 4.25 a & b), Double Chevron sign (c) and Triple Chevron signs (d) can be installed. If Single Chevron signs are to be used for roads operating at or more than 100km/h, a larger size single chevron (b) shall be used.

Figure 4.25 Chevrons, Sign W24



4.6.2 Road Narrowing

The signs available for indicating road narrowing on one or both sides are shown in [Figure 4.26](#) and [Figure 4.27](#) respectively.

The sign should be erected where a reduction in carriageway width presents a danger to the road users.

Figure 4.26 Road Narrows on One Side, Sign W25



Figure 4.27 Road Narrows on Both Sides, Sign W26



4.6.3 Divided Carriageway

The "Road Divides" sign shown in [Figure 4.28](#) is used at the beginning of a divided carriageway to indicate a central median ahead.

The sign shown in [Figure 4.29](#) is used to indicate the end of a divided carriageway. The sign may be accompanied by a plate showing the distance to the merge.

The sign shown in [Figure 4.30](#) should be erected at or as near as possible to the beginning of the two-way traffic and it may be repeated after 100 metres.

Figure 4.28 Road Divides, Sign W27



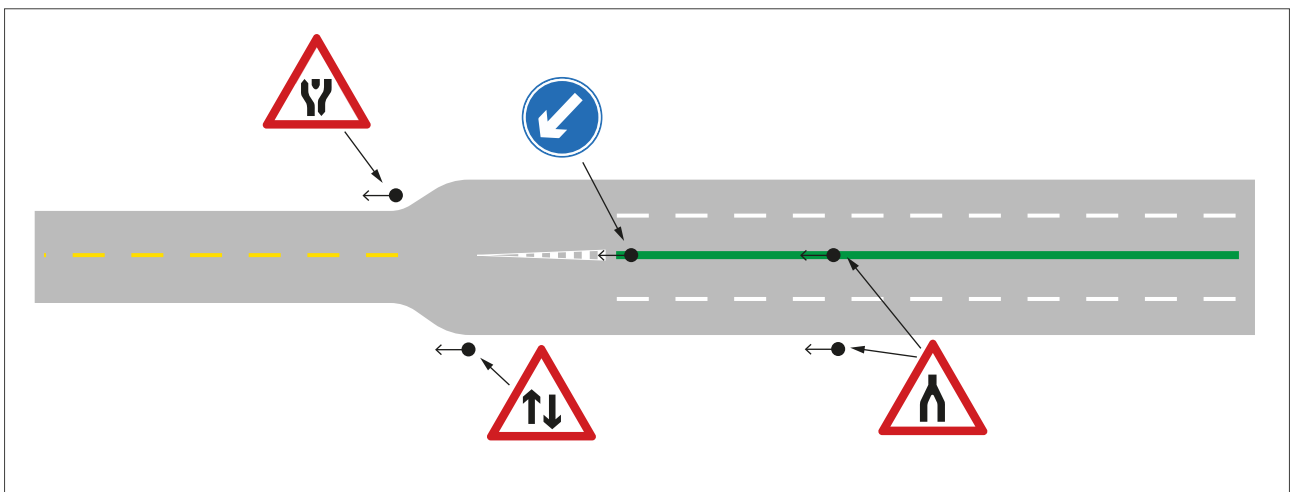
Figure 4.29 End of Divided Carriageway, Sign W28



Figure 4.30 Two Way Traffic, Sign W29



Figure 4.31 Signing at Transition To and From Divided Carriageway



4.6.4 Steep Gradient

The signs prescribed for indicating steep gradients are shown in Figure 4.32 (Descent) and Figure 4.33 (Ascent).

The descent sign shown in Figure 4.32 should be used where the gradient is greater than 5 % or 1 in 20. The sign can be repeated on the hill where the gradient steepens noticeably.

On a fairly long descent there may be portions that are steep and others that are appreciably less so and below the 5% or 1 in 20 criteria. In this case it may be advisable to treat the steeper portions as separate hills.

The ascent sign shown in Figure 4.33 should only be used where the gradient exceeds 10 % or 1 in 10.

Figure 4.32 Steep Descent Ahead, Sign W30

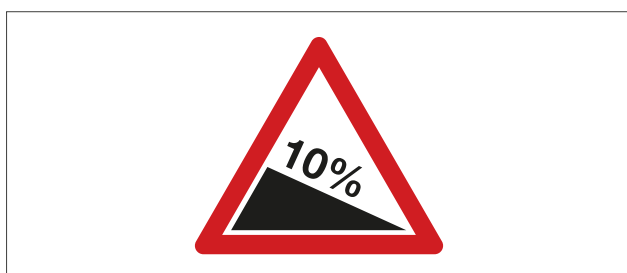


Figure 4.33 Steep Ascent Ahead, Sign W31



4.6.5 Restricted Headroom

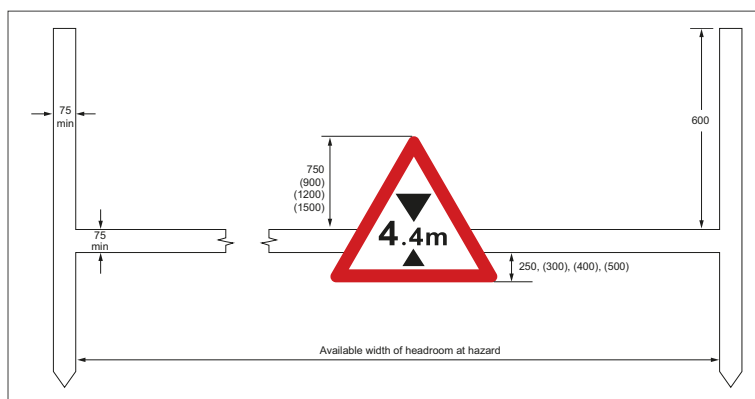
The sign illustrated in Figure 4.34 should be used to indicate restricted headroom over the carriageway.

This sign should also be mounted, where possible, on the structure itself to give additional warning. In the case of rail over road bridges this sign is fixed by the Rail Authority. If the headroom varies across the width of the carriageway (e.g., at an arch bridge) the sign should be supplemented by the 'goal post' marking shown in Figure 4.35, indicating the width over which the clearance height is available.

Figure 4.34 Restricted Headroom, Sign W32



Figure 4.35 'Goal Post' Marking at Hazard, Sign W33



In the case of rail over-bridges, the clearance height should be agreed with the Rail Authority Area Engineer.

4.6.6 Overhead Electric Cables

The sign warning of overhead electric cables is presented in [Figure 4.36](#). It must always be accompanied by the supplementary plate shown in [Figure 4.37](#) which should have black text on a white background.

These signs are associated with the overhead electrified railway cables of Urban Rail Systems, and they should be used in advance of all level crossings on these lines.

Measurements appearing on the supplementary plate should be agreed with the Rail Authority.

Figure 4.36 Overhead Electric Cables, Sign W34

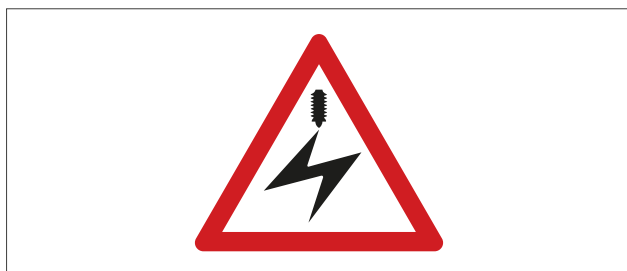


Figure 4.37 Safe Height Plate, Sign W35



4.6.7 Level Crossings

Signs warning of level crossings ahead are shown in [Figure 4.38](#) to [Figure 4.40](#).

The sign shown in [Figure 4.38](#) may also be associated with level crossings operated by 'Lights and Bells' only.

Automatic level crossings other than 'Lights and Bells' crossing are equipped with barriers (half barriers or full barriers) and lights. The signs associated with such crossings are shown in [Figure 4.40](#) to [Figure 4.42](#).

Refer to the Road Marking Manual (RDM Volume 6, Part 1) for appropriate road markings.

Before undertaking any works or alteration to sign layouts, consultation should take place with the relevant railway authority.

Figure 4.38 Level Crossing Ahead, Unguarded by Gates or Lifting Barriers, Sign W36

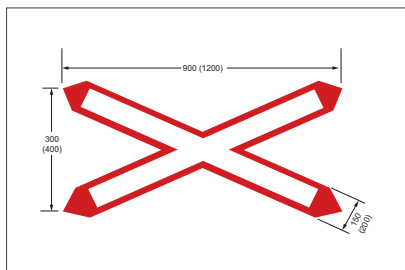


Figure 4.39 Level Crossing Ahead, Guarded by Gates or Lifting Barriers, Sign W37

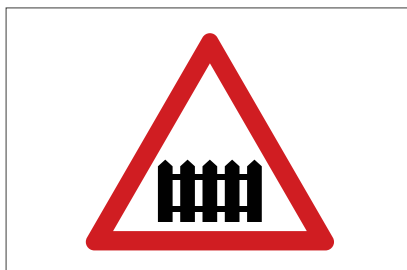
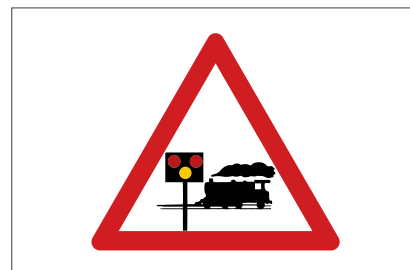
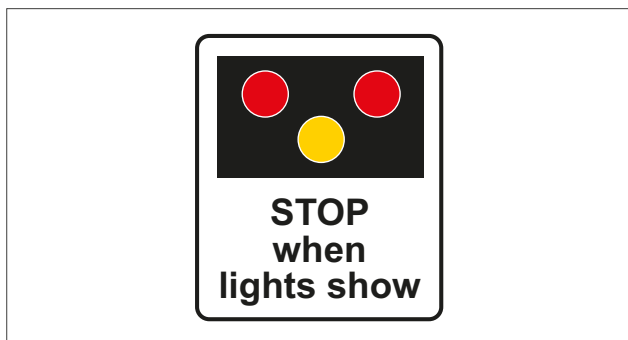
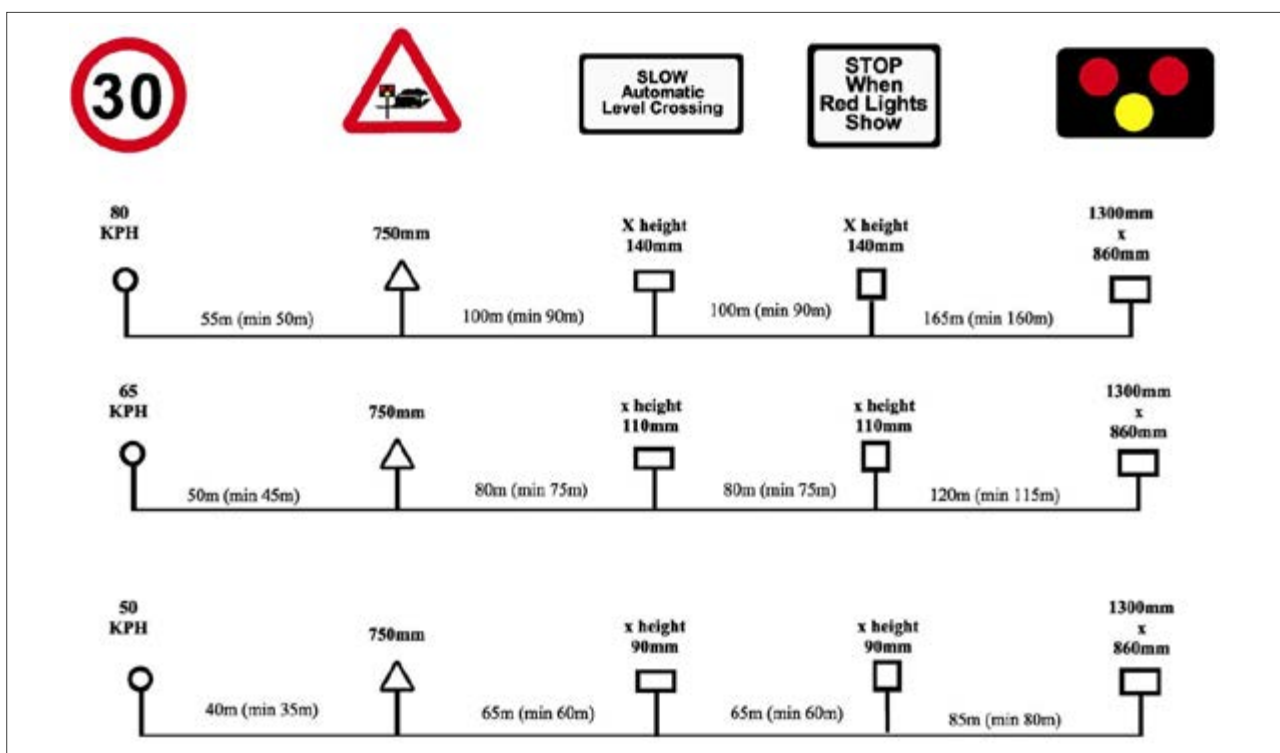


Figure 4.40 Level Crossing Ahead with Lights and Barriers, Sign W38



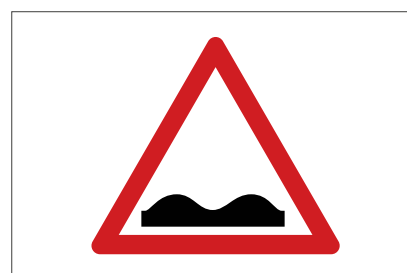
The spacing and distances of signs from the level crossing are indicated in [Figure 4.43](#). Where local site conditions preclude full compliance with these guidelines, departures may be acceptable provided that safety standards are not impaired.

Figure 4.41 Stop When Red Lights Show, Sign W39**Figure 4.42** Slow Automatic Level Crossing, Sign W40**Figure 4.43** Spacing and Distance of Signs from Level Crossing

4.6.8 Rise and Dips

Localised sharp rises or dips in the road surface are indicated by three types of warning sign which are shown in [Figure 4.44](#), [Figure 4.45](#) and [Figure 4.46](#).

The sign shown in [Figure 4.46](#) should be used to indicate a series of sharp rises and dips or an uneven road. It is also used to indicate dangers arising from irregularities in the road surface which would impair the control of vehicles at their normal speed.

Figure 4.44 Sharp Rise Ahead e.g., Hump-Back Bridge, Sign W41**Figure 4.45** Sharp Dip Ahead, Sign W42**Figure 4.46** Series of Bumps and Dips Ahead, Sign W43

4.6.9 Slippery Road

The 'Slippery Road Ahead' sign displayed in [Figure 4.47](#) is intended for use where the danger of vehicles skidding is higher than normal.

For signs on slippery roads at road works sites refer to [Chapter 10](#) on Road Works.

Figure 4.47 Slippery Road Ahead, Sign W44



Figure 4.48 Unprotected Canal, River or Quay Ahead, Sign W45



4.6.10 River, Canal or Quayside

Wherever a public road approaches a water hazard such as an unprotected river, canal or quayside, the sign shown in [Figure 4.48](#) should be used.

4.6.11 Traffic Signals

The 'Traffic Signals Ahead' sign is shown in [Figure 4.49](#). Refer to Traffic Signals Manual (RDM Volume 6, Part 3) when using the 'Traffic Signals Ahead' sign.

Figure 4.49 Traffic Signals Ahead, Sign W46



The sign should always be used on approaches to signals on roads with a speed limit of 50 km/h or more. It is not necessary on roads restricted to 30 km/h or less unless the visibility of the first primary signal is less than 45 m. The sign should not be used where visibility is only impaired by waiting vehicles, as such can be overcome by regulation.

For temporary traffic signs at road works, refer to [Chapter 10](#), Road Works.

4.6.12 Schools and Children

There are three signs that are prescribed to give advance warning of the presence of schools and children.

1. 'School Ahead' sign shown in [Figure 4.50](#).
2. 'School Children Crossing Ahead' sign shown in [Figure 4.51](#).
3. Children Crossing (in Residential areas) sign shown in [Figure 4.52](#).

Figure 4.50 School Ahead, Sign W47

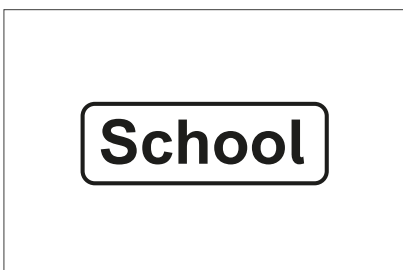


Figure 4.51 School Children Crossing Ahead, Sign W48



Figure 4.52 Children Crossing (in residential areas), Sign W49



The 'School Ahead' sign should be used to indicate that there is a school, children's playground or other place frequented by large numbers of children.

Alternatively, the sign can be used with twin amber lights mounted directly below the [Figure 4.51](#). This should be done either where an adult or junior school warden service operates or where prevailing circumstances such as vehicle speeds or highway alignment constitute a significant danger to children crossing.

The twin flashing amber lights should be on a black background. The lenses should have a minimum diameter of 200mm and flash at a rate of between 60-80 flashes a minute. Flashes in each amber light should overlap so that one light is always showing when in operation.

Where an adult or junior school warden service is in operation, the flashing lights sequence should only be activated during the warden's relevant duty periods. In other cases, the lights should only be activated for the minimum periods necessary to cover child crossing movements during those times when the children are entering or leaving the school.

Either of the signs shown in [Figure 4.50](#) and [Figure 4.51](#) should be used in the vicinity of schools but not both simultaneously. Both signs should be sited in accordance with the distances given in [Table 4.1](#). The 'School Children Crossing' sign with flashing amber lights below sign shown in [Figure 4.51](#) should not be erected within 100 m of traffic signals or pedestrian complexes where children may otherwise cross in safety.

The sign illustrated in [Figure 4.52](#) should be used to alert road users to the danger of children crossing roads in residential areas rather than schools and their associated playgrounds.

The sign should only be provided at the entrances to roads of a primarily residential character with continuous housing frontage. Signs should be provided at the entrances to housing estates from main traffic routes.

Figure 4.53 Accompanied Horses or Ponies, Sign W50

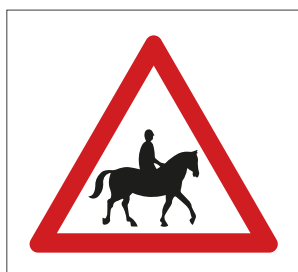


Figure 4.54 Cattle or Farm Animals, Sign W51



Figure 4.55 Sheep, Sign W52



Figure 4.56 Wild Animals, Sign W53



4.6.13 Animal Crossing

The warning signs prescribed to indicate the possibility of animals crossing the road are:

1. Accompanied horses or ponies.
2. Cattle or farm animals.
3. Sheep.
4. Deer or wild animals.

These signs are shown in [Figure 4.53](#) to [Figure 4.56](#).

The 'Accompanied Horses or Ponies' sign ([Figure 4.53](#)) should only be used where traffic volumes exceed 300 vehicles per day and where at least 5 accompanied horses or ponies regularly cross or use the stretch of road in a day.

1

2

3

4

Warning Signs

The cattle or farm animal sign (Figure 4.54) may be used at certain locations where livestock are brought across or likely to be encountered along the stretches of the road. The following should be considered before the provision of such a sign:

1. The location should be in regular use for the above purpose.
2. There should be evidence of an accident record or the location should be considered hazardous.
3. The available sight distance should be less than that required for the 85th percentile speed (see Table 4.1).

The sign depicting a sheep, Figure 4.55 may be used where sheep are likely to be found crossing or straying onto the road, e.g., commonage.

The sign depicting a deer, Figure 4.56 may be used where wild animals are likely to be encountered.

4.6.14 Cross Winds

The sign shown in Figure 4.57 should be used where crosswinds occur regularly and may cause road users to lose control of their vehicle. The sign can also be used when crosswinds constitute a risk of overturning high sided goods vehicles. Risk prone areas will include lengthy or elevated sections of viaducts spanning high sided valleys or in areas where no natural protection from crosswinds exists.

The 'Crosswinds' sign can be supplemented by the distance plate shown in Figure 4.57 to indicate the length of road affected.

4.6.15 Pedestrians

The 'Pedestrian Crossing Ahead' warning sign is illustrated in Figure 4.58. It should be used in connection with uncontrolled crossings or at other pedestrian crossings where visibility of the crossing is impaired.

Refer to the Road Marking Manual (RDM Volume 6, Part 1) for appropriate markings for pedestrian crossings.

4.6.16 Tunnel

The sign shown in Figure 4.59 may be used in advance of a tunnel. Where the height varies across the width of the tunnel, the 'goalpost' sign shown in Figure 4.35 should be used.

4.6.17 Falling Rocks

The sign shown in Figure 4.60 should be used where there is a danger of rocks falling onto a stretch of road. A supplementary plate to show the distance of the stretch of road likely to be affected can be used (See Figure 4.24).

Figure 4.57 Crosswinds, Sign W54

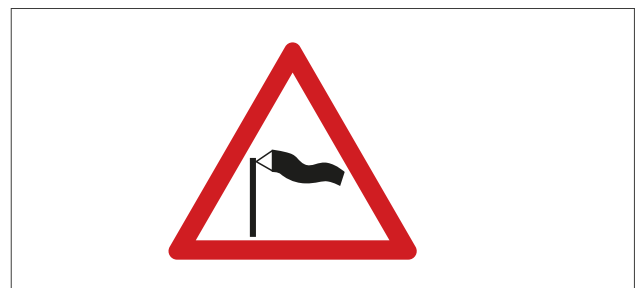


Figure 4.58 Pedestrian Crossing Ahead, Sign W55



Figure 4.59 Tunnel Ahead, Sign W56



Figure 4.60 Falling Rocks, Sign W57



4.6.18 Low Flying Aircraft

The 'Low-Flying Aircraft' sign shown in Figure 4.61 is for use on roads skirting or in the vicinity of airfields. It is used to warn of low flying aircraft or sudden aircraft noise which may startle road users.

4.6.19 Drive on the Left

The sign shown in Figure 4.62 should be used at exits from ports and airports. It may also be used where hazards may exist on exits from long-stay tourist camps and resorts. The sign should be duplicated on the right-hand side of the road.

Figure 4.61 Low Flying Aircraft, Sign W58



Figure 4.62 Keep Left, Sign W59



4.7 Roadside Delineators

Two types of delineators are available: flexible and rigid. It is a matter for the road authority to determine which type should be used in terms of suitability and value for money.

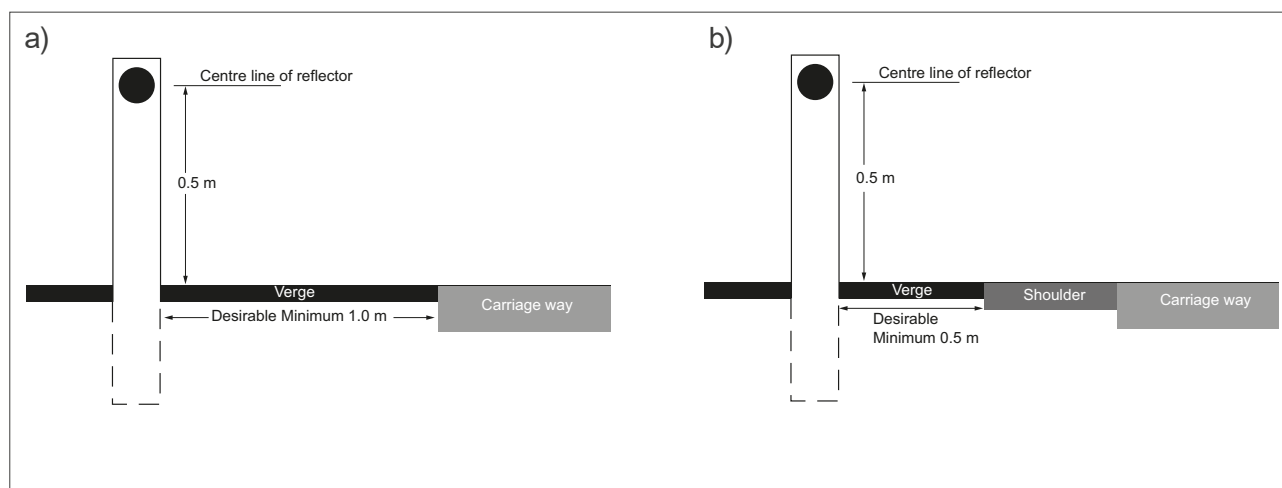
All types of delineators should incorporate a 60mm diameter colour cube reflector of the colour specified.

Placement should be as indicated in Figure 4.63(a) and (b). Where drainage trenches interfere with the recommended placement, the delineators may be placed on either side of the trench at the discretion of the Engineer.

Delineators can be used to indicate the edge of the carriageway on embankments, mountain roads and other points where special danger exists. They should also be used to indicate the place where the carriageway suddenly narrows, such as around obstructions near the kerb (e.g., a bridge parapet). In these cases, the reflectors may be affixed to the structure rather than a separate post.

Details of the location and colour of delineators used on road schemes are given in the Road Marking Manual (RDM Volume 6, Part 1).

Figure 4.63 Delineators on Road with/without Shoulder



4.8 Mounting of Warning Signs

Warning signs in rural areas should normally be mounted at minimum of 1.5m above the carriageway level. Where spray is excessive and likely to soil the sign, the higher mounting height should be used.

Where signs are erected above footways, a minimum height of 2.1m above the pavement is recommended. If supplementary plates are used then the mounting height should be measured to the bottom of the plate.

Plates should be separated from the sign or another plate by a distance equal to the 'x'- height of the lettering. Alternatively, both may be combined on a grey backing plate with 3 stroke width (s/w) clearance all round (see [Section 5.2.7](#) for more details on the 'x'-height and stroke width).

Further details on sign mounting are given in Chapter 2, [Section 2.4](#).

4.8.1 Mounting More Than One Sign on a Post

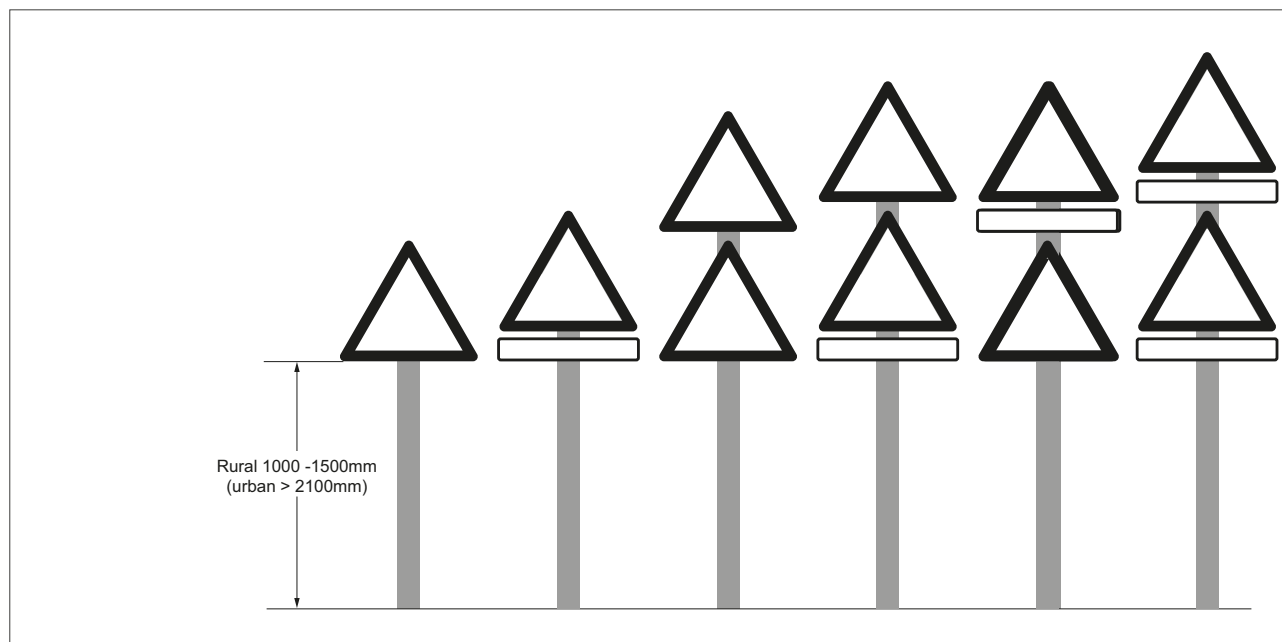
[Figure 4.64](#) shows all the allowable combinations of signs and plates that can be mounted on a single post. Normally no more than two signs should be mounted together. The combination of the sign and its associated plate can be considered as one sign for this purpose.

Warning signs should not be mounted with STOP or GIVE WAY regulatory signs. If mounted with other types of signs, warning signs should always appear on top.

In the event that assemblies of more than one warning sign are created, then the signs indicating hazards encountered first should be placed uppermost.

No assemblies should exceed 3.75 m above ground level. All proposed assemblies should be critically examined to ensure that the intended warning is clear.

Figure 4.64 Assembly Combinations of Warning Signs and Supplementary Plates



4.9 Additional Warning Signs

Some additional warning signs that can be used, as per requirement, are shown in Figure 4.65 to Figure 4.75.

Figure 4.65 Road Widens, Sign W60

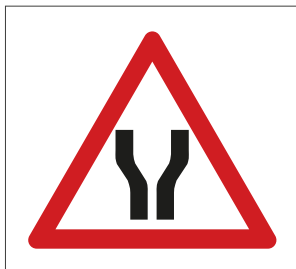


Figure 4.66 Narrow Bridge Ahead, Sign W61



Figure 4.67 Gap in the Median, Sign W62



Figure 4.68 Built Up Area, Sign W63

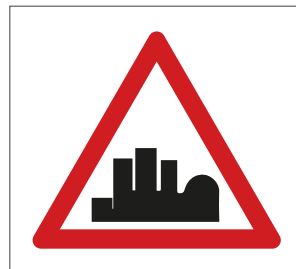


Figure 4.69 Speed Breaker, Sign W64



Figure 4.70 Rumble Strip, Sign W65



Figure 4.71 Ferry, Sign W66



Figure 4.72 Queue Ahead, Sign W67



Figure 4.73 Object Hazard (Left), Sign W68

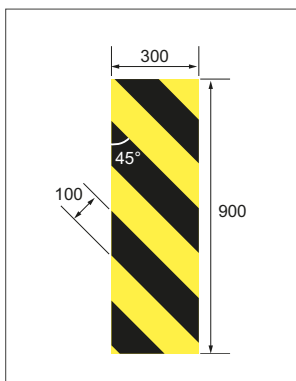


Figure 4.74 Object Hazard (Right), Sign W69

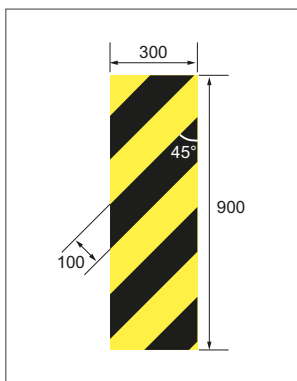


Figure 4.75 Two way Hazard Marker, Sign W70

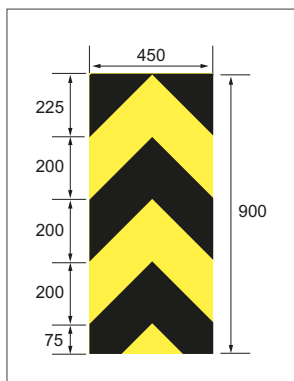


Figure 4.76 Old/Disabled People Crossing, Sign W71



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Warning Signs

5 Informatory Signs

5.1 Introduction

This chapter describes the Informatory Signs for use on various classes of roads.

Directional Signs for Expressway are given in Chapter 8. Other informatory signs for Facilities and Persons with Disabilities are given in Chapters 6 and 7 respectively.

Directional information signs belong to one of three major groups:

1. Advance directional signs giving road users information about destinations available from a junction ahead (Figure 5.1).
2. Direction signs giving route information at a junction (Figure 5.2).
3. Route confirmatory signs giving confirmation of present route selection and information about the destinations ahead after a road junction (Figure 5.3).

Figure 5.1 Advance Direction Sign



Figure 5.2 Direction Sign



Figure 5.3 Route Confirmatory Sign



On Arterial or Trunk roads (Class A, B & UA), directional signs include advance direction, direction and route confirmatory signs. On other routes it is not necessary to use all three types. Details are given in Section 5.2 on the principles of directional signing and the circumstances under which each type of directional sign should be provided.

The guidelines contained in this chapter describe the technical standards which should be adopted in the majority of situations. Some relaxation of the standards may be accepted for specific junctions because of geometry or for other reasons. A layout of signs for a particular junction should satisfy these basic requirements:

1. The colour of directional signs shall be appropriate to the route indicated. The relationship between the road hierarchy and colour coding of signs is defined in Section 5.2.
2. A destination, once shown on a sign, should appear on all subsequent signs along the route until either the destination is reached or a road turning off to the destination has passed.
3. When routes have numbers, they should always be indicated. The consistency of route number signing is particularly important to enable through traffic on main routes to negotiate the dense urban areas of the larger cities and towns.

5.2 General Principles of Design

The messages should be clear and legible so that the driver has no doubt about which route to take to reach the destination.

5.2.1 The Road Hierarchy and Colour Coding

The foreground and background colours of directional information signs depend on the classification of the road along which a destination is reached.

Table 2.1 in Chapter 2 shows the classification of roads in Kenya. The colour schemes of directional signs for different road classes are shown in Table 5.1.

Table 5.1 Colour Schemes for Directional Signs

Type of Road (Functional)	Class of Road	Background Colour	Colour of Text, Border, Arrows and Chevrons	Colour of Route Number
Arterial or Trunk	A, B, UA, UC	Green	White	Yellow
Collector	C, D, UC	White	Black	Black
Local	E – W, UL	White	Black	Black

5.2.2 Provision of Directional Signs

It is not necessary to provide all three types of directional signs at every junction. The provision of signs depends upon the type of junction.

Table 5.2 below gives guidance as to the signs which should be provided at the different types of junctions. Judgement should be used in specific cases.

Table 5.2 When to Provide Directional Signs

Junction Type	Type of Sign											
	Advanced directional sign on			Direction signs pointing along			Confirmatory signs on exit			Route Marker signs at exit on		
Road Class	A	C	L	A	C	L	A	C	L	A	C	L
A - A	Y	-	-	Y	-	-	Y	-	-	Y	-	-
A - C	Y	Y	-	Y	Y	-	N	N	-	Y	Y	-
A - L	N	-	N	Y	-	Y	N	-	N	N	N	-
C - C	-	N*	-	-	Y	-	-	N	-	-	N	-
C - L	-	N	N	-	Y	Y	-	N	N	-	N	N
L - L	-	-	N	-	-	Y	-	-	N	-	-	N

Note: A – Class A, B & UA roads, C – Class C, D & UC roads, L – Class E to W & UL roads

* Except important junctions leading to major centres of population or Classes A and B Roads.

The selection of the destinations to be displayed is determined by the classification of the route to be signed. Table 5.3 indicates the type of destinations to be displayed on all directional signs according to the classification of the route. Specific rules exist for route confirmatory signs which can show more than two destinations and details of these are given in the next section.

When the destinations have been selected, they should be arranged in the order of distance, with the furthest at the top of the list of place names for each direction.

Table 5.3 Destinations to be Shown

Road Class	Advance Direction Sign	Direction Sign
A	Terminal destination	Same as in Advance Direction Sign, add name of the nearest town
B	Terminal destination or major intermediate town	Same as in Advance Direction sign, add name of the nearest town
C	Nearest significant town or village	Same as in Advance Direction Sign, add name of the local destination, where warranted
Minor Roads		Nearest town or village

5.2.3 Terminal Destinations

These are the towns, cities or ports at the terminals of International and National Trunk roads or at the terminals of the extended route in the case of cross border routes. A figure showing terminal destinations for arterial and trunk routes (classes A, B & UA roads) is given in [Figure A.1-2](#) of [Appendix 1](#).

5.2.4 Other Destinations

These destinations are regional towns or towns of sizeable population. Such locations need not be on the actual route if they have been by-passed. In the case of Class B roads, where no significant location exists between the junction to be signed and the end of the route, the nearest significant location on the route to which it joins should be displayed. Tourist resorts or places of scenic or historic interest can be shown as 'significant destinations' on directional signs.

5.2.5 Alphabets

Transport typeface in Sans Serif font shall be followed for alphabets. Signs shall be in sentence case where the first letter of a word is a capital letter and the rest are small letters. All capital letters can be used in special cases for proper names or acronyms or if the Engineer deems it should be used. Names in italics can be used to emphasise a place name.

5.2.6 Common Dimensions

The 'x' -height and stroke width (s/w) are the common dimensions used when designing sign faces. The "x" height is the height of the lower case 'x' in the English alphabet. The stroke width is the dimension used when specifying clearances between the different elements of the sign face. It is equal to one quarter of the 'x'- height.

All upper and lower case, letters, numerals and associated characters are placed on individual "tiles" to assist in the correct spacing of text. The tile height is twice the 'x' - height.

The ratio, therefore, of stroke width to 'x'-height to tile height is 1:4:8.

The tile shows the horizontal clearances in the total space occupied by each character of the type face. Each character has its own tile width and this is expressed as a percentage of the 'x'- height.

[Figures A1-3 to A1-6](#) in [Appendix 1](#) show how the characters are placed on tiles. They also indicate the special tiles of lesser width for the letters T, V or Y when they are followed by a, e, g, o, r, and u. A specially increased tile width for the upper case 'W', to be used when it is followed by any other upper-case letter, is also defined.

The size of a sign is determined by its 'x'-height. The recommended 'x'-heights appropriate for different speed categories of road are also given in [Table A1-1](#) in [Appendix 1](#). The sign face is designed by arranging the different elements of the sign according to dimensions of borders, tiles and symbols and the recommended clearances stipulated later in this section.

The overall sign size can be calculated by applying the selected 'x' -height.

5.2.7 Letter and Block Spacing

Each character available in script is illustrated mounted on a tile. The English script is mounted on square or rectangular tiles that stand upright. The Local Language script is inclined at 15 degrees. Letters in the Local Language stand on slanted tiles.

Words in either language are formed by putting tile edges together. The tile width for each character is designed to ensure the correct spacing between the characters. The assembly of a number of tiles to form a word or a number of words, place names or route numbers is called a block. Spacing is measured between tiles. The spacing between words is given in Table 5.4.

Table 5.4 Spacing Between Words, Route Numbers and Distances

Description of Horizontal Clearance	Dimension S/W
Between related words	3
Between letters and directly associated numerals (A1, C99, 500 m)	1
Between place names and associated route numbers	3
Between place names and associated distances	3
Between different route numbers on same line	3

For bilingual signs, Local Language script should always be positioned above its English equivalent. A clearance between the top of the English tiles and the bottom of the Local Language tiles should be maintained. This clearance normally is 0.5 s/w but varies according to the type of sign and the destination displayed. Bilingual blocks are formed by justifying both sets of tiles to the left. The Local Language tiles should be offset so that the mid-point of the slanted side of the Local Language tile is in line with the left side of the English tile.

A sample text is shown in Figure 5.5 Legend Panel Design.

Figure 5.4 Sample Text



5.2.8 Patches and Legend Panels

Patches and legend panels are used on signs to indicate the status of routes reached directly or indirectly from a junction ahead.

A legend panel should be used on advance direction signs where a route leaving the junction ahead has a different status to the road on which a sign is placed. The legend panel should contain both place name and route number and will have a colour scheme in accordance with Table 5.1.

Legend panels should be designed so that the space between tile edges and the edge of the panel is 2 s/w. The legend panel corners should be set at radius to 1 s/w. A white border of 0.5 s/w is added to the darkly coloured legend panels if placed on dark backgrounds. Patches show the route number.

The design of legend panels is illustrated in Figure 5.5. Figure 5.6 shows patch design.

Figure 5.5 Legend Panel Design

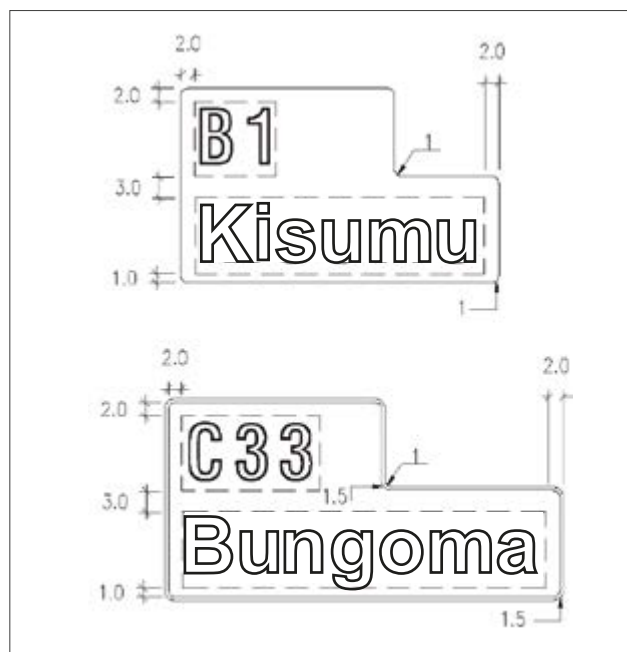
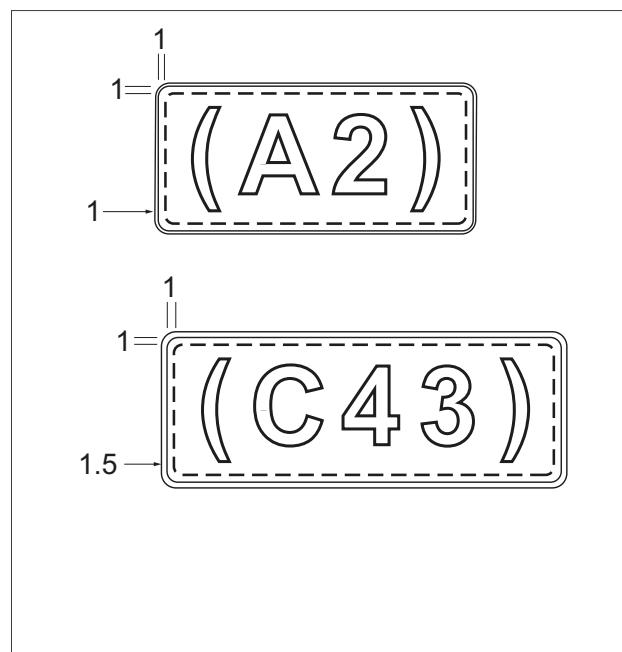


Figure 5.6 Patch Design



5.3 Sign Face Design

5.3.1 Advance Sign Design

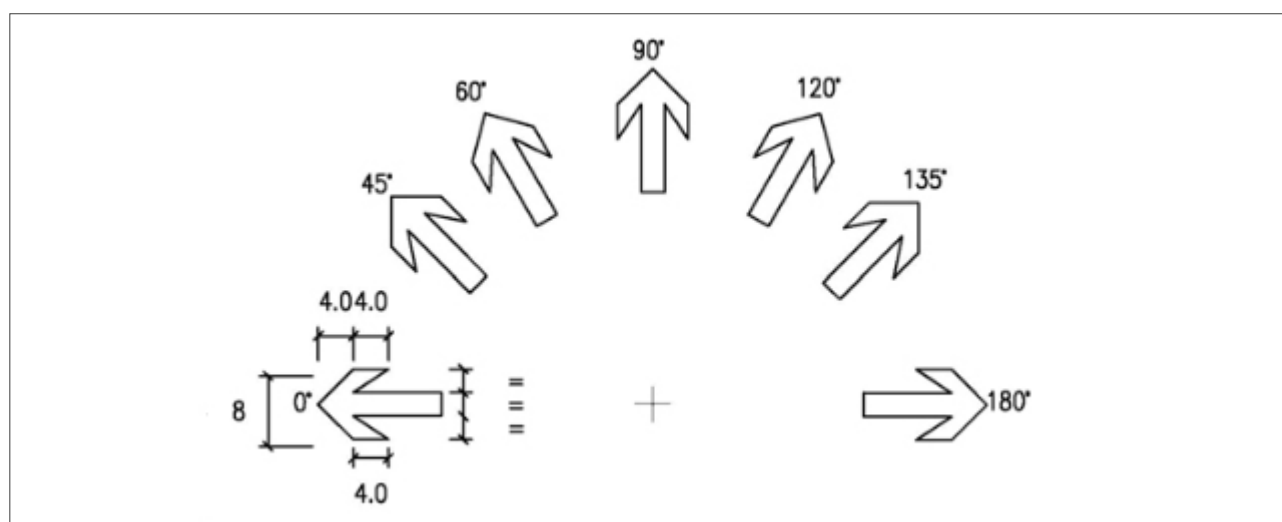
There are four basic types of advance direction sign:

1. Stack type (Sign I1).
2. Route Direction (Sign I2).
3. Map type (Sign I3).
4. Lane Destination (Sign I4).

5.3.2 Stack Type Design

Stack type signs have directional panels indicating the route number and destinations for each direction. Colour of each directional panel is according to the route indicated. The direction is indicated by an appropriately orientated arrow. The standard design and orientation of the arrows is shown in Figure 5.7.

Figure 5.7 Design and Orientation of Arrows



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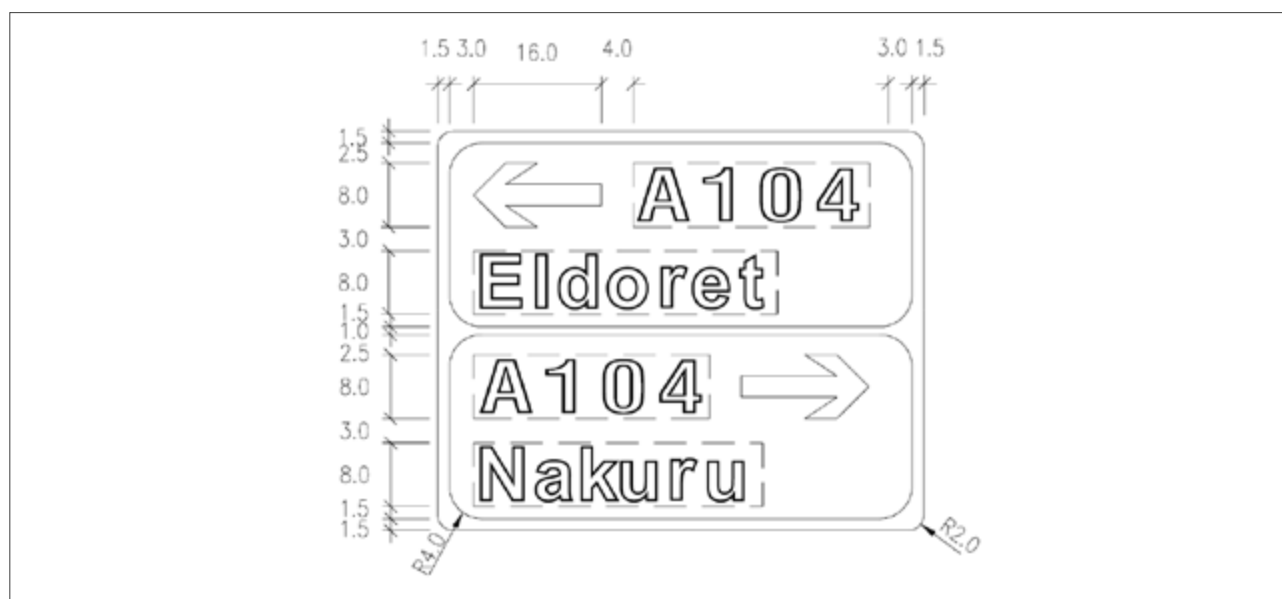
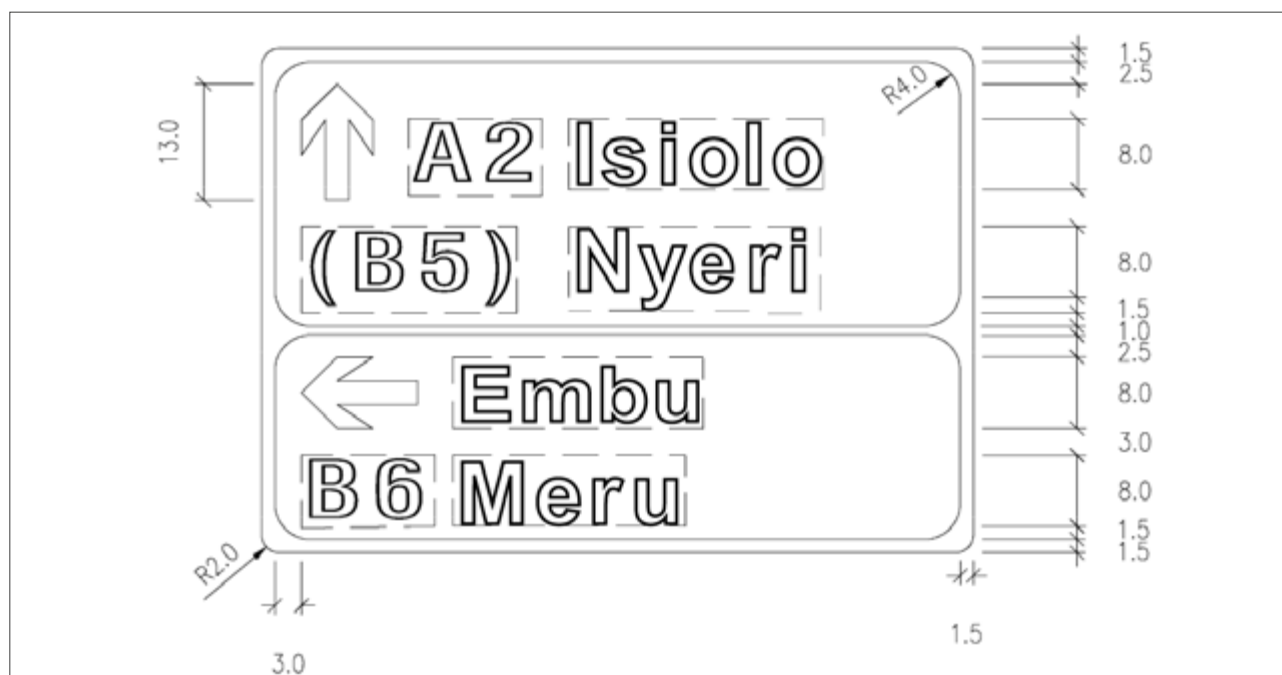
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Informative Signs

Stack type signs should be designed in accordance with the rules defined in [Table 5.5](#) and demonstrated in [Figure 5.8](#) to [Figure 5.10](#). When it is necessary to show more than one destination for each direction, the second destination should appear underneath the first. Where this destination is reached by another route after the junction, the route number should be shown in brackets underneath the arrow and route number for the first destination. If both destinations are reached via the same route, then the route number and arrow should be centred vertically alongside the two place names.

Figure 5.9 Alternative Design of Stack Type Sign (Sign I1-B)**Figure 5.10** Alternative Design of Stack Type Sign (Sign I1-C)

5.3.3 Route Direction Signs

In some urban areas it may not be feasible to accommodate stack type signs. In these cases, route direction signs can be used. These signs are similar to stack type signs. They have directional panels which are coloured according to the route indicated and use the same arrows as stack type signs.

Route direction signs do not show any destinations, but they display route numbers of routes that may be reached directly or indirectly from the junction ahead. All the route numbers are un-bracketed.

Table 5.6 indicates the design rules normally used for route direction signs. Where more than one route is available in any direction the route numbers are enclosed by boxes and are 6 s/w high. Where a single route number is shown for a direction the gaps to borders are increased from 3 s/w to 4 s/w and the numbers are 8 s/w high.

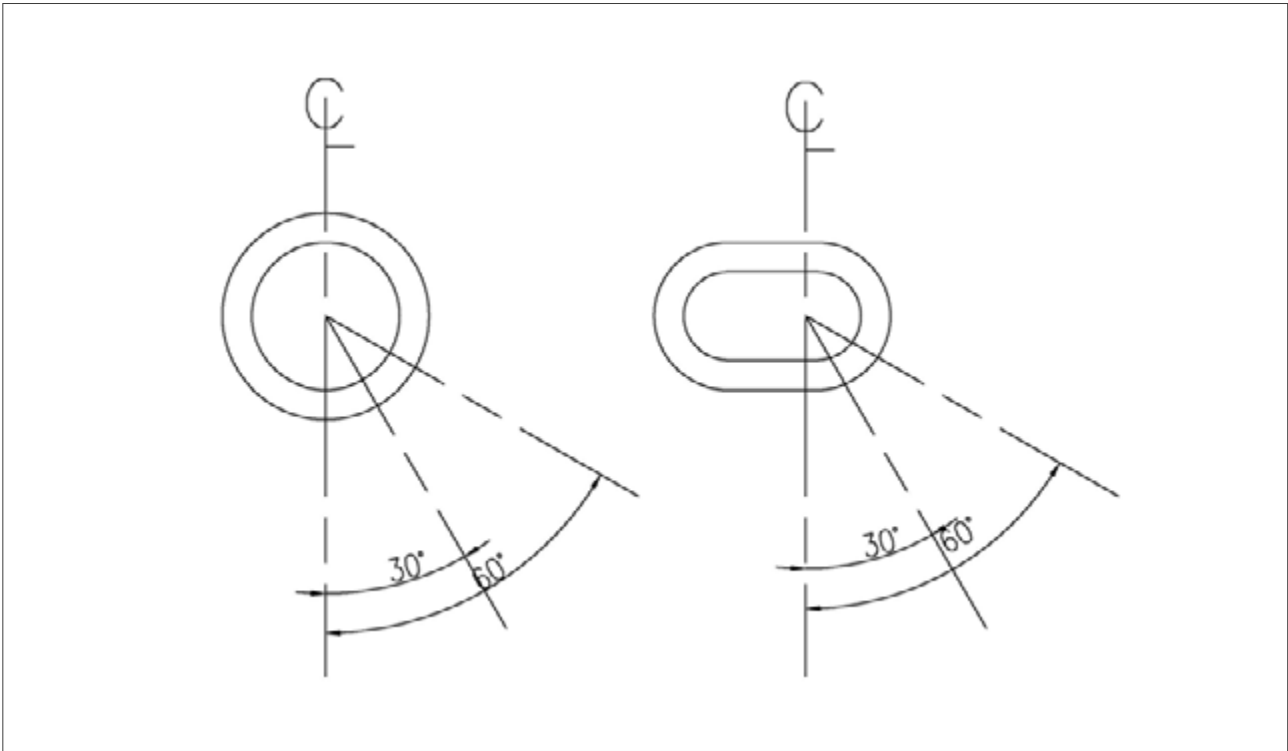
Table 5.7 Dimensions for Roundabout Route Symbols

Dimension	S/W
Width of Route Symbol	4.0
Inner Diameter of Roundabout	14.0
Outer Diameter of Roundabout	22.0
Exit Arm Length	5.0
Entry Arm Length (Minimum)	8.0
Exit Stub Arm Length	3.0
Width of Arrow Head	3.0
Distance of Entry Route Symbol from Bottom Border	2.0
Distance of Arrow (Inner Angle) from Centre of Exit arm	2.5

There should always be a break in any roundabout symbol to the right-hand side of the entry arm. This is illustrated in Figure 5.13 and is formed by:

1. Measuring angles of 30° and 60° anti-clockwise from the centre of the circular symbol.
2. Forming the ends of the break using the lines formed by these angles when drawn across the lower portion of the circular symbol.

Figure 5.13 Forming Break in Roundabout Symbol



Once the route symbols have been designed the rest of the information can be added to the sign. Table 5.8 indicates the design rules to be used when designing roundabout signs. For the straight ahead arms the route number may be placed on either side of the arrow for optimum layout (Figure 5.14). Then the destinations may be positioned above and centrally justified with the route number. Another arrangement, such as positioning the route number directly above the arrow head, may sometimes produce a more effective design (Figure 5.16).

Table 5.8 Dimensions for Map Type Signs of Roundabouts

Dimension	S/W
Border Width	1.5
Inner Radius of Border	4.0
Outer Radius of Border	2.0
Horizontal Gaps to Side Border	3.0
Vertical Gaps to Bottom Border	2.0
Vertical Gap to Top Border	3.0
Vertical Gap Between English and Local Version Place name	0.5
Vertical Gap Between Different Place names	3.0
Horizontal Gap Between Arrow and Route Number (or legend panel)	3.0
Vertical Gaps Between Route Number and Place name	3.0
Horizontal Gap Between Side Destination Place names and the Furthest Horizontal extent of Roundabout Symbol	2.0

For side destinations the route number should normally be centred alongside the arrow head. In the interests of economic and balanced sign design, the destination(s) may appear above or below the route number. The number of arms will normally dictate the type of design that is suitable. In all cases information for different exits should not be closer than 12 s/w to each other.

Exits from the roundabout that cannot be represented accurately by horizontal or vertical arms should be shown by arms at angles of 45°, 135°, 225°, or 315° from the vertical. In the latter case the break in the roundabout symbol should then occur between 330° and 350° from the vertical.

Figure 5.14 to Figure 5.16 below show examples of map type signs for roundabouts.

The distances to the junction may be indicated in tiles of 75% normal size at the bottom of the sign adjacent to the entry arm.

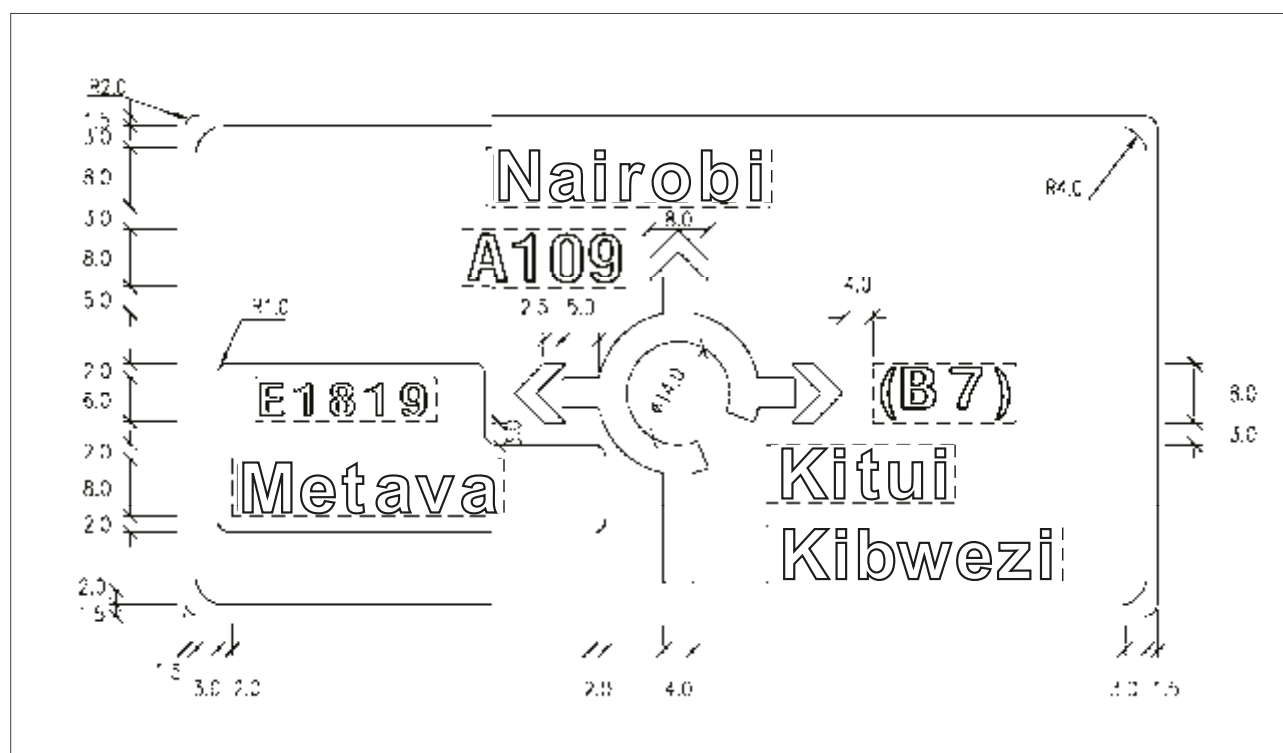
Figure 5.14 Design of Roundabout Map Type Sign (Sign I3-a)

Figure 5.15 Design of Roundabout Map Type Sign (Sign I3-b)

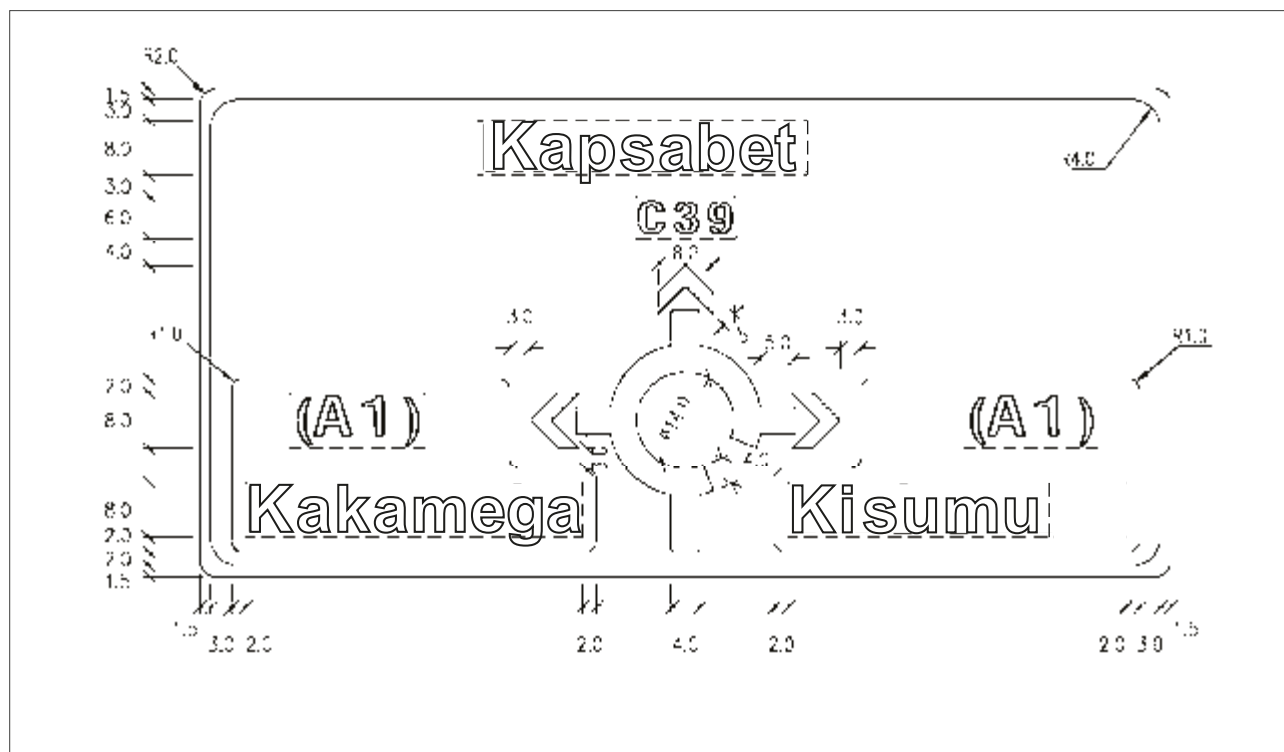
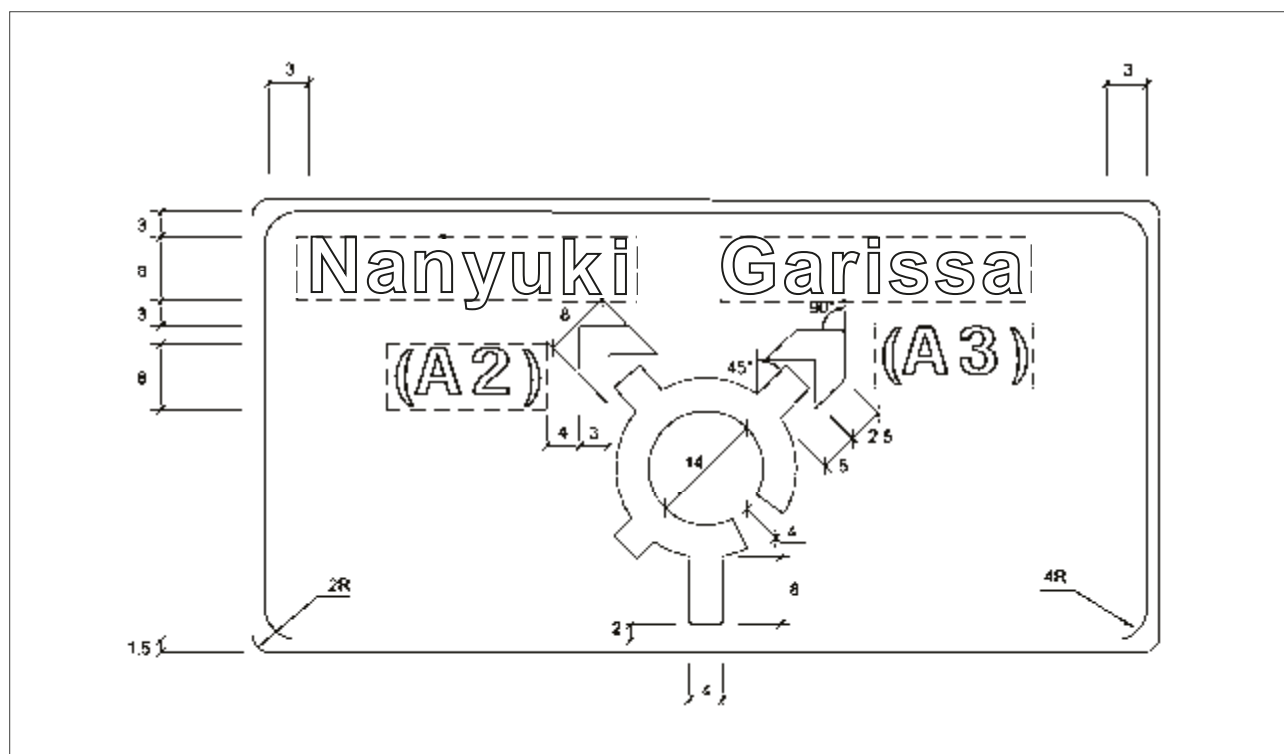


Figure 5.16 Design of Roundabout Map Type Sign (Sign I3-c)



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Informatory Signs

5.3.5 Lane Destination Signs

Lane destination signs should only be used before junctions where movements can only be made from specific lanes. In urban areas, lane destination signs will be of substantial assistance in guiding vehicles into the most appropriate lane at junction approaches. Lane destination signs reflect the colour schemes for different classifications of road on the legend panels.

Lane destination signs use arrows and lane markings to guide motorists into the correct lane for their destination on approach to junctions. The dimensions that are to be used when designing lane destination signs are given in Table 5.9. The arrows are the same as those used on stack type signs although their standard length is 18 s/w. Arrows at 690, 900 and 1200 from the vertical can be used in addition to curved versions for left or right turns. These are shown in the example layouts in Figure 5.17 and Figure 5.18.

Table 5.9 Dimensions for Lane Destination Signs

Dimension	Multiple of S/W
Border Width	1.5
Inner Radius of Border	4.0
Outer Radius of Border	2.0
Vertical Gap to Top Border	3.0
Vertical Gap to Bottom Border	2.5
Horizontal Gap to Side Border	3.0
Lane Line Width	1.0
Length of Lane Line Segments	8.0
Length of Gap Between Lane Line Segments	4.0
Vertical Gap Above or Below Lane Lines	1.5
Link Line Width	1.0
Vertical Gap Above Link Line	1.5
Vertical Gap Below Link Line	2.5

Figure 5.17 Design of Lane Destination Sign (Sign I4-a)

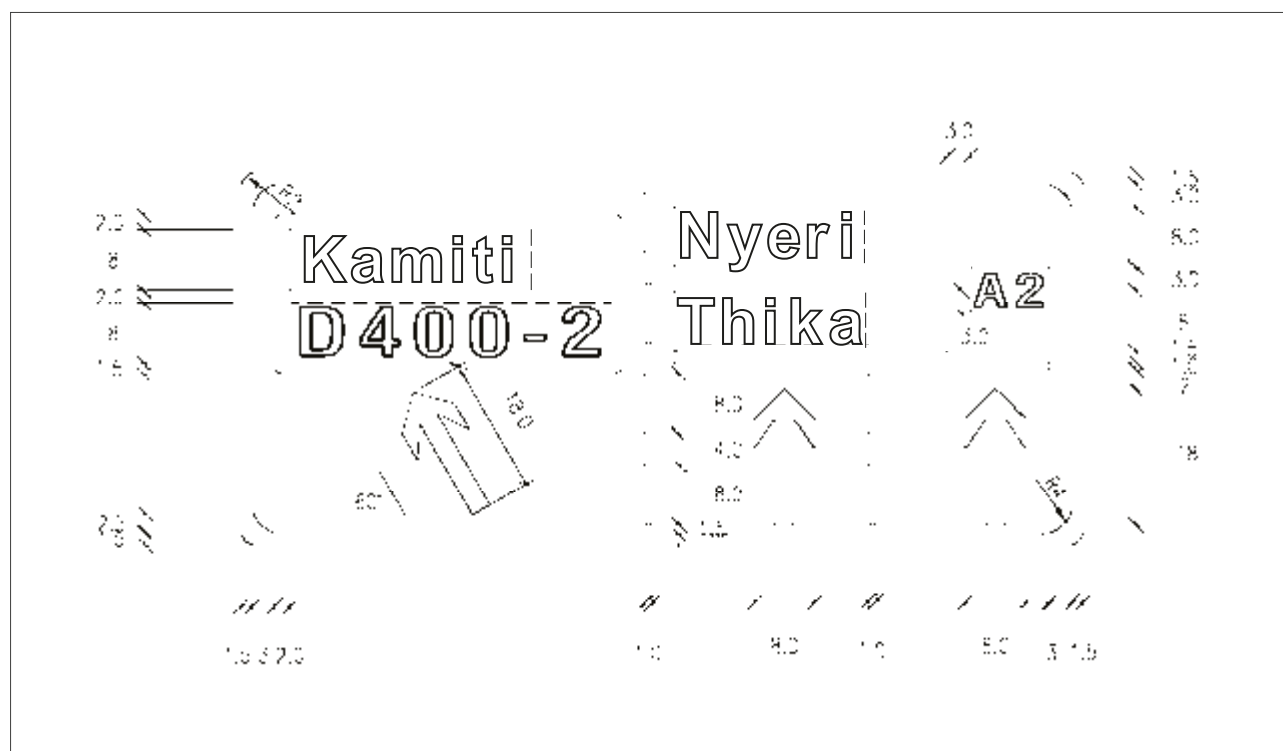
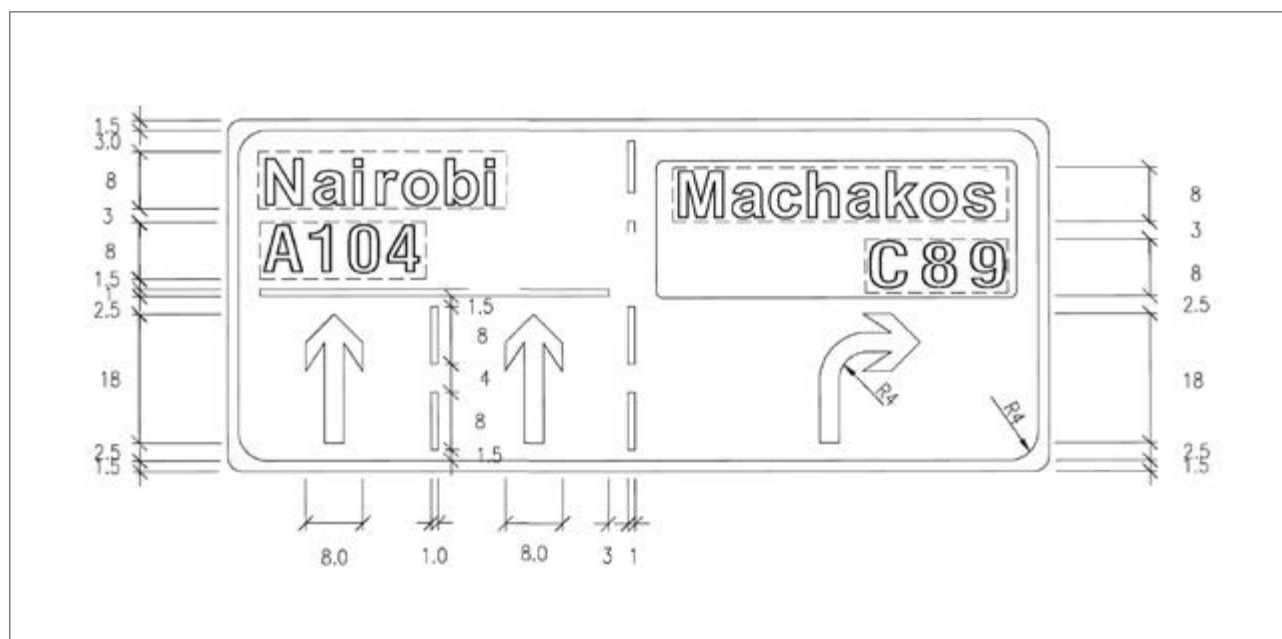


Figure 5.18 Design of Lane Destination Sign (Sign I4-b)

A link line is a horizontal line separating destinations from the lane lines or arrows below. A link line should be used only if two or more lanes are related to the same destination(s) or can be reached via more than one lane.

Single arrows referring to a destination ahead should be centred on the legend to which they refer (in the case of curved arrows their full width should be centred). However, when two arrows relate to the same destination, each arrow should be centred between the adjacent lane lines or border.

Lanes leading to the same destination should be depicted as having equal width. No single lane width should be more than twice the width of the narrowest lane. Where the lengths of legend are so different that they exceed the above ratio the width of the narrowest lane must be increased. When this is done the horizontal spacing rules do not change except that the gaps to the side border are increased (the legend being centred horizontally on the sign). Alternatively, the largest destination in the widest lane may be condensed, indented, or abbreviated.

Where a lane line is truncated at the top of the sign, the minimum length should be 3 s/w. Where this cannot be achieved, the line should be omitted.

Route numbers should normally appear underneath destinations and be justified left.

Route numbers may be justified right, if they are associated with right turn lanes. When two destinations are shown in the same direction, the route number may appear alongside the destinations.

5.3.6 Direction Signs

Direction signs are located at the road junction itself and point along exits from the junction. They perform two main functions:

1. They indicate the location of the junction.
2. They show the destinations on the routes indicated.

Direction signs should be positioned at the junction so that they point as clearly as possible to the route to which they refer. Preferably they should be on the nearside of the carriageway. It is important that direction signs are correctly sized and are clearly visible to approaching drivers. Table A.1-1 in Appendix 1 indicates the recommendations for sizes and clear visibility distance for different speed categories of the road.

The basic dimensions to be used when designing direction signs are shown in Table 5.10. Some dimensions vary according to the number of destinations.

Dimension	S/W
Border Width	1.5
Inner Border Radius (1)	1.0
Outer Border Radius	2.5
Outer Border Radius of Top and Bottom Corners at Pointed End (1)	2.0
Outer Border Radius of Point (1)	1.5
Horizontal Gap Between Place names and Side Border	2.5
Chevron Width (1)	4.0
Gap Between Chevron and Top Border	1.5
Gap Between Chevron and Side Border	2.5
Gap Between Chevron and Bottom Border	1.5
Horizontal Gap Between Place names and Route Numbers and Chevron	3.0
Vertical Gap Between Separate Destinations	1.5
Vertical Gap Between Place names and Top border: One Destination	5.0
Vertical Gap Between Place names and Top border: Two Destinations	2.5
Vertical Gap Between Place names and Bottom border: One Destination	5.0
Vertical Gap Between Place names and Bottom border: Two Destinations	1.5

2. Radii greater than those shown in [Figure 5.20](#) to [Figure 5.25](#) can be used on a case-by-case basis, with the approval of the Engineer.

Direction signs pointing along Class A and B Roads should show the route number and terminal destination. Direction signs indicating the next town may show the distance in km. Examples of these types of signs are shown in [Figure 5.19](#) which illustrate the positioning of the route number in relation to the destinations shown. Directional signs are numbered as I5.

[illegible]

5.3.8 Direction Signs for Collector and Local Roads

Direction signs pointing along class C, D and other minor routes may have to show distances as well as the route number and destinations. If a route confirmatory sign exists on these roads after the junction it is not necessary to indicate distances on the direction sign (see Figure 5.23). The use of more than one destination on bilingual signs on these routes is not recommended.

When distances are shown they are always accompanied by the abbreviation "km" which should be shown in lower case English tiles at 75% of the normal size. The position of the "km" and distances vary according to the number and language of destinations shown and are demonstrated on the signs in Figure 5.24 to Figure 5.25. The route number is always indicated at the bottom of the sign and aligned with the start of the text.

Figure 5.20 Design of Directional Sign for Class C Sign I5-b

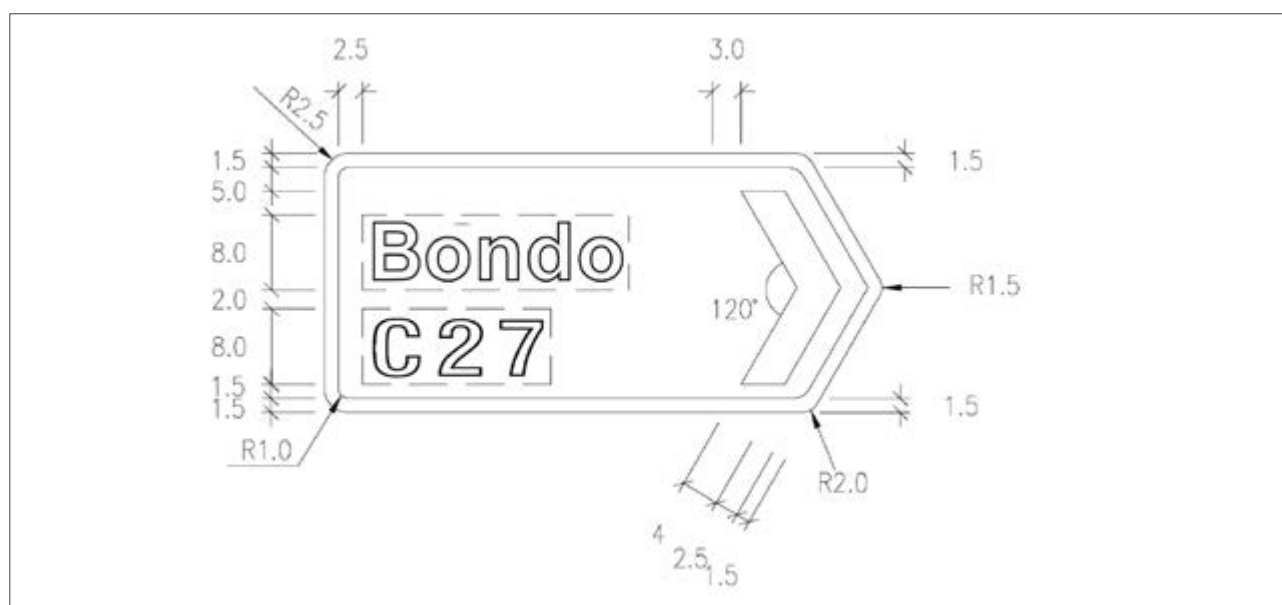


Figure 5.21 Design of Directional Sign for Class C Showing Distance Sign I5-c

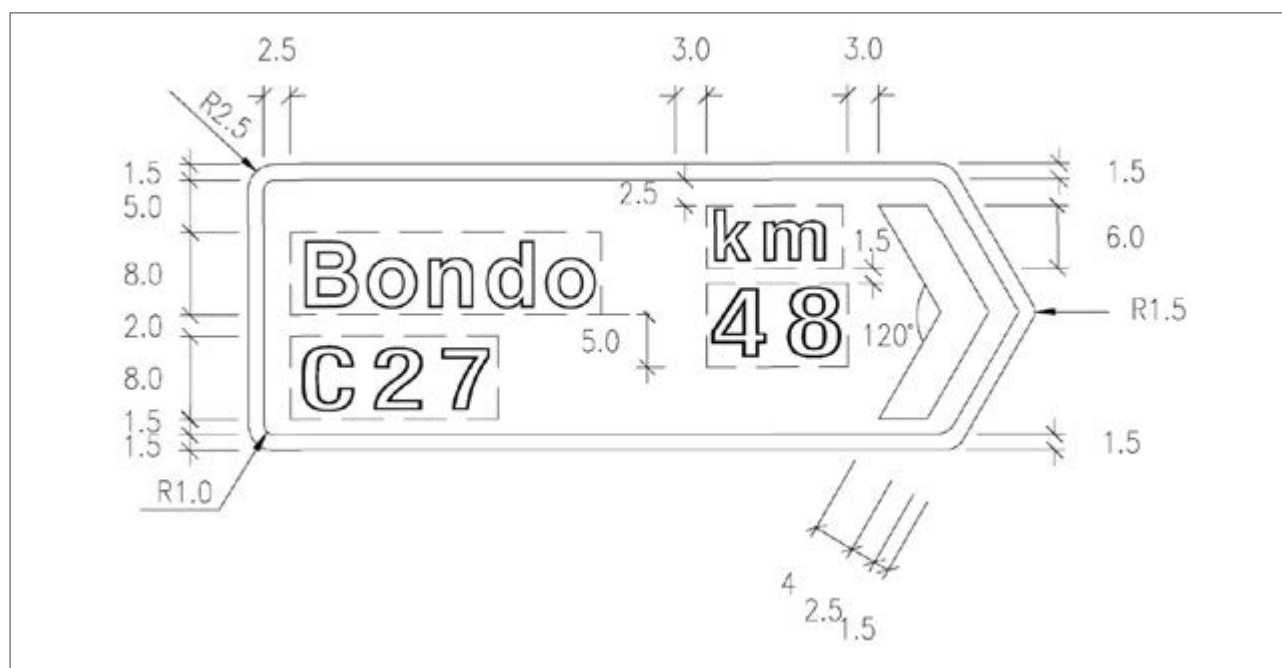
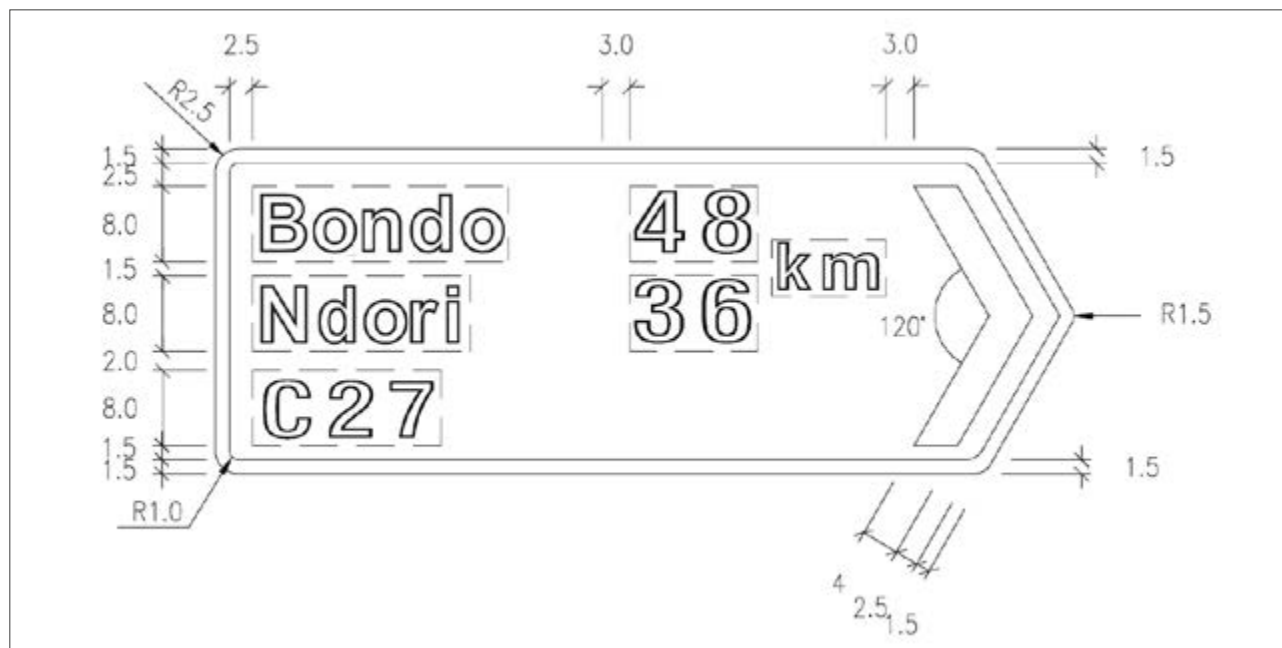
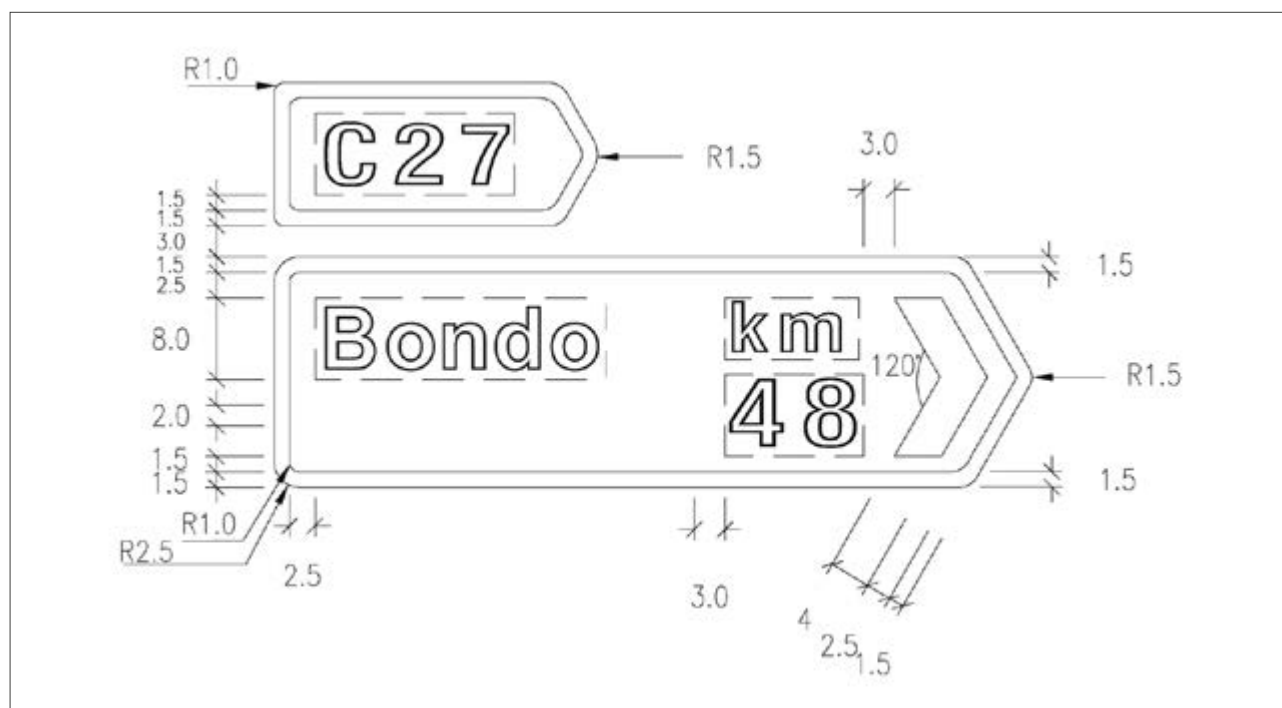


Figure 5.22 Design of Direction Sign for Class C Road (Two Destination) Sign I5-d

Where existing signs, which do not incorporate a route number, are to be left in place, a separate sign indicating the route number should be provided above the main sign as indicated in [Figure 5.26](#). Alternatively, this route number sign may be mounted below the main sign.

Figure 5.23 Design of Direction Sign for Class C Road (Route number above main sign) Sign I5-e

For high level direction signs mounted on a single pole and where distances must be shown, the chevron can be omitted to reduce the length of the sign.

Where the route number has been increased to more than standard capital size e.g. 6 or 8 stroke widths, the width of the tiles shall be increased proportionally.

As shown in [Figure 5.23](#), an additional separation of 1 Stroke width (at 'x' height of sign) is provided between the tile of the capital letter and the tile of the first numeral.

5.3.9 Route Confirmatory Signs

Route Confirmatory signs (I6) may be located after road junctions. They serve two purposes:

1. To confirm to road users that they have taken their intended route.
2. To give additional information about the road ahead.

Confirmatory signs can show more than two destinations ahead but they must duplicate the destinations indicated previously on advance direction signs and direction signs for the route. The additional destinations may be towns of lesser importance on the route and need not be included on subsequent direction signing until they become relevant.

Significant destinations that can be reached by turning off the route from a junction ahead can be displayed in brackets.

Route confirmatory signs are coloured over their entire area according to the status of the road on which they are located. This is regardless of whether destinations reached by turning off onto another classification of route are mentioned.

Route confirmatory signs must be correctly sized and be clearly visible to approaching drivers if the information is to be read and understood. Recommendations on size and clear visibility distance of route confirmatory signs are provided in Table A1-1 in Appendix 1.

There are two types of route confirmatory signs:

1. Route number with place names and respective distances.
2. Route marker.

The first type has the route number placed centrally at the top of the sign followed by a list of destinations arranged in distance order with the furthest appearing at the top.

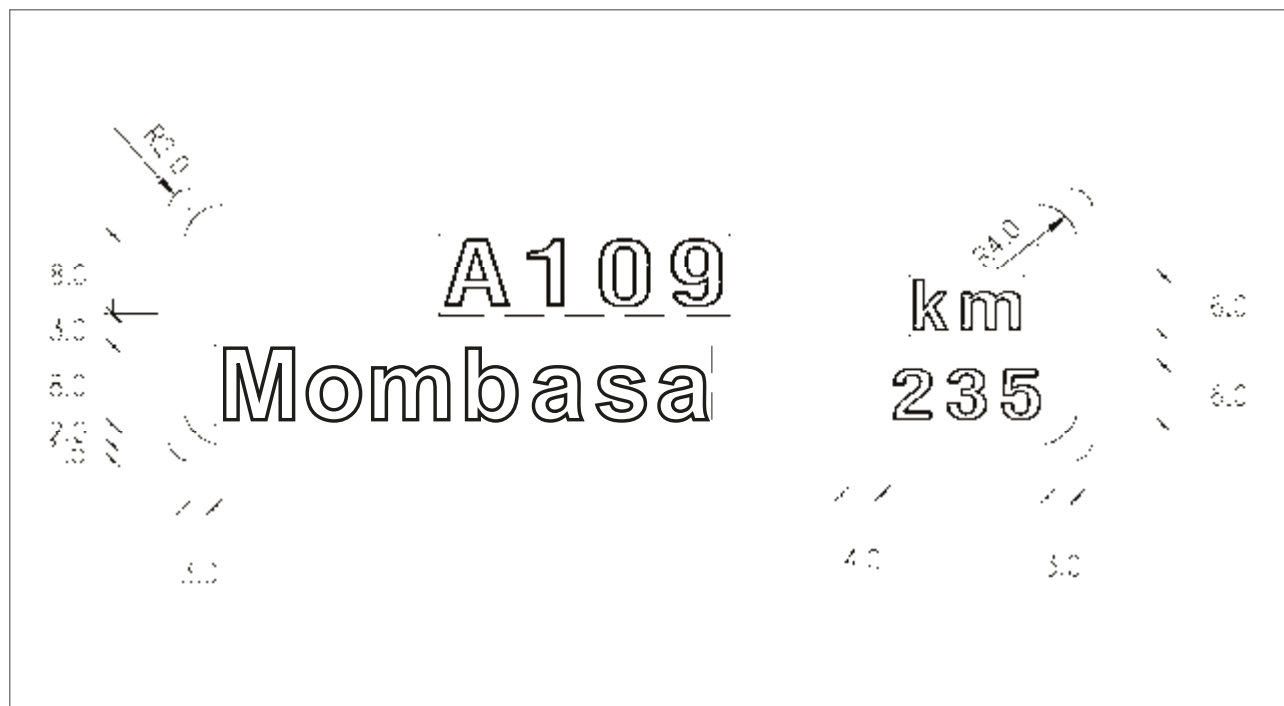
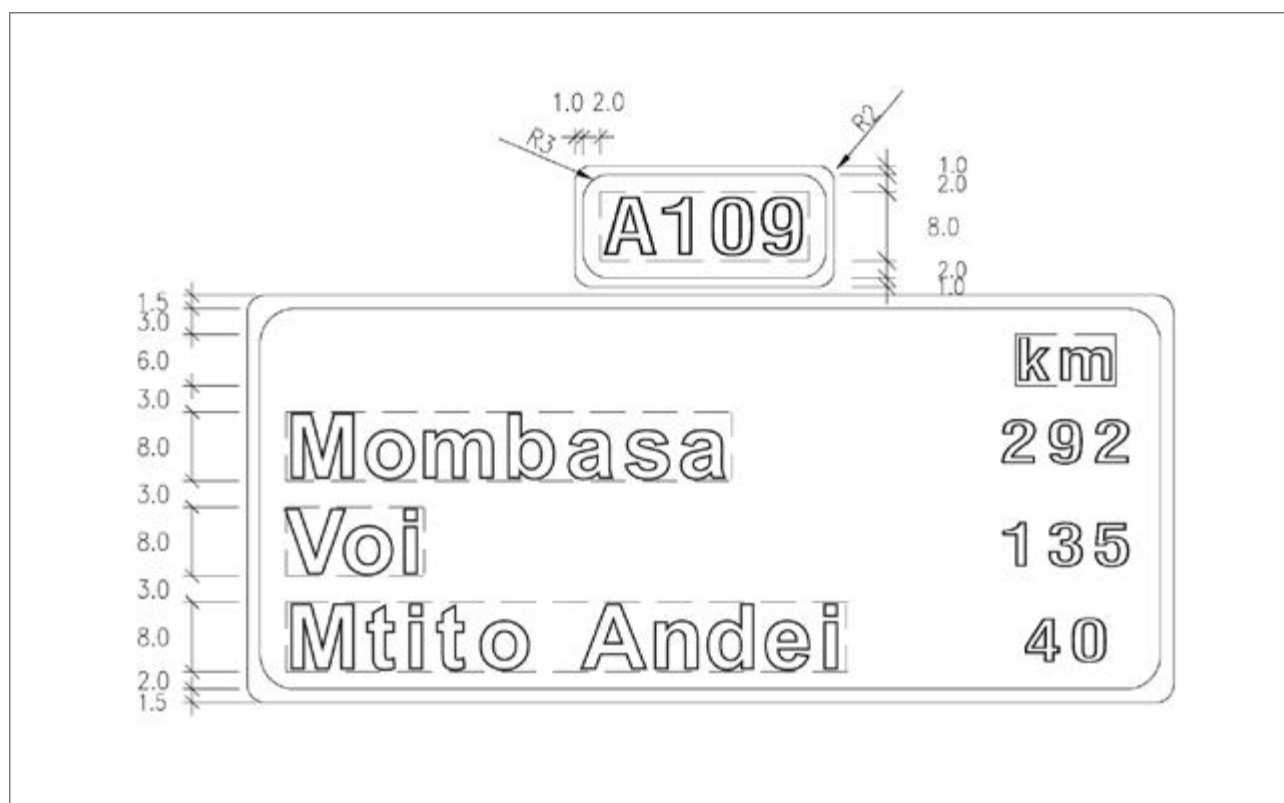
Distances in kilometres should be shown adjacent to each respective destination. Any destinations located off the main route, such as terminal destinations on spur routes, should be shown in brackets. The position of bracketed destinations on confirmatory signs is governed by the distance to the junction at which the road user must turn off the main route to reach that destination, irrespective of the overall distance to the destination itself.

Where only one destination is shown, the route number appears within the sign. When multiple destinations are shown, the route number is displayed separately on a route marker sign which is centred 1 s/w above the main sign. (All the dimensions used are defined in Table 5.11.

On bilingual signs, the top of the letters in "km" are aligned with the top of the letters of the Local Language version of the first place name indicated. The "km" tiles appear at the top of the sign centred over the top distance mentioned.

Table 5.11 Dimensions for Route Confirmatory Signs

Dimensions	Multiples of S/W	
	Main Sign	Route Marker
Sign Border Width	1.5	1.0
Inner radius of border	4.0	3.0
Outer radius of border	2.0	2.0
Gap between route number and side borders	–	2.0
Gap between route number and top or bottom borders	–	2.0
Gap to top border	3.0	–
Gap to side border	3.0	–
Horizontal gap between place names and distances	4.0	–

Figure 5.24 Design of Route Confirmatory Sign with One Destination (Sign I6-a)**Figure 5.25** Design of Route Confirmatory Sign with Three Destinations (Sign I6-b)

6 Other Informatory Signs

6.1 Introduction

Other Information signs are those that provide guidance information on facilities rather than route selection or regulations.

Other information signs are categorised into five groups:

1. Signs displaying civic or geographic information such as the name of a town or river (Sign I7).
2. Signs indicating alternative or diversionary routes (I8).
3. Signs indicating facilities available ahead that is of interest to road users generally (I9).
4. Signs indicating tourist facilities available ahead (I10).
5. Sign to indicate truck escape ramp (E1).

All informatory signs proposed in this manual begin with I, except signs for escape ramp which begins with E.

An example of each of these sign types is presented in Figure 6.1 to Figure 6.5.

The directional signs defined here should normally be mounted separately from those defined in Chapter 5.

The design of the signs should conform to the design rules covered in Chapter 5 for directional information signs. Sign size will be governed by the 'x' -height in the normal way as described in Chapter 5.

Figure 6.1 Town or Village, Sign I7



Figure 6.2 Alternative Route for High Vehicle, Sign I8

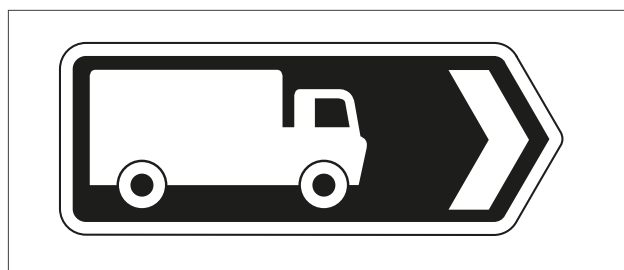


Figure 6.3 Car Park with Facilities for Disabled, Sign I9



Figure 6.4 Tourist Facility, Sign I10



Figure 6.5 Escape Ramp, Sign E1



6.2 Geographical Information Sign

6.2.1 Town or Village Signs

The name of a town or village may be indicated by a sign (I7) as shown in Figure 6.6. In most cases the place name should be indicated in English language. It can be supplemented by local language.

Figure 6.6 Town or Village, Sign I7



These signs should be coloured as appropriate to the status of the road on which they are located i.e., white letters and borders on a green background for a national route or black letters and borders on a white background for a regional or local route. These signs should not appear on Expressways.

6.2.2 County Boundary Signs

County boundary signs, as illustrated in Figure 6.7, indicate the name of the county and its crest if desired. All script shown on county boundary signs should appear in both Local Language and English languages. All county boundary signs should have black text and borders on a white background.

Figure 6.7 County Boundary, Sign I10



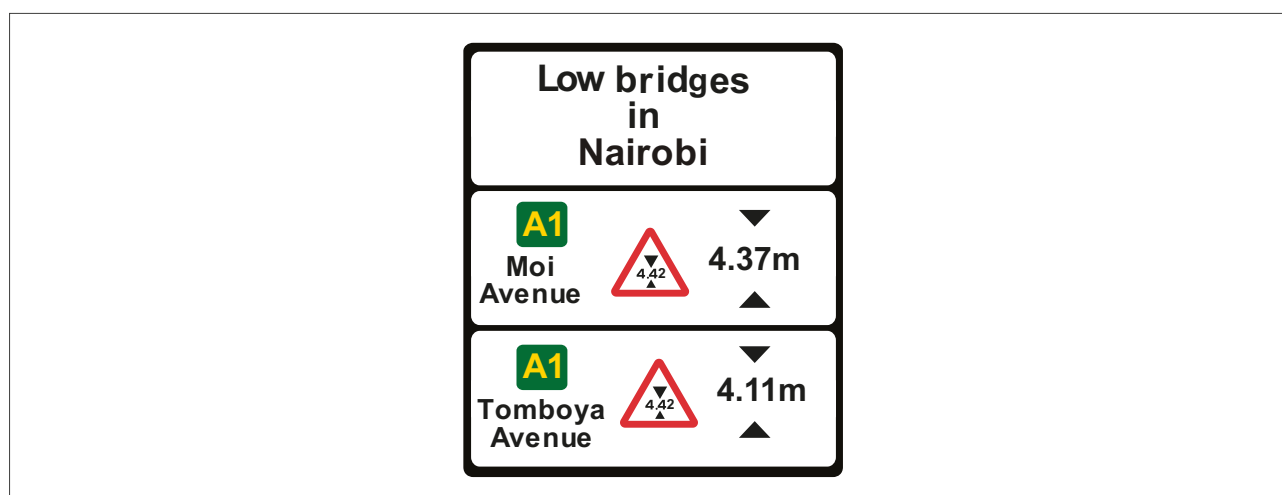
6.3 Advisory Informative Signs

Advisory information signs provide details of alternative and diversionary routes.

6.3.1 Advance Sign for Low Clearances at Bridges

Advance information of restricted height clearances may be provided by the sign I11 shown in Figure 6.8.

Figure 6.8 Advance Sign for Low Clearance, Sign I11



This sign should be designed like a stack type sign (see Chapter 5). It should incorporate Triangular warning signs of 300 mm size, and the height restriction should be shown beside the Triangular symbol. The height shown should be the same as that indicated on the warning signs located immediately before the hazard. A separate panel should be provided to display the location of the height restriction(s).

The sign should be sited in advance of an alternative route.

6.3.2 Slow Lanes

On long inclines, it is sometimes necessary to provide a "Slow Lane" to accommodate slow moving vehicles. The sign I12 shown in Figure 6.9 should be used up to 300m before the start of the slow lane. This sign has black symbols, text, and borders on a white background. For the associated road markings see Road Markings Manual (RDM Volume 6, Part 1).

Figure 6.9 Slow Lane, Sign I12

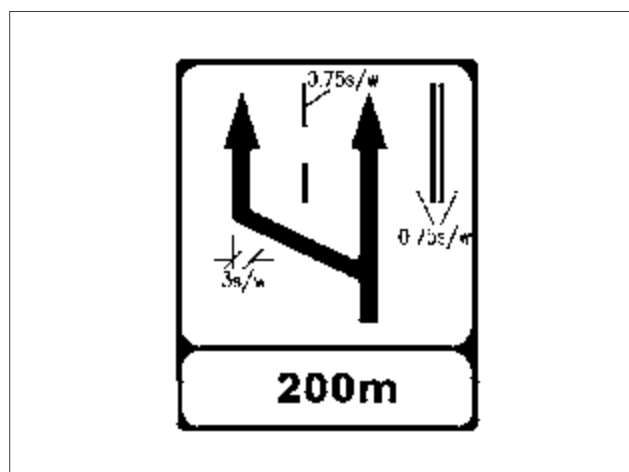
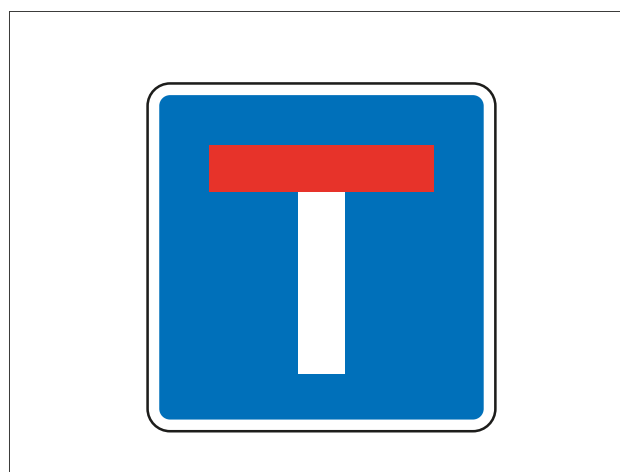


Figure 6.10 No Through Way, Sign I13



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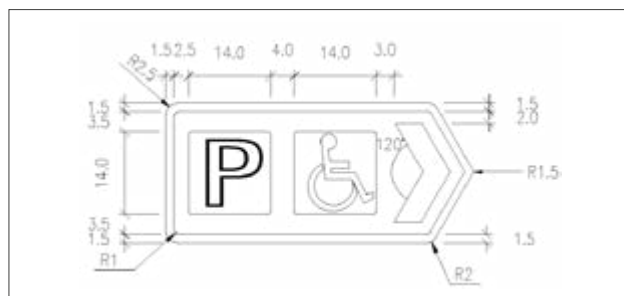
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Figure 6.12 Lay-by, Sign I15



Figure 6.14 Disabled Car Park, Sign I17



6.2.2 Hospital Signs

The main feature of all signs indicating hospitals should be the "H" symbol illustrated in Figure 6.15.

The hospital may be indicated up to 500m ahead by the sign shown in Figure 6.15. In the immediate vicinity of the hospital, the sign shown in Figure 6.16 should be erected to face oncoming traffic. Both signs should have white symbols, lettering, and borders on a blue background.

Figure 6.15 Hospital Ahead, Sign I18



Figure 6.16 Hospital, Sign I19



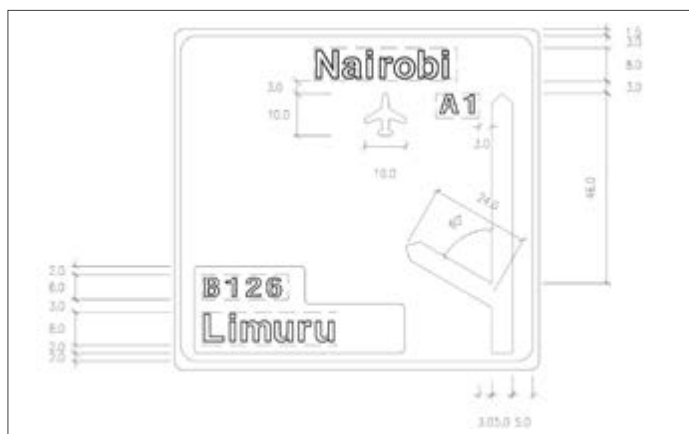
6.4.3 Airport Signs

Airports should be signed using the aeroplane symbol shown in Figure 6.17. This symbol should normally be incorporated onto the directional signs shown in Chapter 2 and will assume the relevant colour scheme. For example, an airport located on a national route should be shown with a white symbol on a green background, as shown in Figure 6.17. Repeat signs showing only the symbol will normally be black on a white background.

Figure 6.17 Airport Symbol, Sign I20



Figure 6.18 Directional Sign Incorporating Airport Symbol, Sign I20



6.4.4 Ferry Terminal Signs

Ferry terminal signs incorporate the ferry symbol shown in Figure 6.19 which should assume the colour scheme of the route along which the ferry terminal is reached. The ferry terminal symbol should normally be accompanied by the relevant name such as Likoni. This is shown in Figure 6.19 where the ferry does not accommodate trucks, the truck symbol may be replaced by a car.

Figure 6.19 Ferry Symbol, Sign I21

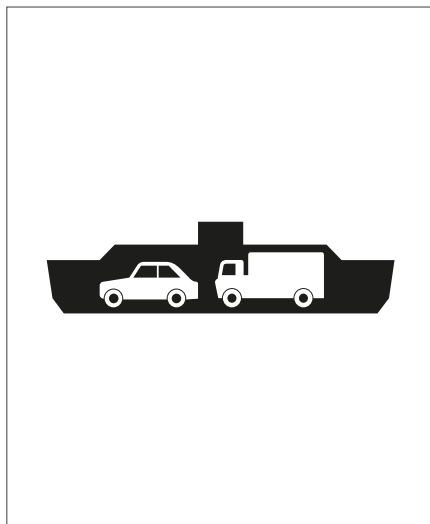
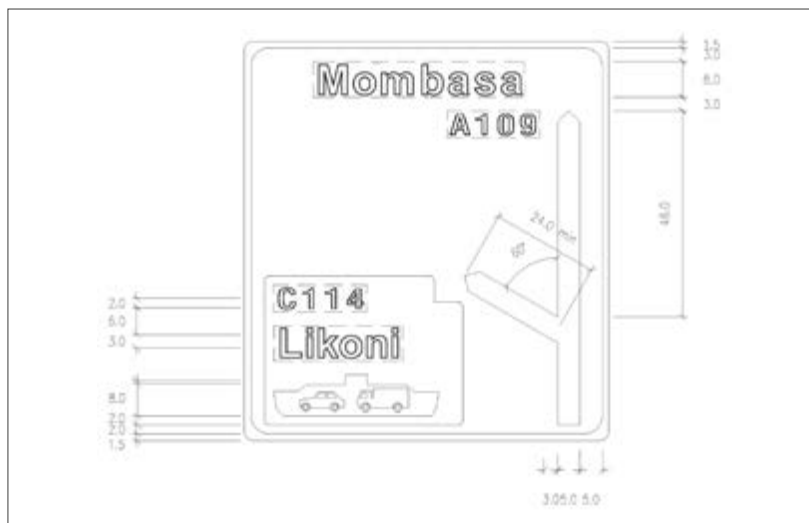


Figure 6.20 Directional Sign Incorporating Ferry Symbol, Sign I21



6.4.5 Industrial Estate Signs

Signs to industrial estates follow the same rules as ferry terminal and airport signs as previously described. The industrial estate symbol is shown in Figure 6.21 and should be incorporated onto signs as shown in Figure 6.22. The symbol may be shortened by the omission of one (1) roof peak.

Figure 6.21 Industrial Estate Symbol, Sign I22

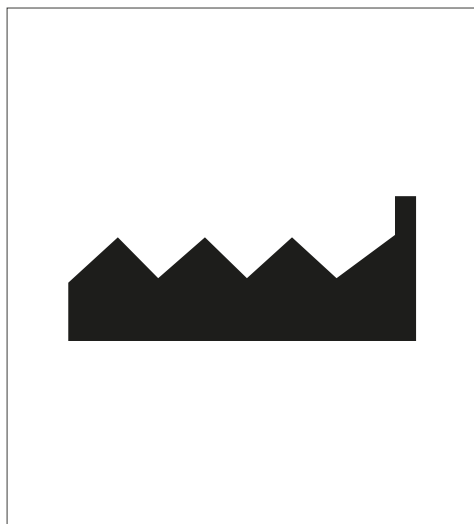
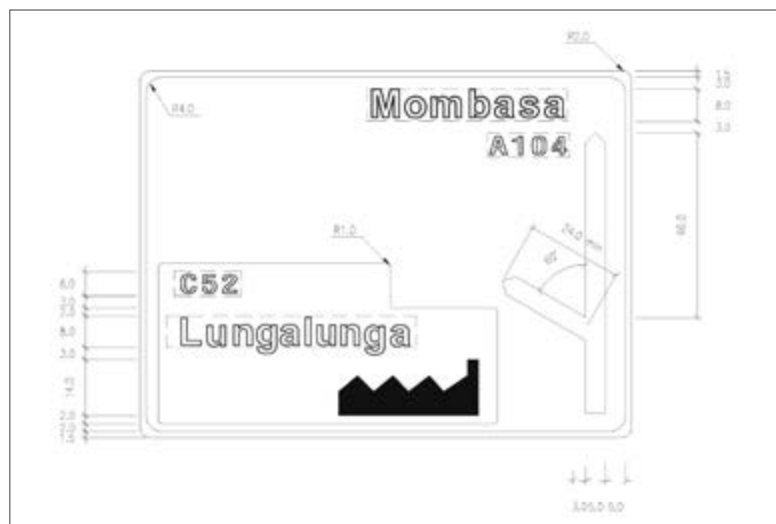


Figure 6.22 Directional Sign Incorporating Industrial Estate Symbol, Sign I22



6.5 Signs to Tourist Attractions

Tourist signs are distinctive brown signs with white text used to direct drivers to a tourist destination. Sample of tourist signs are given in [Appendix 2](#).

The basic sign designs should conform to the normal design rules for traffic signs described in [Chapter 5](#). It should feature white legend, borders, arrows, and chevrons on a brown background. Reflectorising is recommended on national routes and where attractions are open after dark.

Particular care must be taken with the siting of signs so as to avoid clutter and intrusion. Advance direction signs to major tourist attractions should be combined with those to other destinations by using brown panels on map or stack-type signs. In the case of direction signs, separate signs should be used for tourist attractions.

6.6 Additional Facilities Signs

Additional useful facilities signs are shown in [Figure 6.23](#) to [Figure 6.38](#). These signs are for eating places, refreshment, resting place, toilet, first aid, petrol pump, etc.

Figure 6.23 Eating Place, Sign I23



Figure 6.24 Light Refreshments, I24



Figure 6.25 Resting Place, Sign I25

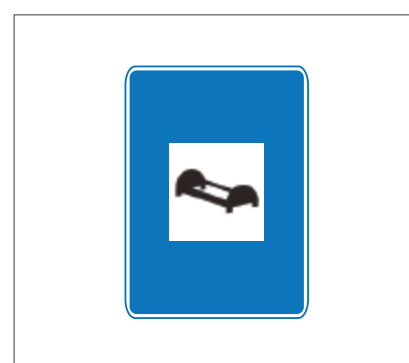


Figure 6.26 Toilet, Sign I26

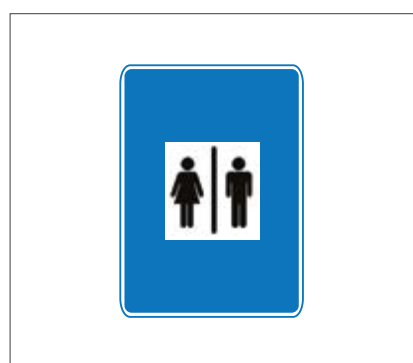


Figure 6.27 Fuel Pump, Sign I27

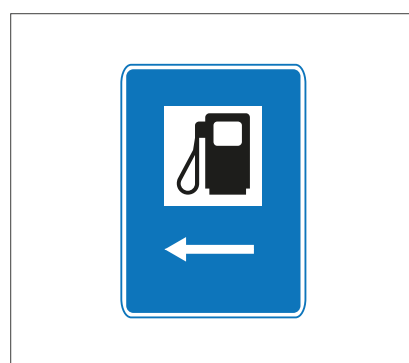


Figure 6.28 Pedestrian Over Bridge, I28



Figure 6.29 Pedestrian Subway, Sign I29



Figure 6.30 Picnic Site, Sign I30



Figure 6.31 Camp Site, Sign I31



6.7 Signs for Escape Ramp

Truck Escape Ramps (TERs) are generally used in two sets of circumstances: on long mountain grades in rural areas, and on short, steep hills likely to be in areas of dense traffic and development. The latter are often situations where accidents involving fatalities and serious property damage have happened or are probable.

The signs to be used for escape ramps are given in [Figure 6.39](#) and [Figure 6.40](#) and typical layouts of truck escape ramp signs are given at the end of the [Appendix 2](#).

Figure 6.32 Escape Ramp Sign at Ramp, Sign I32



Figure 6.33 Direction of Ramp, Sign I33



7 Signs for Persons with Disabilities

7.1 The International Symbol of Accessibility

The International Symbol of Accessibility (ISA) is also known as the international wheelchair symbol. It is used as an information sign with a blue background and an image of a person using a wheelchair overlaid in white. The wheelchair figure should always be seen facing right as shown in Figure 7.1.

All signs for Persons with Disabilities (PWD) recommended in this manual start with letters PD.

7.2 Size, Shape and Colour

The signs for PWD shall be rectangular and have a blue background. A white symbol shall be displayed to indicate the facility. The size of the normal sign shall be 600 mm x 600 mm and the symbol shall be as shown in Figure 7.1 and Figure 7.2.

7.3 Parking Information

The parking area for PWD should be indicated using wheelchair symbol, as shown in Figure 7.2. The direction of the parking should be indicated using an arrow, see Figure 7.3

7.4 Ramped Entrance to Subway/Over Bridge

These signs shall inform the persons with disabilities about the ramp facility to enter a pedestrian subway /foot over bridge. These signs shall be rectangular and have a blue background, with a white ramp to indicate the facility, as shown in Figure 7.4.

7.5 Telephone and Toilet Facility

The sign to indicate telephone facilities should be as shown in Figure 7.5. Toilet Facility sign is shown in Figure 7.6.

7.6 Way Finding Sign for Disabled

The way finding sign should be as shown in Figure 7.7.

Figure 7.1 International Symbol of Accessibility, Sign P1



Figure 7.2 Parking Information, Sign P2



Figure 7.3 Parking Areas, Sign PD3**Figure 7.4** Ramped Entrance, Sign PD4**Figure 7.5** Telephone Facilities, Sign PD5**Figure 7.6** Toilet Facilities, Sign PD6**Figure 7.7** Wayfinding, Sign PD7**Figure 7.8** Old People Crossing, Sign W71

7.7 Old / Disabled People Crossing

This sign for old / disabled people crossing is shown in [Figure 7.8](#). It is the same as warning sign W71.

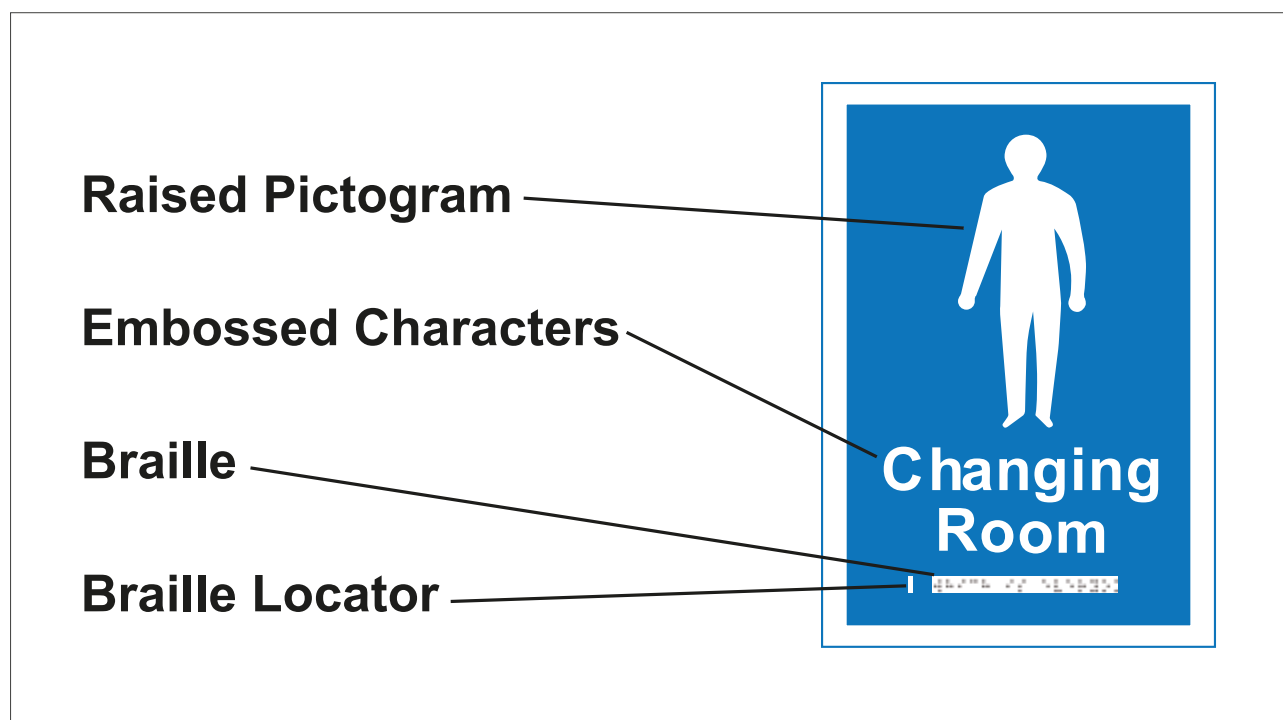
7.8 Signs for Visually Challenged and Persons with Hearing Impairment

Signs in Braille should be included for directional and way finding signs. For easy accessibility, the Braille signs should be placed at 1400 to 1600mm height. The Braille face must be raised by at least 5mm to enable reading by touch.

Braille, embossed letters, raised pictograms and tactile features should be incorporated in signs to assist the visually challenged. The information should contain pictogram, text, and Braille. If sign includes a map, then every element of the map should be raised. Such maps should be placed at appropriate height so that people in standing and sitting position can access the maps for tactile reading.

Braille signs must be placed at junction crossing, adjacent to audible signs. Braille signs should not replace tactile / raised text messages as may visually challenged people do not use Braille as their primary form of communication. An example of raised message with Braille is shown in **Figure 7.9**.

Figure 7.9 Sign for Visually Challenged



Audible elements of signs are helpful for persons with limited visibility. Pedestrian crossings with traffic lights should be accompanied by audible elements.

For persons with hearing impairment, the signs are in pictorial form, which have universal recognition. The sizes of signs are such that those can be easily read from a distance, depending on the approach speed.

For assistance of the deaf pedestrians, the push button unit for crossing can be of vibratory / buzzing nature, to alert the user when walk sign comes on.

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Signs for Persons with Disabilities

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Signs for Persons with Disabilities

8 Expressway Signs

8.1 Introduction

This chapter describes the design of signs on expressways and the design of signs that direct traffic to expressways from other roads.

The provision of signs on expressways presents fewer problems than for other roads because:

1. There are no junctions at grade except at the terminals.
2. There are standard road layouts for entering and leaving expressways.
3. The classes of traffic using expressways are restricted.

Drivers of vehicles on expressways must be given adequate early indication of junctions and the end of the expressway.

Whether signs are on expressways or indicate expressways from other roads, the design rules normally conform to those covered in [Chapter 5](#) for directional information signs. A distinctive colour scheme of white lettering, symbols and borders on a blue background is used.

All signs for expressways recommended in this manual start with the letters IM.

Expressway signs include advance direction signs, direction signs, and route confirmation signs, examples of which are shown in [Figure 8.1](#) to [Figure 8.3](#). They also include some other information signs which are used in specific situations such as the 'End of Expressway' sign shown in [Figure 8.4](#).

Figure 8.1 Advance Direction, Sign IM1



Figure 8.2 Direction Sign, Sign IM2



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Expressway Signs

Figure 8.3 Route Confirmation, Sign IM3



Figure 8.4 End of Expressway, Sign IM4



8.2 General Principles

This section defines the general principles of design for expressway signs. It should be read in conjunction with the corresponding section of [Chapter 5](#) which covers design rules applicable to directional information signs.

8.2.1 Colour Coding

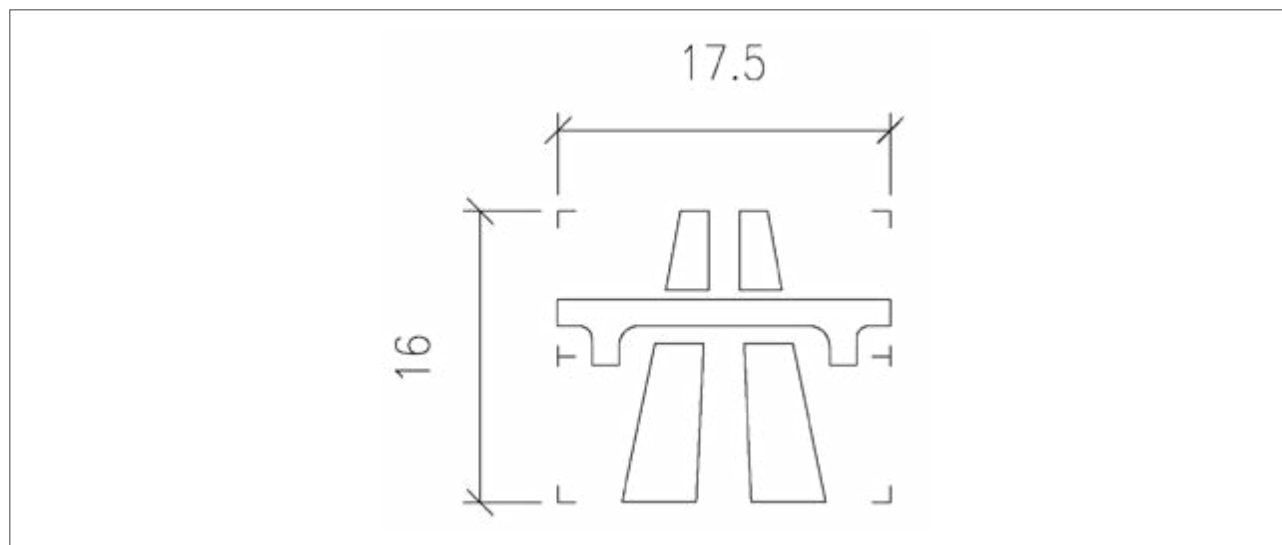
Signs located on the main carriageway of expressways are normally white on a blue background. They do not contain any patches or panels of other colours except where a junction number is indicated, which should be shown as white on a black patch.

At the terminals of the expressway, signs on the mainline should retain a blue background but connecting national or regional roads should be indicated by appropriately coloured legend panels. On expressway exit slips leading to other routes, the background colour of map type signs will be appropriate to the route with the higher classification.

For signs located on service roads, slip roads and other roads that lead to an expressway, the colour of the signs on the road shall be as per the class of the road. However, the route number of the expressway that can be reached by following that road should be shown in brackets on a blue patch. Same principle will apply for roads leading out of an expressway.

Direction signs on other roads approaching an expressway junction should show the expressway symbol illustrated in [Figure 8.5](#) on a legend panel. Direction signs located at the same junction pointing down the expressway or a slip road should also include the expressway symbol but should be white on a blue background.

Expressways along with entry and exit ramps to and from the expressway are to have the colour of signs of expressways. The service roads and other roads to access the expressway ramps are to have the sign colour of the category of that road, till the junction with the ramp.

Figure 8.5 Expressway Symbol, Sign IM4

Signs demonstrating the system of colour coding are illustrated in the [Appendix 3](#).

8.2.2 The Selection of Destinations

The selection of destinations on expressway signs should be determined in accordance with the principles set out in [Chapter 5](#).

8.2.3 Alphabets

Where expressway route numbers are indicated, the alphabets shall be used as shown in [Figure A.1-5](#) in Appendix 1.

The size of the alphabets is defined in terms of the 'x' -height which is described in [Chapter 5](#). The 'x' -heights recommended for use on signs on different speed categories of road are given in [Table A.1-1](#) in the Appendix 1.

8.3 Sign Face Design

The sign face design is for two categories of signs:

1. Signs located on national or regional roads that indicate expressways.
2. Signs located on expressways.

The size of signs on the expressway shall be 1200 mm.

8.3.1 Signs on Trunk or Collector Roads Indicating Expressways

Appropriate advance information should be provided to direct the users to the expressway entry.

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Expressway Signs

8.3.2 Patches

Where it is appropriate to indicate the direction to an expressway from a distant junction, the expressway route number (enclosed in brackets) should be placed on a blue patch on the main background of the sign (see Figure 8.6 and Figure 8.7).

Blue patches containing expressway numbers should use the expressway alphabet and can be placed on any of the advance direction or direction signs defined in Chapter 5. The design of patches is described in Section 5.2 of Chapter 5.

Patches should usually be placed to the right of the main route number indicated for the direction. They should usually be centred horizontally. Clearances shall be measured from the patch like a normal route number. To minimising the sign size, a patch may be placed underneath the main route number and centred vertically.

Figure 8.6 and Figure 8.7 show examples of signs incorporating expressway patches.

Figure 8.6 Sign Incorporating Expressway Patch, Sign IM5

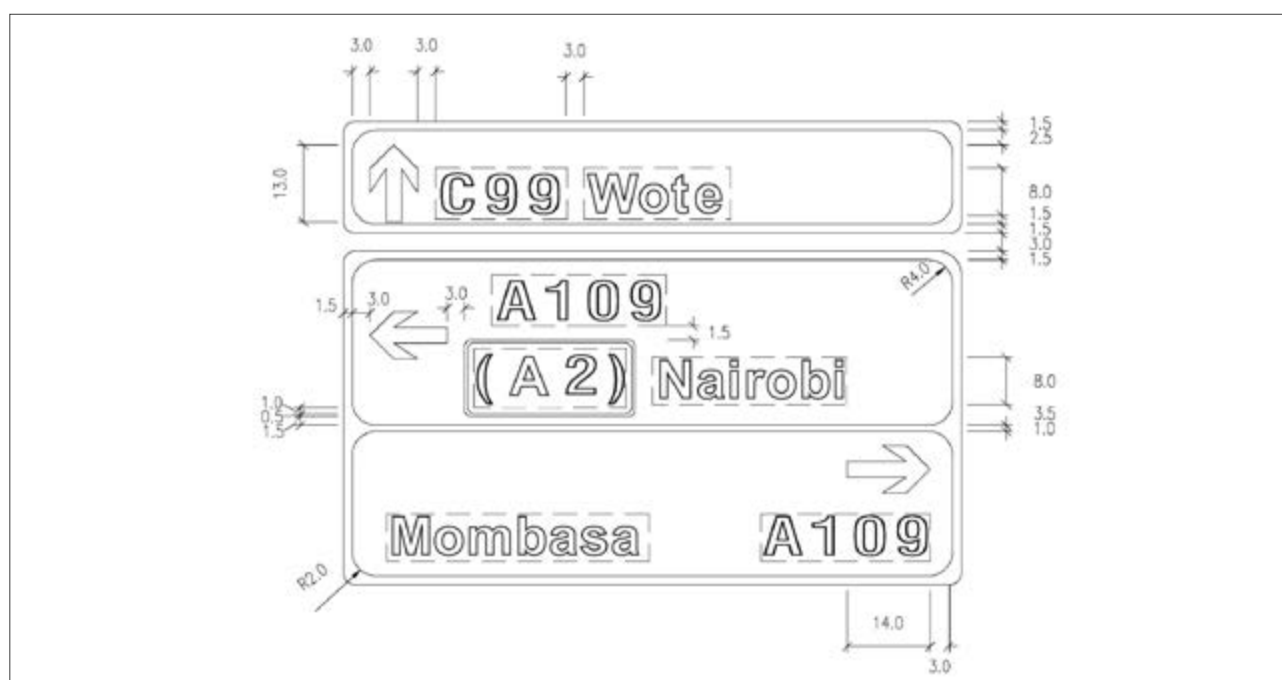
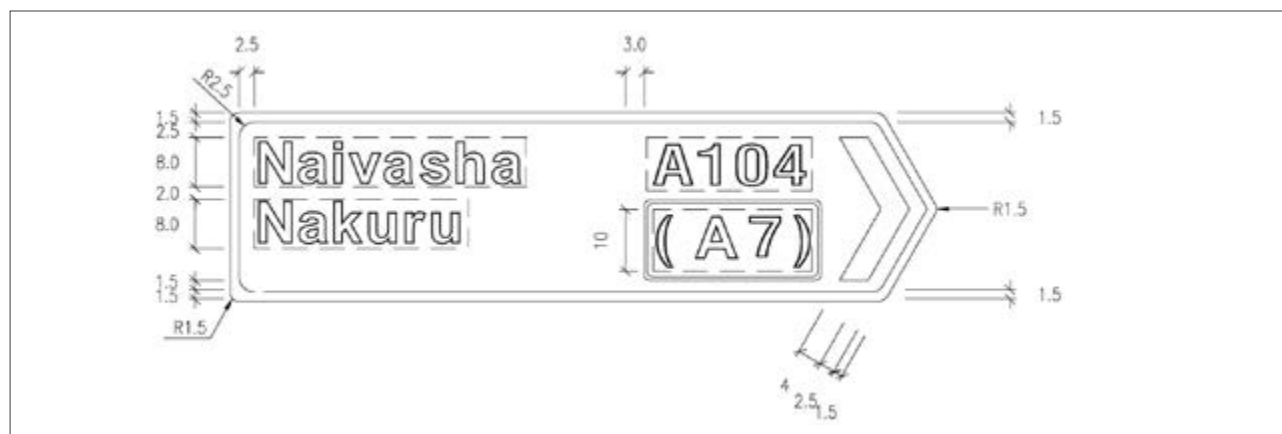


Figure 8.7 Sign Incorporating Expressway Patch, Sign IM6



8.3.3 Legend Panels

Expressway legend panels should be used on map type advance direction signs on roads (other than expressway) which have junctions with a expressway. The junction arm indicated by the legend panel should be the start of the expressway or an expressway slip road. The expressway destinations, route number and the expressway symbol may be shown on the blue panel.

Legend Panels should be designed as described in Section 5.2 of Chapter 5. The relative positioning of the expressway symbol, route number and destinations may be varied to optimise the sign design. Alternative layouts of legend panels are illustrated in Figure 8.8 to Figure 8.10.

Figure 8.8 Intermediate Junction Signing from Connecting National Route, Sign IM7

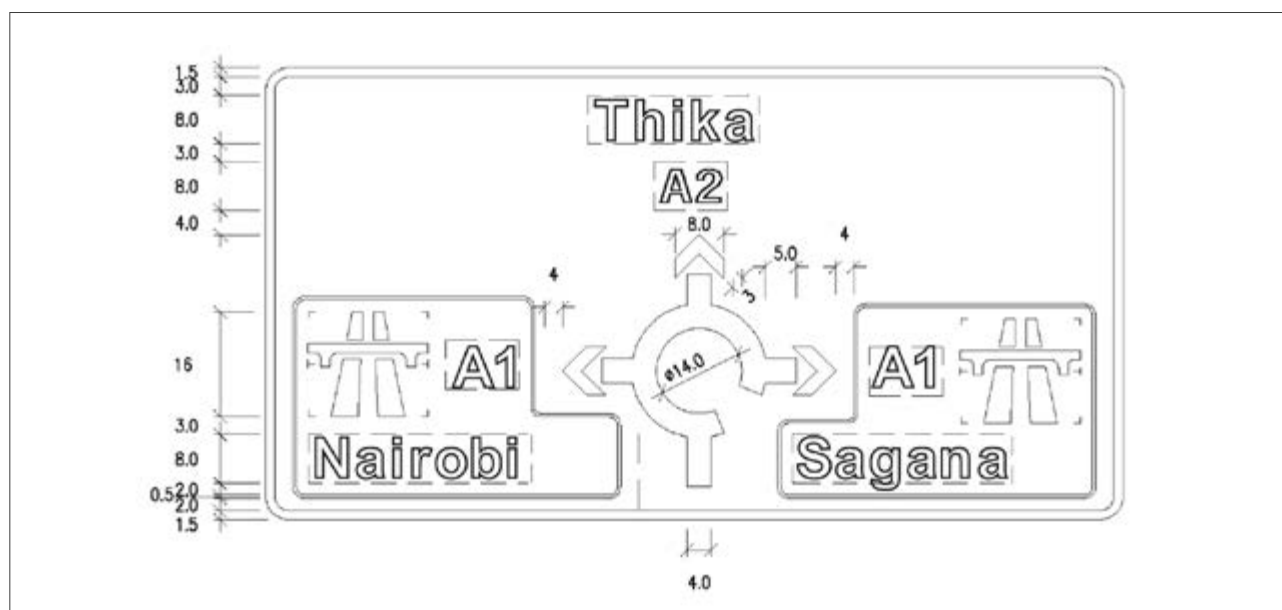
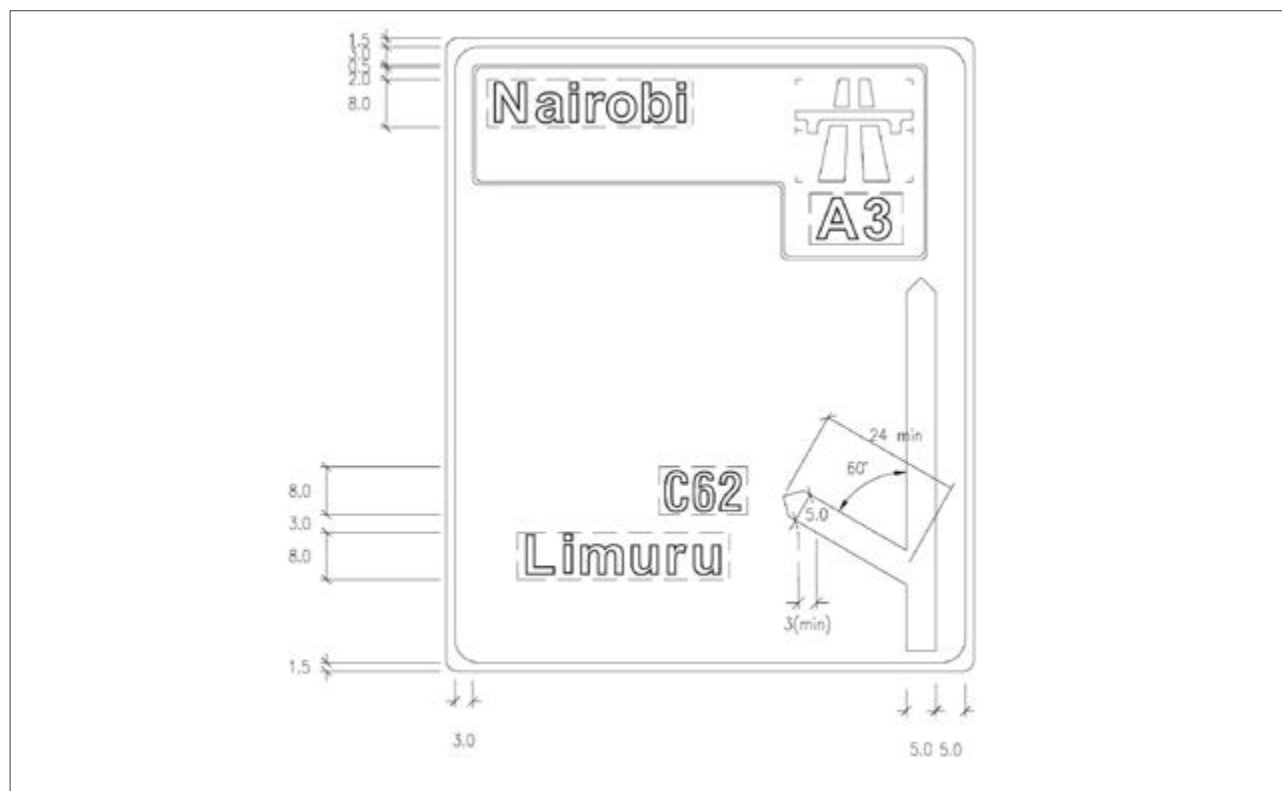


Figure 8.9 Signing from a Terminal Roundabout, Sign IM8



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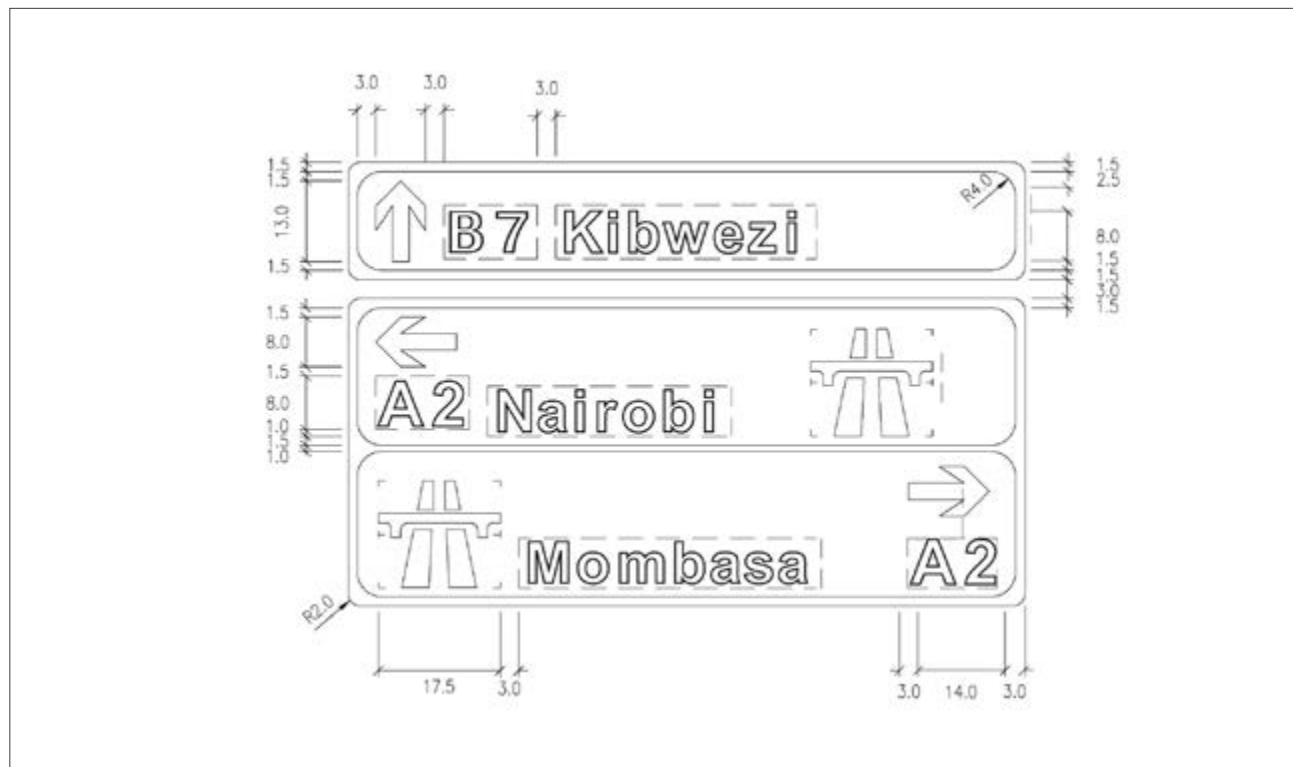
Expressway Signs

Map type signs incorporating blue expressway panels are shown in [Figure 8.8](#) and [Figure 8.9](#). [Figure 8.8](#) shows how an intermediate junction on the expressway should be signed from a connecting national road. [Figure 8.9](#) shows how the start of the expressway should be signed from an at-grade roundabout. [Figure 8.10](#) indicates the sign used where the road ahead changes to an expressway.

Stack type advance direction signs can be used to indicate expressways. While map type advance direction signs incorporate blue legend panels to indicate the expressway (as shown in [Figure 8.10](#)), stack type signs have separate coloured panels for each class of road. When a route of expressway status is shown, the normal design rules apply, except that the expressway symbol has also to be shown.

For left and straight-ahead directions, the symbol should appear to the right of the destinations mentioned. For a right turn the symbol should appear on the left. In each case normal clearances should be used to position the symbol which should sit 1 s/w above the baseline of the tiles of the English place name. An example sign is illustrated in [Figure 8.10](#). A coloured version of this sign is shown in the [Appendix 3](#).

Figure 8.10 Expressway Status Sign on National Route, Sign IM10



8.3.4 Direction Signs

Direction signs located on roads other than expressway but pointing directly onto an expressway should be white on a blue background. The expressway symbol should be placed at the opposite end of the sign to the chevron as indicated in Figure 8.11.

Figure 8.11 Direction Sign from National or Regional Road Sign IM 11



8.4 Signs on Expressway

All directional signs on expressways are in the blue and white colour scheme. On exit slip roads, map type signs should have a background colour appropriate to the higher classification of joining roads and appropriate coloured legend panels. Stack type signs on exit slip roads should have full coloured directional panels appropriate to the route indicated.

The "entry to expressway" sign (Figure 8.13) indicates the commencement point of an expressway. An additional sign (Figure 8.12) may be required to indicate the restrictions on traffic entering the expressway.

Note: This sign should be used in advance of the start of the expressway. Variations include "Expressway Left" and "Expressway Right".

Figure 8.12 Expressway Ahead, Sign IM12

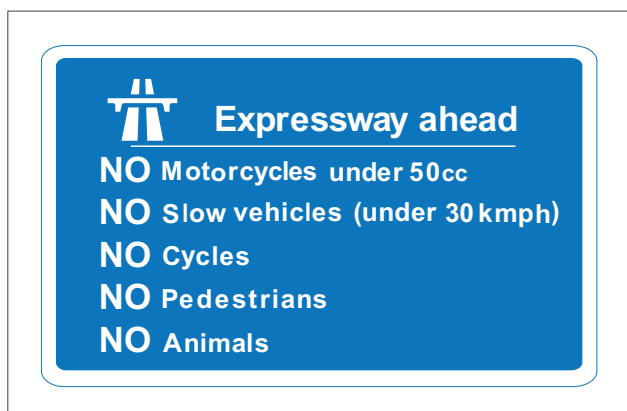
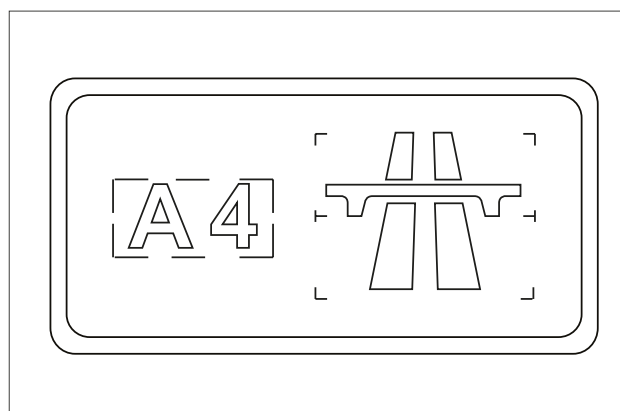
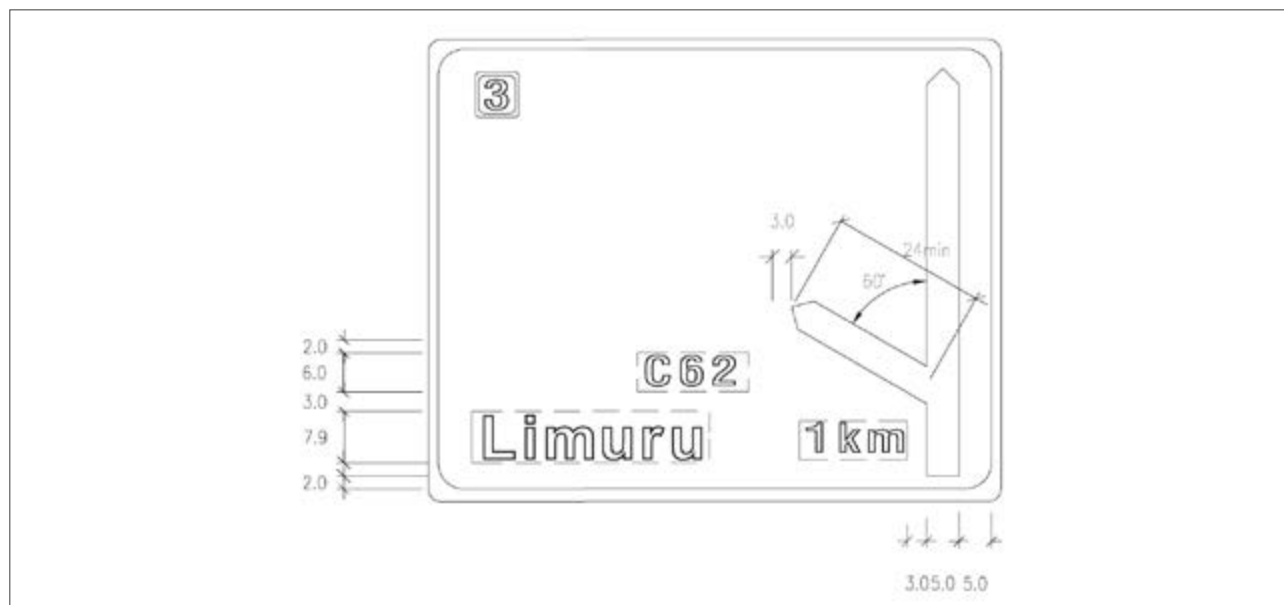
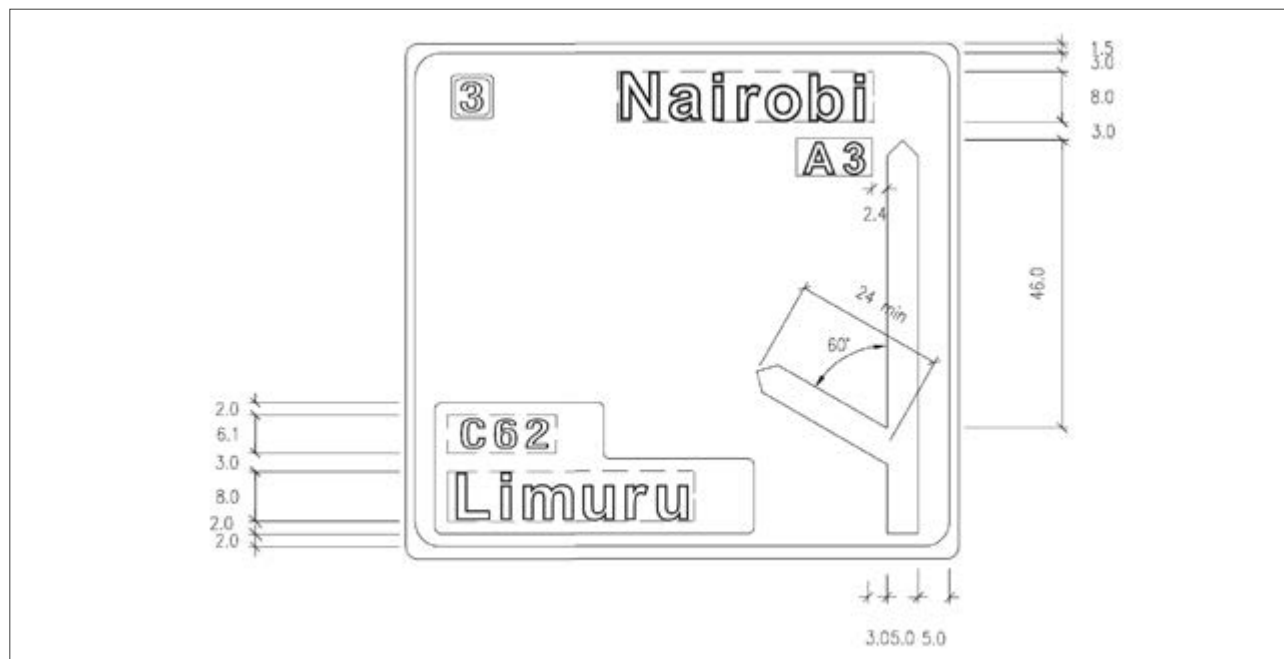


Figure 8.13 Entry to Expressway, Sign IM13

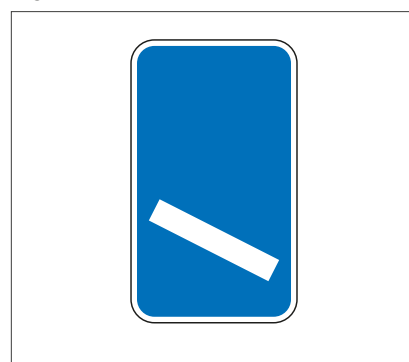


On the approach to intermediate junctions on the expressway, a map type advance direction sign should be sited at 1km from the start of the slip road taper. The design should incorporate the junction number and the distance as indicated in Figure 8.14.

A final advance direction sign should be sited at 500m from the start of the exit slip road taper. This sign is shown in Figure 8.15 and should not indicate the distance. It should be sited to avoid obscuring the 300 m countdown marker shown in Figure 8.16 or avoid being obscured itself by any over-bridges.

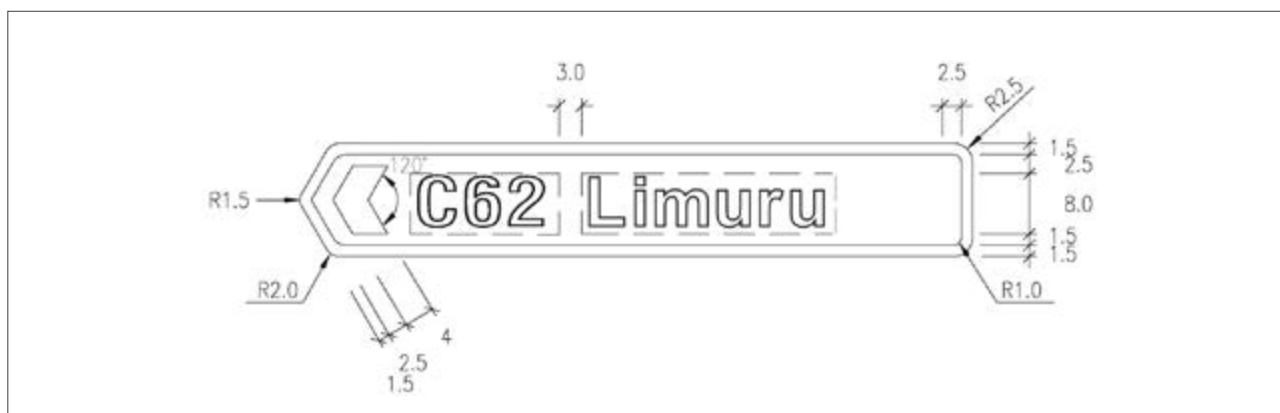
Figure 8.14 Advance Direction Sign for Intermediate Expressway Junction, Sign IM14**Figure 8.15** Final Advance Direction Sign for Intermediate Expressway Junction, Sign IM15

The countdown marker posts illustrated in [Figure 8.16](#) to [Figure 8.18](#) should be used at 300m, 200m and 100 m from the start of the slip road taper.

Figure 8.16 300 m Countdown Marker, Sign IM16**Figure 8.17** 200 m Countdown Marker, Sign IM17**Figure 8.18** 100 m Countdown Marker, Sign IM18

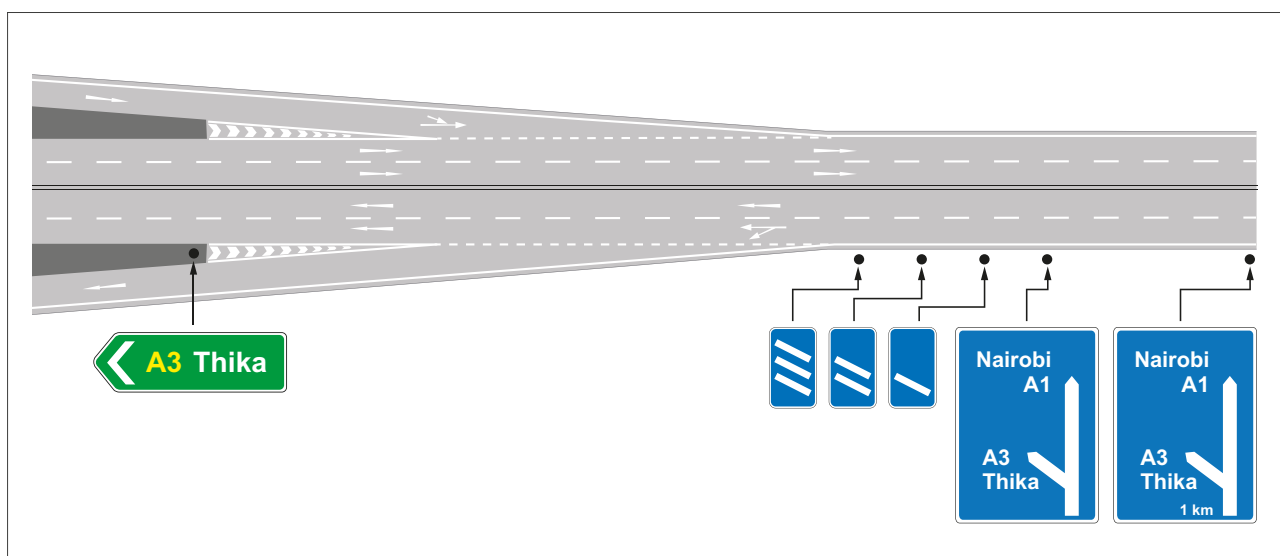
A direction sign should be sited on the nose of the division between the main carriageway and the exit slip road. It should show the major route or destination already signed on the advance direction sign for that junction. The sign should have white borders, chevron, and text on a blue background. An example is shown in [Figure 8.19](#).

Figure 8.19 Direction Sign for Expressway Exit Slip Road, Sign IM19



The relative positioning of all signs on the expressway at an intermediate junction is indicated in [Figure 8.20](#).

Figure 8.20 Position of Expressway Signs at an Intermediate Junction



On exit slip roads, map type advance direction signs take the background colour appropriate to the higher classified route at the junction. Information relating to other routes is shown on legend panels (see [Figure 8.21](#)).

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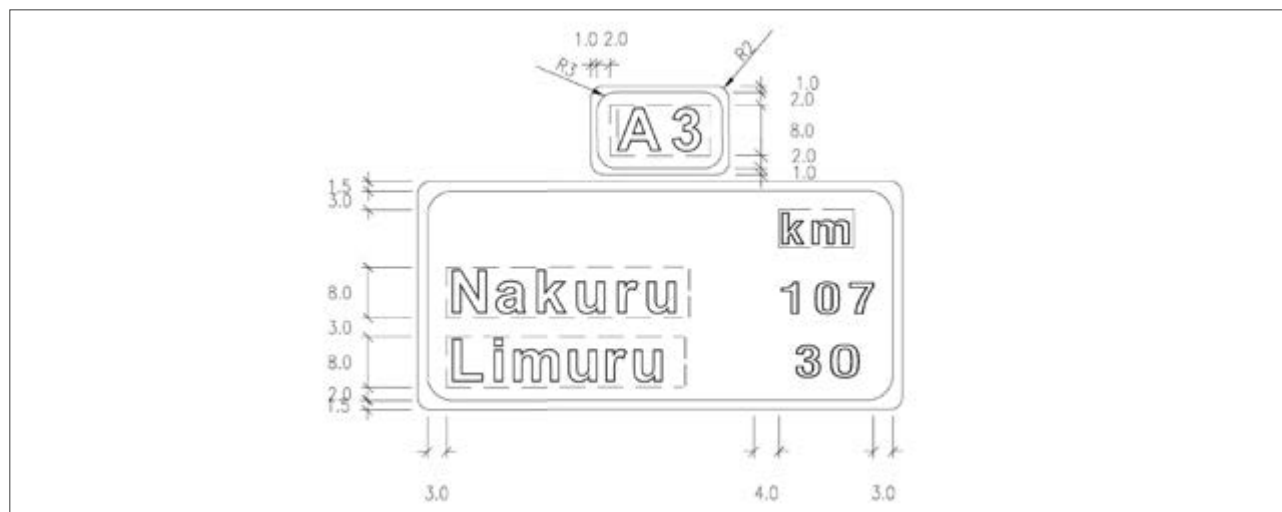
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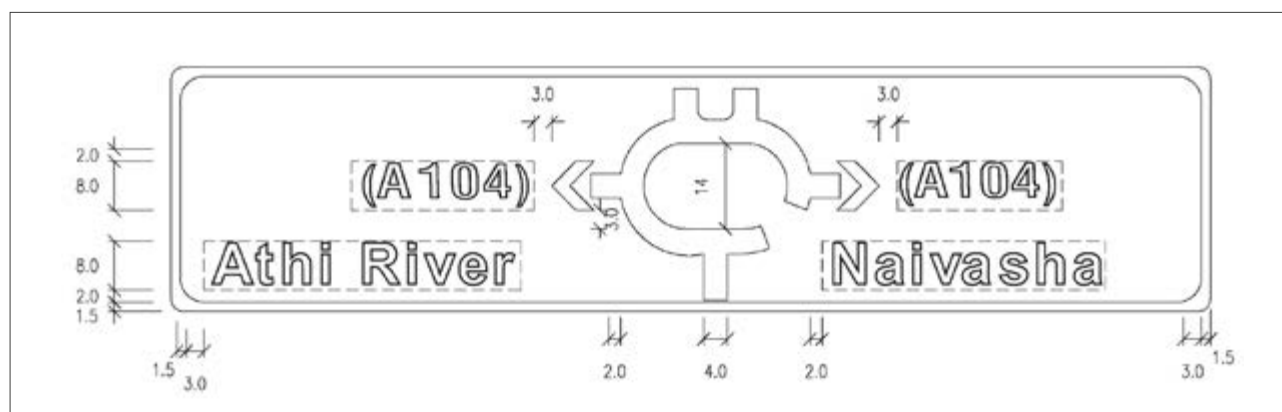
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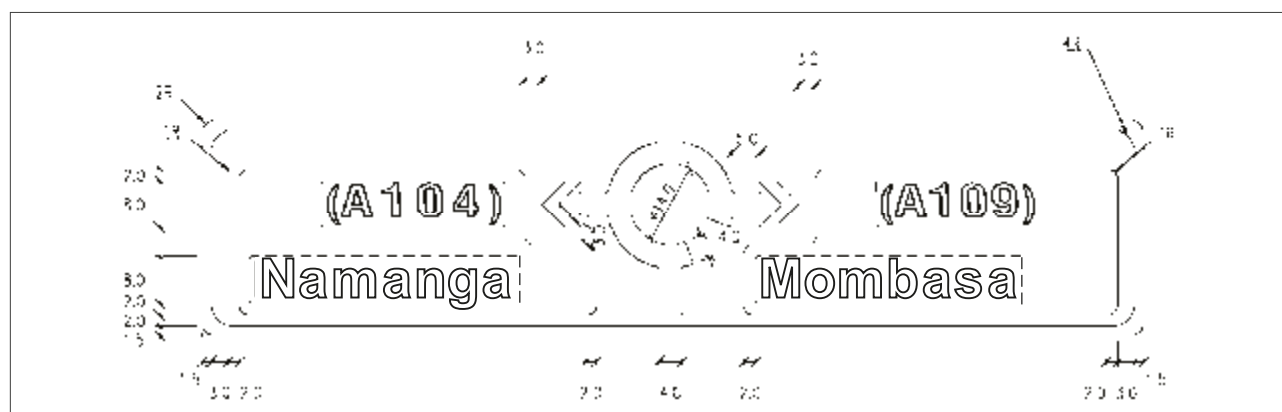
Expressway Signs

Figure 8.21 Map Type advance direction sign on Expressway Exit Slip Road, Sign IM20

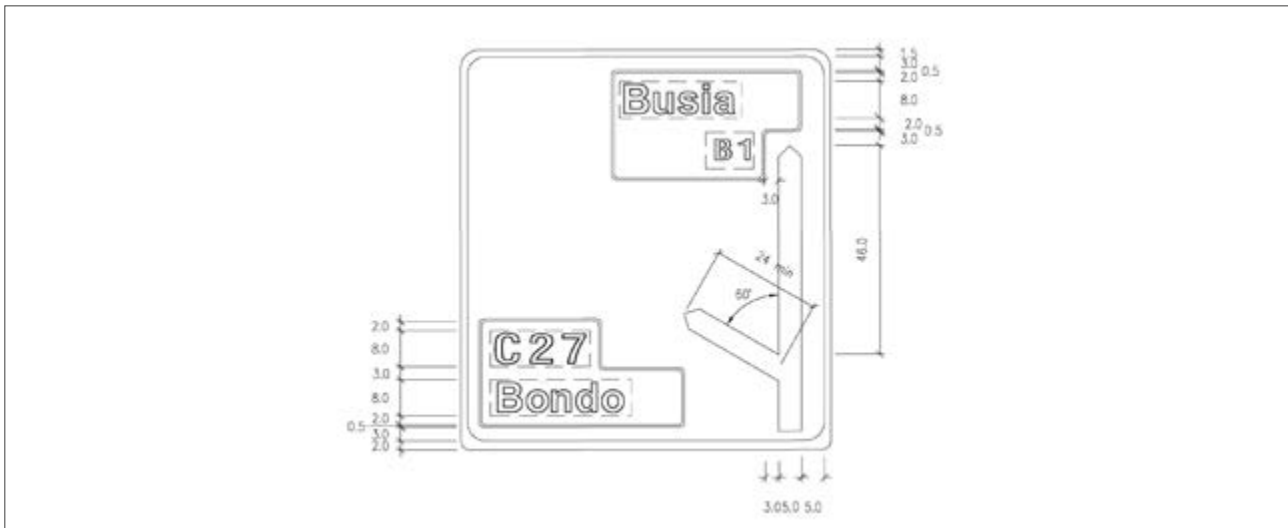
A route confirmation sign may be provided 350 m beyond the acceleration lane of an entry slip road (see Figure 8.22). Where interchanges are at close intervals, these signs may be omitted.

Figure 8.22 Route Confirmation Sign for Placement Beyond Entry Slip Road, Sign IM21

When an expressway ends at an at-grade terminal roundabout, advance direction signs should be sited at 1 km and 0.5 km from the roundabout. Roads leading from the junction should be indicated on coloured legend panels appropriate to their classification. Countdown markers should be placed on both sides of the approaching carriageway (see Figure 8.23).

Figure 8.23 Countdown Marker Arrangement for Both Sides, Sign IM22

Where an expressway leads straight into a road of another classification at a grade separated junction, the junction will be signed as normal. Details of the joining roads appear on the blue background in legend panels, coloured according to their classification. See Figure 8.24.

Figure 8.24 Sign Arrangement for entry to Roads of different Classification at Grade, Sign IM23

The sign shown in Figure 8.27 should be placed on both sides of the carriageway, at the point where the expressway regulations cease to have effect. The signs shown in Figure 8.25 and Figure 8.26 should also be placed on both sides of the carriageway in advance of the sign in Figure 8.27 at the distances shown.

Figure 8.25 End of Expressway 1 km, Sign IM24**Figure 8.26** End of Expressway 500 m, Sign IM25**Figure 8.27** End of Expressway, Sign IM26

8.4.1 Gantry Signs on Expressways

Overhead gantry signs are used in urban areas where traffic volume and speed are high. Gantry signs are useful where there are multiple lanes and it is difficult to read the roadside signs while driving, because of the multiple lanes of traffic. Overhead gantry signs can direct the drivers to take for the proper lanes for their intended destination.

Gantry signs can be considered where:

1. Lane indication is necessary for the appropriate destinations ahead.
2. The geometric configuration at the junction (e.g. horizontal or vertical curve) requires it.
3. There are 3 traffic lanes or more in one direction.

There are two types of gantry sign:

1. One for the "lane-drop" situation where the left lane of the approach carriageway forms the off-slip lane and does not continue through the junction.
2. One for the "non-lane-drop" situation where the number of lanes on the approach carriageway continue through the junction.

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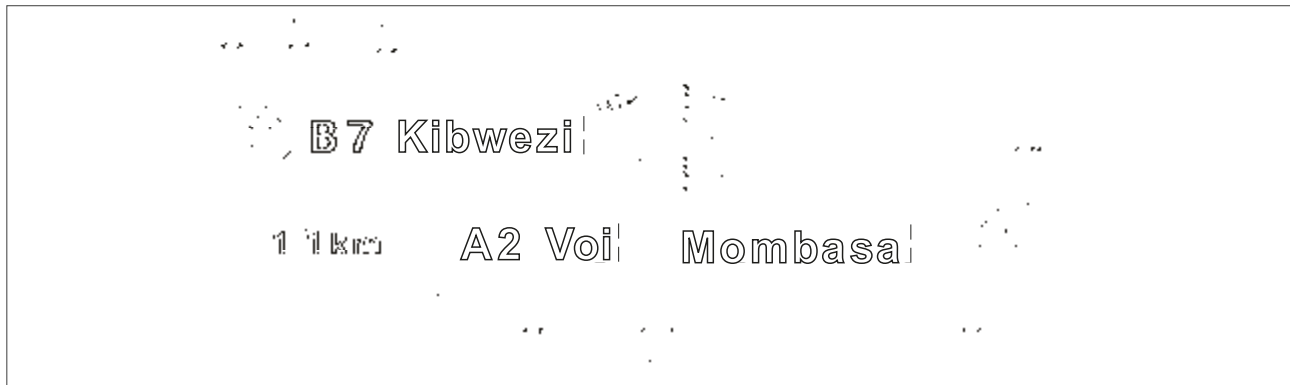
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Expressway Signs

8.4.2 Non-Lane-Drop Signs

The non-lane-drop sign is similar in design to standard stack type signs. The assembly consists of two main signs, one above the other. The lower sign with the ahead arrow should be centred over the main running carriageway whilst the upper sign is offset so that the inclined arrow is not directly above the lower sign. An example sign is shown in Figure 8.28.

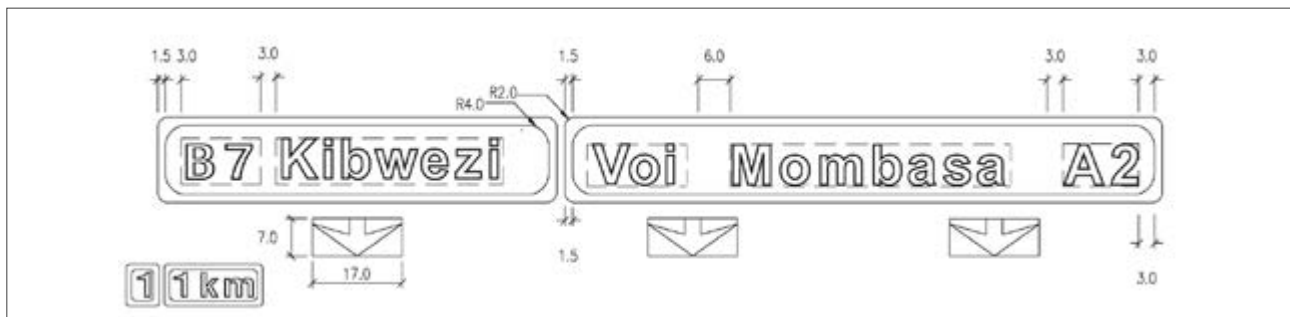
Figure 8.28 Non Lane-Drop Signing – Sign on the Bottom, Sign IM27



8.4.3 Lane-Drop Signs

Lane-drop signs are also similar in design to stack type signs although directional arrows are replaced by lane arrows located below the main signs. Each route should be shown on a separate directional panel to give the appearance of two panels side by side. These separate signs should be aligned over the appropriate traffic lanes (see Figure 8.29).

Figure 8.29 Lane-Drop Signing, Sign IM28



9 Guidelines for Signs on Urban Roads

9.1 General

The signing system on city roads should give clear information to the users. Usually, in city environment traffic signs get lost amongst the advertisement hoardings and roadside activities. The pedestrians, cyclists and PWD should be given clear directions so that they can use the roadways safely.

Signs are designed to be legible and read in time to react. Sign design should be based on:

1. Long visibility distances.
2. Large lettering and symbols.
3. Short messages for quick comprehension.

9.2 Colour of Signs

The colour of all types of signs except direction information signs shall be same as that shown in [Chapter 5](#). The direction information signs shall have white lettering, border and arrows on blue background. In case of facility signs, a black symbol shall be displayed within white square in blue background.

9.3 Messages on Overhead and Shoulder Mounted Signs

The messages on all signboards shall be bilingual - local language and English.

9.4 Warrants for Sign Installation

The placement of regulatory and warning signs shall follow the guidelines mentioned in [Chapters 3](#) and [4](#). Distances of installation shall be based on the speeds and clear visibility distance as mentioned in [Table 4.2](#) and Appendix 1.

9.5 Size of Signs

For expressway, the minimum size of signs shall be 1200 mm. For all other signs, three sizes shall be used: small (600 mm), normal (750 mm) and large (900 mm).

9.6 Siting of the Signs with Respect to Carriageway

The placement of the signs shall be as per [Section 2.4](#). Where site condition demands any deviation, the competent authority may decide after due consultation with traffic engineering specialists.

9.7 Size of Letters

The size of letters shall be as per [Table A.1-1](#) in the Appendix 1. For supplementary plates attached with facility signs, regulatory signs or warning signs, the recommended letter size is 100 mm. Text with 100 to 125 mm letter size shall be used in the supplementary plates.

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9.8 Guidelines for Informatory Signs Installation

General guidelines given in [Chapters 5](#) and [6](#) for information signs shall be adopted for installation of signs in city roads as well.

9.8.1 Gantry Mounted Signs

Gantry mounted signs are discussed in [Section 8.3.6](#). These are appropriate for a location ahead of a flyover / interchange in an urban setting.

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Guidelines for Signs on Urban Roads

10 Signs for Roadworks

10.1 Introduction

This chapter describes the signs recommended for use at roadworks. The first section describes the design of signs which are recommended for use at both urban and rural roadworks. [Section 10.27](#) deals specifically with works at urban locations and [Section 10.28](#) deals with rural locations.

The layouts shown are to demonstrate the principles to be used in the treatment of various road work situations. Uniformity of signs help the road user in understanding what to expect. Uniformity requires that similar hazards be treated similarly.

Detour signs and warning signs associated with roadworks should have a unique colour board - orange-yellow in place of the traditional golden-yellow. Suitable colours for the different messages are:

- BLUE for regulatory messages.
- RED for prohibitory messages.
- BLACK / WHITE for informatory messages.

Suitable colour shades for roadwork signs are as per British Standard Colour Shades given in BS 381 C Colour Charts and Munsell Colour System. [Table 10.1](#) gives the reference number for each colour.

Table 10.1 Colour Shades for Roadwork Signs

Colour	BS 381 C Number	Munsell Reference
Blue	109	2.5 PB 3/7
Green	262	0.65 BG 2.84/8.45
Orange	557	2.5 YR 6/14
Red	537	7.5 R 4.5/16
Yellow	356	10 YR 7.5/12

Flashing and Steady-state lamps should be in light-yellow.

All roadwork signs recommended in this manual start with RW.

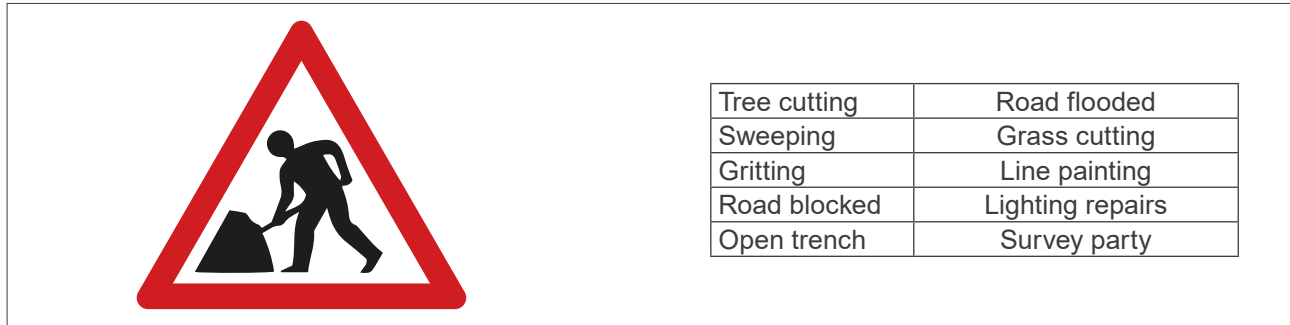
10.2 Recommended Signs for Road works

Recommended signs are given in the following sections. Dimensions of signs are given in the [Appendix 4](#). Supplementary information plates may be used with the signs, indicating the distance to the hazard or activity ahead. For stability on the ground, triangular warning signs may be mounted on a rectangular frame.

10.2.1 Roadworks Ahead

This sign is shown in Figure 10.1. It is to be used in advance of all roadworks and should be the first sign a driver sees on approaching roadworks. It may include a supplementary plate indicating the nature of the work. Examples of such plates are:

Figure 10.1 Roadworks Ahead, Sign RW1



10.2.2 Uneven Surface

This sign should be used where it becomes necessary to direct traffic over a temporary uneven surface. The sign is shown in Figure 10.2.

Figure 10.2 Uneven Surface, Sign RW2



Figure 10.3 Slippery Road Ahead, Sign RW3



10.2.3 Slippery Road Ahead

This sign is appropriate where the road has become slippery due to roadworks e.g. mud deposited on the surface. When this sign is used in a position away from the current roadworks, it should have a yellow background. The sign is shown in Figure 10.3.

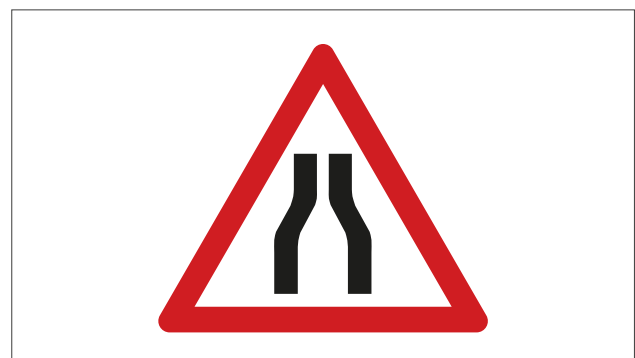
10.2.4 Road Narrows

Road narrowing signs are shown in Figure 10.4 and Figure 10.5.

Figure 10.4 If narrowing is from left side, the left line should be shown as narrowing, Sign RW4



Figure 10.5 Road Narrows, Sign RW5



10.2.5 Manual Traffic Control Ahead

These signs indicate the presence of manual traffic control either by flagman or Stop/Go battens. The signs are shown in Figure 10.6.

10.2.6 Traffic Signals Ahead

This sign indicates the presence of traffic control by means of temporary signals. It is shown in Figure 10.7.

Figure 10.6 Manual Traffic Control Ahead, Sign RW 6



Figure 10.7 Traffic Signals Ahead, Sign RW 7



10.2.7 Two-way Traffic Ahead

This sign is used where a one-way street is converted to two-way operation for the purpose of carrying out roadworks.

This sign should also be used when a road which has been converted to one-way operation is restored to two-way traffic following completion of roadworks. The period immediately following the re-opening of the road to two-way traffic can be hazardous because drivers have become accustomed to one-way operation. The sign should be left in position for a period of two to three weeks and then removed. When the two-way sign is used in this context, it should have a yellow background. The sign is shown in Figure 10.8.

Figure 10.8 Two-Way Traffic Ahead, Sign RW8

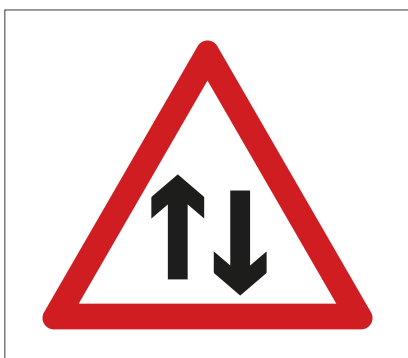
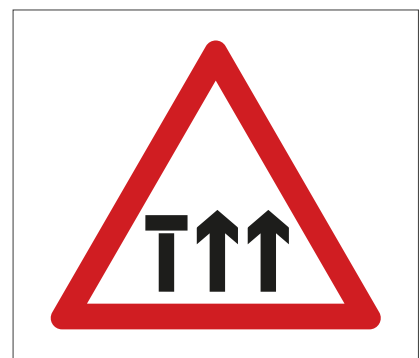


Figure 10.9 Left Lane closed for two-lanes in one direction, Sign RW 9



Figure 10.10 Left Lane closed for three-lanes in one direction, Sign RW10



10.2.8 Signs for Lane Closures Ahead

Closure of the left, centre or right lanes in multi-lane roadways are shown in Figure 10.9 to Figure 10.14.

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Signs for Roadworks

Figure 10.11 Centre Lane closed for three-lanes in one direction Sign RW11



Figure 10.12 Right Lane closed for three-lanes in one direction Sign RW12

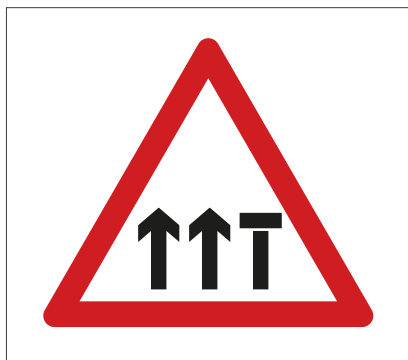


Figure 10.13 Left Lane closed two right lanes diverted, Sign RW13



Figure 10.14 Left Through Centre Lane, Closed Right Lane Diverted, Sign RW14



Figure 10.15 Major Road Works Ahead, Sign RW15



10.2.9 Major Road Works Ahead

This sign should be used ahead of major works and should continue for a period of months. It should be mounted on poles, properly set into the ground, not in drums. The sign shown in [Figure 10.15](#) should be used in association with this.

Figure 10.16 No Through Road, Sign RW16



Figure 10.17 Road Closed, Sign RW17



10.2.10 No Through Road

If a road is blocked ahead, and through nature of the road changes because of construction, the sign shown in [Figure 10.16](#) is to be used.

10.2.11 Road Closed

This sign shown in [Figure 10.17](#) is to be used in emergency situations such as trees across the road, etc.

10.3 Delineation of Roadworks

The most commonly used devices to guide traffic around roadworks are cones and drums.

Cones are transportable and can be quickly deployed, however they are unstable in windy conditions and are easily knocked over by traffic. Whereas Drums are stable and present a solid appearance to drivers. However, they occupy more road space than cones and require a considerable amount of maintenance.

The guidelines for using cone or drum are:

1. Short Term Roadworks.
2. Minor works which can be completed within one day and works of a mobile nature are best protected by cones. Examples of such works are:
 - a. Routine road maintenance
 - b. Manhole inspection
 - c. Public lighting maintenance
 - d. Long Term Roadworks:

For excavations that are to be left open for more than one day, it is recommended to use drums, with barrier tape strung between them and the lead-in and exit tapers marked by cones. In urban areas, gaps between barrels should not be greater than 1.5 m across each end and not greater than 3 m longitudinally. The spacing of cones should be in accordance with Table A.4-1 and A.4-2 for urban works and Table A.4-3, A.4-4 and A.4-5 of Appendix 4, for rural works.

Figure 10.18 Drum

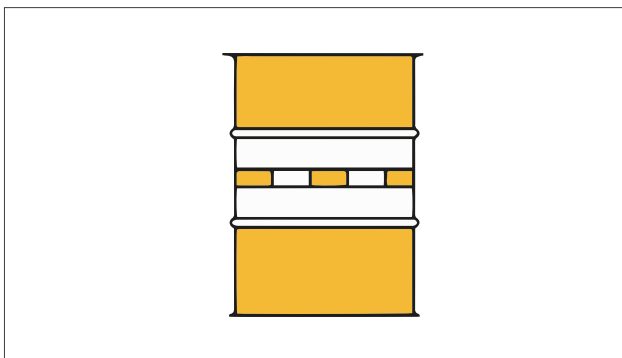
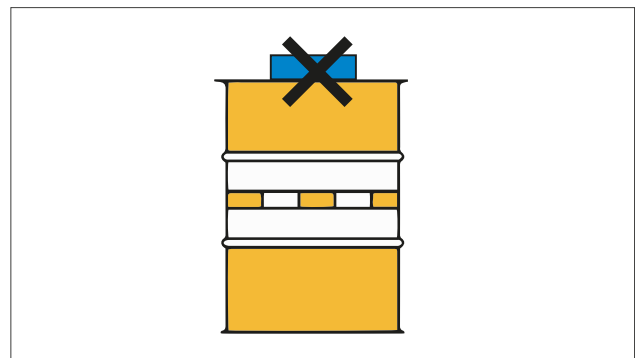


Figure 10.19 Do Not Weigh Down a Drum Using a Block



10.3.1 Drums

Drums should be painted with orange and white horizontal bands as illustrated in Figure 10.18. They should have an undercoat and gloss coat.

Drums should not be weighed down by placing a heavy weight such as a concrete block on top of it as shown in Figure 10.19. This is a hazardous practice.

The recommended method of weighting drums is to remove the tops and fill them to a quarter full with sand or gravel. They must not be overfilled as this would make them too heavy and a source of hazard to the road users.

The practice of placing planks across drums as shown in Figure 10.20 is not recommended.

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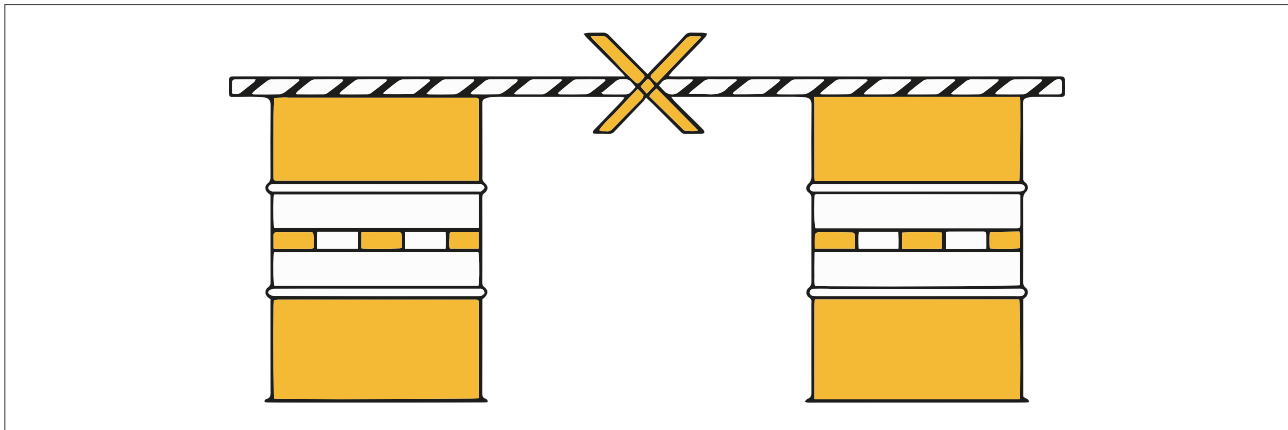
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Signs for Roadworks

Figure 10.20 Do Not Place a Plank Across Drums

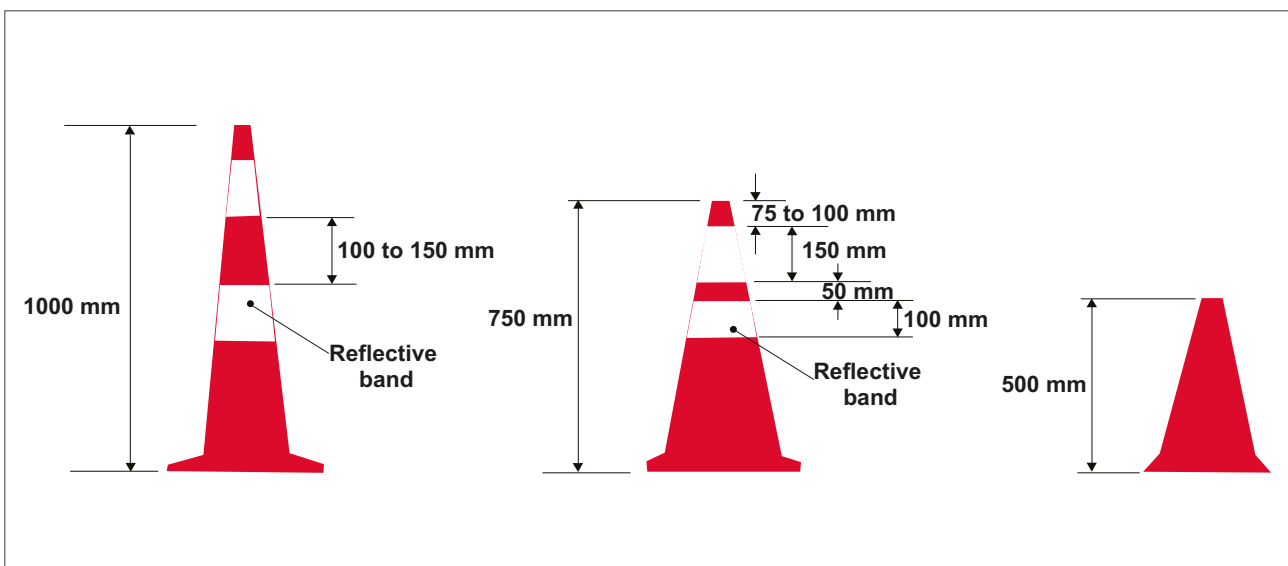
An alternative method of creating a continuous barrier around roadworks is by adding barrier tape along with drums. This tape should be of reflective fluorescent strips or a continuous red and white plastic tape.

Barrier tape should not be used to span large gaps between drums. The maximum spacing is 3 m. Tape should be always kept taut. Barrier tape is not suitable for protecting deep excavations.

10.3.2 Cones

Traffic cones may be of the height of 500 mm, 750 mm and 1000 mm and of 300 to 500 mm in diameter or in a square shape. They shall be of brilliant red/orange/yellow and shall be made of a material that can be struck without causing damage to the impacting vehicle. **Figure 10.21** shows the traffic cones of 500 mm, 750 mm and 1000 mm.

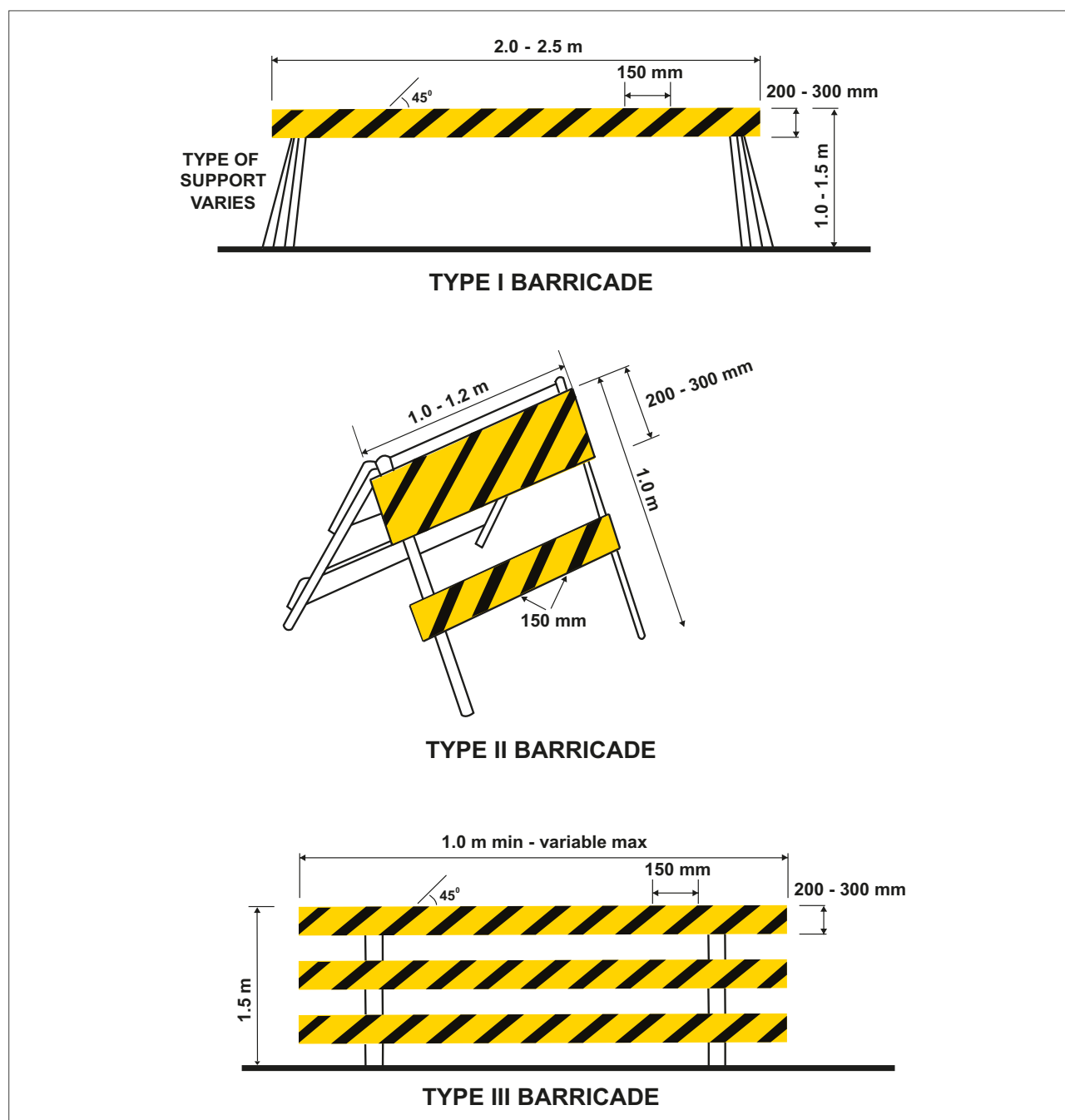
As a general rule, cones between 750 mm to 1m high should be used. Additional stability may be obtained by surrounding the base with sandbags - NOT concrete blocks.

Figure 10.21 Traffic Cones

10.3.3 Barricades

Barricades are intended to provide containment without significant deflection or deformation when hit by a vehicle. They can re-direct errant vehicles along the barrier. Barricades can be portable or permanent. Portable barricades should be stable under adverse weather conditions. Figure 10.22 shows standard barricades. Type I and II are portable, Type III is permanent.

Figure 10.22 Standard Barricades



10.3.4 Lamps

Lamps should always be used along the length of an obstruction during the hours of darkness. Lamps should have yellow lenses and show a steady light in both directions. Flashing lights should not be used along the length of the works. These should be used only at the extremities to indicate the presence of a hazard. Lamps should have a retro-reflective ring around the lens or have reflectors fitted to the body of the lamp.

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10.3.5 Keep Left / Keep Right / Pass Either Side

The signs direct drivers to keep left, right or pass either side. They consist of white arrows on a blue reflective background with a white border, as shown in [Figure 10.23](#) to [Figure 10.25](#). The use of supplementary message signs is not recommended.

This type of sign should not be used on a footpath as guidance for pedestrians.

Figure 10.23 Keep Left, Sign RW21

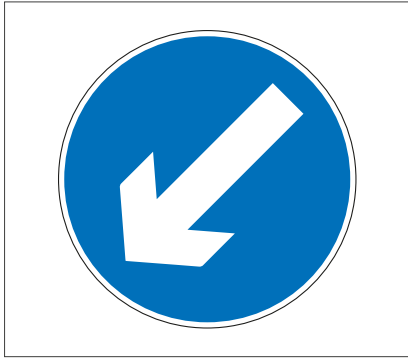


Figure 10.24 Keep Right, Sign RW22



Figure 10.25 Pass On Either Side, Sign RW23



10.3.6 Chevron

The chevron illustrated in [Figure 10.26](#) indicates a sharp change in direction and is used to guide traffic around roadworks in situations where it is not possible to provide a lead-in taper of cones, such as at a junction.

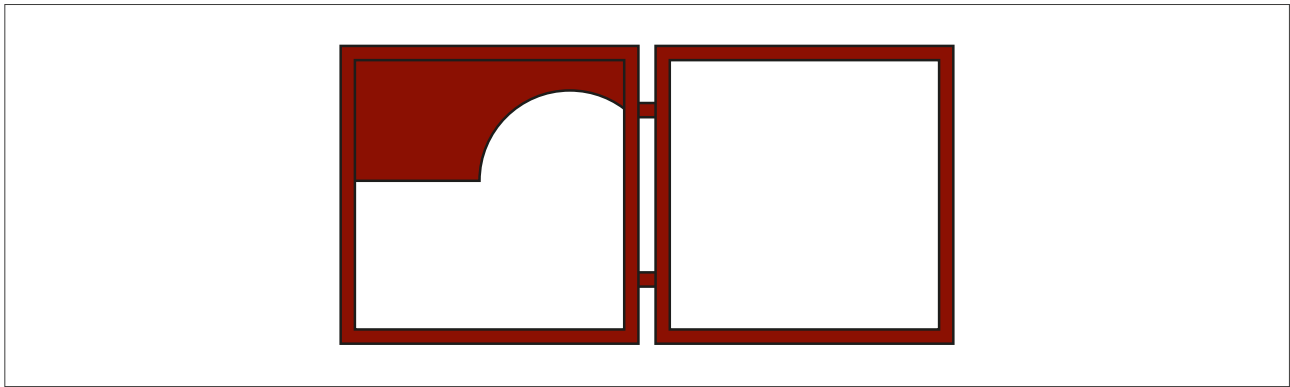
Figure 10.26 Chevron Sign RW24



10.4 Protection of Pedestrians at Roadworks

Excavations on footpaths can constitute a serious hazard for pedestrians, particularly for the visually impaired and mobility impaired. The following guidelines are recommended for their protection:

1. Where a footpath is obstructed by roadworks, the entire area obstructed should be surrounded by a protective barrier. The barrier should be substantial enough to prevent a person from tripping over it. It should also be capable of giving positive guidance around the obstruction for a visually impaired person. Barriers should consist of identical frames hinged together. Frames are made from square hollow section of steel with a wire mesh welded to it. Each frame is approximately 1m square. A number of these barriers are hinged together as shown in [Figure 10.27](#).
2. Such barriers are ideal for a small opening where the barriers can be arranged in a closed box fashion. They are unsuitable for a long length due to their inherent instability in this configuration. There are patented plastic barriers available in the market which can be used for long length.
3. When the footpath is completely blocked, an alternative passage must be provided on the carriageway. This passage must be kept clear of obstructions such as tools and timber etc. and be protected from passing traffic by means of cones or barrels. Examples of the deployment of signs and barriers are shown in the [Appendix 4](#).

Figure 10.27 Protective Pedestrian Barrier Sign RW25

10.4.1 Pedestrians Signs

The sign shown in **Figure 10.28** should be used when roadworks on a pedestrian crossing make the crossing unusable. They should be attached to the signal or beacon poles and face the carriageway. Signals or beacons should also be switched off.

The sign shown in **Figure 10.29** should be used with an appropriate arrow where a new temporary route is made for pedestrians and the new route is not obvious.

When the direction of vehicular traffic flow at a pedestrian crossing is altered, the sign shown in **Figure 10.30** with appropriate wording (Look Left/Look Right) should be provided.

Figure 10.28 Crossing Not in Use, Sign RW26**Figure 10.29** Pedestrians Keep Right, Sign RW27**Figure 10.30** Pedestrians Look Left, Sign RW28

10.5 Traffic Control at Road Works

To maintain two-way traffic operation at roadworks sites, an unobstructed width of at least 5.5 m is required. Where this width is not available, the width should be restricted to not more than 3.7 m to discourage two-way operations. The minimum width required to carry a single lane of traffic is normally 3 m, but car only traffic can be maintained at 2.5 m.

On minor roads with light traffic, the single lane can operate without specific traffic control. However, the following criteria must be met for this method to work safely:

1. Both ends of the site must be inter-visible.
2. The length of single lane operations should not exceed 50m.

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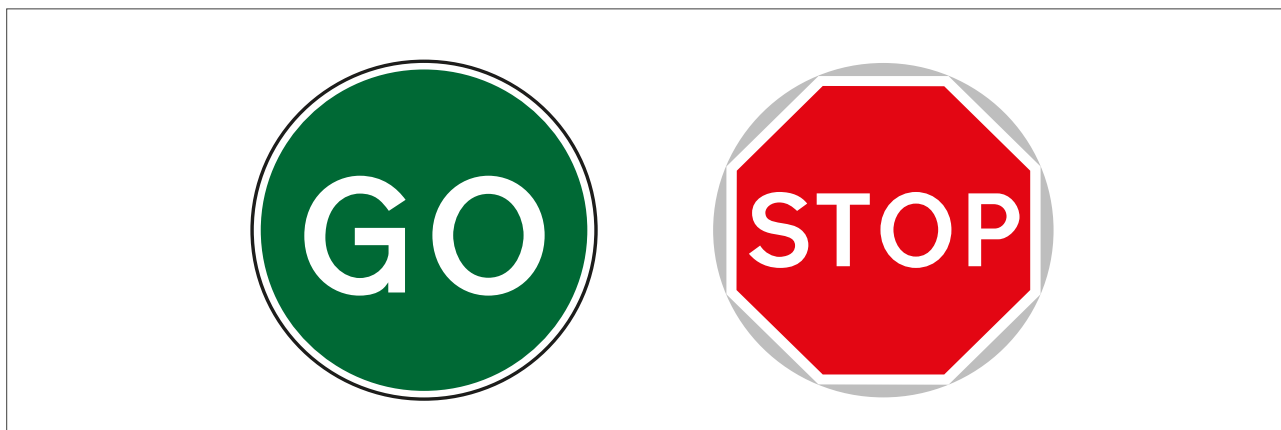
Signs for Roadworks

If traffic volume increase, some form of traffic control is required. The basic type of control is by using flagmen or STOP/GO signs as shown in **Figure 10.31**. These consist of two signs mounted back-to-back on a pole so that they can be easily rotated to show in either direction.

Roadworks up to 20 m in length can generally be controlled by one person using a STOP/GO sign. When the length is greater, it is recommended to have this sign at each end.

Where the length is greater than 100 m, it becomes difficult to control traffic manually. A more sophisticated method like temporary traffic signal is recommended.

Figure 10.31 Stop/Go, Sign RW29



10.5.1 Temporary Traffic Signals

When the length of the single lane is too long for manual control or when control is required during the hours of darkness, temporary traffic signals may be used.

Where traffic is approximately equal in both directions, fixed time operation can be used. Where flow is tidal, like on a major commuter route, timings must be adjusted during the day to cater to the varying traffic flows. Signals can also be operated in vehicle actuated mode.

10.6 Diversions

An alternative to temporary traffic signal is one-way operation by diverting the traffic in the opposite direction to another route.

When traffic is diverted proper guidance is required. The end of diversion should also be clearly shown. The signs shown in **Figure 10.32** to **Figure 10.35** are recommended for diversion.

Figure 10.32 Diversion ('m') Ahead, Sign RW30



Figure 10.33 Diversion Right, Sign RW31



Figure 10.34 Diversion Left, Sign RW32



Figure 10.35 Diverted Traffic Straight Ahead, Sign RW33**Figure 10.36** Diverted Traffic Right, Sign RW34**Figure 10.37** End of Diversion, Sign RW35

Advance warnings of diversion ahead are provided by the sign shown in [Figure 10.32](#). Distance of 100m, 200m may be used as appropriate.

One of the signs shown in [Figure 10.34](#), to [Figure 10.35](#) should be used, with an appropriate arrow, where there is only one alternative route.

Signs with arrows pointing to the appropriate direction are to be used at all decision points along the diversion.

The sign shown in [Figure 10.37](#) is to be used with an appropriate arrow to indicate a return to the original route.

Signs shown in [Figure 10.38](#) and [Figure 10.39](#) are used on divided roads where one carriageway is converted temporarily to two-way operation during road works. They indicate that traffic should cross over to the other carriageway.

[Figure 10.40](#) is to be used when the two-way movement gets separated ahead. [Figure 10.41](#) is to be used when the separation of two-way movement ends ahead.

Figure 10.38 Crossover to Left, Sign RW36**Figure 10.39** Crossover to Right, Sign RW37**Figure 10.40** Divided Road Ahead, Sign RW38**Figure 10.41** End of Divided Road, Sign RW39**Figure 10.42** No Overtaking, Sign RW40**Figure 10.43** End of Construction, Sign RW41

The sign shown in [Figure 10.42](#) indicates that overtaking is prohibited. The end of the prohibition is indicated by the same sign with a supplementary plate as shown in [Figure 10.43](#).

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Signs for Roadworks

10.7 Deployment of Roadwork Signs

The signing of roadworks should be carefully planned. The appropriate layout shown in Appendix 4, may be used to determine the number of cones, signs etc. required. If the situation is not specifically covered, the sign requirement can be determined by the applying the same principles contained in the layouts. The main criteria are:

1. Adequate advance warning.
2. Lead-in taper, where possible.
3. Safe guidance past roadworks.
4. Protection of works personnel.

10.7.1 Procedure for Setting Out Signs

1. A responsible person should be designated with the task of setting out signs and cones etc. He/she should be responsible for the maintenance of the signs throughout the duration of the works, ensuring that they are cleaned regularly and that signs and cones which are removed to allow access for site traffic are immediately replaced.
2. All personnel on site should wear brightly coloured jackets with a combination of fluorescent material (for daytime visibility) and retro-reflective material (for nighttime visibility).
3. The first sign a driver should see on approaching roadworks is the roadworks ahead sign. A supplementary plate may be added to this sign indicating the nature of the works.
4. The setting out of cones and signs etc. should be performed facing oncoming traffic. Setting out should commence with the roadworks sign at the recommended distance from the hazard. Signs and cones should then be set out in sequence, moving back towards the obstruction.
5. Signing should be confined to those signs which are relevant to the situation. Too many signs can detract from the message they are intended to convey. A driver attempting to assimilate several different signs in rapid succession may miss the relevant ones. It is better to use no sign at all than an incorrect one. Motorists who are continually confronted with inappropriate signs will eventually develop total disregard for all signs.
6. It is good practice for the person who set out the signs to drive through the area and check the signs from a motorist's viewpoint. This should be done from all approaches.
7. As work proceeds along the carriageway, signs should be repositioned as required.
8. Permanent signs which are temporarily made redundant by virtue of the roadworks should be covered up.
9. The site should be kept tidy, and all mud carried onto the road should be cleaned off.
10. On completion of roadworks, all roadwork signs should be immediately removed.

10.8 Urban Roadworks

Appropriate locations of signs, cones, etc are shown in **Figures A 4-15 to A 4-28** in Appendix 4. In urban areas it is not always possible to follow the diagrams precisely due to the presence of parked cars, entrances, traffic islands, etc. However, the general principles should be followed as far as possible.

The spacing of signs and length of taper shown in the diagrams are appropriate for speeds of 50 km/h and a hazard width of 3.2 m unless otherwise stated. For different speeds and widths see **Tables A.4-1 and A.4-2**. The descriptions of the figures are given in **Table 10.2**.

Table 10.2 Description of Roadwork Figures

Figure Number	Description	Notes
A 4-15	Work on walkways	
A 4-16	Give and Take	Shuttle working – self explanatory
A 4-17	One way shuttle working with flagman	This layout is suitable for day-time. use only
A 4-18	One way shuttle working with temporary signal	This layout is suitable for day or night-time.
A 4-19	Two-way street – roadworks forcing traffic over centreline	This arrangement should be used in situations where traffic is forced over the centreline by roadworks and where a minimum width of 5.5m is still available to traffic. In these circumstances it is essential to control opposing traffic to avoid head-on collisions.
A 4-20 to A 4-36	Self-explanatory	

Figures A.4-24 to A.4-27 are for use on major commuter routes where the loss of a lane during peak hours can have a serious effect on traffic flow.

10.9 Rural Roadworks

Three situations may arise:

- Fixed works.
- Mobile works.
- Detours.

10.10 Fixed Works

A good traffic layout must be prepared which will be consistent with the plan for the construction work.

10.10.1 Choosing a Layout

Nine possible layouts, which make provision for a similar number of typical work situations, are presented in Figures A 4-28 to A 4-36 in the Appendix 4. Most other work situations can be dealt with application of the same principles. Six of these layouts apply to two-way roads and three to four-lane divided roads.

10.11 Margin for Safety

A buffer space is needed between the point where a traffic lane is closed and the work area. The buffer space provides a margin of safety for both traffic and workers on fast roads.

10.12 Lighting

There are three types of lighting:

1. Floodlighting.
2. Flashing lamps.
3. Steady-state lamps.

Good floodlighting of the work area is essential for the safety and efficiency of the work team in the case of night shift working. The positioning of the lighting units should not cause glare to drivers.

1

Flashing lamps are best suited for hazard identification and may be positioned at prominent points, terminal points, points of change in direction and points of special hazard. Strobe lighting is to be used for works vehicles.

2

Steady-state lamps are most suitable for delineating the edge of the travelled way in the work area.

3

10.13 Reflectorising

Reflectorising of signs, chevrons, barriers, and other devices is desirable. Reflectorising should be such that the work area is adequately marked if the lighting is vandalised or otherwise fails. The use of fluorescent strips alongside the reflectorising on chevrons and barriers adds to their daylight efficacy. The use of Class 1 type reflective material is recommended.

4

5

10.14 Colour

Detour signs and warning signs associated with roadworks should be in a unique colour, orange in place of the traditional golden-yellow. For other signs, blue denotes mandatory, red prohibitory and black-white is for information.

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Flashing and Steady-state lamps should be in light-yellow colour.

8

10.15 One Way Operations

Where site work on two-way roads restricts the available road width, it may become necessary to implement one-way shuttle operation using either traffic signals or flag / batten-men.

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It should be very clear to road users as to whether one-way or two-way operation is in force. If one lane operation is intended, there should clearly be insufficient width for two vehicles. When vehicles are required to wait for green before proceeding, a signed waiting area should be identified such that obstruction is not caused to any opposing vehicle which may first have to clear the restricted right of way. Where a one-way shuttle operation ceases at works, all related signing should be covered.

10

10.16 Implementing Layout Guidelines

The following points augment the illustrated guidelines in layouts [A.4-28](#) to [A.4-36](#) in the Appendix 4.

1. Passing vehicles, having been alerted by the ROAD WORKS signs should be guided through the work area and be correspondingly informed when they are about to leave it.
2. Where a pedestrian footpath is being obstructed, alternative safe provision for pedestrians must be provided.

Prevailing traffic speed and roadway type are the principal considerations in layout. These dictate the distance of the first sign and the spacing of subsequent signs in advance of the hazard. High speed roads require larger signs and, at night, more warning lamps. Relevant guidance is given in [Tables A 4-1](#) to [A 4-5](#) in the Appendix 4.

1. The Triangular sign “MAN-WITH-SHOVEL” should precede every fixed roadworks site.
2. Signs, lamps, and other equipment should be cleaned periodically and should be replaced if damaged. The quality of the message which a sign imparts depends on how clean and smart its appearance is to the public eye.
3. Paraffin lamps should be filled, wick trimmed, batteries of electric lamps replaced, and lenses cleaned before they have lost their standard efficiency.
4. On high-speed roads, a buffer area clear of vehicles and fixed obstructions should, where possible, separate the work area from approaching vehicles.

5. Whenever the nature of the work permits, the work area should be clearly and positively marked off by cones, drums, barriers, garlands, etc.
6. The marked-off area should encompass all works machines, stored material, and debris.
7. Access to the work area should be clearly identifiable, have good sight distance and not cause ambiguity to passing vehicles.
8. The use of STOP-GO battens as illustrated in Layouts A.4-31 and A.4-32 is appropriate to daylight only. The batten-man should wear a fluorescent orange waistcoat and should be trained on how to deploy the batten.
9. Work staff performing any hazardous task outside the railed-off work area should wear a safety waistcoat made in fluorescent orange material and with reflective strips if worn at night.
10. Drums, cones, barriers, and chevrons all have their appropriate uses; they should be fresh and clean in appearance and where appropriate carry reflective bands.
11. A loose weight must never be placed on top of a barrier or drum. Stability may be added to drums by quarter filling with sand or clay, but not water.
12. Brightly coloured tape or catenary marker is a useful aid to marking off a work area but is insufficient protection for an open trench or drop.
13. The lamps shown in Layouts A.4-28 to A.4-30 are included as a reminder only and their placement is merely indicative. The characteristics of each individual site should have a major bearing on the number and positioning of lamp units.
14. Mud and debris dropped by work-vehicles should be frequently cleaned off.
15. Parked vehicles should not obscure any sign or danger area.
16. The site should be kept tidy and orderly; it improves the public image and makes for a safer work environment.
17. Signs and other directions should be covered whenever they are not meant to apply and should be removed immediately their further presence is not necessary.

10.17 Mobile Works

The nature of mobile work requires a different approach of signing and control. Different types of activity require different type of traffic control. Levels of signing and control are needed for various categories of mobile works.

Four levels of warning related are needed. These levels depend on the work type, the amount of traffic and the available clear sight distance. Appropriate levels of warning are indicated in Table 10.3.

10.18 Four Levels of Warning

The four levels of warning to be used for mobile road work are:

1. No signing.
2. Road works sign on vehicles, use of strobe lighting in poor visibility condition.
3. One flagman with or without (2) mentioned above.
4. Fixed signing.

The level of warning to be used for various operations are shown in Table 10.3.

Table 10.3 Level of Warning

Traffic Volume	Level of Warning					
	Heavy		Medium		Light	
Sight Distance	Good	Poor	Good	Poor	Good	Poor
WORK OPERATION						
Snow Clearing	1, 2	1,2	1	1	1	1
Salting/Gritting	1, 2	1, 2	1	1	1	1
Sweeping (Mechanical)	1	1	1	1	1	1
Patching	4	4	2, 3, 4	2,3, 4	1	1
Grass/Hedge Cutting	4	4	2, 3, 4	3, 4	1, 2	3, 4
Line Painting	4	4	2	3	1, 2	1, 3
Overhead Works	4	4	3, 4	4	3, 4	4
Cats Eyes / Delineators	4	4	4	4	4	4
Survey Party	4	4	1	3, 4	1	3, 4
Tree Clearance	4	4	4	4	1	3, 4
Surfacing	4	4	4	4	4	4
Road Opening	4	4	4	4	4	4
Surface Dressing	4	4	4	4	4	4

10.19 Detours

Detour signs should be accompanied by temporary direction signage. Road users should be warned in advance of any detour. A large sign should be placed at the point of commencement. A sign should be placed at every point of possible confusion along a detour and at suitable intervals (say 2 km). A sign should be placed at the point where the detour ends.

Appendix 1: Letter Sizes and Siting

Table A1-1: Letter Sizes and Siting Details of Directional Information Sign

Type of Road	Advance Direction Signs			Direct Signs		Route Confirmatory signs	Route Directions Signs	Route Marker Signs	Town and Village Signs
	x height (mm)	Distance of signs from intersection (m)	Clear minimum visibility distance of sign (m)	x height (mm)	Clear minimum visibility distance of sign (m)	x height (mm)	x height (mm)	x height (mm)	x height (mm)
Expressways	250 (200)	Standard siting	180	200 (150)	140	200 (150)	-	-	-
Dual Carriage roads built to hear motorway standards	200 (150)	Standard siting	180	200 or 150	110	150	-	120 (150)	
Unrestricted dual carriageways and single three-lane carriageways	200 (150)	230 (note 1)	140	(note 2) 150 or 100	100 (75)	150	-	120 (150)	-
Other dual and single Carriageway roads on National Routes	150 (120)	150 (note 1)	110	100 (note 3)	75	120 (150)	100 (80)	120 (100)	100
Other roads	100 (80)	50	60	100 (80) (50)	30	100 (80)	100 (80)	-	100 (80)
Slip roads leading from the through carriageway at grade separated junctions	100 (80)	50 or half way along slip road whichever is less							

- Notes: 1. These distances apply to level roads: they should be decreased on uphill gradients and increased on downhill gradients.
2. The small sizes may be used at junctions where traffic speeds are generally less than on the open road, for example at roundabout exits or where there is real difficulty in siting larger size.
3. In lieu of the 100mm x height a specially reduced 80mm x height can be used if lack of space forbids the use of the larger size: for advance direction signs other than roundabout signs if a map type with 100mm x height is too large a 100mm stack type sign is to be preferred to a 80mm x height map type sign.
4. The route marker sign follows the normal rules for route confirmatory sign: height of 50mm should only be used at Local Road junctions or restrictive locations.

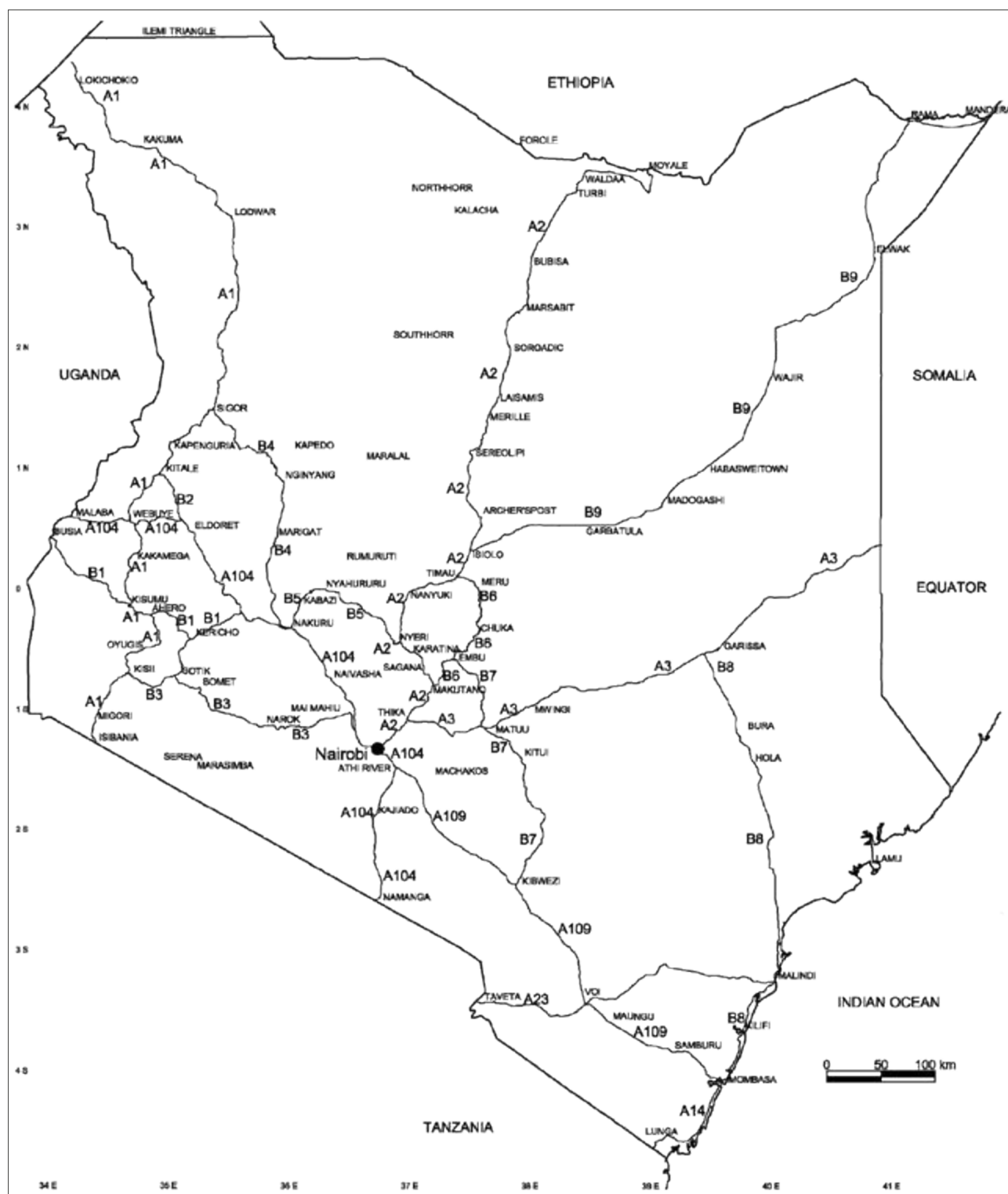
Figure A1-1: Terminal Destinations for Class A and B Roads

Table A1-2: Terminal Destinations for International Trunk Roads (Class A)

International Route Number	Terminal Destinations
A1	Isbania - Kisumu - Kitale - Lodwa - Lokichogio
A2	Nairobi- Isiolo - Marsabit - Moyale
A3	Thika - Garissa - Somalia
A104	Malaba - Nakuru - Nairobi -Athi River - Namanga
A109	Athi River - Voi - Mombasa
A14	Mombasa – Lunga Lunga - Tanga (Tanzania)
A23	Voi - Taveta - Himo (Tanzania)

Table A1-3: Terminal Destinations for National Trunk Roads (Class B)

National Route Number	Terminal Destinations
B1	Busia - Kisumu - Kericho - Londiani
B2	Kitale - Eldoret
B3	Kisii- Narok – Mai Mahiu Turn-off
B4	Nakuru - Sigor
B5	Nakuru - Nyaururu - Nyeri
B6	Makutano - Embu - Meru
B7	Embu - Kitui- Kibwezi
B8	Garissa - Hola - Garsen - Malindi- Mombasa
B9	Isiolo - Wajir - Mandera

Note: For practical purposes it may be necessary in some cases to include the names of destinations other than those listed. See Table 5-3 in Section 5.2 for general guidelines.

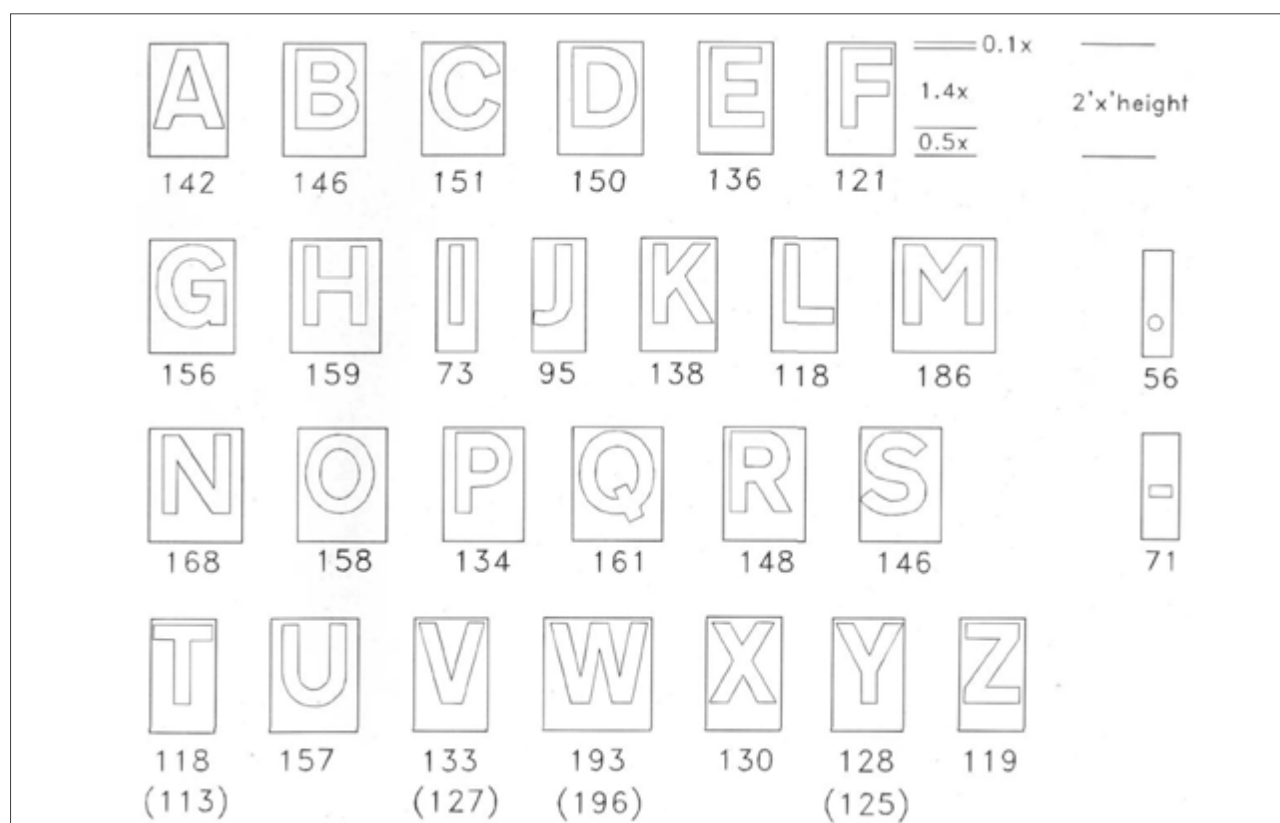
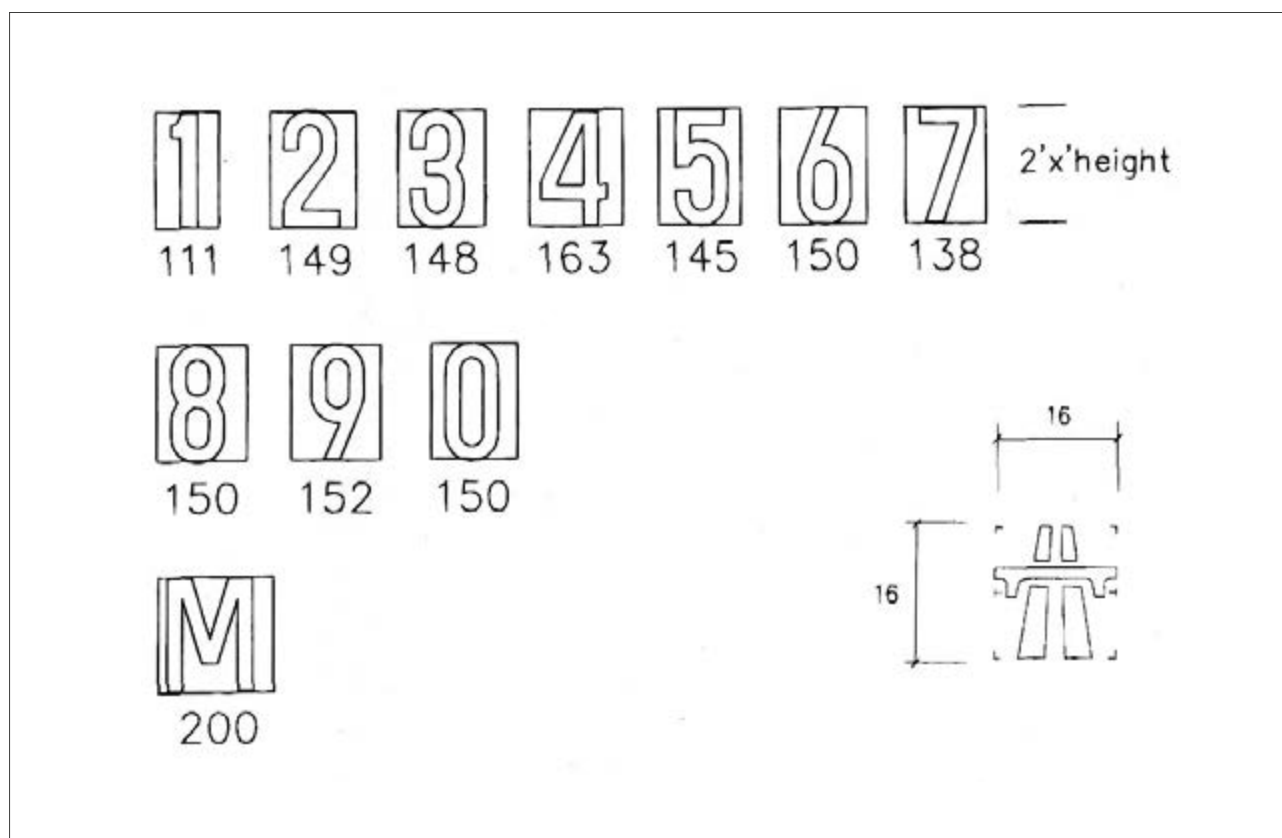
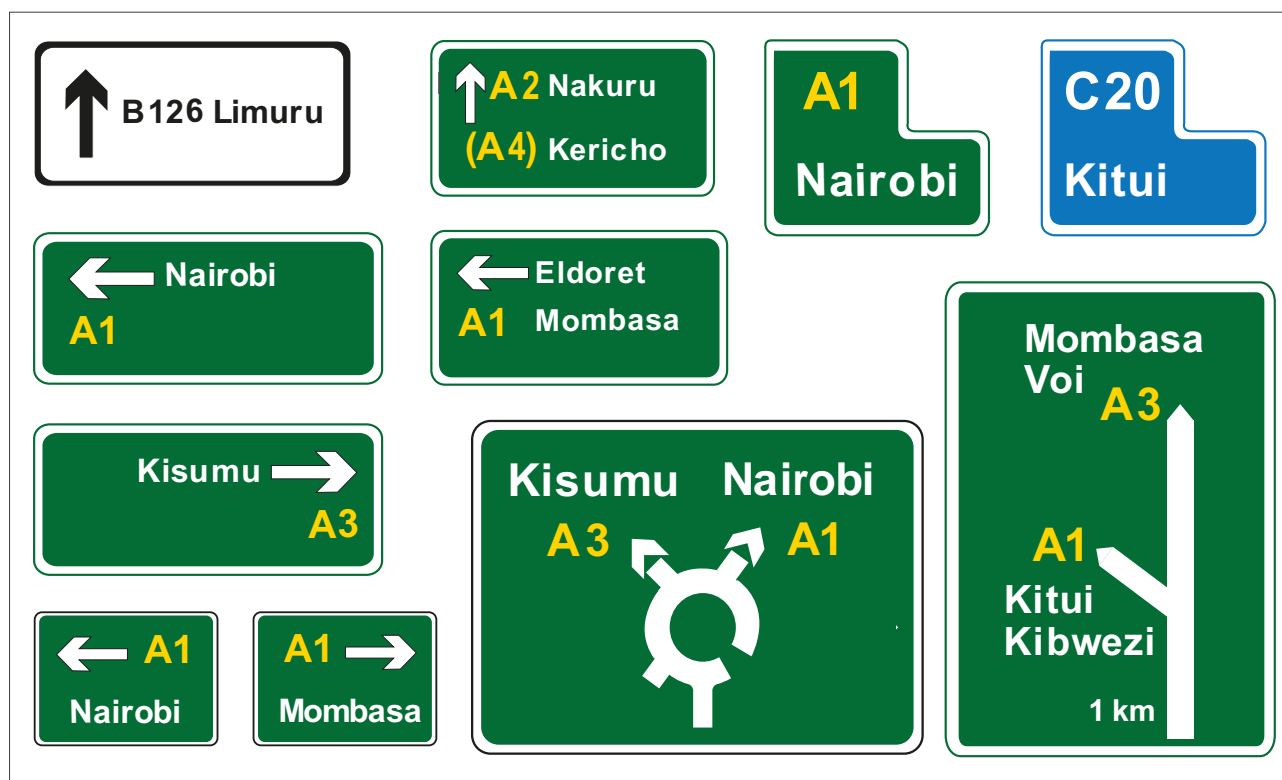
Figure A1-2: Upper Case Alphabet (Tile width expressed as a percentage of X-Height)

Figure A1-3: Lower Case Alphabet (Tile width expressed as a percentage of X-Height)**Figure A1-4:** Numerical (Tile width expressed as a percentage of X-Height)

Figure A1-5: Numerical for Expressway (Tile width expressed as a percentage of X-Height)**Figure A1-6:** Examples of Coloured Sign Faces

Appendix 2: Tourist Signs

Figure A2-1: Tourist Advance Direction Sign



Figure A2-2: Tourist Attraction Direction Sign



Figure A2-3: Sign Showing Distance to Tourist Attraction

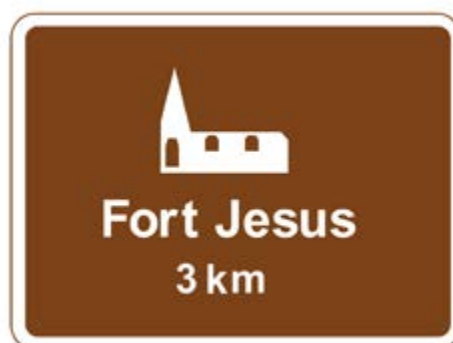


Figure A2-4: Signs to Approved Tourist Information Points

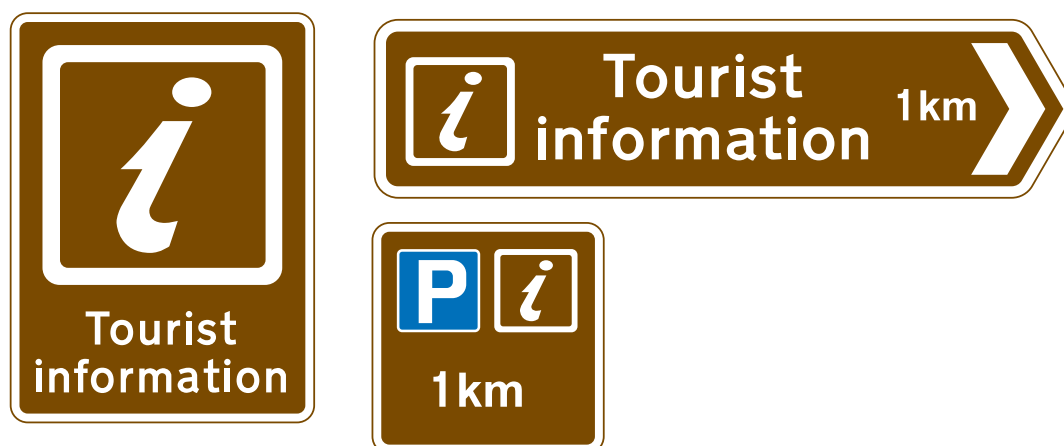


Figure A2-5: Directional Tourist Signs



Figure A2-6: Youth Hostel



Figure A2-7: Pedestrian Sign to Car Park



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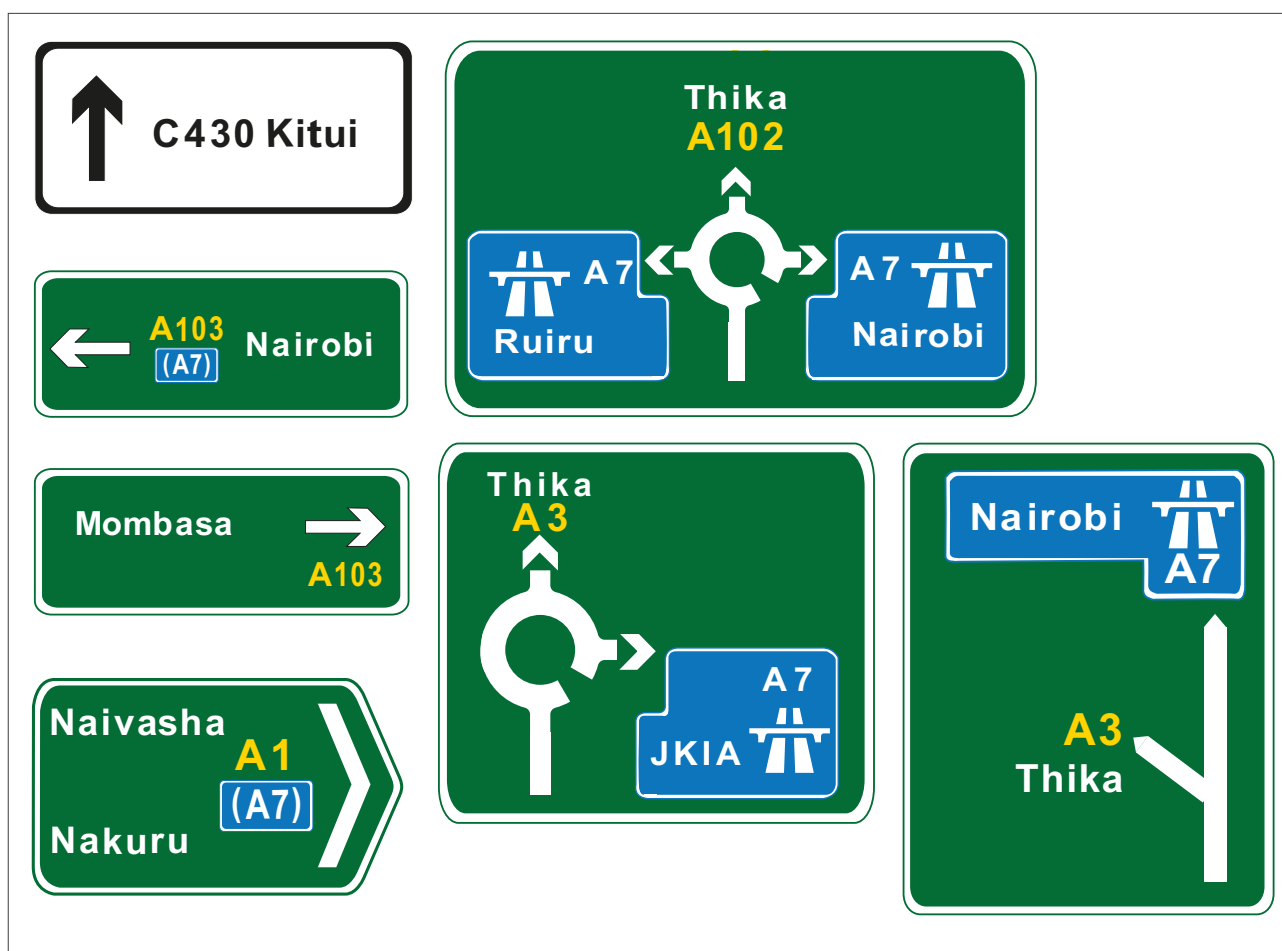
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Appendices

Appendix 3: Expressway Signs

Figure A3-1: Example Signs with Patches



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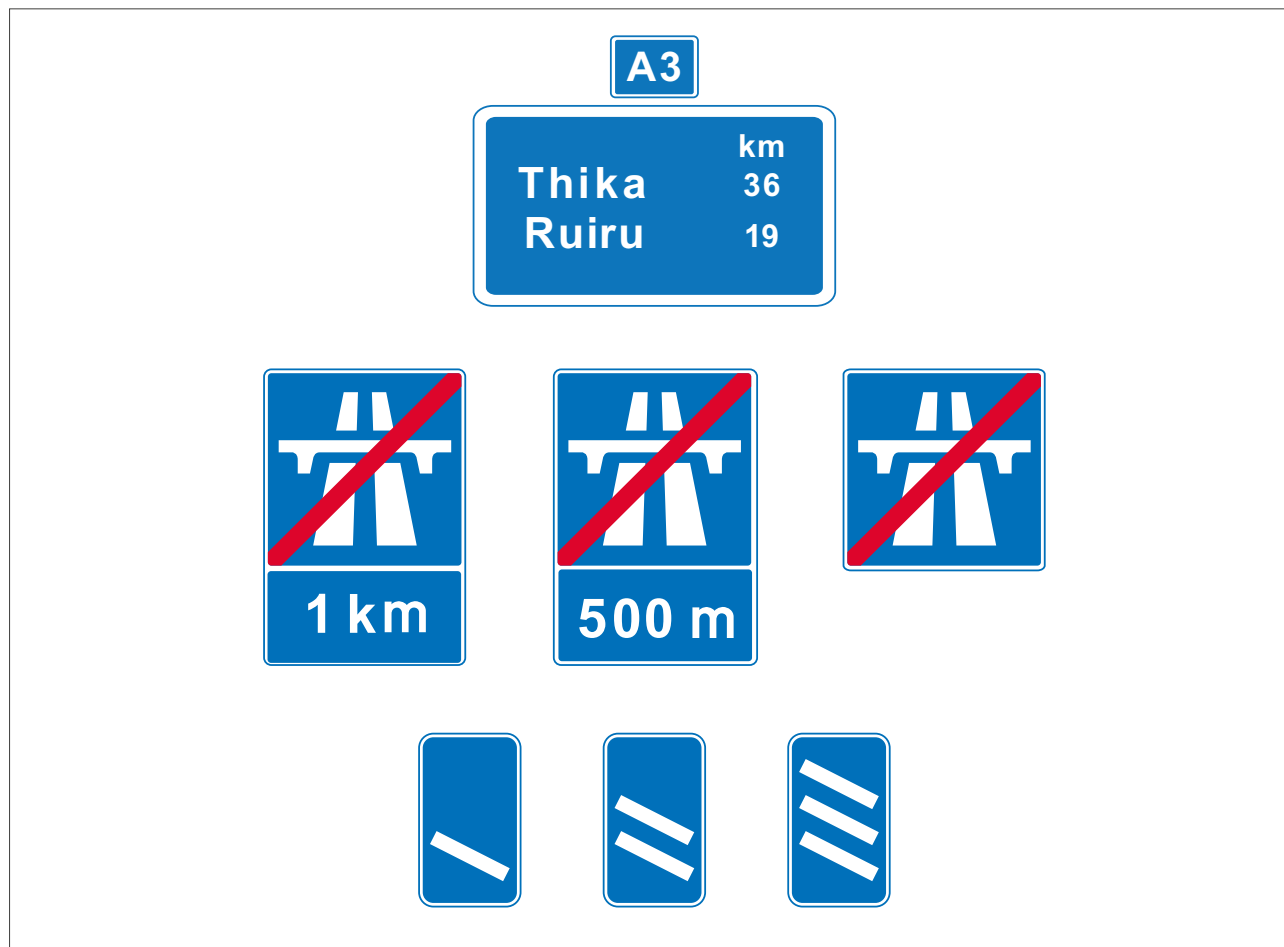
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Figure A3-2: Example Expressway Signs



Appendix 4: Traffic Control for Roadworks

Table A4-1: Lead-in Taper for Urban Situations

Average Speed of Traffic (km/h)	Description	Width of Obstruction		
		2m	3.7m	7.3m
50	Length of taper (m)	25	45	90
	Number of cones	9	16	31
50-65	Length of taper (m)	37	69	138
	Number of cones	13	24	47

Notes:

1. This chart only applies where two-way traffic can be maintained past the works. (For "give and take" operation, all tapers should be at 45 degrees).
2. Cone spacing: 3 m. Exit tapers at 45 degrees.
3. General rule of thumb for areas subject to 50 Km/h speed limit: Taper length = 12 times obstruction width.

Table A4-2a: Siting of Roadworks Signs in Urban Situations

Average Speed of Traffic (km/h)	Siting Distance of First Sign in Advance of Works	Minimum Visibility Distance to First Sign
Up to 50	Length of taper (m)	25
	Number of cones	9
50-65	Length of taper (m)	37
	Number of cones	13

Table A4-2b: Siting of "End of Works" Sign for Urban Situations

Average Speed of Traffic (km/h)	Siting Distance of 'End of Works' Sign Beyond the Works (m)
Up to 50	10-30
50-65	30-40

Table A4-3: Siting of Roadworks Signs in Rural Situations

Average Siting Speed Sign in of Traffic (km/h)	Distance of First Advance of Works (m)	Min. Size of 'Road Works Ahead' Signs (mm)	Minimum Number of Signs in Advance of Road (No.)
< 50	50	600	2
50 - 80	150	600	3
>80 Divided	200 - 500	900	4
>80	400 - 750*	1200	4

* Range intended to allow optimum placement consistent with road alignment

Table A4-4: Lead-in Taper for Rural Situations

Average Speed of Traffic (km/h)	Description	Width of Obstruction		
		2m	3.7m	7.3m
<50	Length of taper (m)	25	45	90
	Number of cones	5	8	16
50-80	Length of taper (m)	50	90	180
	Number of cones	9	16	31
>80	Length of taper (m)	75	135	270
	Number of cones	13	23	46

Table A4-5: Siting "End of Works" Sign for Rural Situations

Average Speed of Traffic (km/h)	Siting Distance of 'End of Works' Sign Beyond the Works
<50	10-30
50-80	30-45
>80	45-90

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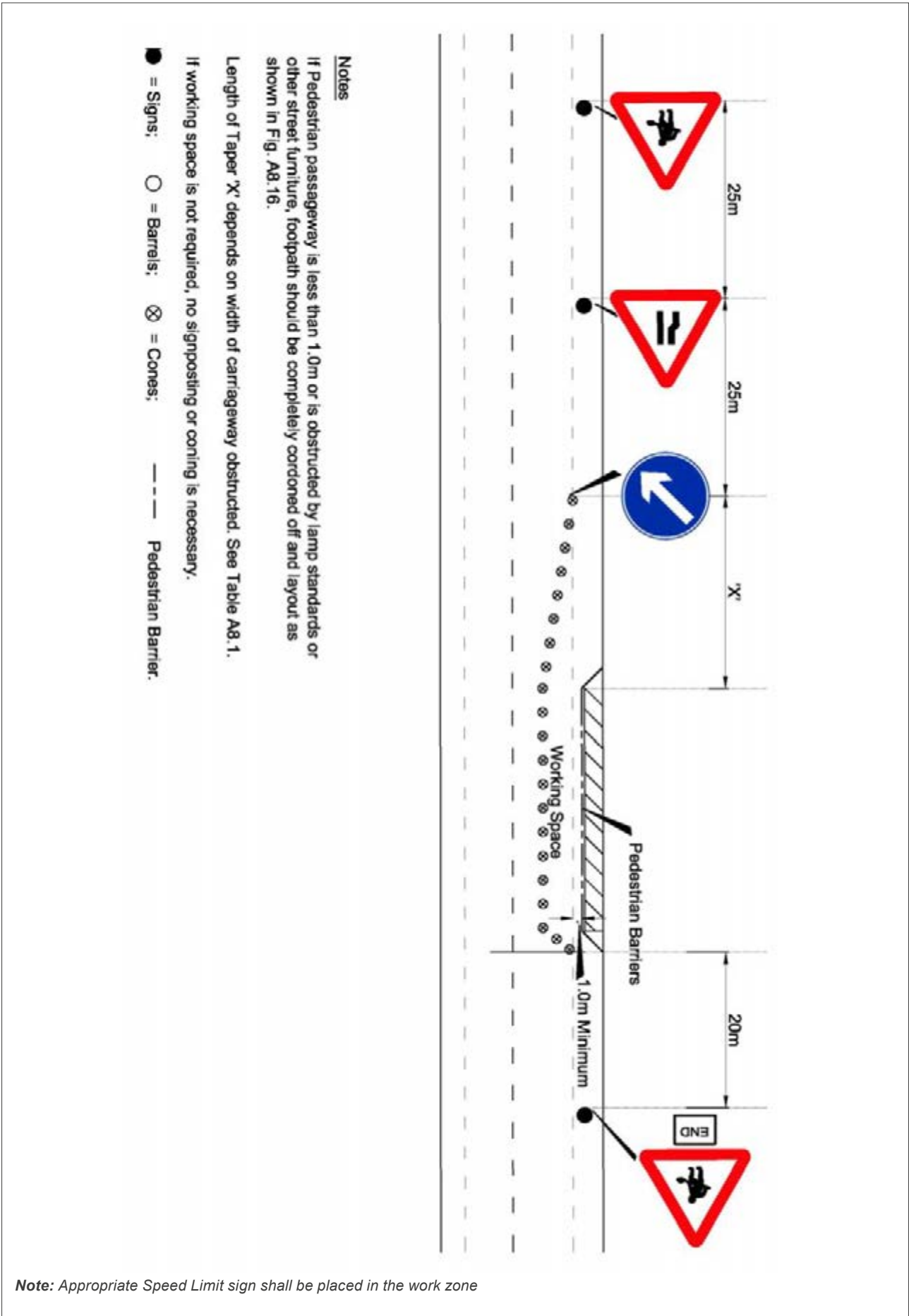
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Appendices

Figure A4-1: Roadworks Partially Obstructing Footpath



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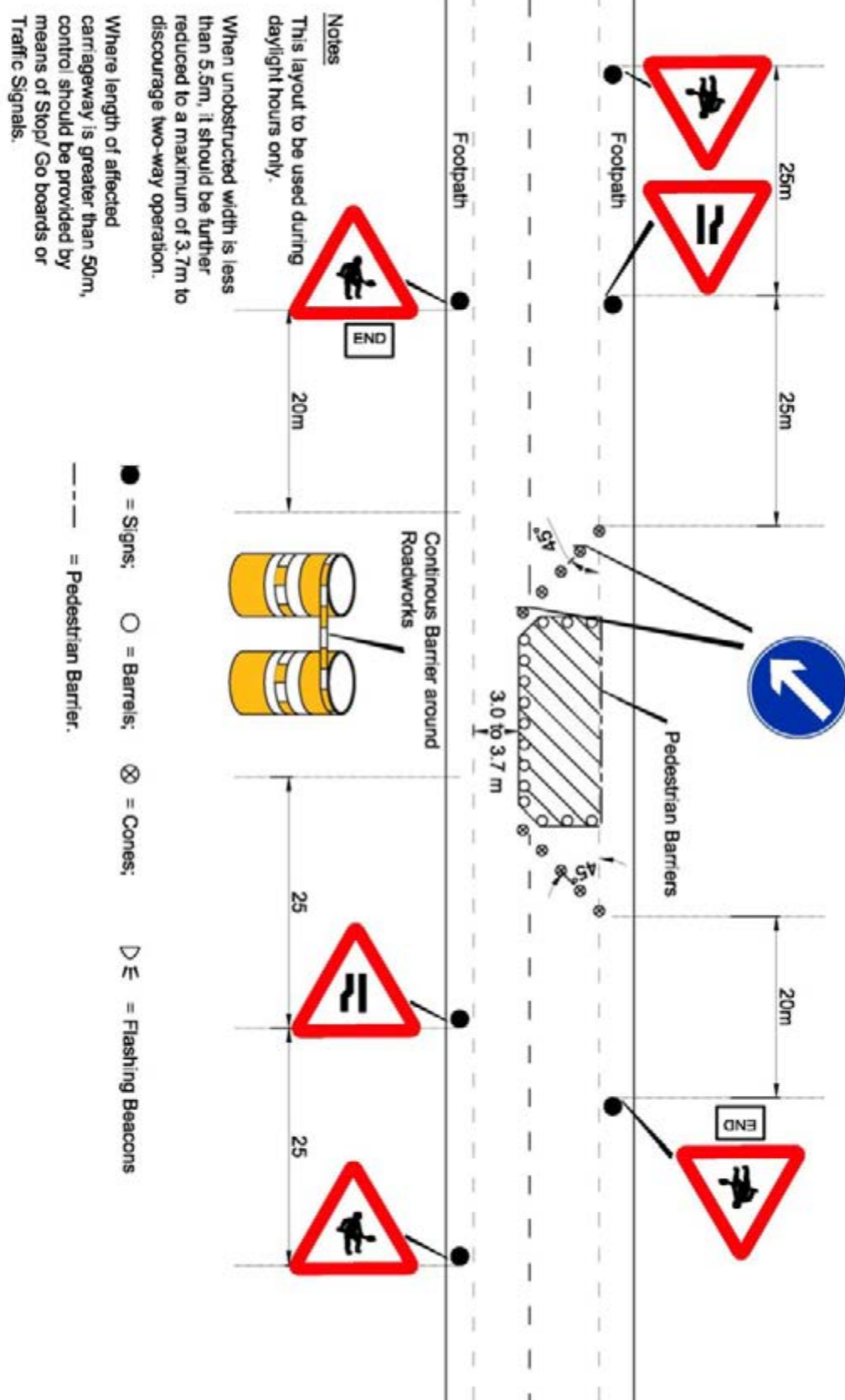
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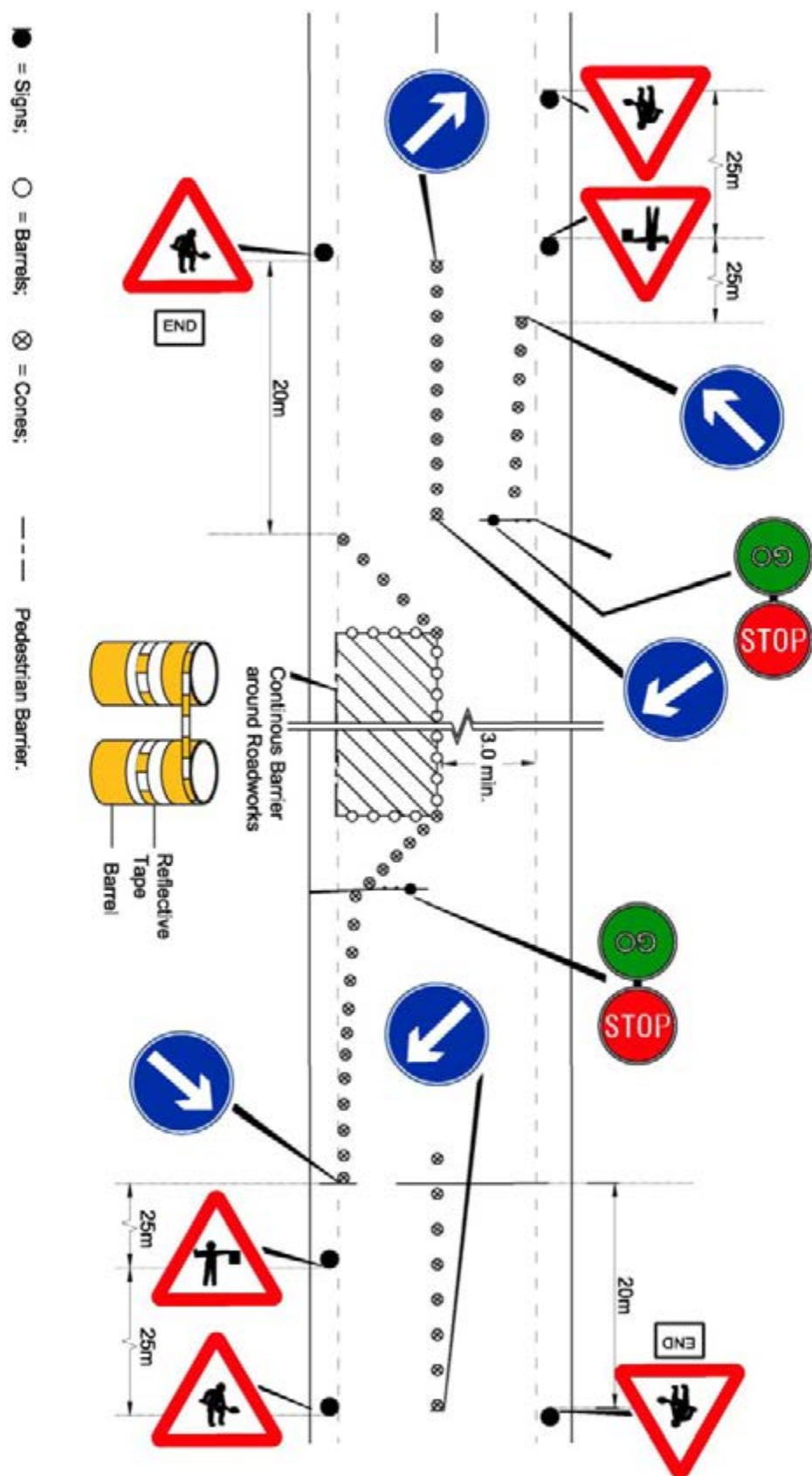
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Appendices

Figure A4-2: 'Give and Take' Operation on Lightly Trafficked Minor Roads (Unobstructed Carriageway Width <5.5 m)

Note: Appropriate Speed Limit sign shall be placed in the work zone

Figure A4-3: One-Way shuttle Working Using Flagmen/ Batten-men (Suitable for Daylight only)



Note: Appropriate Speed Limit sign shall be placed in the work zone

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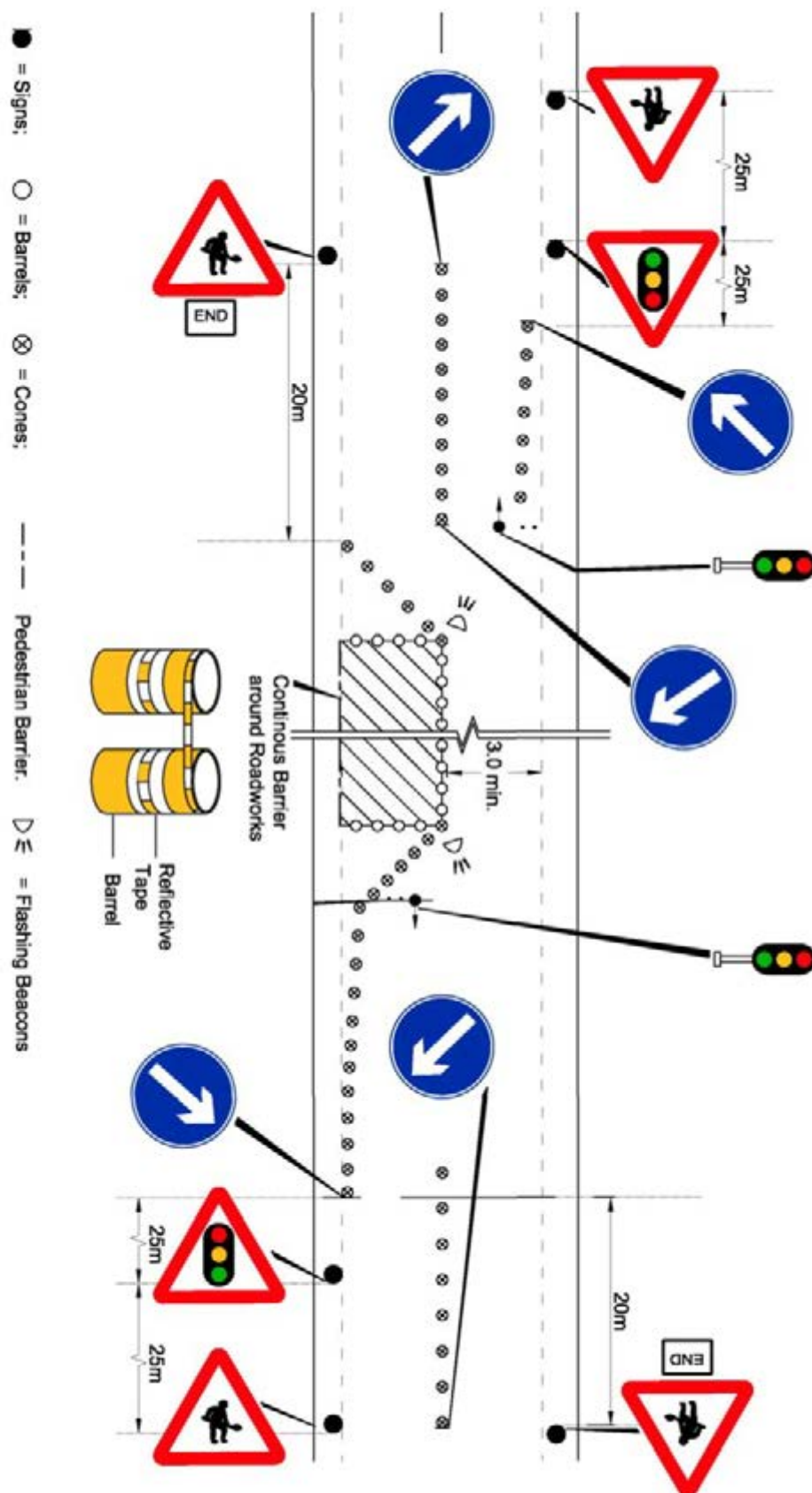
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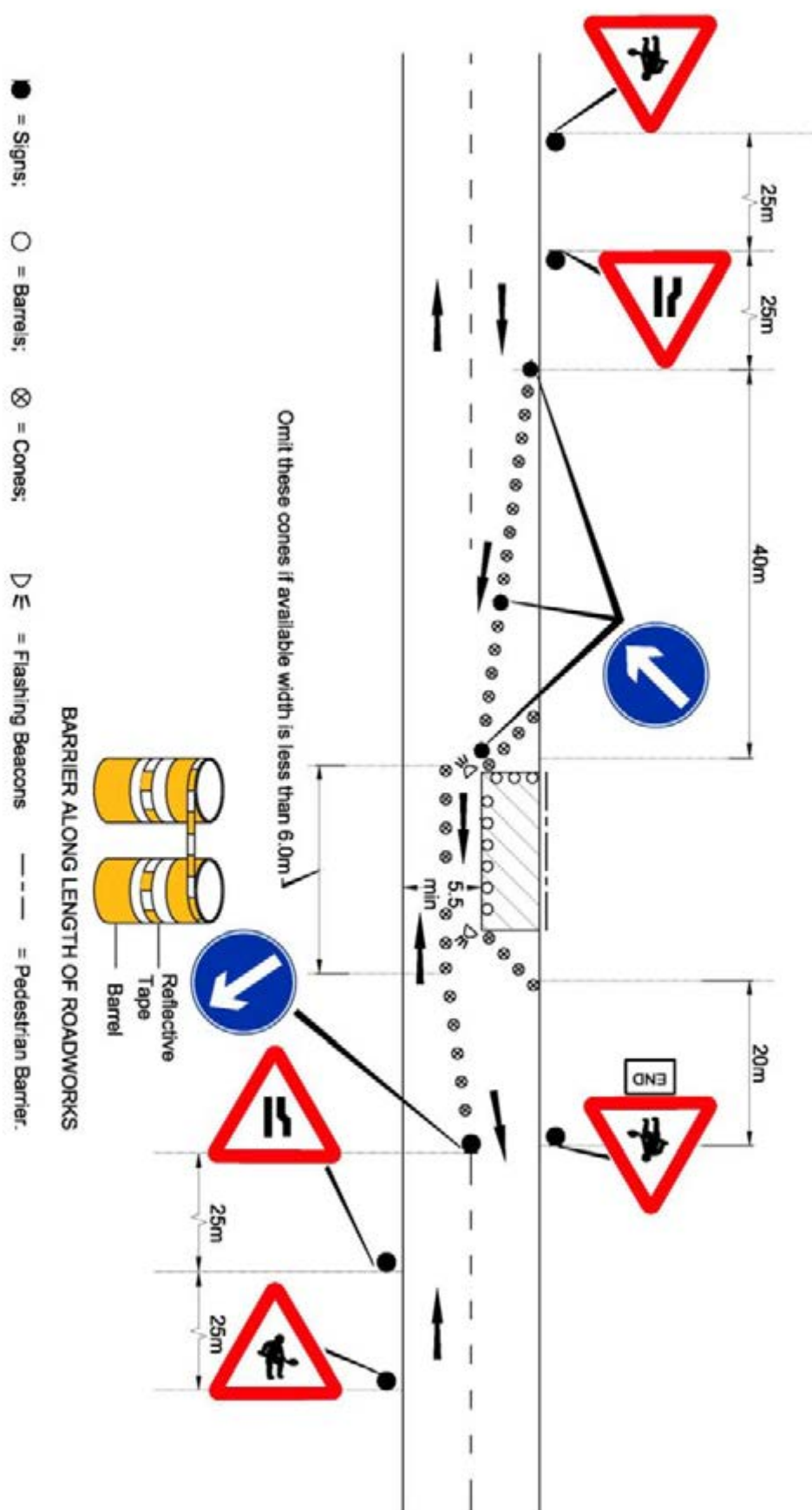
Appendices

Figure A4-4: One-Way shuttle Working Using Temporary Traffic Signals (Suitable for Daylight and Darkness)



Note: Appropriate Speed Limit sign shall be placed in the work zone

Figure A4-5: Two-Way works Forcing Traffic Over Centre Line



Note: Appropriate Speed Limit sign shall be placed in the work zone

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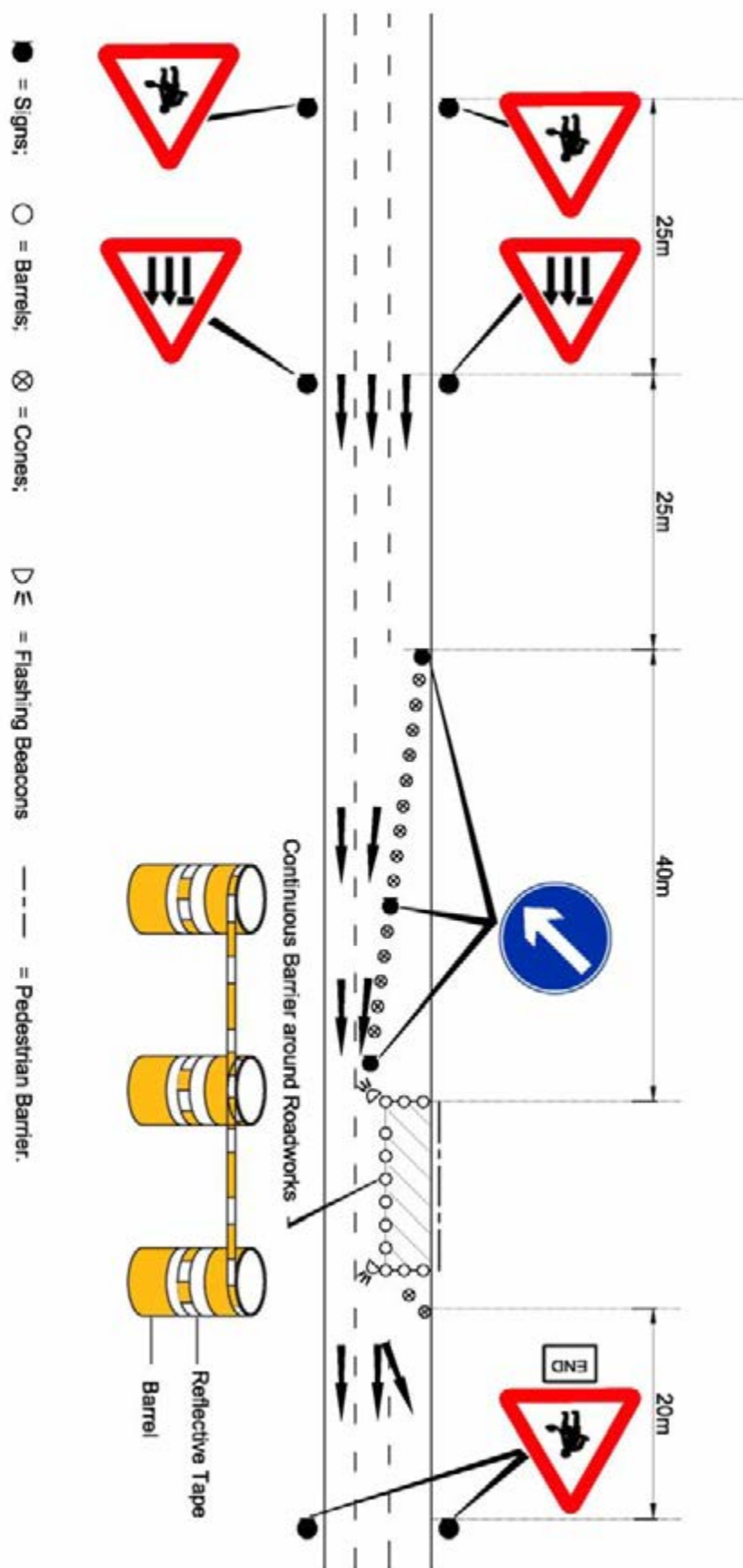
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Appendices

Figure A4-6: Three Lane One-way Street, Works Inside Lane

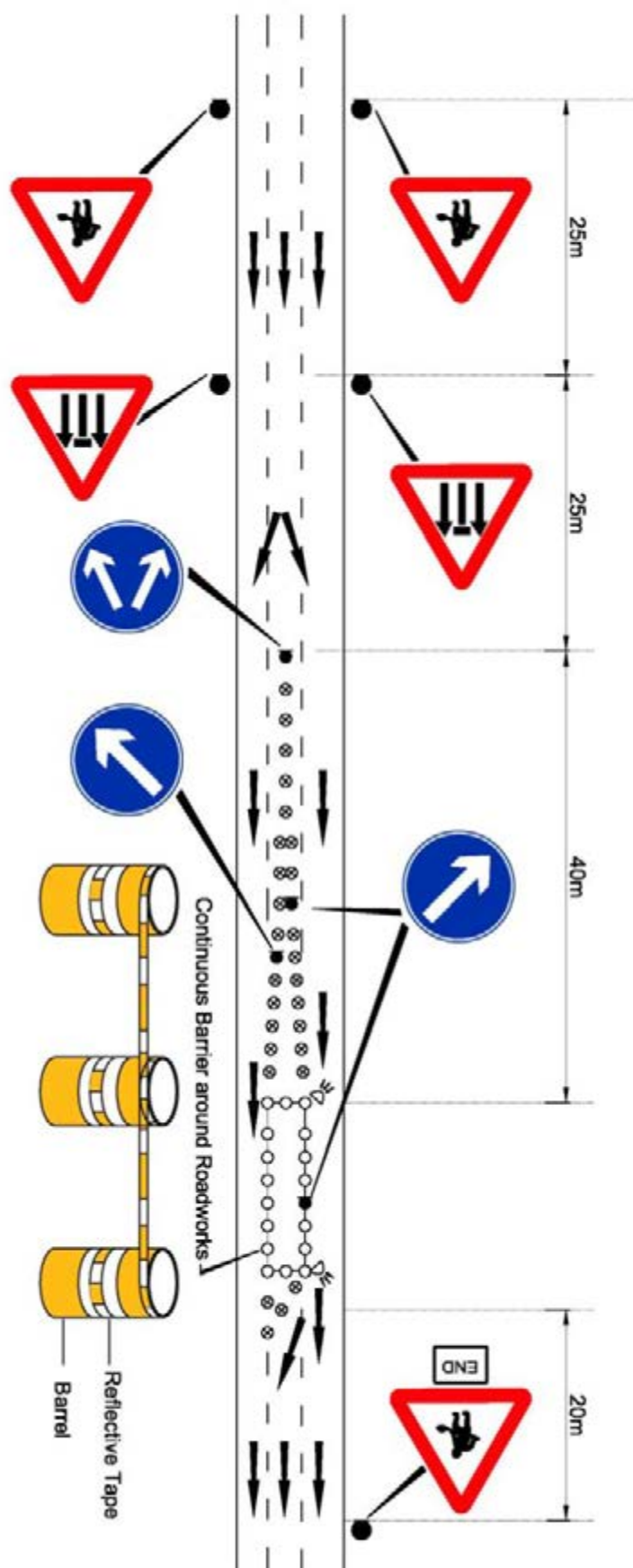


Note: Appropriate Speed Limit sign shall be placed in the work zone

Figure A4-7: Three Lane One-Way Street, Works in Centre Lane

Note: - This arrangement not to be used on dual carriageways.

● = Signs; ○ = Barrels; ⊗ = Cones; D = Flashing Beacons



Note: Appropriate Speed Limit sign shall be placed in the work zone

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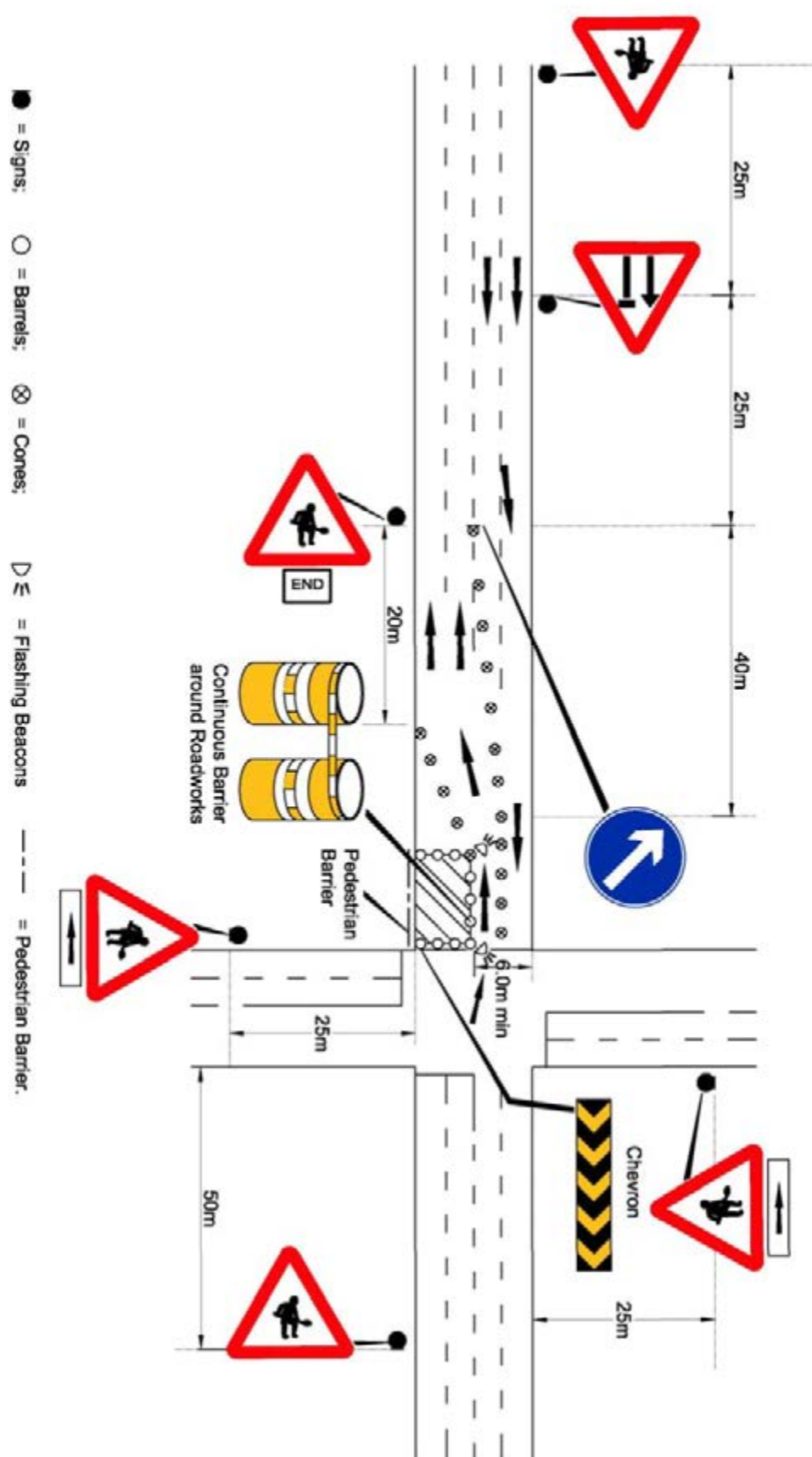
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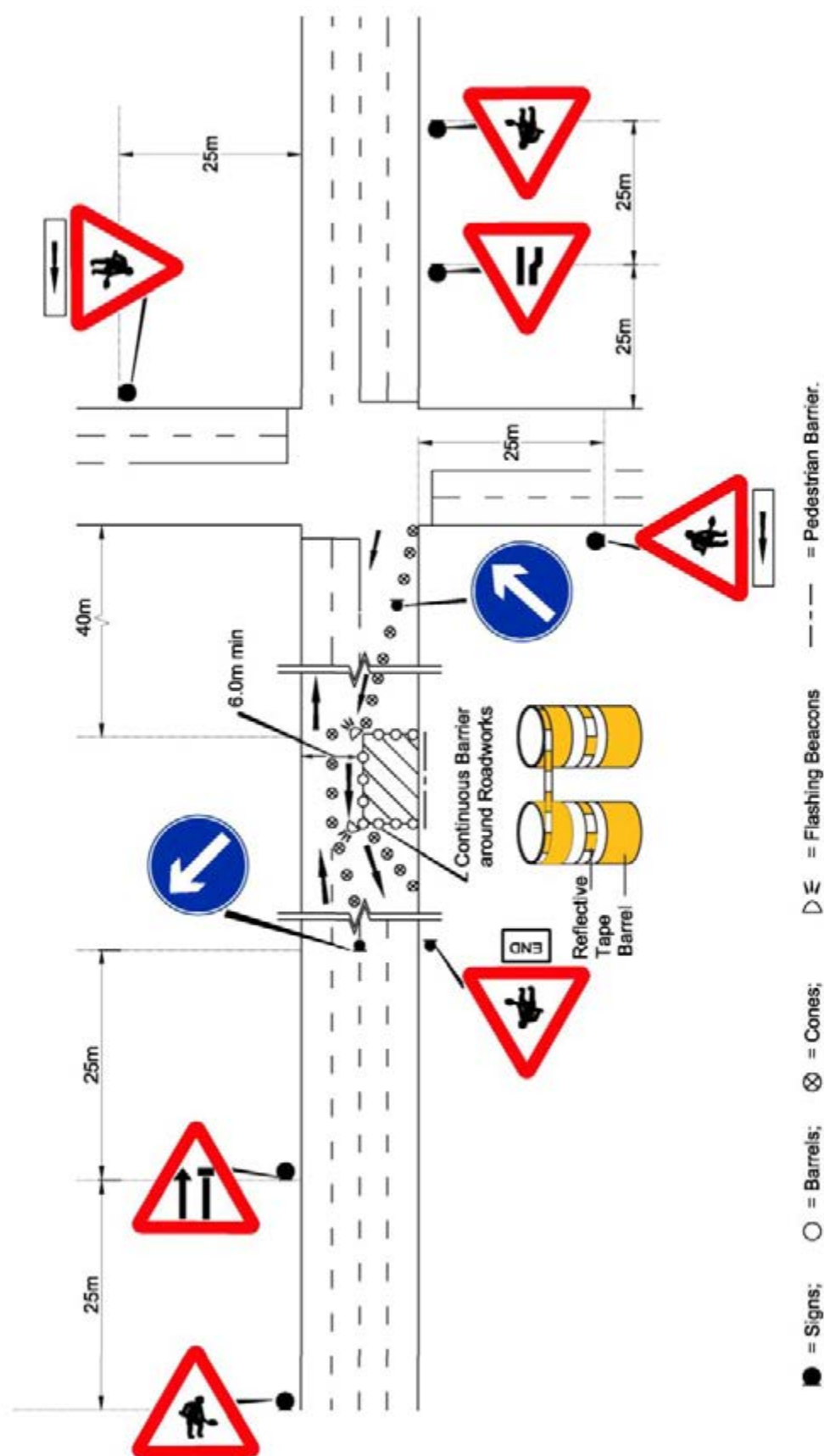
Appendices

Figure A4-8: Four Lane Street, Works in Two Lanes at Junction



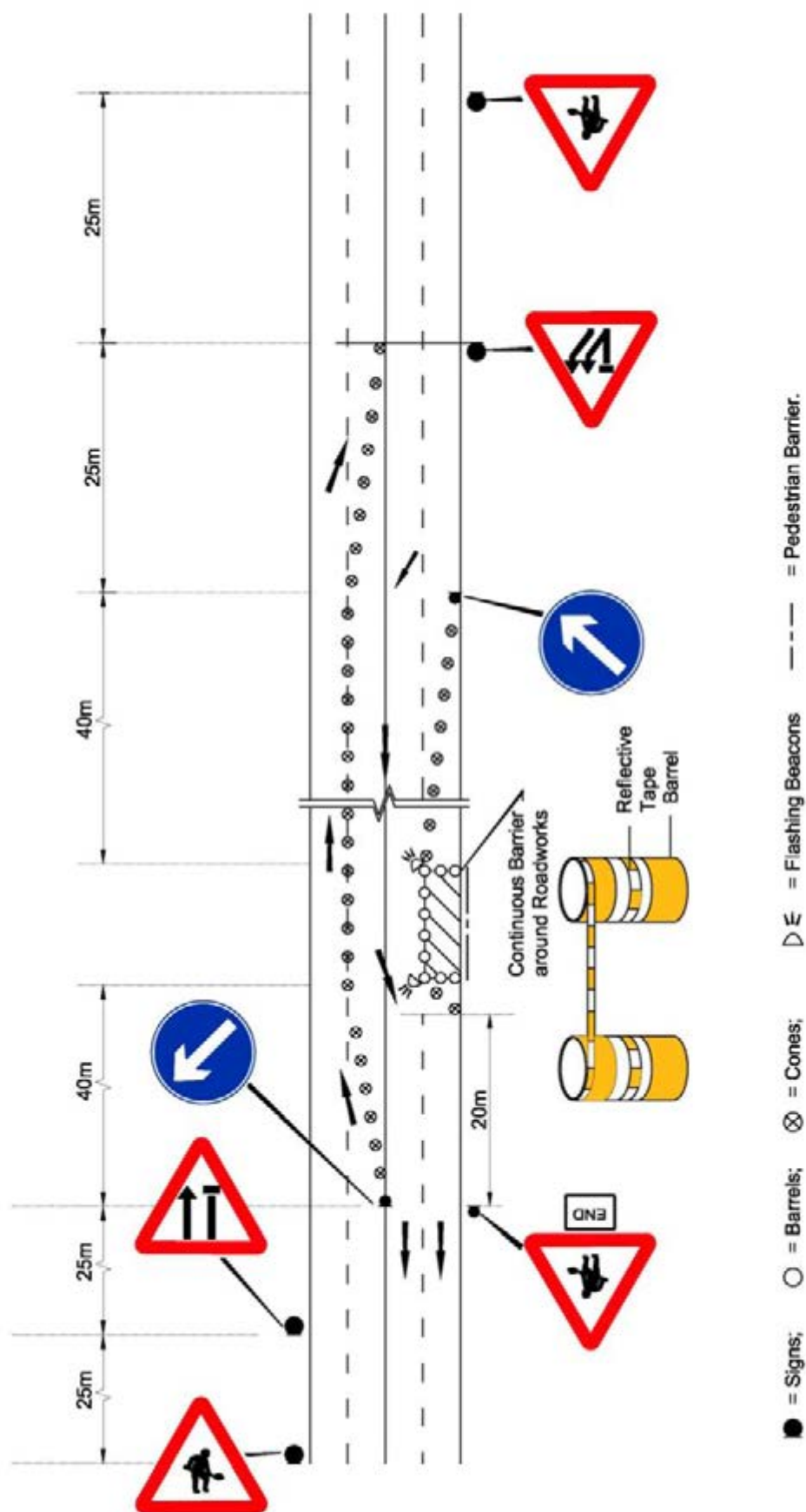
Note: Appropriate Speed Limit sign shall be placed in the work zone

Figure A4-9: Four Lane Street, Works In Two Lanes Near Junction

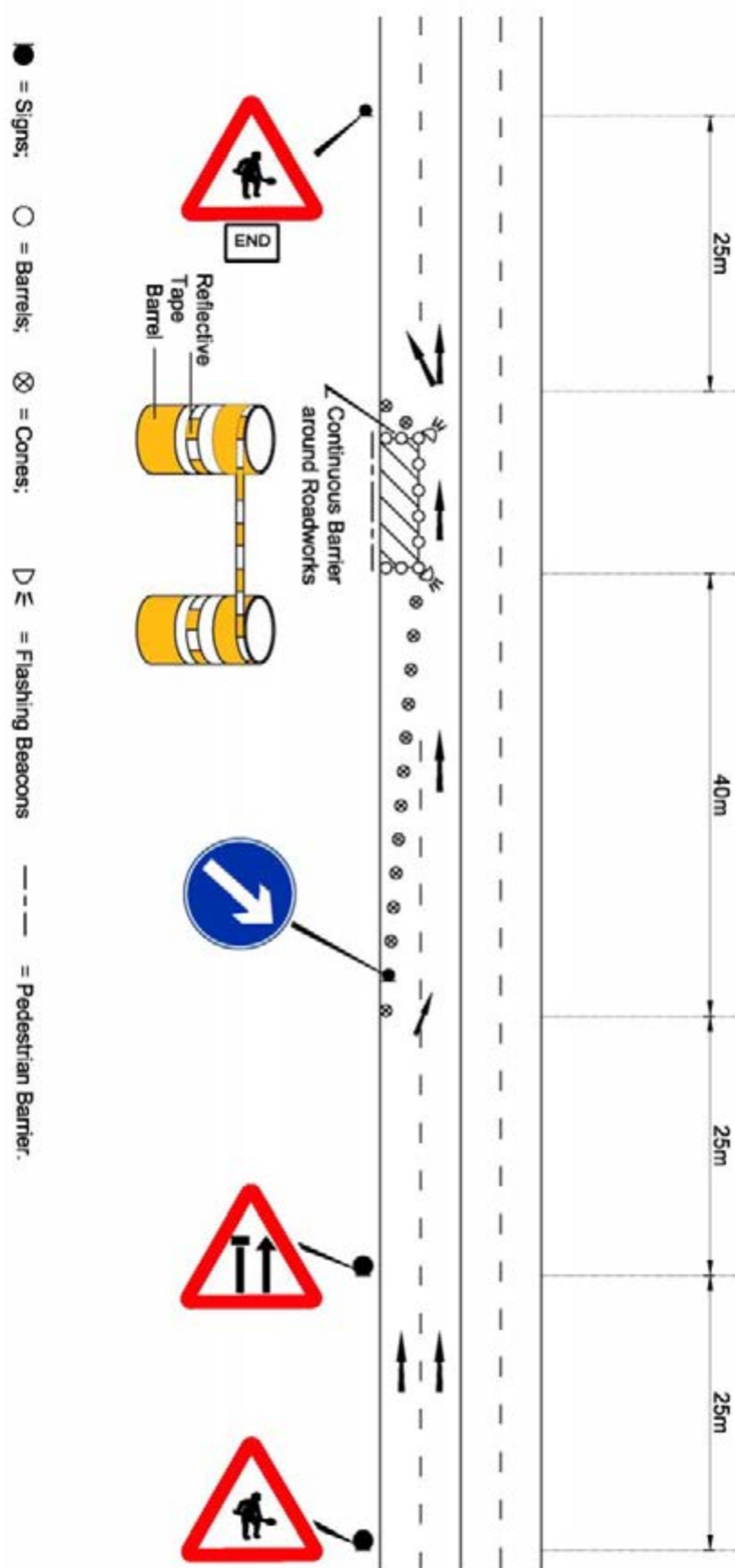


Note: Appropriate Speed Limit sign shall be placed in the work zone

Figure A4-10: Four Lane Street, Works in Inner City Bound Lane, Morning Peak



Note: Appropriate Speed Limit sign shall be placed in the work zone

Figure A4-11: Four Lane Street, Works in Inner City Bound Lane, Off Peak and Evening Peak

Note: Appropriate Speed Limit sign shall be placed in the work zone

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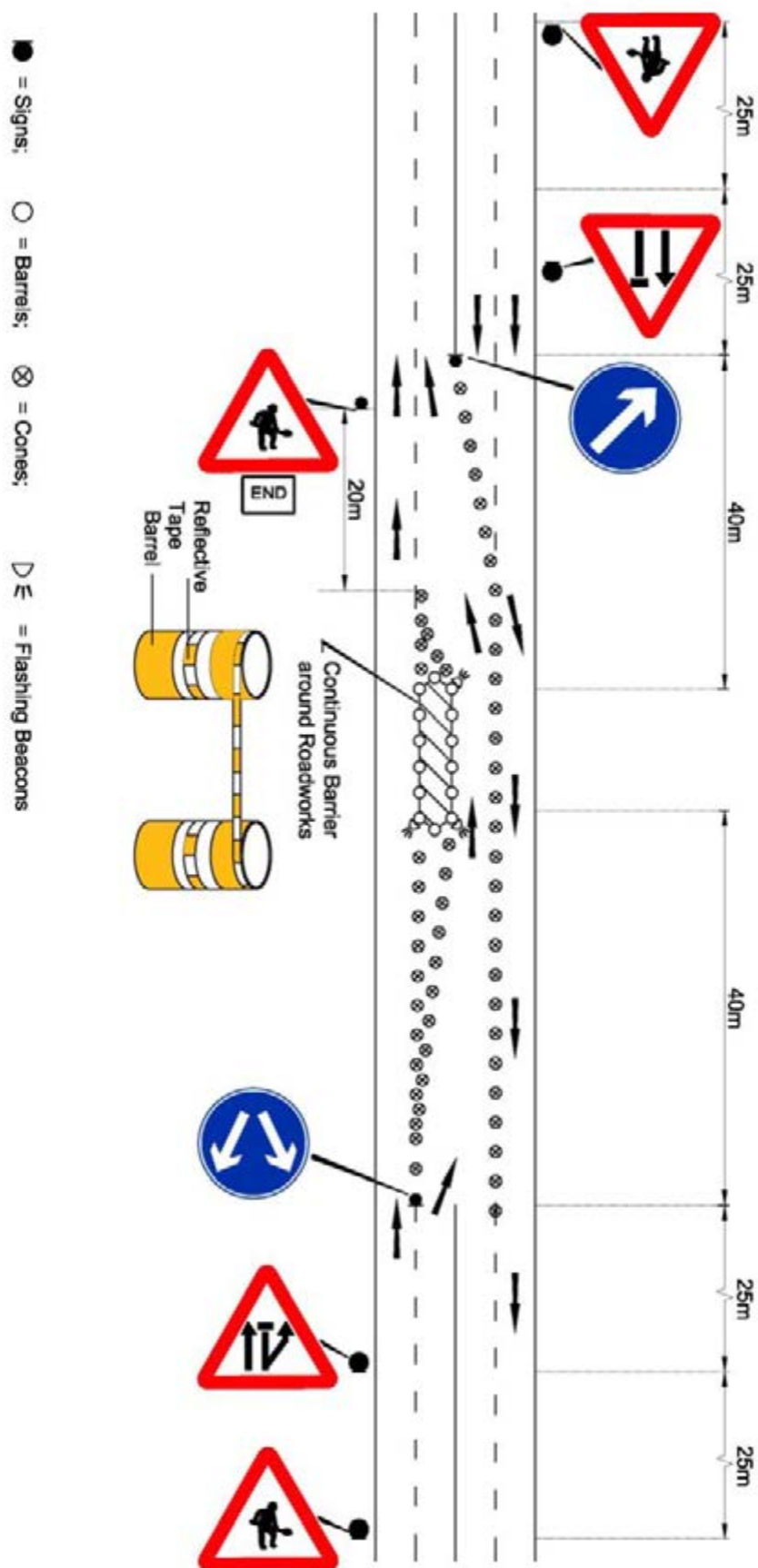
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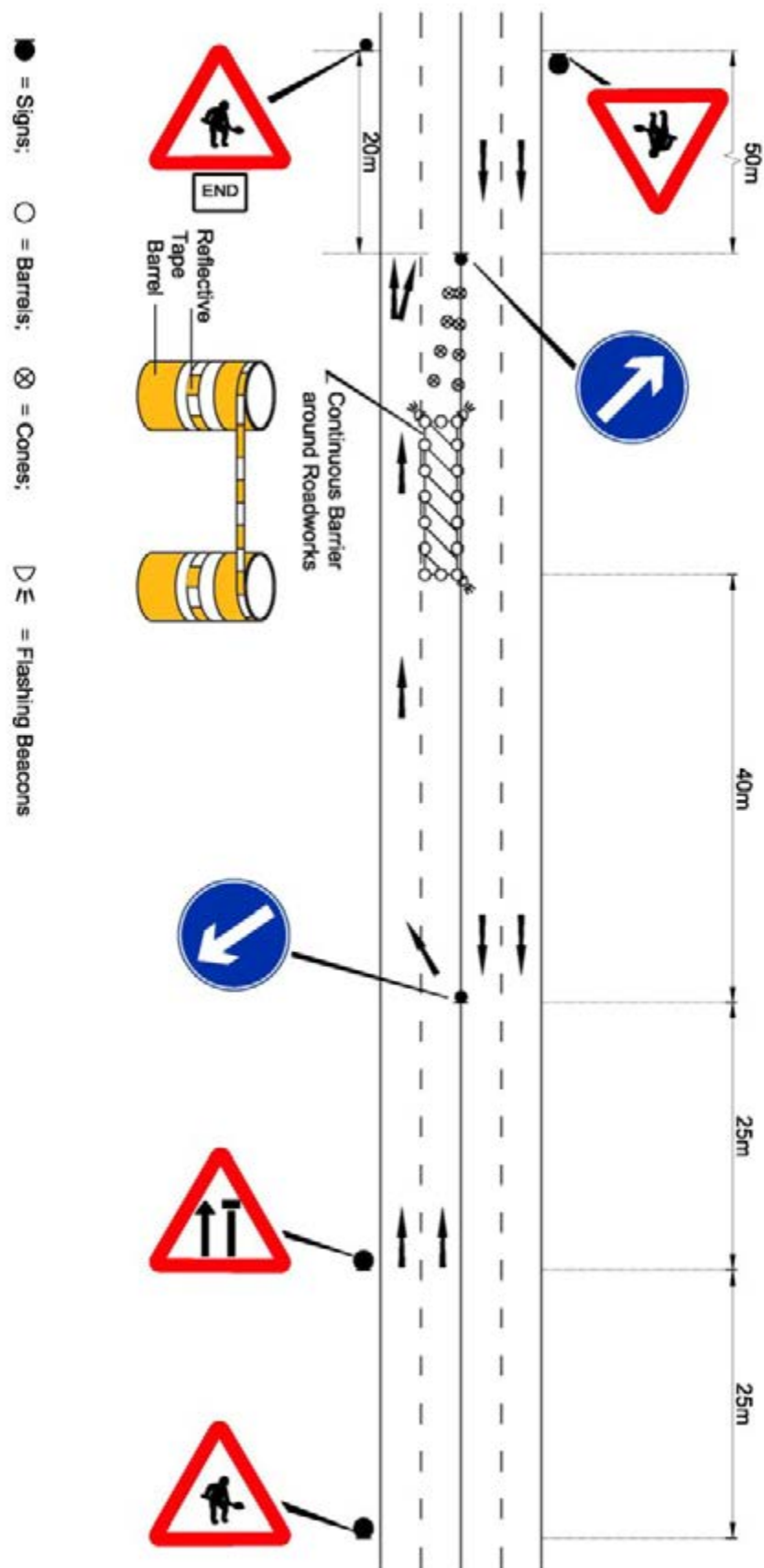
Appendices

Figure A4-12: Four Lane Street, Works in Outer City Bound Lane, Morning Peak



Note: Appropriate Speed Limit sign shall be placed in the work zone

Figure A4-13: Four Lane Street, Works in Outer City Bound Lane, Off Peak and Evening Peak



Note: Appropriate Speed Limit sign shall be placed in the work zone

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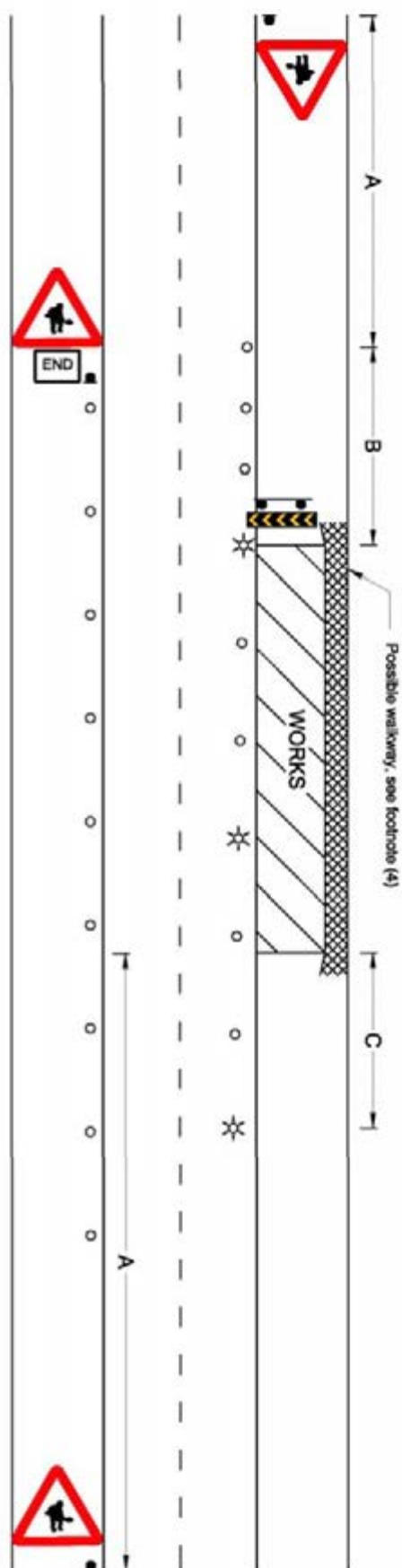
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Appendices

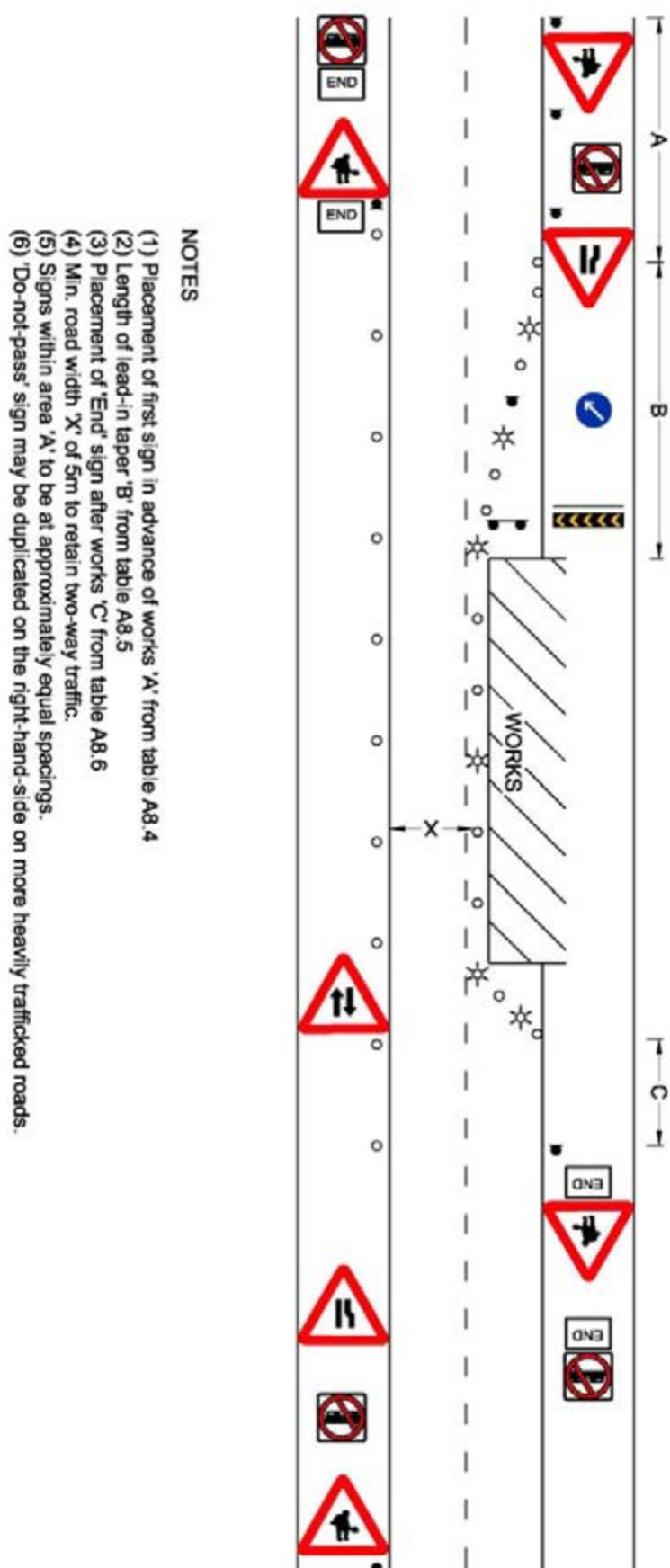
Figure A4-14: Two-Way Traffic Works in Margin/Shoulder/Footpath Only

- NOTES**
- (1) Placement of first sign in advance of works 'A' from table A8.4
 - (2) Length of lead-in taper 'B' from table A8.5
 - (3) Placement of 'End' sign after works 'C' from table A8.6
 - (4) Where a footpath is closed, and alternative of at least 1.2m is required.
 - (5) Where all activity takes place off the carriageway, less stringent standards may be applied.
 - (6) 'Do-not-pass' sign may be used where active works are in progress.



Note: Appropriate Speed Limit sign shall be placed in the work zone

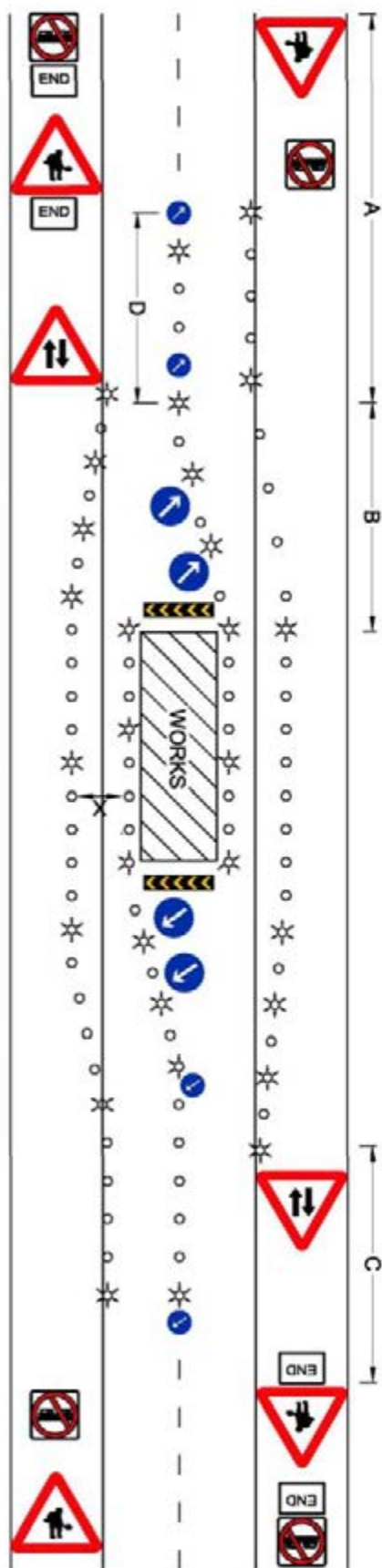
Figure A4-15: Two-Way Traffic Carriageway Encroachments



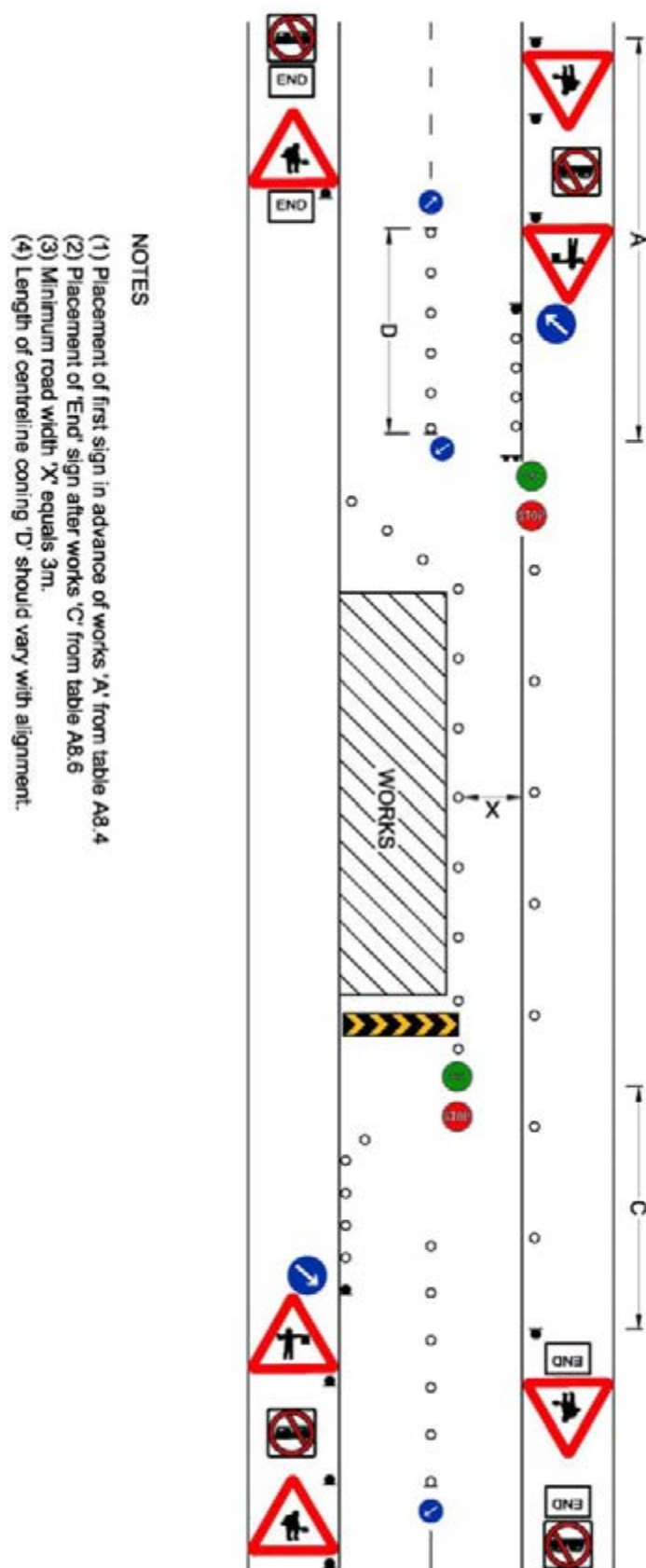
Note: Appropriate Speed Limit sign shall be placed in the work zone

Figure A4-16: Use of Road Shoulder by Traffic

- NOTES**
- (1) Placement of first sign in advance of works 'A' from table A8.4
 - (2) Length of lead-in taper 'B' from table A8.5
 - (3) Placement of 'End' sign after works 'C' from table A8.6
 - (4) Minimum road width 'X' equals 3m.
 - (5) Length of centreline coning 'D' should vary with alignment.



Note: Appropriate Speed Limit sign shall be placed in the work zone

Figure A4-17: One-Way Shuttle Working Using Flagmen/ Batten-men (Suitable for Daylight Only)

Note: Appropriate Speed Limit sign shall be placed in the work zone

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Appendices

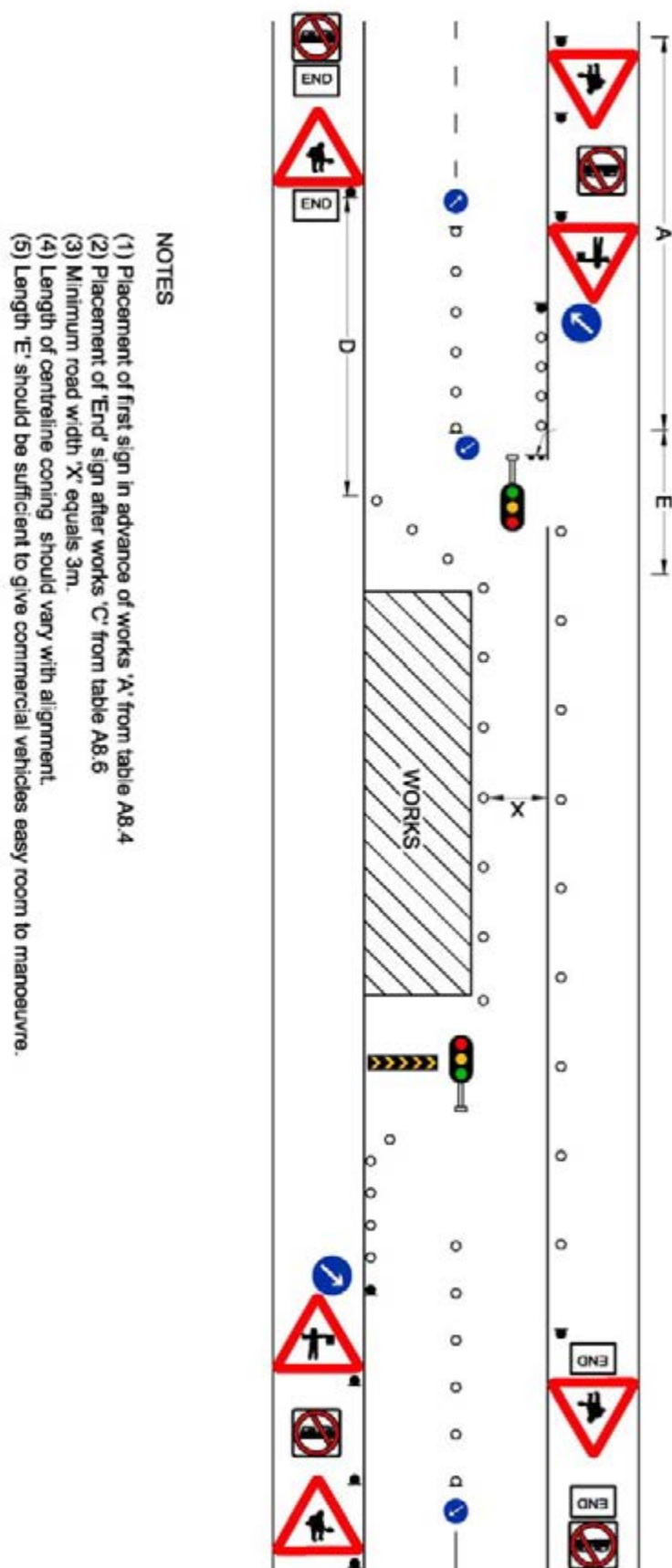
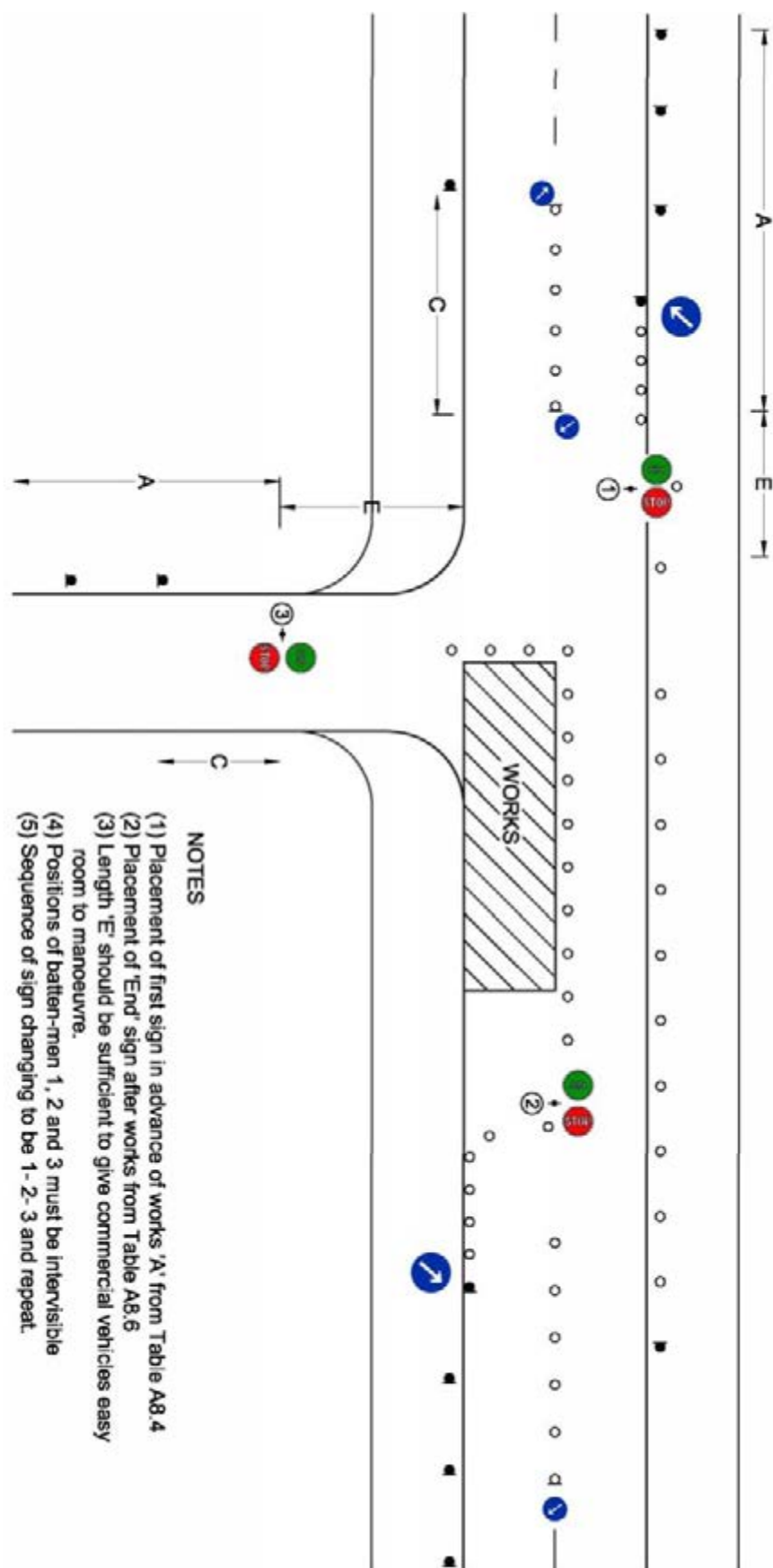
Figure A4-18: One-Way Shuttle Working Using Temporary Traffic Signals (Suitable for Daylight and Darkness)

Figure A4-19: One-Way Shuttle Working at a "T" Junction



Note: Appropriate Speed Limit sign shall be placed in the work zone

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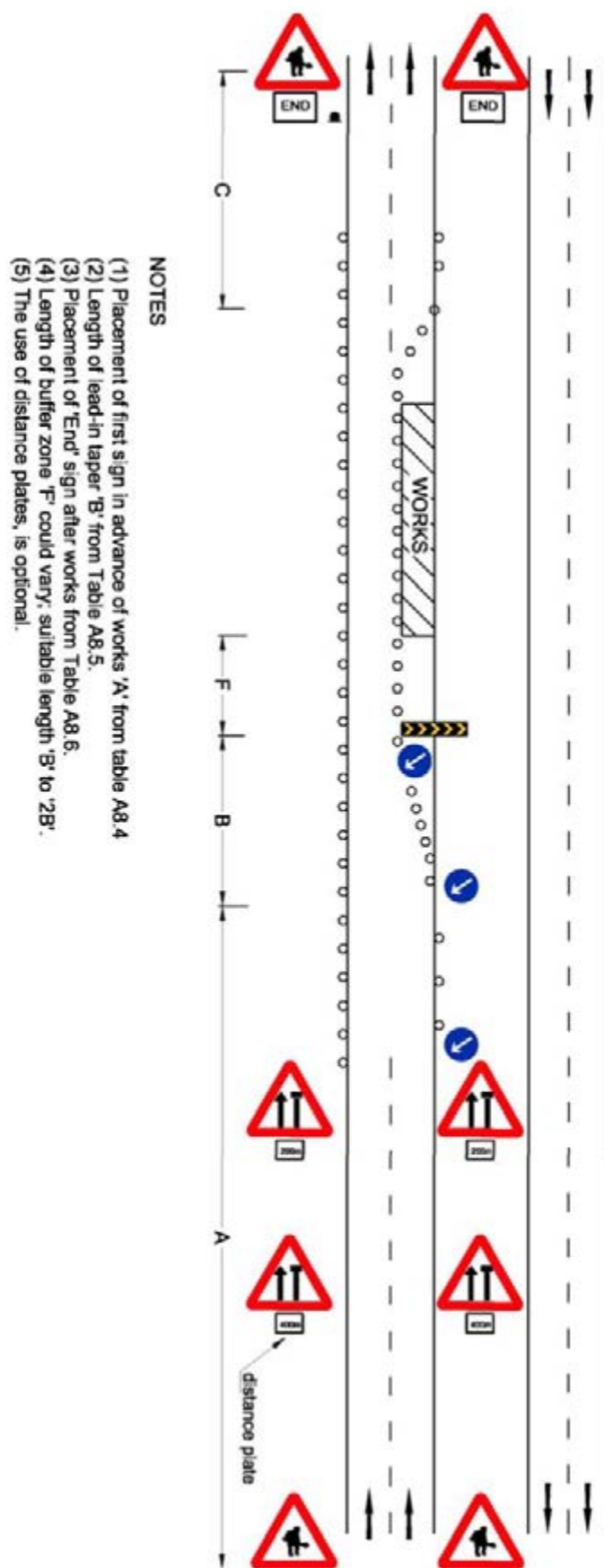
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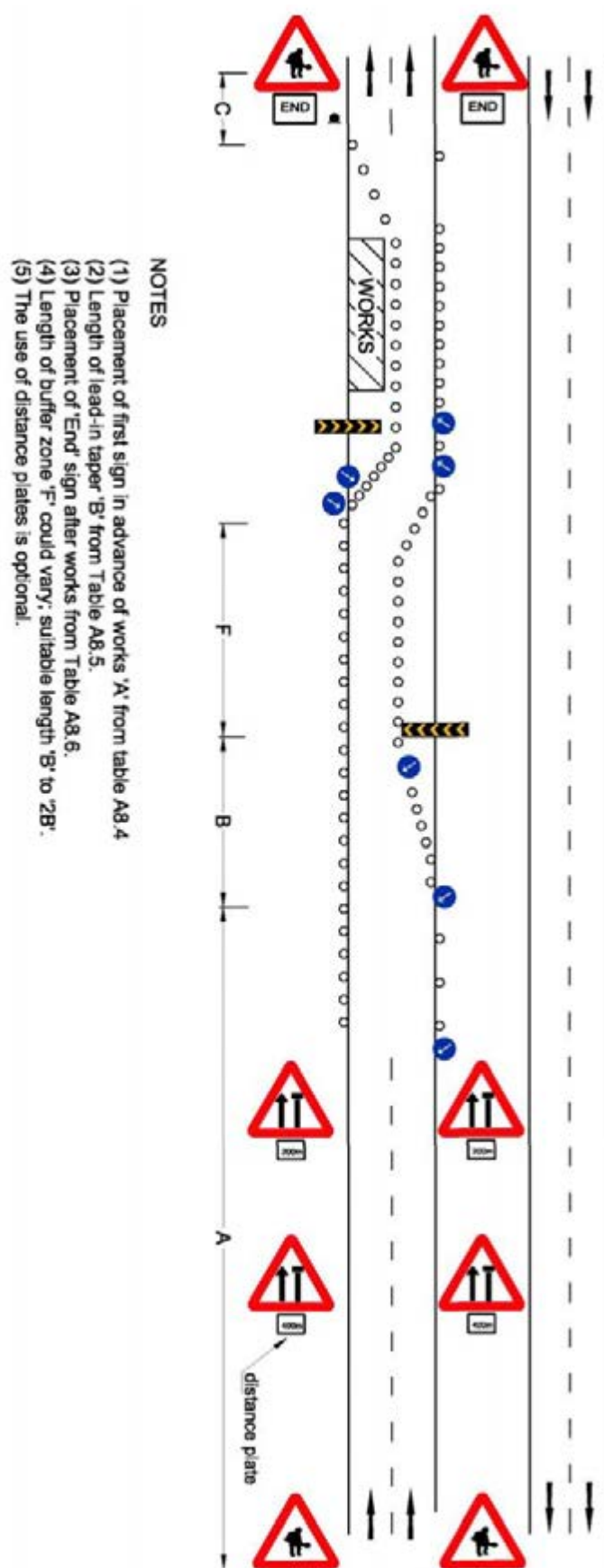
Appendices

Figure A4-20: Dual Carriageway, Works in Fast Lane



Note: Appropriate Speed Limit sign shall be placed in the work zone

Figure A4-21: Dual Carriageway, Works in Slow Lane



Note: Appropriate Speed Limit sign shall be placed in the work zone

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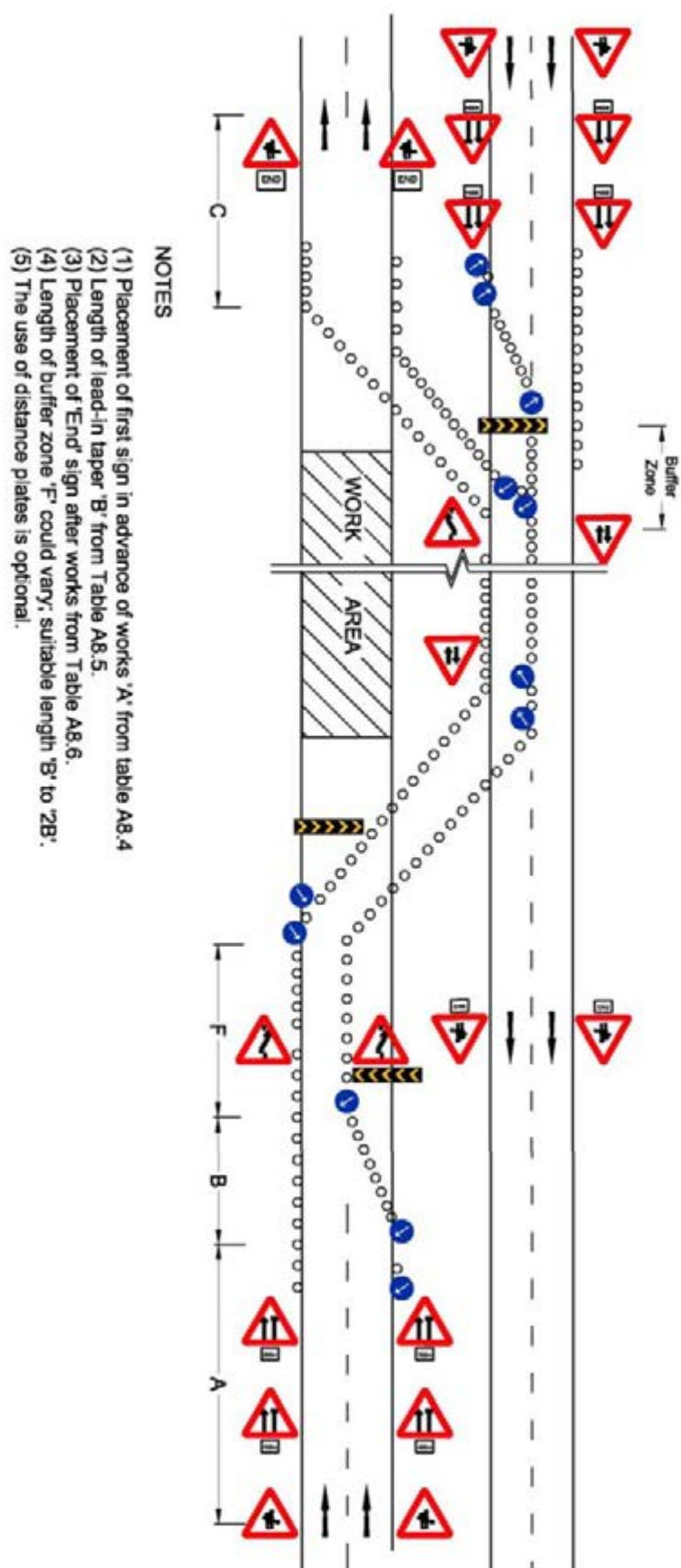
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Appendices

Figure A4-22: Dual Carriageway, One Carriageway Closed



Note: Appropriate Speed Limit sign shall be placed in the work zone

Appendix A5

Catalogue of Signs

Regulatory Signs (R)

Stop, Sign R1



GIVE WAY Sign R2



Regulatory Mandatory Signs (M)

Left Turn, Sign M1



Straight Ahead, Sign M2



Right Turn, Sign M3



Straight Ahead or Left Turn, Sign M4



Keep Left, Sign M5



Keep Right, Sign M6



Turn Left Ahead, Sign M7



Turn Right Ahead, Sign M8



Pass Either Side, Sign M9



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Appendices

Regulatory Prohibitory Signs (P)

No Entry, Sign P1



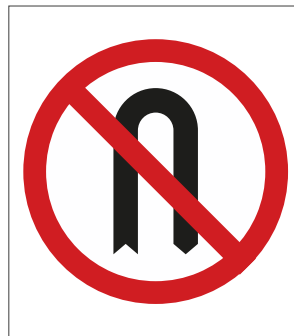
No Right Turn, Sign P2



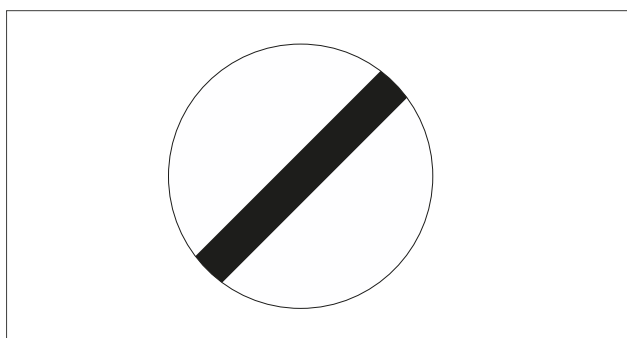
No Left Turn, Sign P3



No U Turn, Sign P4

Built Up Area, Speed Limit
50 km/h, Sign P5Special Area, Speed Limit
70 km/h, Sign P6Special Area, Speed Limit
80 km/h, Sign P7Expressway, Speed Limit
120km/h, Sign P8

End of Maximum Speed Limit, Sign P9



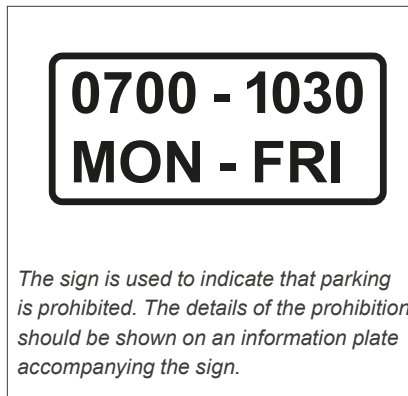
End of Minimum Speed Limit, Sign P10



Parking Permitted, Sign P11



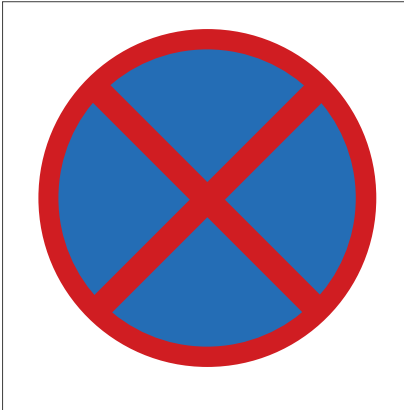
Restriction Plate, Sign P12



Parking Prohibited, Sign P13



No Stopping, Sign P14



Disc Parking Plate, Sign P15



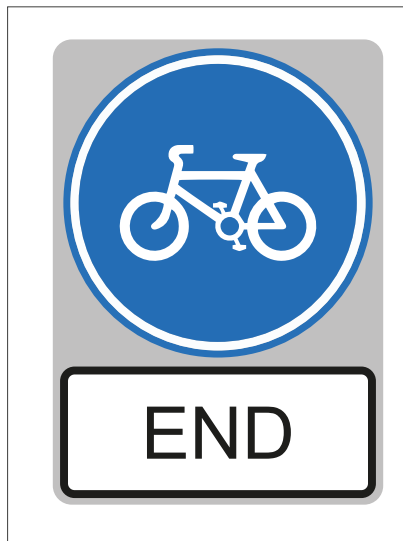
Taxi Stand, Sign P16



Commencement of Cycle Track, Sign P17



End of Cycle Track, Sign P18



Commencement of Cycleway, Sign P19



Shared Pedestrian/Cycle Track, Sign P20



School Warden's Stop Sign, Sign P21



Pedestrianised Street, Sign P22



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Appendices

Weight Restriction, Sign P23



Vehicle Weight Restriction, Sign P24



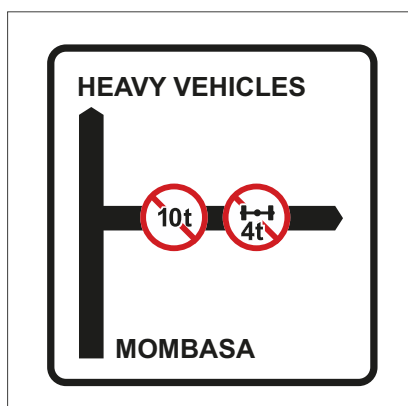
Axle Weight Restriction, Sign P25



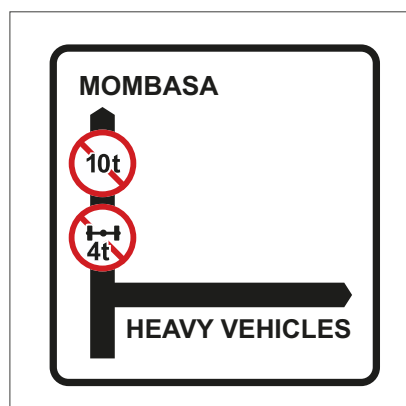
Advance Sign Indicating Weight Restriction Ahead, Sign P26



Indicating Route Subject to Restriction, Sign P27



Sign Indicating Route Subject to Restriction, Sign P28



Height Restriction, Sign P29



Parking Restriction for Heavy Vehicles, Sign P30



End of Restriction, Sign P31



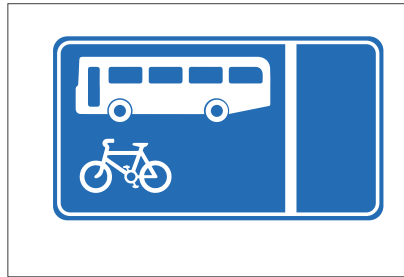
No Overtaking, Sign P32



Advance Sign for With Flow Bus Lane, Sign P33



With-Flow Bus Lane, Sign P34



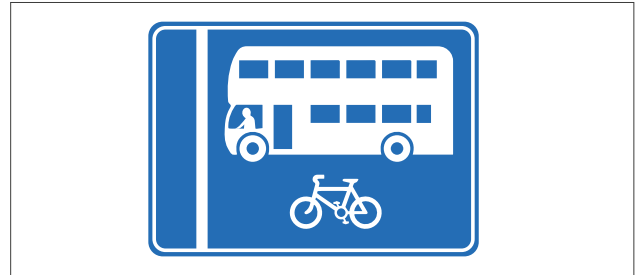
Information Plate, Sign P35



Advance Sign for Offside Bus Lane, Sign P36



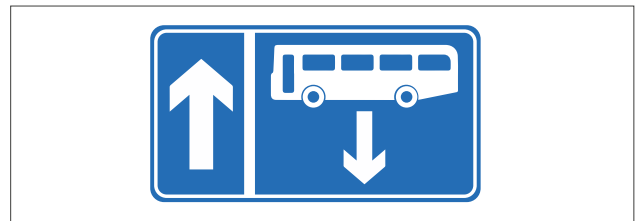
Offside Bus Lane, Sign P37



Information Plate, Sign P38



Contra Flow Bus Lane, Sign P39



Bus Lane, Sign P40



Bus Lane, Sign P41



Bus Lane, Look Right, Sign P42



Bus Lane Look Left, Sign P43



Buses Prohibited, Sign P44



Cars Prohibited, Sign P45



Trucks Prohibited, Sign P46



Tractors Prohibited, Sign P47



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Appendices

Motorcycles Prohibited, Sign P48



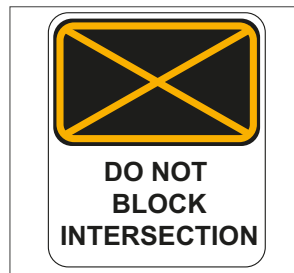
Cycles Prohibited, Sign P49



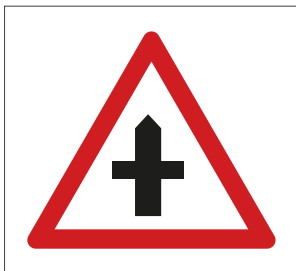
Horns Prohibited, Sign P50



Do Not Block Intersection, Sign P51

**Warning Signs (W)**

Crossroads, Sign W1



Side Road, Sign W2



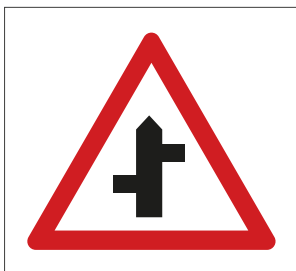
T Junction, Sign W3



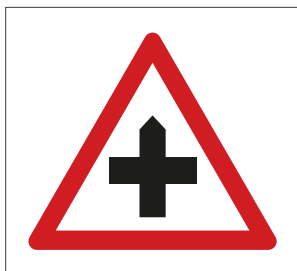
Y Junction, Sign W4



Staggered Crossroads, Sign W5



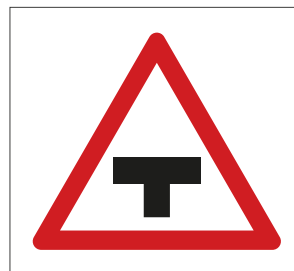
Crossroads, Sign W6



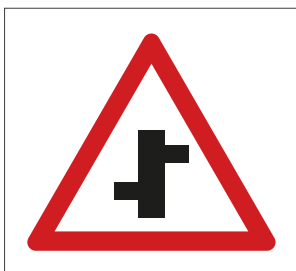
Side Road, Sign W7



T Junction, Sign W8



Staggered Crossroads, Sign W9



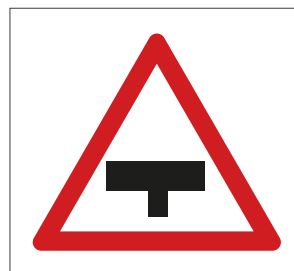
Y Junction, Sign W10



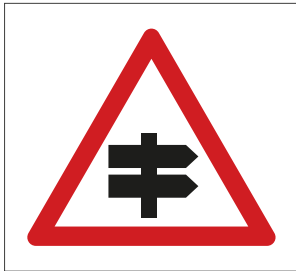
Crossroads, Sign W11



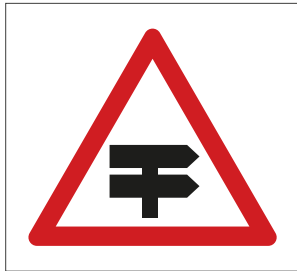
T Junction, Sign W12



Crossroads with Dual Carriageway, Sign W13



T – Junction with Dual Carriageway, Sign W14



Merging Traffic, Sign W15



Merging / Diverging Traffic, Sign W16



Roundabout Ahead,
Sign W17



Mini Roundabout Ahead,
Sign W18



Sharp Turn Ahead (*Left / Right*), Sign W19



Sharp Bend Ahead (*Left / Right*), Sign W20



Series of Sharp Curves Ahead, Sign W21



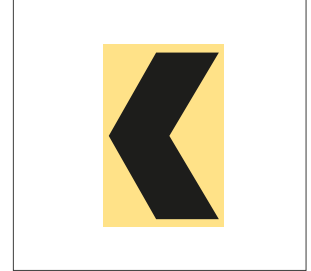
Series of Bends Ahead, Sign W22



Distance Plates, Sign W23



Single Chevron (Normal), Sign W24a



Single Chevron (Large), Sign W24b



Double Chevron, Sign W24c



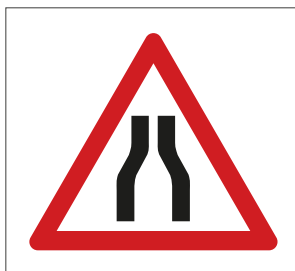
Triple Chevron, Sign W24d



Road Narrows on One Side, Sign W25



Road Narrows on Both Sides, Sign W26



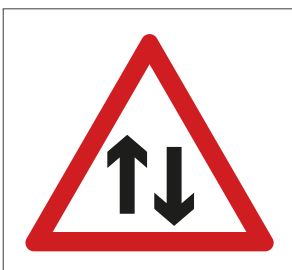
Road Divides, Sign W27



End of Divided Carriage-way, Sign W28



Two Way Traffic, Sign W29



Steep Descent Ahead, Sign W30



Steep Ascent Ahead, Sign W31



Restricted Headroom, Sign W32



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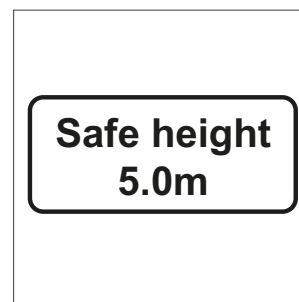
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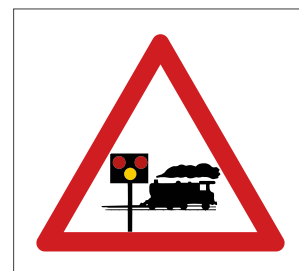
11

Appendices

Safe Height Plate, Sign **W35**



Level Crossing Ahead with Lights and Barriers, Sign W38



Sharp Dip Ahead, Sign **W42**



Traffic Signals Ahead,
Sign **W46**



Children Crossing (in residential areas),
Sign **W49**



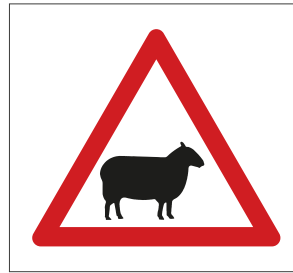
Accompanied Horses or Ponies, Sign **W50**



Cattle or Farm Animals, Sign **W51**



Sheep, Sign **W52**



Wild Animals, Sign **W53**



Crosswinds, Sign **W54**



Pedestrian Crossing Ahead, Sign **W55**



Tunnel Ahead, Sign **W56**



Falling Rocks, Sign **W57**



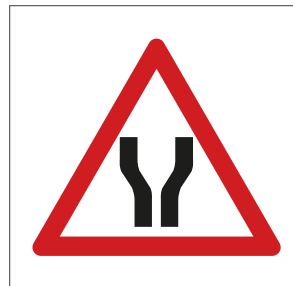
Low Flying Aircraft, Sign **W58**



Keep Left, Sign **W59**



Road Widens, Sign **W60**



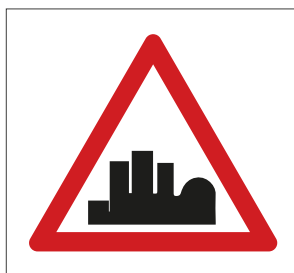
Narrow Bridge Ahead, Sign **W61**



Gap in the Median, Sign **W62**



Built Up Area, Sign **W63**



Speed Breaker, Sign **W64**



Runble Strip, Sign **W65**



Ferry, Sign **W66**



Queue Ahead, Sign **W67**



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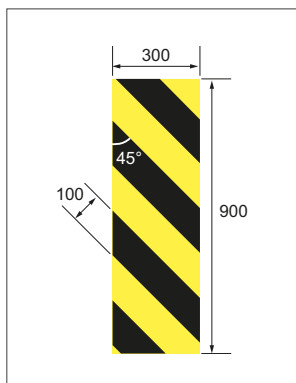
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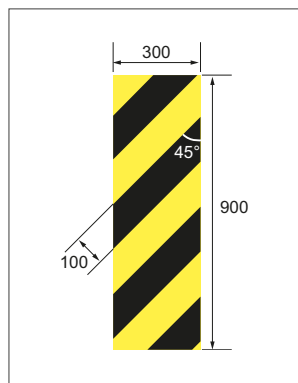
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Appendices

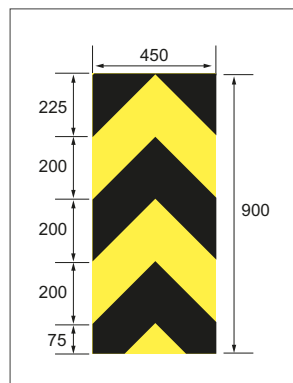
Object Hazard (Left), Sign W68



Object Hazard (Right), Sign W69



Two way Hazard Marker, Sign W70



Old/Disabled People Crossing, Sign W71

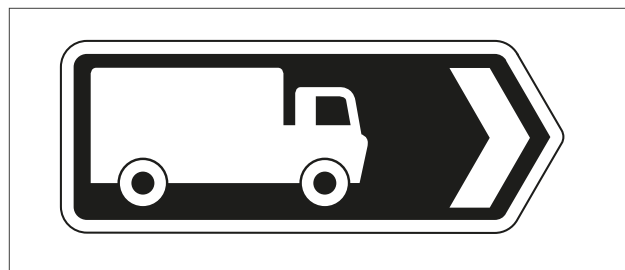


Other Information Signs (I)

Town or Village, Sign I7



Alternative Route for High Vehicle, Sign I8



Car Park with Facilities for Disabled, Sign I9



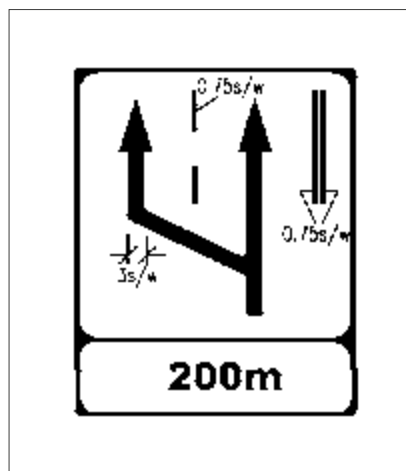
Tourist Facility, Sign I10



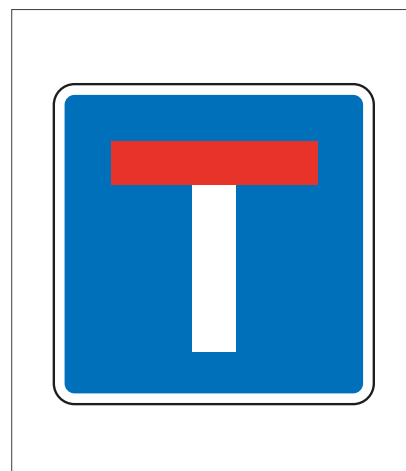
Escape Ramp, Sign E1



Slow Lane, Sign I12



No Through Way, Sign I13



Lay-by Ahead, Sign I14



Lay-by, Sign I15



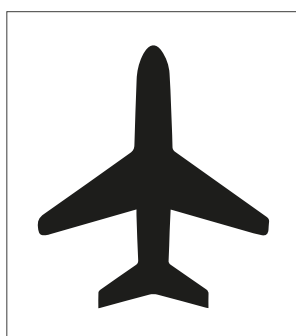
Hospital Ahead, Sign I18



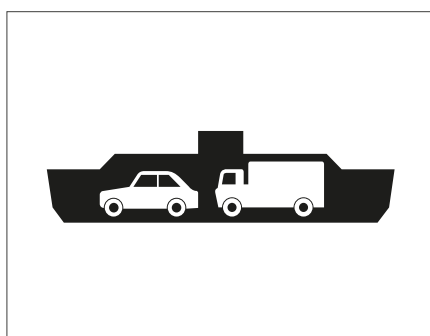
Hospital, Sign I19



Airport Symbol, Sign I20



Ferry Symbol, Sign I21



Industrial Estate Symbol, Sign I22



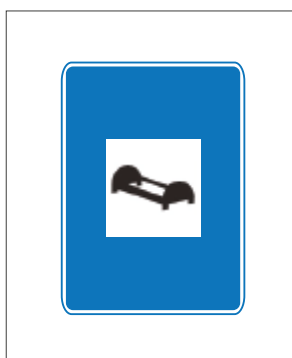
Eating Place, Sign I23



Light Refreshments, Sign I24



Resting Place, Sign I25



Toilet, Sign I26



Fuel Pump, Sign I27



Pedestrian Over Bridge, Sign I28



Pedestrian Subway, Sign I29



Picnic Site, Sign I30



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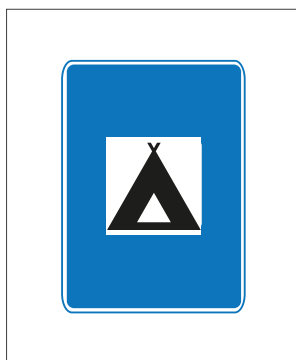
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11

Appendices

Camp Site, Sign I31



Escape Ramp Sign at Ramp, Sign I32



Direction of Ramp, Sign I33



International Symbol of Accessibility, Sign PD1



International Symbol of Accessibility, Sign PD2



Parking Areas, Sign PD3



Ramped Entrance, Sign PD4



Telephone Facilities, Sign PD5



Toilet Facilities, Sign PD6



Wayfinding, Sign PD7



Old People Crossing, Sign W71



Expressway Signs (IM)

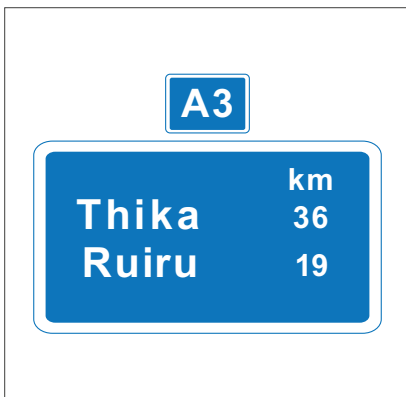
Advance Direction, Sign IM1



Direction Sign, Sign IM2



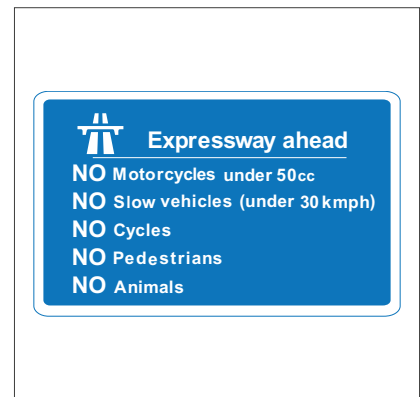
Route Confirmation, Sign IM3



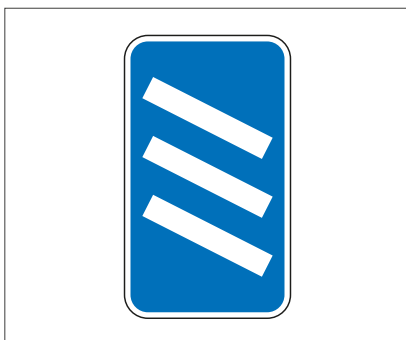
End of Expressway, Sign IM4



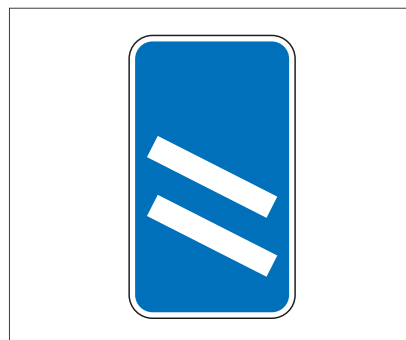
Expressway Ahead, Sign IM12



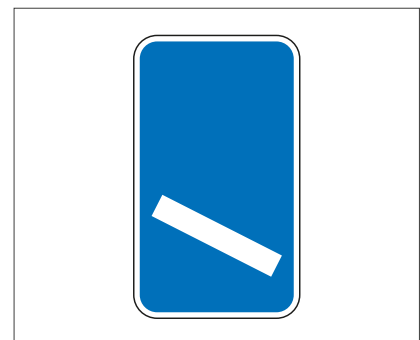
300 m Countdown Marker, Sign IM16



200 m Countdown Marker, Sign IM17



100 m Countdown Marker, Sign IM18



End of Expressway 1 km, Sign IM24



End of Expressway 500 m, Sign IM25



End of Expressway, Sign IM26



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11

Appendices

Roadworks Ahead, Sign **RW1**



Uneven Surface, Sign **RW2**



Slippery Road Ahead, Sign **RW3**



Road Narrows One Side, Sign **RW4**



Road Narrows, Sign **RW5**



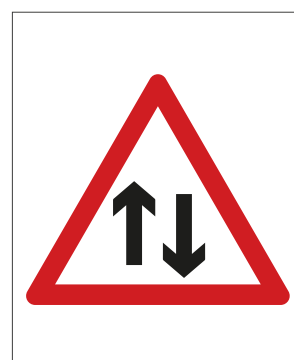
Manual Traffic Control Ahead, Sign **RW 6**



Traffic Signals Ahead, Sign **RW 7**



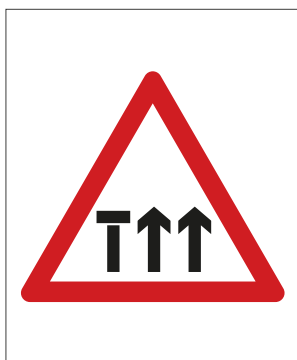
Two-Way Traffic Ahead, Sign **RW8**



Left Lane closed for two-lanes in one direction, Sign **RW 9**



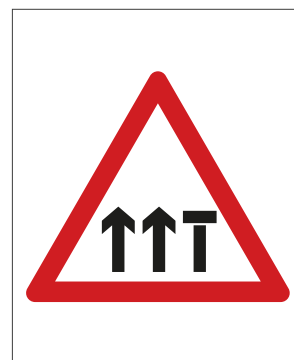
Left Lane closed for three-lanes in one direction, Sign **RW10**



Centre Lane closed for three-lanes in one direction Sign **RW11**



Right Lane closed for three-lanes in one direction Sign **RW12**



Left Lane closed two right lanes diverted, Sign **RW13**



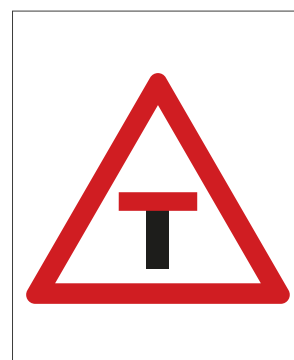
Left Through Centre Lane, Closed Right Lane Diverted, Sign **RW14**



Major Road Works Ahead, Sign **RW15**



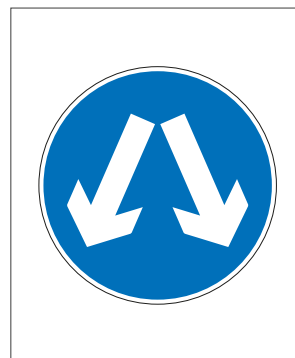
No Through Road, Sign **RW16**



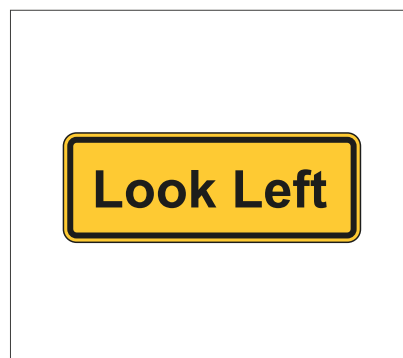
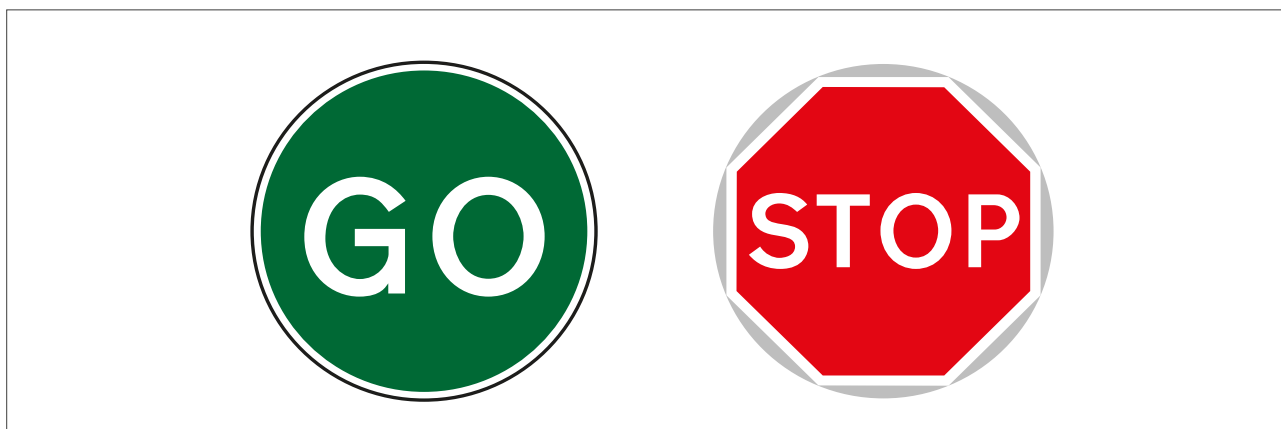
Road Closed, Sign **RW17**

Keep Left, Sign **RW21**

Keep Right, Sign **RW22**

Pass On Either Side, **RW23**

Crossing Not in Use, Sign **RW26**

Pedestrians Keep Right, Sign **RW27**

Pedestrians Look Left, Sign **RW28**

Stop/Go, Sign **RW29**

Diversion ('m') Ahead, Sign **RW30**

Diversion Right, Sign **RW31**

Diversion Left, Sign **RW32**


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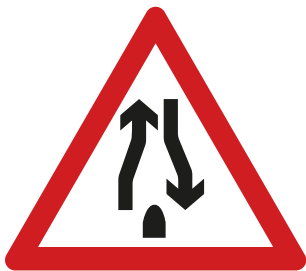
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11

Appendices

Diverted Traffic Straight Ahead, Sign **RW33**Diverted Traffic Right, Sign **RW34**End of Diversion, Sign **RW35**Crossover to Left, Sign **RW36**Crossover to Right, Sign **RW37**Divided Road Ahead, Sign **RW38**End of Divided Road, Sign **RW39**No Overtaking, Sign **RW40**End of Construction, Sign **RW41**

Appendix 6

Specifications & Standards

The specifications and standards for various parts of signs are given in [Table A6-1](#).

Table A6-1: Materials and Specifications

Part	Material	Specification / Standard
Plates & Support	Structural steel	KS 572
	Steel	KS EAS 134
	Galvanised Iron	KS 572
	Reinforced concrete	KS 1933
	Wooden post	KS 1608
Substrate / base material	Stainless Steel	ASTM A 240
	Carbon steel, Mild steel plates	ASTM A 516, grade 60 & 70
	Alloy steel plates	ASTM A387 Gr.2, 5, 9, 11, 12 & 22
	Aluminium	KS ISO 6362
	Aluminium Composite Materials (ACM)	ASTM D903, ASTM E638, ASTM C393
	Fibre Reinforced Plastic (FRP)	ISO 527
Retro reflective material	Retro reflective sheet	KS 814, ASTM D4956 Type XI
Fasteners	Nuts, bolts, washers	ASTM A 516, ASTM A193, ASTM A194
Support / footing	Concrete	KS 95 for M25
	Reinforcing steel	KS EAS 412-2

Standards referred in the manual are given in [Table A6-2](#).

Note: The year is not mentioned. The latest version is to be referred.

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Table A6-2: List of Standards / Specifications

Sl. No.	Standard	Name
1	ASTM A 240	Standard Specification for Chromium and Chromium-Nickel Stainless Steel Plate, Sheet and Strip for Pressure Vessels and General Applications
2	ASTM A 516	Standard Specification for Pressure Vessel Plates, Carbon Steel, for Moderate- and Lower-Temperature Service
3	ASTM A193	Alloy steel and stainless-steel bolting materials for high temperature or high-pressure service.
4	ASTM A194	Standard Specification for Carbon and Alloy Steel Nuts for Bolts for High Pressure or High Temperature Service
5	ASTM A387	Standard Specification for Pressure Vessel Plates, Alloy Steel, Chromium-Molybdenum plates.
6	ASTM C393	Standard Test Method for Core Shear Properties of Sandwich Constructions by Beam Flexure
7	ASTM D 256	Standard Test Methods for Determining the Izod Pendulum Impact Resistance of Plastics
8	ASTM D 790	Standard Test Methods for Flexural Properties of Unreinforced and Reinforced Plastics and Electrical Insulating Materials
9	ASTM D3039	Standard test method for determination of the tensile properties of fiber-reinforced composites
10	ASTM D4956	Standard Specification for Retroreflective Sheeting for Traffic Control
11	ASTM D903	Standard Test Method for Peel or Stripping Strength of Adhesive Bonds
12	ASTM E638	Tensile strength
13	ASTM E638	Standard Test Method for Calibration of Heat Transfer Rate Calorimeters Using a Narrow-Angle Blackbody Radiation Facility
14	ASTM E8	Standard Test Methods for Tension Testing of Metallic Materials
15	ASTM F436	Specification for Hardened Steel Washers, carbon steel, steel, washers, weathering steel, Hardened steel washers
16	BS 381	Colour Charts and Munsell Colour System
17	BS 381C	Colour Chart
18	CIE 1931	International Commission on Illumination
19	EN 12899	Fixed, vertical road traffic signs - Part 1: Fixed signs
20	ISO 527	Plastics -Determination of tensile properties
21	KS 1608	Specification for hardwood poles, droppers, laths, guardrail post and spacer blocks
22	KS 1933	Concrete poles for telephone, power and lighting purposes-Specification.
23	KS 412-2	Steel for the reinforcement of concrete-Part 2: Ribbed bars
24	KS 572	Specification for hot-rolled structural steel sections
25	KS 814	Specification for retroreflective sheeting materials for traffic signs
26	KS 95 for M25	Specification for natural aggregates for use in concrete
27	KS EAS 134	Standard specifies the dimensions and sectional properties of cold rolled steel sections
28	KS EAS 412-2	Steel for the reinforcement of concrete-Part 2: Ribbed bars
29	KS ES 572	Specification for hot-rolled structural steel sections
30	KS ISO 6362	Wrought aluminium and aluminium alloys — Extruded rods/bars, tubes and profiles

1
2
3
4
5
6
7
8
9
10
11
Appendices

1

2

3

4

5

6

7

8

9

10

11

Appendices

1
2
3
4
5
6
7
8
9
10
11
Appendices

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State Department for Roads
Ministry of Roads and Transport
Transcom House, Ngong Road
P.O. Box 30260 – 00100
NAIROBI

State Department for Roads
Ministry of Roads and Transport
Works Building
Ngong Road
P.O. Box 30260 – 00100
NAIROBI